

DRD_2025_Lesson_1

CHUNK 1: Let's start

Let's print version information about R:

```
sessionInfo()

## R version 4.4.0 (2024-04-24)

## Platform: aarch64-apple-darwin20
## Running under: macOS 15.4.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.0
##
## locale:
##  [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Rome
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## loaded via a namespace (and not attached):
##  [1] digest_0.6.37      R6_2.6.1           fastmap_1.2.0       xfun_0.51
##  [5] cachem_1.1.0       knitr_1.50          htmltools_0.5.8.1
rmarkdown_2.29
##  [9] lifecycle_1.0.4    cli_3.6.4           sass_0.4.9
jquerylib_0.1.4
## [13] compiler_4.4.0     rstudioapi_0.17.1  tools_4.4.0
evaluate_1.0.3
## [17] bslib_0.9.0        yaml_2.3.10         rlang_1.1.5
jsonlite_1.9.1
```

To check the objects that are present in your workspace:

```
ls()

## character(0)
```

It is good practice to remove all the variables from the workspace (unless you need them). You can do this in 2 ways: 1) Go to Workspace → Erase workspace or 2) digit

```
rm(list=ls())
```

To know the current working directory

```
getwd()

## [1] "/Users/francesco/Library/CloudStorage/Dropbox/DRD_2025/1"
```

To set the new working directory

```
setwd("~/Dropbox/DRD_2025/1")
```

```
getwd()
```

```
## [1] "/Users/francesco/Library/CloudStorage/Dropbox/DRD_2025/1"
```

CHUNK 2: Vectors

Let's create a vector named V which contains the integer number 1

```
V1 <- 1
```

```
V1
```

```
## [1] 1
```

Let's create a vector named V1 which contains the integer numbers from 1 to 5

```
V1 <- c(1,2,3,4,5)
```

```
V1
```

```
## [1] 1 2 3 4 5
```

or you can use

```
V1 <- c(1:5)
```

```
V1
```

```
## [1] 1 2 3 4 5
```

Another example

```
V1 <- c(1:500)
```

V1

```
# #      [1]     1     2     3     4     5     6     7     8     9    10    11    12    13    14    15  
16   17   18
```

[illegible][illegible][illegible][illegible]

```
##      [91]  91  92  93  94  95  96  97  98  99 100 101 102 103 104 105
106 107 108
```

```
## [109] 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123
124 125 126
```

```
## [127] 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141
142 143 144
```

```
## [145] 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159
160 161 162
```

```
## [163] 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177
178 179 180
```

```
## [181] 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195
196 197 198
```

```
## [199] 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213
214 215 216
```

```
## [217] 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231
232 233 234
```

```
## [235] 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249
250 251 252
```

```
## [253] 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267
268 269 270
```

```
## [271] 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285
286 287 288
```

```
## [289] 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303
304 305 306
```

```
## [307] 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321
322 323 324
```

```
## [325] 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339
340 341 342
```

```
## [343] 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357
358 359 360
```

```
## [361] 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375
376 377 378
## [379] 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393
394 395 396
## [397] 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411
412 413 414
## [415] 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429
430 431 432
## [433] 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447
448 449 450
## [451] 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465
466 467 468
## [469] 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483
484 485 486
## [487] 487 488 489 490 491 492 493 494 495 496 497 498 499 500
```

Remember: you cannot create an object called “1”

```
1 <- c(1,2,3,4,5)

## Error in 1 <- c(1, 2, 3, 4, 5): invalid (do_set) left-hand side to
assignment
```

Let's explore the V1 object

```
length(V1)
## [1] 500

class(V1)
## [1] "integer"

str(V1)
## int [1:500] 1 2 3 4 5 6 7 8 9 10 ...

summary(V1)
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.0   125.8   250.5   250.5   375.2   500.0
```

```
head(V1)
## [1] 1 2 3 4 5 6

table(V1)
## V1
```

```
## 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
18 19 20
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37
38 39 40
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77
78 79 80
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97
98 99 100
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117
118 119 120
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137
138 139 140
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157
158 159 160
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177
178 179 180
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197
198 199 200
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217
218 219 220
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237
238 239 240
```

```
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257
258 259 260
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277
278 279 280
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297
298 299 300
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317
318 319 320
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337
338 339 340
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357
358 359 360
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377
378 379 380
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397
398 399 400
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417
418 419 420
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437
438 439 440
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457
458 459 460
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
```

```
## 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477
478 479 480
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
## 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497
498 499 500
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
```

Now we want to create a vector named V2 which contains decimal numbers from 0 to 2. We will use the function “seq”

```
?seq
```

```
V2 <- seq(from=0,to=10,by=0.1)
V2
```

```
## [1] 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0 1.1
1.2 1.3 1.4
## [16] 1.5 1.6 1.7 1.8 1.9 2.0 2.1 2.2 2.3 2.4 2.5 2.6
2.7 2.8 2.9
## [31] 3.0 3.1 3.2 3.3 3.4 3.5 3.6 3.7 3.8 3.9 4.0 4.1
4.2 4.3 4.4
## [46] 4.5 4.6 4.7 4.8 4.9 5.0 5.1 5.2 5.3 5.4 5.5 5.6
5.7 5.8 5.9
## [61] 6.0 6.1 6.2 6.3 6.4 6.5 6.6 6.7 6.8 6.9 7.0 7.1
7.2 7.3 7.4
## [76] 7.5 7.6 7.7 7.8 7.9 8.0 8.1 8.2 8.3 8.4 8.5 8.6
8.7 8.8 8.9
## [91] 9.0 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 10.0
```

```
length(V2)
```

```
## [1] 101
```

```
class(V2)
```

```
## [1] "numeric"
```

```
str(V2)
```

```
## num [1:101] 0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 ...
```

```
summary(V2)
```

```
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0 2.5 5.0 5.0 7.5 10.0
```

```
head(V2)
```

```
## [1] 0.0 0.1 0.2 0.3 0.4 0.5
```

```
table(V2)
```

```
## V2
```

```
## 0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1 1.1 1.2 1.3 1.4 1.5 1.6  
1.7 1.8 1.9
```

```
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
1 1 1
```

```
## 2 2.1 2.2 2.3 2.4 2.5 2.6 2.7 2.8 2.9 3 3.1 3.2 3.3 3.4 3.5 3.6  
3.7 3.8 3.9
```

```
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
1 1 1
```

```
## 4 4.1 4.2 4.3 4.4 4.5 4.6 4.7 4.8 4.9 5 5.1 5.2 5.3 5.4 5.5 5.6  
5.7 5.8 5.9
```

```
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
1 1 1
```

```
## 6 6.1 6.2 6.3 6.4 6.5 6.6 6.7 6.8 6.9 7 7.1 7.2 7.3 7.4 7.5 7.6  
7.7 7.8 7.9
```

```
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
1 1 1
```

```
## 8 8.1 8.2 8.3 8.4 8.5 8.6 8.7 8.8 8.9 9 9.1 9.2 9.3 9.4 9.5 9.6  
9.7 9.8 9.9
```

```
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
1 1 1
```

```
## 10
```

```
## 1
```

Now we want to create a vector named V3 which contains the names of the numbers from “one” to “ten”.

```
V3 <-
```

```
c("one", "two", "three", "four", "five", "six", "seven", "eight", "nine", "ten")
```

```
V3
```

```
## [1] "one" "two" "three" "four" "five" "six" "seven" "eight"  
"nine"
```

```
## [10] "ten"
```

```
length(V3)
```

```
## [1] 10
```

```
class(V3)
```



```
## [1] "character"

str(V3)

## chr [1:10] "one" "two" "three" "four" "five" "six" "seven" "eight"
## "nine" ...

summary(V3)

##      Length      Class      Mode
##      10 character character
```

```
head(V3)

## [1] "one"  "two"  "three" "four"  "five"  "six"

table(V3)

## V3

## eight  five  four  nine   one seven  six   ten three  two
##      1     1     1     1     1     1     1     1     1     1
```

What does happen if we do not use the quotes?

```
V3 <- c(one,two,three)

## Error: object 'one' not found
```

Let's try in this way: create 3 objects and concatenate them:

```
one <- 1

two <- 2
three <- 3
V3 <- c(one,two,three)
V3
```

```
## [1] 1 2 3

length(V3)

## [1] 3

class(V3)

## [1] "numeric"

str(V3)

## num [1:3] 1 2 3

summary(V3)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	1.0	1.5	2.0	2.0	2.5	3.0

```
head(V3)
```

```
## [1] 1 2 3
```

```
table(V3)
```

```
## V3
```

```
## 1 2 3
```

```
## 1 1 1
```

Now we play a bit with datatypes. First of all, we create a vector of characters, named V4

```
V4 <- c("one", "two", "three", "one", "one", "two")
```

```
V4
```

```
## [1] "one" "two" "three" "one" "one" "two"
```

```
length(V4)
```

```
## [1] 6
```

```
class(V4)
```

```
## [1] "character"
```

```
str(V4)
```

```
## chr [1:6] "one" "two" "three" "one" "one" "two"
```

```
summary(V4)
```

```
## Length Class Mode
```

```
## 6 character character
```

```
head(V4)
```

```
## [1] "one" "two" "three" "one" "one" "two"
```

```
table(V4)
```

```
## V4
```

```
## one three two
```

```
## 3 1 2
```

An now we create a vector named V5 by converting V4 from datatype=character to datatype=factor

```
V4
```

```
## [1] "one" "two" "three" "one" "one" "two"
```

```
V5 <- factor(V4)
```

```
V5
```

```
## [1] one two three one one two
```

```
## Levels: one three two
```

Note that the levels of the factor are ordered according to the alphabetical order. We can reorder the levels:

```
V5 <- factor(V5, levels=c("one","two","three"))
```

```
length(V5)
```

```
## [1] 6
```

```
class(V5)
```

```
## [1] "factor"
```

```
str(V5)
```

```
## Factor w/ 3 levels "one","two","three": 1 2 3 1 1 2
```

```
summary(V5)
```

```
## one two three
```

```
## 3 2 1
```

```
head(V5)
```

```
## [1] one two three one one two
```

```
## Levels: one two three
```

```
table(V5)
```

```
## V5
```

```
## one two three
```

```
## 3 2 1
```

```
levels(V5)
```

```
## [1] "one" "two" "three"
```

```
length(levels(V5))
```

```
## [1] 3
nlevels(V5)
## [1] 3
```

We can go from factors to numbers...

```
V5
## [1] one two three one one two
## Levels: one two three
```

```
V6 <- as.numeric(V5)
```

```
V6
```

```
## [1] 1 2 3 1 1 2
V5
## [1] one two three one one two
## Levels: one two three
```

...or from factors to characters

```
V5
## [1] one two three one one two
## Levels: one two three
```

```
V7 <- as.character(V5)
```

```
V7
```

```
## [1] "one" "two" "three" "one" "one" "two"
V4
## [1] "one" "two" "three" "one" "one" "two"
```

Indexing a vector

Let's go back to our object V1, which is a vector including integers from 1 to 500

```
V1
```

[illegible]

```
## [397] 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411
412 413 414
## [415] 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429
430 431 432
## [433] 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447
448 449 450
## [451] 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465
466 467 468
## [469] 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483
484 485 486
## [487] 487 488 489 490 491 492 493 494 495 496 497 498 499 500
```

Now we want to consider only the 20th element of this vector

```
V1[20]
## [1] 20
```

I can index a vector of any datatype:

```
V2
## [1] 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0 1.1
1.2 1.3 1.4
## [16] 1.5 1.6 1.7 1.8 1.9 2.0 2.1 2.2 2.3 2.4 2.5 2.6
2.7 2.8 2.9
## [31] 3.0 3.1 3.2 3.3 3.4 3.5 3.6 3.7 3.8 3.9 4.0 4.1
4.2 4.3 4.4
## [46] 4.5 4.6 4.7 4.8 4.9 5.0 5.1 5.2 5.3 5.4 5.5 5.6
5.7 5.8 5.9
## [61] 6.0 6.1 6.2 6.3 6.4 6.5 6.6 6.7 6.8 6.9 7.0 7.1
7.2 7.3 7.4
## [76] 7.5 7.6 7.7 7.8 7.9 8.0 8.1 8.2 8.3 8.4 8.5 8.6
8.7 8.8 8.9
## [91] 9.0 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 10.0
```

```
V2[30]
## [1] 2.9
```

```
V5
## [1] one two three one one two
```

```
## Levels: one two three
```

```
V5[6]
```

```
## [1] two
```

```
## Levels: one two three
```

```
V5[20]
```

```
## [1] <NA>
```

```
## Levels: one two three
```

CHUNK 3: Operators

I want to subset V1 in order to retain only the elements that satisfy the condition “higher that 31”

```
V1
```

```
## [1] 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15  
16 17 18
```

```
## [19] 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33  
34 35 36
```

```
## [37] 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51  
52 53 54
```

```
## [55] 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69  
70 71 72
```

```
## [73] 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87  
88 89 90
```

```
## [91] 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105  
106 107 108
```

```
## [109] 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123  
124 125 126
```

```
## [127] 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141  
142 143 144
```

```
## [145] 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159  
160 161 162
```

```
## [163] 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177  
178 179 180
```

```
## [181] 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195  
196 197 198
```

```
## [199] 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213  
214 215 216
```

```
## [217] 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231  
232 233 234
```

```
## [235] 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249  
250 251 252
```

```
## [253] 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267  
268 269 270
```

```
## [271] 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285
286 287 288
## [289] 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303
304 305 306
## [307] 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321
322 323 324
## [325] 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339
340 341 342
## [343] 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357
358 359 360
## [361] 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375
376 377 378
## [379] 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393
394 395 396
## [397] 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411
412 413 414
## [415] 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429
430 431 432
## [433] 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447
448 449 450
## [451] 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465
466 467 468
## [469] 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483
484 485 486
## [487] 487 488 489 490 491 492 493 494 495 496 497 498 499 500
```

```
V1_>31 <- V1[V1>31]

## Error: object 'V1_' not found
```

Ops...I have tried to create an object with a not correct name. Let's try again:

```
V1_greater31 <- V1[V1>31]
```

```
V1_greater31
```

```
##      [1]  32  33  34  35  36  37  38  39  40  41  42  43  44  45  46
47  48  49
```

```
##    [19]  50  51  52  53  54  55  56  57  58  59  60  61  62  63  64
65  66  67
```

```
##    [37]  68  69  70  71  72  73  74  75  76  77  78  79  80  81  82
83  84  85
```

```
##    [55]  86  87  88  89  90  91  92  93  94  95  96  97  98  99 100
101 102 103
```

```
##    [73] 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118
119 120 121
```


[91] 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136
137 138 139
[109] 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154
155 156 157
[127] 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172
173 174 175
[145] 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190
191 192 193
[163] 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208
209 210 211
[181] 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226
227 228 229
[199] 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244
245 246 247
[217] 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262
263 264 265
[235] 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280
281 282 283
[253] 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298
299 300 301
[271] 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316
317 318 319
[289] 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334
335 336 337
[307] 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352
353 354 355
[325] 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370
371 372 373
[343] 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388
389 390 391
[361] 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406
407 408 409
[379] 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424
425 426 427
[397] 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442
443 444 445
[415] 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460
461 462 463
[433] 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478
479 480 481
[451] 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496
497 498 499
[469] 500

```
summary(V1_greater31)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	32	149	266	266	383	500

Now we want to check if two vectors, V8 and V9, are identical:

```
V8 <- c(1:10)
```

```
V9 <- c(1:10)
```

```
V8==V9
```

```
## [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

I can store the result of the operation in a new object, V10

```
V10 <- V8==V9
```

```
V10
```

```
## [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

```
str(V10)
```

```
## logi [1:10] TRUE TRUE TRUE TRUE TRUE TRUE TRUE ...
```

```
summary(V10)
```

```
## Mode TRUE
```

```
## logical 10
```

```
table(V10)
```

```
## V10
```

```
## TRUE
```

```
## 10
```

Let's compare two character vectors:

```
V11 <- c("A", "B", "D", "F", "L")
```

```
V12 <- c("Q", "B", "D", "A", "L")
```

```
V11==V12
```

```
## [1] FALSE TRUE TRUE FALSE TRUE
```

```
V13 <- V11==V12
```

```
V13
```

```
## [1] FALSE TRUE TRUE FALSE TRUE
```

```
table(V13)
```

```
## V13
```

```
## FALSE TRUE
```

```
##      2    3
```

Note that if the two vectors do not have the same length, we achieve a warning message:

```
V11 <- c("A", "B", "D", "F", "L")
```

```
V12_long <- c("Q", "B", "D", "A", "L", "Z")
```

```
V11==V12_long
```

```
## Warning in V11 == V12_long: longer object length is not a multiple  
of shorter
```

```
## object length
```

```
## [1] FALSE TRUE TRUE FALSE TRUE FALSE
```

Now we want to know if two vectors are different:

```
V11!=V12
```

```
## [1] TRUE FALSE FALSE TRUE FALSE
```

```
V14 <- V11!=V12
```

```
table(V14)
```

```
## V14
```

```
## FALSE TRUE
```

```
##      3    2
```

And now we want to subset the vector V11 on the basis of the condition that V11 is equal to V12. Note that the condition is checked for each position of the two vectors

```
V11
```

```
## [1] "A" "B" "D" "F" "L"
```

```
V12
```

```
## [1] "Q" "B" "D" "A" "L"
```

```
V15 <- V11[V11==V12]
```

```
V15
```

```
## [1] "B" "D" "L"
```

Now we want to know the elements of V11 which are included in V12, independently of their position

```
V11 %in% V12
```

```
## [1] TRUE TRUE TRUE FALSE TRUE
```

```
V15 <- V11[V11 %in% V12]
```

```
V15
```

```
## [1] "A" "B" "D" "L"
```

Now we want to know the elements of V11 which are not included in V12, independently of their position

```
V15 <- V11[V11 !%in% V12]
```

```
## Error in parse(text = input): <text>:1:16: unexpected '!'
```

```
## 1: V15 <- V11[V11 !  
##                               ^
```

Ops, something is wrong! Let's try in this way:

```
V15 <- V11[!V11 %in% V12]
```

CHUNK 4: Dataframes

We use the function “read.table” to load a small dataframe from a txt file:

```
SampleSheet <-  
read.table("~/Dropbox/DRD_2025/1/SampleSheet.txt", sep="\t", header=T)
```

```
SampleSheet
```

```
##   Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive  
CD8T
```

```
## 1           A      W      1.867548           1.724855  239.4476  523.9770  
0.05597054  
## 2           B      C      1.841516           2.663802  265.9878  536.2501  
0.13529055  
## 3           C      W      1.335753          -2.658338  320.4341  671.8478  
0.18487413
```

```
## 4          D      C      1.983669          3.613387  308.4593  521.9668
0.03500575
## 5          E      W      2.022408          -4.468693  279.8390  554.8088
0.04006602
## 6          F      C      1.982639          -2.470729  328.3630  655.6088
0.01607650
##          CD4T          NK          Bcell          Mono          Gran Sex Age
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072  0  51
## 2 0.1265136 0.038188822 0.08120917 0.130451761 0.5077724  0  44
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582  1  53
## 4 0.0911106 0.033064802 0.02328309 0.137029687 0.6811102  1  53
## 5 0.2089572 0.022393107 0.05841506 0.054481111 0.6208538  0  44
## 6 0.2114697 0.000000000 0.04797884 0.067280646 0.6727830  0  52
```

```
class(SampleSheet)
```

```
## [1] "data.frame"
```

```
dim(SampleSheet)
```

```
## [1]  6 14
```

```
length(SampleSheet)
```

```
## [1] 14
```

```
str(SampleSheet)
```

```
## 'data.frame':  6 obs. of  14 variables:
```

```
## $ Sample_Name      : chr  "A" "B" "C" "D" ...
## $ Group            : chr  "W" "C" "W" "C" ...
## $ PlasmaBlast      : num  1.87 1.84 1.34 1.98 2.02 ...
## $ CD8pCD28nCD45RAn: num  1.72 2.66 -2.66 3.61 -4.47 ...
## $ CD8.naive         : num  239 266 320 308 280 ...
## $ CD4.naive         : num  524 536 672 522 555 ...
## $ CD8T              : num  0.056 0.1353 0.1849 0.035 0.0401 ...
## $ CD4T              : num  0.1908 0.1265 0.3821 0.0911 0.209 ...
## $ NK               : num  0.03318 0.03819 0.00617 0.03306 0.02239
...
## $ Bcell            : num  0.0192 0.0812 0.0569 0.0233 0.0584 ...
## $ Mono             : num  0.05506 0.13045 0.00693 0.13703 0.05448
...
## $ Gran             : num  0.656 0.508 0.376 0.681 0.621 ...
## $ Sex              : int   0 0 1 1 0 0
## $ Age              : int   51 44 53 53 44 52
```

```
head(SampleSheet)
```

##	Sample_Name	Group	PlasmaBlast	CD8pCD28nCD45RAn	CD8.naive	CD4.naive
----	-------------	-------	-------------	------------------	-----------	-----------

## 1	A	W	1.867548	1.724855	239.4476	523.9770
0.05597054						
## 2	B	C	1.841516	2.663802	265.9878	536.2501
0.13529055						
## 3	C	W	1.335753	-2.658338	320.4341	671.8478
0.18487413						
## 4	D	C	1.983669	3.613387	308.4593	521.9668
0.03500575						
## 5	E	W	2.022408	-4.468693	279.8390	554.8088
0.04006602						
## 6	F	C	1.982639	-2.470729	328.3630	655.6088
0.01607650						

##	CD4T	NK	Bcell	Mono	Gran	Sex	Age
## 1	0.1908321	0.033184511	0.01915430	0.055055601	0.6564072	0	51
## 2	0.1265136	0.038188822	0.08120917	0.130451761	0.5077724	0	44
## 3	0.3820950	0.006167623	0.05693339	0.006932121	0.3764582	1	53
## 4	0.0911106	0.033064802	0.02328309	0.137029687	0.6811102	1	53
## 5	0.2089572	0.022393107	0.05841506	0.054481111	0.6208538	0	44
## 6	0.2114697	0.000000000	0.04797884	0.067280646	0.6727830	0	52

summary(SampleSheet)

##	Sample_Name	Group	PlasmaBlast
----	-------------	-------	-------------

##	Length:6	Length:6	Min. :1.336	Min. :-
	4.4687			
##	Class :character	Class :character	1st Qu.:1.848	1st Qu.:-
	2.6114			
##	Mode :character	Mode :character	Median :1.925	Median :-
	0.3729			
##			Mean :1.839	Mean :-
	0.2660			
##			3rd Qu.:1.983	3rd Qu.:
	2.4291			
##			Max. :2.022	Max. :
	3.6134			

##	CD8.naive	CD4.naive	CD8T	CD4T
##	Min. :239.4	Min. :522.0	Min. :0.01608	Min. :0.09111
##	1st Qu.:269.5	1st Qu.:527.0	1st Qu.:0.03627	1st Qu.:0.14259
##	Median :294.1	Median :545.5	Median :0.04802	Median :0.19989
##	Mean :290.4	Mean :577.4	Mean :0.07788	Mean :0.20183
##	3rd Qu.:317.4	3rd Qu.:630.4	3rd Qu.:0.11546	3rd Qu.:0.21084
##	Max. :328.4	Max. :671.8	Max. :0.18487	Max. :0.38210

```
##           NK           Bcell           Mono           Gran
## Min.      :0.00000   Min.      :0.01915   Min.      :0.006932   Min.
:0.3765
## 1st Qu.:0.01022   1st Qu.:0.02946   1st Qu.:0.054625   1st
Qu.:0.5360
## Median :0.02773   Median :0.05246   Median :0.061168   Median
:0.6386
## Mean    :0.02217   Mean    :0.04783   Mean    :0.075205   Mean
:0.5859
## 3rd Qu.:0.03315   3rd Qu.:0.05804   3rd Qu.:0.114659   3rd
Qu.:0.6687
## Max.    :0.03819   Max.    :0.08121   Max.    :0.137030   Max.
:0.6811
##           Sex           Age
## Min.      :0.0000   Min.      :44.00
## 1st Qu.:0.0000   1st Qu.:45.75
## Median :0.0000   Median :51.50
## Mean    :0.3333   Mean    :49.50
## 3rd Qu.:0.7500   3rd Qu.:52.75
## Max.    :1.0000   Max.    :53.00
```

Important note: - BEFORE R4.0.0, when using read.table() or data.frame(), R converted strings to factors by default. - STARTING FROM R4.0.0, when using read.table() or data.frame(), R no longer converts strings to factors by default. <https://cran.r-project.org/doc/manuals/r-devel/NEWS.html> <https://developer.r-project.org/Blog/public/2022/02/16/stringsasfactors/> We can set stringsAsFactors = TRUE to convert strings to factors when reading a dataframe

```
SampleSheet2 <-
read.table("~/Dropbox/DRD_2025/1/SampleSheet.txt", sep="\t", header=T,
stringsAsFactors = T)
```

```
str(SampleSheet2)
```

```
## 'data.frame':      6 obs. of  14 variables:
## $ Sample_Name      : Factor w/ 6 levels "A","B","C","D",...: 1 2 3 4
5 6
## $ Group            : Factor w/ 2 levels "C","W": 2 1 2 1 2 1
## $ PlasmaBlast      : num  1.87 1.84 1.34 1.98 2.02 ...
## $ CD8pCD28nCD45RAn: num  1.72 2.66 -2.66 3.61 -4.47 ...
## $ CD8.naive         : num  239 266 320 308 280 ...
## $ CD4.naive         : num  524 536 672 522 555 ...
## $ CD8T             : num  0.056 0.1353 0.1849 0.035 0.0401 ...
## $ CD4T             : num  0.1908 0.1265 0.3821 0.0911 0.209 ...
## $ NK               : num  0.03318 0.03819 0.00617 0.03306 0.02239
...
## $ Bcell            : num  0.0192 0.0812 0.0569 0.0233 0.0584 ...
```

```
## $ Mono          : num  0.05506 0.13045 0.00693 0.13703 0.05448
...
## $ Gran          : num  0.656 0.508 0.376 0.681 0.621 ...
## $ Sex           : int   0 0 1 1 0 0
## $ Age           : int   51 44 53 53 44 52
```

Let's check again the str of SampleSheet...we can specifically convert the column "Sex" to factor:

```
str(SampleSheet)

## 'data.frame':    6 obs. of  14 variables:

## $ Sample_Name   : chr  "A" "B" "C" "D" ...
## $ Group         : chr  "W" "C" "W" "C" ...
## $ PlasmaBlast   : num  1.87 1.84 1.34 1.98 2.02 ...
## $ CD8pCD28nCD45RAn: num  1.72 2.66 -2.66 3.61 -4.47 ...
## $ CD8.naive     : num  239 266 320 308 280 ...
## $ CD4.naive     : num  524 536 672 522 555 ...
## $ CD8T          : num  0.056 0.1353 0.1849 0.035 0.0401 ...
## $ CD4T          : num  0.1908 0.1265 0.3821 0.0911 0.209 ...
## $ NK            : num  0.03318 0.03819 0.00617 0.03306 0.02239
...
## $ Bcell        : num  0.0192 0.0812 0.0569 0.0233 0.0584 ...
## $ Mono         : num  0.05506 0.13045 0.00693 0.13703 0.05448
...
## $ Gran         : num  0.656 0.508 0.376 0.681 0.621 ...
## $ Sex          : int   0 0 1 1 0 0
## $ Age          : int   51 44 53 53 44 52
```

```
SampleSheet$Sex <- factor(SampleSheet$Sex)
```

```
str(SampleSheet)

## 'data.frame':    6 obs. of  14 variables:

## $ Sample_Name   : chr  "A" "B" "C" "D" ...
## $ Group         : chr  "W" "C" "W" "C" ...
## $ PlasmaBlast   : num  1.87 1.84 1.34 1.98 2.02 ...
## $ CD8pCD28nCD45RAn: num  1.72 2.66 -2.66 3.61 -4.47 ...
## $ CD8.naive     : num  239 266 320 308 280 ...
## $ CD4.naive     : num  524 536 672 522 555 ...
## $ CD8T          : num  0.056 0.1353 0.1849 0.035 0.0401 ...
## $ CD4T          : num  0.1908 0.1265 0.3821 0.0911 0.209 ...
## $ NK            : num  0.03318 0.03819 0.00617 0.03306 0.02239
...
## $ Bcell        : num  0.0192 0.0812 0.0569 0.0233 0.0584 ...
```



```
## $ Mono          : num  0.05506 0.13045 0.00693 0.13703 0.05448
...
## $ Gran          : num  0.656 0.508 0.376 0.681 0.621 ...
## $ Sex           : Factor w/ 2 levels "0","1": 1 1 2 2 1 1
## $ Age           : int   51 44 53 53 44 52
```

We can also use exploratory functions on specific columns:

```
summary(SampleSheet$CD8.naive)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  239.4   269.5   294.1   290.4   317.4   328.4
```

```
table(SampleSheet$Group)
```

```
##
```

```
## C W
```

```
## 3 3
```

```
class(SampleSheet$Group)
```

```
## [1] "character"
```

```
summary(SampleSheet$Group)
```

```
##      Length      Class    Mode
##           6 character character
```

```
SampleSheet$Group <- factor(SampleSheet$Group)
```

```
table(SampleSheet$Group)
```

```
##
```

```
## C W
```

```
## 3 3
```

```
class(SampleSheet$Group)
```

```
## [1] "factor"
```

```
summary(SampleSheet$Group)
```

```
## C W
```

```
## 3 3
```

```
levels(SampleSheet$Group)
```

```
## [1] "C" "W"
```

Indexing a dataframe

We want to select the first 2 columns of the dataframe. Remember: on the left of the comma you operate on rows, on the right of the comma you operate on columns

```
SampleSheet[,1:2]
```

```
##   Sample_Name Group
```

```
## 1          A      W
## 2          B      C
## 3          C      W
## 4          D      C
## 5          E      W
## 6          F      C
```

If we select only 1 column we do not have anymore a dataframe...

```
SampleSheet[,1]
```

```
## [1] "A" "B" "C" "D" "E" "F"
```

...unless we set drop=F

```
SampleSheet[,1,drop=F]
```

```
##   Sample_Name
```

```
## 1          A
## 2          B
## 3          C
## 4          D
## 5          E
## 6          F
```

If we know the name of the column of interest, we can directly indicate it; we can do this in different ways

```
SampleSheet$Sample_Name
```

```
## [1] "A" "B" "C" "D" "E" "F"
```

```
SampleSheet[, "Sample_Name"]
```

```
## [1] "A" "B" "C" "D" "E" "F"
```

```
SampleSheet[,]$Sample_Name
```

```
## [1] "A" "B" "C" "D" "E" "F"
```

We can operate also on rows...

```
SampleSheet[1,]  
##   Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive  
CD8T  
## 1           A      W      1.867548           1.724855  239.4476   523.977  
0.05597054  
##           CD4T           NK      Bcell           Mono           Gran Sex Age  
## 1 0.1908321 0.03318451 0.0191543 0.0550556 0.6564072 0 51
```

... or at the same time on rows and columns:

```
SampleSheet[1:3,1:2]  
##   Sample_Name Group  
## 1           A      W  
## 2           B      C  
## 3           C      W
```

```
SampleSheet[1:3,"Sample_Name"]  
## [1] "A" "B" "C"  
SampleSheet[1:3,]$Sample_Name  
## [1] "A" "B" "C"  
SampleSheet[1:3,3:5]  
##   PlasmaBlast CD8pCD28nCD45RAn CD8.naive  
## 1      1.867548           1.724855  239.4476  
## 2      1.841516           2.663802  265.9878  
## 3      1.335753          -2.658338  320.4341
```

Order a dataframe

We can order the rows of a dataframe according a certain column using the `order()` function.
For example, to order according to the Age:

```
head(SampleSheet)  
##   Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive  
CD8T
```

```
## 1      A      W      1.867548      1.724855  239.4476  523.9770
0.05597054
## 2      B      C      1.841516      2.663802  265.9878  536.2501
0.13529055
## 3      C      W      1.335753      -2.658338  320.4341  671.8478
0.18487413
## 4      D      C      1.983669      3.613387  308.4593  521.9668
0.03500575
## 5      E      W      2.022408      -4.468693  279.8390  554.8088
0.04006602
## 6      F      C      1.982639      -2.470729  328.3630  655.6088
0.01607650
##          CD4T          NK          Bcell          Mono          Gran Sex Age
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072  0  51
## 2 0.1265136 0.038188822 0.08120917 0.130451761 0.5077724  0  44
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582  1  53
## 4 0.0911106 0.033064802 0.02328309 0.137029687 0.6811102  1  53
## 5 0.2089572 0.022393107 0.05841506 0.054481111 0.6208538  0  44
## 6 0.2114697 0.000000000 0.04797884 0.067280646 0.6727830  0  52
```

...from the smallest to the largest:

```
SampleSheet <- SampleSheet[order(SampleSheet$Age),]
```

```
head(SampleSheet)
```

```
##      Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive
CD8T
## 2      B      C      1.841516      2.663802  265.9878  536.2501
0.13529055
## 5      E      W      2.022408      -4.468693  279.8390  554.8088
0.04006602
## 1      A      W      1.867548      1.724855  239.4476  523.9770
0.05597054
## 6      F      C      1.982639      -2.470729  328.3630  655.6088
0.01607650
## 3      C      W      1.335753      -2.658338  320.4341  671.8478
0.18487413
## 4      D      C      1.983669      3.613387  308.4593  521.9668
0.03500575
##          CD4T          NK          Bcell          Mono          Gran Sex Age
## 2 0.1265136 0.038188822 0.08120917 0.130451761 0.5077724  0  44
## 5 0.2089572 0.022393107 0.05841506 0.054481111 0.6208538  0  44
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072  0  51
## 6 0.2114697 0.000000000 0.04797884 0.067280646 0.6727830  0  52
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582  1  53
```

```
## 4 0.0911106 0.033064802 0.02328309 0.137029687 0.6811102 1 53
```

```
SampleSheet$Age
```

```
## [1] 44 44 51 52 53 53
```

...from the largest to the smallest:

```
SampleSheet <- SampleSheet[order(-SampleSheet$Age),]
```

```
head(SampleSheet)
```

```
## Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive  
CD8T
```

```
## 3 C W 1.335753 -2.658338 320.4341 671.8478  
0.18487413  
## 4 D C 1.983669 3.613387 308.4593 521.9668  
0.03500575  
## 6 F C 1.982639 -2.470729 328.3630 655.6088  
0.01607650  
## 1 A W 1.867548 1.724855 239.4476 523.9770  
0.05597054  
## 2 B C 1.841516 2.663802 265.9878 536.2501  
0.13529055  
## 5 E W 2.022408 -4.468693 279.8390 554.8088  
0.04006602  
## CD4T NK Bcell Mono Gran Sex Age  
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582 1 53  
## 4 0.0911106 0.033064802 0.02328309 0.137029687 0.6811102 1 53  
## 6 0.2114697 0.000000000 0.04797884 0.067280646 0.6727830 0 52  
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072 0 51  
## 2 0.1265136 0.038188822 0.08120917 0.130451761 0.5077724 0 44  
## 5 0.2089572 0.022393107 0.05841506 0.054481111 0.6208538 0 44
```

```
SampleSheet$Age
```

```
## [1] 53 53 52 51 44 44
```

CHUNK 5: Subsetting a dataframe

We want retrieve only the rows for which the column "Group" is equal to "W" (remember that now this column is a factor)

```
class(SampleSheet$Group)
```

```
## [1] "factor"
levels(SampleSheet$Group)
## [1] "C" "W"
SampleSheet[SampleSheet$Group==W,]
## Error: object 'W' not found
```

Ops! Without quotes, W is read by R as an object, and in the Console there is not an object called W. Let's try in this way:

```
SampleSheet[SampleSheet$Group=="W",]
##   Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive
##   CD8T
## 3           C      W      1.335753          -2.658338   320.4341   671.8478
##   0.18487413
## 1           A      W      1.867548           1.724855   239.4476   523.9770
##   0.05597054
## 5           E      W      2.022408          -4.468693   279.8390   554.8088
##   0.04006602
##           CD4T           NK           Bcell           Mono           Gran Sex Age
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582   1  53
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072   0  51
## 5 0.2089572 0.022393107 0.05841506 0.054481111 0.6208538   0  44
```

```
SampleSheet_W <- SampleSheet[SampleSheet$Group=="W",]
```

```
SampleSheet_W
```

```
##   Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive
##   CD8T
## 3           C      W      1.335753          -2.658338   320.4341   671.8478
##   0.18487413
## 1           A      W      1.867548           1.724855   239.4476   523.9770
##   0.05597054
## 5           E      W      2.022408          -4.468693   279.8390   554.8088
##   0.04006602
##           CD4T           NK           Bcell           Mono           Gran Sex Age
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582   1  53
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072   0  51
## 5 0.2089572 0.022393107 0.05841506 0.054481111 0.6208538   0  44
```

```
SampleSheet
```

```
##      Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive
CD8T

## 3      C      W      1.335753      -2.658338      320.4341      671.8478
0.18487413
## 4      D      C      1.983669      3.613387      308.4593      521.9668
0.03500575
## 6      F      C      1.982639      -2.470729      328.3630      655.6088
0.01607650
## 1      A      W      1.867548      1.724855      239.4476      523.9770
0.05597054
## 2      B      C      1.841516      2.663802      265.9878      536.2501
0.13529055
## 5      E      W      2.022408      -4.468693      279.8390      554.8088
0.04006602
##      CD4T      NK      Bcell      Mono      Gran Sex Age
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582 1 53
## 4 0.0911106 0.033064802 0.02328309 0.137029687 0.6811102 1 53
## 6 0.2114697 0.000000000 0.04797884 0.067280646 0.6727830 0 52
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072 0 51
## 2 0.1265136 0.038188822 0.08120917 0.130451761 0.5077724 0 44
## 5 0.2089572 0.022393107 0.05841506 0.054481111 0.6208538 0 44
```

```
dim(SampleSheet_W)
```

```
## [1] 3 14
```

Alternatively:

```
condition <- SampleSheet$Group=="W"
```

```
condition
```

```
## [1] TRUE FALSE FALSE TRUE FALSE TRUE
```

```
SampleSheet_W <- SampleSheet[condition,]
```

```
SampleSheet_W
```

```
##      Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive
CD8T

## 3      C      W      1.335753      -2.658338      320.4341      671.8478
0.18487413
## 1      A      W      1.867548      1.724855      239.4476      523.9770
0.05597054
## 5      E      W      2.022408      -4.468693      279.8390      554.8088
0.04006602
```

```
##           CD4T           NK           Bcell           Mono           Gran Sex Age
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582 1 53
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072 0 51
## 5 0.2089572 0.022393107 0.05841506 0.054481111 0.6208538 0 44
```

```
dim(SampleSheet_W)
```

```
## [1] 3 14
```

I can subset on both rows and columns at the same time: see this example...

```
SampleSheet_W <-
SampleSheet[SampleSheet$Group=="W",c("Sample_Name", "Group")]
```

```
SampleSheet_W
```

```
## Sample_Name Group
```

```
## 3          C      W
## 1          A      W
## 5          E      W
```

```
dim(SampleSheet_W)
```

```
## [1] 3 2
```

...and this example

```
SampleSheet_W <- SampleSheet[SampleSheet$Group=="W",c(1,2,5)]
```

```
SampleSheet_W
```

```
## Sample_Name Group CD8.naive
```

```
## 3          C      W 320.4341
## 1          A      W 239.4476
## 5          E      W 279.8390
```

```
dim(SampleSheet_W)
```

```
## [1] 3 3
```

Let's check the str of SampleSheet_W:

```
str(SampleSheet_W)
```

```
## 'data.frame': 3 obs. of 3 variables:
```

```
## $ Sample_Name: chr "C" "A" "E"
```



```
## $ Group      : Factor w/ 2 levels "C","W": 2 2 2
## $ CD8.naive   : num  320 239 280
```

There is something strange...I have only “W”, but R remembers also the “C”! I can avoid this by using the droplevels function:

```
SampleSheet_W <- droplevels(SampleSheet_W)
```

```
dim(SampleSheet_W)
```

```
## [1] 3 3
```

```
str(SampleSheet_W)
```

```
## 'data.frame': 3 obs. of 3 variables:
```

```
## $ Sample_Name: chr  "C" "A" "E"
## $ Group      : Factor w/ 1 level "W": 1 1 1
## $ CD8.naive   : num  320 239 280
```

Another example of subsetting rows according to a condition:

```
SampleSheet[SampleSheet$CD8.naive>300,]
```

```
## Sample_Name Group PlasmaBlast CD8pCD28nCD45Ran CD8.naive CD4.naive
CD8T
```

```
## 3      C      W      1.335753      -2.658338  320.4341  671.8478
0.18487413
## 4      D      C      1.983669      3.613387  308.4593  521.9668
0.03500575
## 6      F      C      1.982639      -2.470729  328.3630  655.6088
0.01607650
##      CD4T      NK      Bcell      Mono      Gran Sex Age
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582 1 53
## 4 0.0911106 0.033064802 0.02328309 0.137029687 0.6811102 1 53
## 6 0.2114697 0.000000000 0.04797884 0.067280646 0.6727830 0 52
```

```
SampleSheet_CD8.naive_300 <- SampleSheet[SampleSheet$CD8.naive>300,]
```

```
str(SampleSheet_CD8.naive_300)
```

```
## 'data.frame': 3 obs. of 14 variables:
```

```
## $ Sample_Name : chr  "C" "D" "F"
## $ Group      : Factor w/ 2 levels "C","W": 2 1 1
## $ PlasmaBlast : num  1.34 1.98 1.98
## $ CD8pCD28nCD45Ran: num  -2.66 3.61 -2.47
## $ CD8.naive    : num  320 308 328
## $ CD4.naive    : num  672 522 656
```

```
## $ CD8T      : num  0.1849 0.035 0.0161
## $ CD4T      : num  0.3821 0.0911 0.2115
## $ NK        : num  0.00617 0.03306 0
## $ Bcell     : num  0.0569 0.0233 0.048
## $ Mono      : num  0.00693 0.13703 0.06728
## $ Gran      : num  0.376 0.681 0.673
## $ Sex       : Factor w/ 2 levels "0","1": 2 2 1
## $ Age       : int   53 53 52
```

```
SampleSheet_CD8.naive_300 <- droplevels(SampleSheet_CD8.naive_300)
```

```
str(SampleSheet_CD8.naive_300)
```

```
## 'data.frame':   3 obs. of  14 variables:
```

```
## $ Sample_Name      : chr  "C" "D" "F"
## $ Group             : Factor w/ 2 levels "C","W": 2 1 1
## $ PlasmaBlast       : num  1.34 1.98 1.98
## $ CD8pCD28nCD45RAn : num  -2.66 3.61 -2.47
## $ CD8.naive         : num  320 308 328
## $ CD4.naive         : num  672 522 656
## $ CD8T              : num  0.1849 0.035 0.0161
## $ CD4T              : num  0.3821 0.0911 0.2115
## $ NK                : num  0.00617 0.03306 0
## $ Bcell             : num  0.0569 0.0233 0.048
## $ Mono              : num  0.00693 0.13703 0.06728
## $ Gran              : num  0.376 0.681 0.673
## $ Sex               : Factor w/ 2 levels "0","1": 2 2 1
## $ Age               : int   53 53 52
```

Now we want to subset the dataframe to retain only the samples whose name i stored in the vector v_A_C_D:

```
v_A_C_D <- c("A", "C", "D")
```

```
SampleSheet
```

```
##   Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive
##   CD8T
## 3           C    W    1.335753         -2.658338   320.4341   671.8478
##           0.18487413
## 4           D    C    1.983669          3.613387   308.4593   521.9668
##           0.03500575
## 6           F    C    1.982639         -2.470729   328.3630   655.6088
##           0.01607650
## 1           A    W    1.867548          1.724855   239.4476   523.9770
##           0.05597054
```

```
## 2          B          C      1.841516          2.663802  265.9878  536.2501
0.13529055
## 5          E          W      2.022408          -4.468693  279.8390  554.8088
0.04006602
##          CD4T          NK          Bcell          Mono          Gran Sex Age
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582  1  53
## 4 0.0911106 0.033064802 0.02328309 0.137029687 0.6811102  1  53
## 6 0.2114697 0.000000000 0.04797884 0.067280646 0.6727830  0  52
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072  0  51
## 2 0.1265136 0.038188822 0.08120917 0.130451761 0.5077724  0  44
## 5 0.2089572 0.022393107 0.05841506 0.054481111 0.6208538  0  44
```

```
SampleSheet$Sample_Name %in% v_A_C_D
```

```
## [1] TRUE TRUE FALSE TRUE FALSE FALSE
```

```
SampleSheet_A_C_D <- SampleSheet[SampleSheet$Sample_Name %in% v_A_C_D,]
```

```
SampleSheet_A_C_D
```

```
## Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive
CD8T
```

```
## 3          C          W      1.335753          -2.658338  320.4341  671.8478
0.18487413
## 4          D          C      1.983669          3.613387  308.4593  521.9668
0.03500575
## 1          A          W      1.867548          1.724855  239.4476  523.9770
0.05597054
##          CD4T          NK          Bcell          Mono          Gran Sex Age
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582  1  53
## 4 0.0911106 0.033064802 0.02328309 0.137029687 0.6811102  1  53
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072  0  51
```

```
str(SampleSheet_A_C_D)
```

```
## 'data.frame': 3 obs. of 14 variables:
```

```
## $ Sample_Name : chr "C" "D" "A"
## $ Group : Factor w/ 2 levels "C","W": 2 1 2
## $ PlasmaBlast : num 1.34 1.98 1.87
## $ CD8pCD28nCD45RAn: num -2.66 3.61 1.72
## $ CD8.naive : num 320 308 239
## $ CD4.naive : num 672 522 524
## $ CD8T : num 0.185 0.035 0.056
## $ CD4T : num 0.3821 0.0911 0.1908
## $ NK : num 0.00617 0.03306 0.03318
## $ Bcell : num 0.0569 0.0233 0.0192
```

```
## $ Mono          : num  0.00693 0.13703 0.05506
## $ Gran          : num  0.376 0.681 0.656
## $ Sex           : Factor w/ 2 levels "0","1": 2 2 1
## $ Age           : int   53 53 51
```

```
SampleSheet_A_C_D <- droplevels(SampleSheet_A_C_D)
str(SampleSheet_A_C_D)
```

```
## 'data.frame':   3 obs. of  14 variables:
## $ Sample_Name   : chr   "C" "D" "A"
## $ Group         : Factor w/ 2 levels "C","W": 2 1 2
## $ PlasmaBlast   : num   1.34 1.98 1.87
## $ CD8pCD28nCD45RAn: num  -2.66 3.61 1.72
## $ CD8.naive     : num   320 308 239
## $ CD4.naive     : num   672 522 524
## $ CD8T          : num   0.185 0.035 0.056
## $ CD4T          : num   0.3821 0.0911 0.1908
## $ NK           : num   0.00617 0.03306 0.03318
## $ Bcell        : num   0.0569 0.0233 0.0192
## $ Mono         : num   0.00693 0.13703 0.05506
## $ Gran         : num   0.376 0.681 0.656
## $ Sex          : Factor w/ 2 levels "0","1": 2 2 1
## $ Age         : int    53 53 51
```

CHUNK 6: Save options

To write a txt file (or a csv file, specifying the comma as separator)

```
write.table(SampleSheet_A_C_D, file="SampleSheet_A_C_D.txt", sep="\t")
```

Let's try to open the file that you have created and check how it looks like. Then, try to save the same file setting row.names=F. What is the difference?

```
write.table(SampleSheet_A_C_D, file="SampleSheet_A_C_D_rownamesF.txt", sep="\t", row.names=F)
```

Saving data into R data formats can reduce considerably the size of large files. The objects can be read back from the file at a later date by using specific functions.

Saving in Rds format:

```
saveRDS(SampleSheet_A_C_D, file = "SampleSheet_A_C_D.rds")
```

Note that if you save your data as rds, it will be possible to restore the object under a different name .

```
rm(list=ls())
```

```
readRDS("SampleSheet_A_C_D.rds")
```

```
##   Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive
CD8T
## 3           C    W    1.335753          -2.658338   320.4341   671.8478
0.18487413
## 4           D    C    1.983669           3.613387   308.4593   521.9668
0.03500575
## 1           A    W    1.867548           1.724855   239.4476   523.9770
0.05597054
##           CD4T           NK           Bcell           Mono           Gran Sex Age
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582   1  53
## 4 0.0911106 0.033064802 0.02328309 0.137029687 0.6811102   1  53
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072   0  51
```

```
SampleSheet_A_C_D <- readRDS("SampleSheet_A_C_D.rds")
```

```
SampleSheet_A_C_D
```

```
##   Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive
CD8T
## 3           C    W    1.335753          -2.658338   320.4341   671.8478
0.18487413
## 4           D    C    1.983669           3.613387   308.4593   521.9668
0.03500575
## 1           A    W    1.867548           1.724855   239.4476   523.9770
0.05597054
##           CD4T           NK           Bcell           Mono           Gran Sex Age
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582   1  53
## 4 0.0911106 0.033064802 0.02328309 0.137029687 0.6811102   1  53
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072   0  51
```

```
new <- readRDS("SampleSheet_A_C_D.rds")
```

```
new
```

```
##   Sample_Name Group PlasmaBlast CD8pCD28nCD45RAn CD8.naive CD4.naive
CD8T
## 3           C    W    1.335753          -2.658338   320.4341   671.8478
0.18487413
```

```
## 4          D      C      1.983669          3.613387  308.4593  521.9668
0.03500575
## 1          A      W      1.867548          1.724855  239.4476  523.9770
0.05597054
##          CD4T          NK          Bcell          Mono          Gran Sex Age
## 3 0.3820950 0.006167623 0.05693339 0.006932121 0.3764582  1  53
## 4 0.0911106 0.033064802 0.02328309 0.137029687 0.6811102  1  53
## 1 0.1908321 0.033184511 0.01915430 0.055055601 0.6564072  0  51
```

```
ls()
```

```
## [1] "new"          "SampleSheet_A_C_D"
```

Saving in Rdata format:

```
save(SampleSheet_A_C_D, file="SampleSheet_A_C_D.RData")
```

Note that if you save your data as RData, when you will load it in the workspace the object cannot be restored under different name, but the original object name will be automatically used.

```
rm(list=ls())
```

```
load("SampleSheet_A_C_D.RData")
ls()
```

```
## [1] "SampleSheet_A_C_D"
```

```
rm(list=ls())
```

```
new <- load("SampleSheet_A_C_D.RData")
```

The function `save.image()` is just a short-cut for 'save my current workspace', i.e., `save(list = ls(all.names = TRUE), file = ".RData", envir = .GlobalEnv)`. It is also what happens with `q("yes")`.

```
save.image("Lesson_1_final_part.RData")
```

CHUNK 7: Exercises:

- Load the SampleSheet file
- Order the rows according to the CD4T column (from the smallest to the largest)
- How many males and how many females do you have?
- How many rows have CD8T <0.05?
- What is the sex of the subjects with CD8T <0.05?
- What is the sex of the subjects with CD8.naive <300?
- What are the mean and the median age of males and females?

CHUNK 8: A larger dataframe: the Infinium 27k manifest

The file is in a spreadsheet format . Let's import in R the first 20 rows of the file

```
Infinium27 <-  
read.table("~/Dropbox/DRD_2025/1/Infinium27.txt", sep="\t", header=T, nrow  
=20)
```

```
dim(Infinium27)
```

```
## [1] 20 35
```

```
head(Infinium27)
```

##	IlmnID	Name	IlmnStrand	AddressA_ID		
## 1	cg00000292	cg00000292	TOP	990370		
## 2	cg00002426	cg00002426	TOP	6580397		
## 3	cg00003994	cg00003994	TOP	7150184		
## 4	cg00005847	cg00005847	BOT	4850717		
## 5	cg00006414	cg00006414	BOT	6980731		
## 6	cg00007981	cg00007981	TOP	5260689		
##				AlleleA_ProbeSeq	AddressB_ID	
## 1	AAACATTAATTACCAACCACTCTTCCAAAAACACTTACCATTAAAACCA				6660678	
## 2	AATATAATAACATTACCTTACCCATCTTATAATCAAACCAAACAAAAACA				6100343	
## 3	AATAATAATAATACCCCTATAATACTAACTAACAAACATACCCTCTTCA				7150392	
## 4	TACTATAATACACCCTATATTTAAACACTAACTTACCCCATTAATAACA				1260113	
## 5	CTCAAAACCAAACAAACAAACCCCAATACTAATCATTAATAAAATCA				4280093	
## 6	CCAAATTCAAAAACTACAATAAACTAACTATAAACAAATAATCACA				6860356	
##				AlleleB_ProbeSeq	GenomeBuild	Chr
##						
## 1	AAACATTAATTACCAACCGCTCTTCCAAAAACACTTACCATTAAAACCG				36	16
	28797601					
## 2	AATATAATAACATTACCTTACCCGTCTTATAATCAAACCAAACGAAAACG				36	3
	57718583					
## 3	AATAATAATAATACCCCTATAATACTAACTAACAAACATACCCTCTTCG				36	7
	15692387					
## 4	TACTATAATACACCCTATATTTAAACACTAACTTACCCCATTAATAACG				36	2
	176737319					
## 5	CTCGAAACCGAACAACAAACAAACCCCAATACTAATCGTTAATAAAATCG				36	7
	148453770					
## 6	CCAAATTCAAAAACTACGACTAAACTAACTATAAACAAATAATCACG				36	11
	93502242					
##	Floudy	Species	Source	SourceVersion	SourceStrand	
## 1	diploid	Homo sapiens	NCBI:RefSeq	36.1	TOP	

```

## 2 diploid Homo sapiens NCBI:RefSeq          36.1          TOP
## 3 diploid Homo sapiens NCBI:RefSeq          36.1          BOT
## 4 diploid Homo sapiens NCBI:RefSeq          36.1          BOT
## 5 diploid Homo sapiens NCBI:RefSeq          36.1          BOT
## 6 diploid Homo sapiens NCBI:RefSeq          36.1          BOT
##
##                                     SourceSeq
## 1 CGGCCTCAATGGTAAGTGTCCCTTGGAAGAGCGGCTGGTAATTAATGCCC
## 2 CGCTCTCGTCTGGTTTGATCACAAGACGGGCAAGGTAATGTCACCACATT
## 3 GGTGGTGGTGGTGCCCCCTGTGATGCTGGCTGGCAAACATGCCCTCTTCG
## 4 TACTGTAATGCACCCTGTATTTAAGGCACTGGGCTTGCCCCATTAAAGCG
## 5 CTCGGAAACCGAGCAGGGCAAAACCCAGTGCTGATCGTTAGTGGGATCG
## 6 CCAAGTTCAAGGGGCTGCGACTGGAGCTGGCTGTGGACAAGATGGTCACG
##
TopGenomicSeq
## 1
TGGGGTGAGTGAGACCACGGGCCTCACCCCGGACCAAGTTAAGCGGAATCTGGAGAAATA [CG] GCCTCAA
TGGTAAGTGTCCCTTGGAAGAGCGGCTGGTAATTAATGCCCTCCTGCACCCCC
## 2
CCGCTGTCGACCAGCGCAGAATAATGCCACTTTTGATTGCAAAGTGCTATCAAGGAACCA [CG] CTCTCGT
CTGGTTTGATCACAAGACGGGCAAGGTAATGTCACCACATTGTCCAGCGGCAT
## 3
GGTGGTGGTGGTGGTGGTGGTGGTGCCCCCTGTGATGCTGGCTGGCAAACATGCCCTCTT [CG] TTGGGGT
ATCCCGCGATTATGCAAGATGAGGAAGAAGTAGAGAGCTCGGGGTAAGACATA
## 4
CAGATAACTCAATACTGTAATGCACCCTGTATTTAAGGCACTGGGCTTGCCCCATTAAAG [CG] CCATAAA
TTTGAAGGCCAATGATCGGTTTTTCATGTAACGGGTGGTACTTCATACTGAAGT
## 5
GAACCGGCCCAGCTCGGAAACCGAGCAGGGCAAAACCCAGTGCTGATCGTTAGTGGGAT [CG] CGCCTGT
GAATAGCCACTGCCCTCCAGCCTGGGCAACAGCCAGACCCCGTCTGTTTAATA
## 6
AGCCCACGGAGCCCAAGTTCAAGGGGCTGCGACTGGAGCTGGCTGTGGACAAGATGGTCA [CG] TGCATTG
CGGTGGGGCTGCCCCCTGCTGCTCATCTCGCTGGCCTTCGCGCAGGAGATCTCG
##   Next_Base Color_Channel TSS_Coordinate Gene_Strand      Gene_ID
Symbol
## 1      T              Red        28797310          +  GeneID:487
ATP2A1
## 2      T              Red        57718214          +  GeneID:7871
SLMAP
## 3      T              Red        15692819          -  GeneID:4223
MEOX2
## 4      C              Grn        176737051         +  GeneID:3232
HOXD3
## 5      C              Grn        148454441         +  GeneID:57541
ZNF398
## 6      T              Red        93501742          +  GeneID:24145
PANX1

```


	Synonym	Accession	GID
## 1	ATP2A; SERCA1;	NM_173201.2	GI:47132613
## 2	SLAP; KIAA1601;	NM_007159.2	GI:56550042
## 3	GAX; MOX2;	NM_005924.3	GI:55956906
## 4	HOX4; HOX1D; HOX4A; Hox-4.1; MGC10470;	NM_006898.4	GI:23510372
## 5	P51; P71; ZER6; KIAA1339;	NM_020781.2	GI:25777702
## 6	MRS1; UNQ2529; MGC21309;	NM_015368.3	GI:39995063

##

Annotation

1 isoform a is encoded by transcript variant a;
sarcoplasmic/endoplasmic reticulum calcium ATPase 1; calcium pump 1; SR
Ca(2+)-ATPase 1; calcium-transporting ATPase sarcoplasmic reticulum
type; fast twitch skeletal muscle isoform; endoplasmic reticulum class
1/2 Ca(2+) ATPase; go_component: membrane; go_component: integral to
membrane; go_component: sarcoplasmic reticulum; go_component: smooth
endoplasmic reticulum; go_function: ATP binding; go_function: hydrolase
activity; go_function: nucleotide binding; go_function: calcium ion
binding; go_function: magnesium ion binding; go_function: calcium-
transporting ATPase activity; go_function: hydrolase activity; acting
on acid anhydrides; catalyzing transmembrane movement of substances;
go_process: metabolism; go_process: cation transport; go_process:
proton transport; go_process: calcium ion transport; go_process:
regulation of striated muscle contraction

2

Sarcolemmal-associated protein; go_component: integral to plasma
membrane; go_component: smooth endoplasmic reticulum; go_function:
unfolded protein binding; go_process: protein folding; go_process:
muscle contraction

3

growth arrest-specific homeo box; go_component: nucleus; go_function:
transcription factor activity; go_process: circulation; go_process:
development; go_process: regulation of transcription; DNA-dependent

4

homeobox protein Hox-D3; Hox-4.1; mouse; homolog of; homeo box D3;
go_component: nucleus; go_function: transcription factor activity;
go_process: morphogenesis; go_process: regulation of transcription;
DNA-dependent

5

isoform b is encoded by transcript variant 2; zinc finger DNA binding
protein ZER6; zinc finger-estrogen receptor interaction; clone 6; zinc
finger DNA binding protein p52/p71; go_component: nucleus; go_function:
DNA binding; go_function: zinc ion binding; go_function: metal ion
binding; go_function: transcriptional activator activity; go_process:
transcription; go_process: regulation of transcription; DNA-dependent

```
## 6
innexin; go_component: membrane; go_component: gap junction;
go_component: integral to membrane
##
Product Distance_to_TSS
CPG_ISLAND
## 1 ATPase; Ca++ transporting; fast twitch 1 isoform a 291
TRUE
## 2 sarcolemma associated protein 369
TRUE
## 3 mesenchyme homeo box 2 432
TRUE
## 4 homeobox D3 268
FALSE
## 5 zinc finger 398 isoform b 671
TRUE
## 6 pannexin 1 500
TRUE
## CPG_ISLAND_LOCATIONS MIR_CPG_ISLAND MIR_NAMES V34 V35
## 1 16:28797486-28797825 NA NA NA NA
## 2 3:57716811-57718675 NA NA NA NA
## 3 7:15691512-15693551 NA NA NA NA
## 4 NA NA NA NA
## 5 7:148453584-148455804 NA NA NA NA
## 6 11:93501124-93502564 NA NA NA NA
```

Now we import the entire dataframe:

```
Infinium27 <-
read.table("~/Dropbox/DRD_2025/1/Infinium27.txt", sep="\t", header=T)
dim(Infinium27)
```

```
## [1] 27723 35
```

```
head(Infinium27)
```

##	IlmnID	Name	IlmnStrand	AddressA_ID	AlleleA_ProbeSeq	AddressB_ID
## 1	cg00000292	cg00000292	TOP	990370		
## 2	cg00002426	cg00002426	TOP	6580397		
## 3	cg00003994	cg00003994	TOP	7150184		
## 4	cg00005847	cg00005847	BOT	4850717		
## 5	cg00006414	cg00006414	BOT	6980731		
## 6	cg00007981	cg00007981	TOP	5260689		
##						
## 1	AAACATTAATTACCAACCACTCTTCCAAAAACACTTACCATTAAAACCA					6660678
## 2	AATATAATAACATTACCTTACCCATCTTATAATCAAACCAACAAAAACA					6100343
## 3	AATAATAATAATACCCCTATAATACTAACTAACAACATACCCTCTTCA					7150392

[illegible]

5
GAACCGGCCCCAGCTCGGAAACCGAGCAGGGCAAACCCCCAGTGCTGATCGTTAGTGGGAT [CG] CGCCTGT
GAATAGCCACTGCCCTCCAGCCTGGGCAACAGCCAGACCCCGTCTGTTTAATA

6
AGCCCACGGAGCCCAAGTTCAAGGGGCTGCGACTGGAGCTGGCTGTGGACAAGATGGTCA [CG] TGCATTG
CGGTGGGGCTGCCCCCTGCTGCTCATCTCGCTGGCCTTCGCGCAGGAGATCTCG

##	Next_Base	Color_Channel	TSS_Coordinate	Gene_Strand	Gene_ID
----	-----------	---------------	----------------	-------------	---------

## 1	T	Red	28797310	+	GeneID:487
------	---	-----	----------	---	------------

ATP2A1

## 2	T	Red	57718214	+	GeneID:7871
------	---	-----	----------	---	-------------

SLMAP

## 3	T	Red	15692819	-	GeneID:4223
------	---	-----	----------	---	-------------

MEOX2

## 4	C	Grn	176737051	+	GeneID:3232
------	---	-----	-----------	---	-------------

HOXD3

## 5	C	Grn	148454441	+	GeneID:57541
------	---	-----	-----------	---	--------------

ZNF398

## 6	T	Red	93501742	+	GeneID:24145
------	---	-----	----------	---	--------------

PANX1

##	Synonym	Accession	GID
## 1	ATP2A; SERCA1;	NM_173201.2	GI:47132613
## 2	SLAP; KIAA1601;	NM_007159.2	GI:56550042
## 3	GAX; MOX2;	NM_005924.3	GI:55956906
## 4	HOX4; HOX1D; HOX4A; Hox-4.1;	MGC10470; NM_006898.4	GI:23510372
## 5	P51; P71; ZER6; KIAA1339;	NM_020781.2	GI:25777702
## 6	MRS1; UNQ2529; MGC21309;	NM_015368.3	GI:39995063

Annotation
1 isoform a is encoded by transcript variant a;
sarcoplasmic/endoplasmic reticulum calcium ATPase 1; calcium pump 1; SR
Ca(2+)-ATPase 1; calcium-transporting ATPase sarcoplasmic reticulum
type; fast twitch skeletal muscle isoform; endoplasmic reticulum class
1/2 Ca(2+) ATPase; go_component: membrane; go_component: integral to
membrane; go_component: sarcoplasmic reticulum; go_component: smooth
endoplasmic reticulum; go_function: ATP binding; go_function: hydrolase
activity; go_function: nucleotide binding; go_function: calcium ion
binding; go_function: magnesium ion binding; go_function: calcium-
transporting ATPase activity; go_function: hydrolase activity; acting
on acid anhydrides; catalyzing transmembrane movement of substances;
go_process: metabolism; go_process: cation transport; go_process:
proton transport; go_process: calcium ion transport; go_process:
regulation of striated muscle contraction

2
Sarcolemmal-associated protein; go_component: integral to plasma
membrane; go_component: smooth endoplasmic reticulum; go_function:

```

unfolded protein binding; go_process: protein folding; go_process:
muscle contraction
## 3
growth arrest-specific homeo box; go_component: nucleus; go_function:
transcription factor activity; go_process: circulation; go_process:
development; go_process: regulation of transcription; DNA-dependent
## 4
homeobox protein Hox-D3; Hox-4.1; mouse; homolog of; homeo box D3;
go_component: nucleus; go_function: transcription factor activity;
go_process: morphogenesis; go_process: regulation of transcription;
DNA-dependent
## 5
isoform b is encoded by transcript variant 2; zinc finger DNA binding
protein ZER6; zinc finger-estrogen receptor interaction; clone 6; zinc
finger DNA binding protein p52/p71; go_component: nucleus; go_function:
DNA binding; go_function: zinc ion binding; go_function: metal ion
binding; go_function: transcriptional activator activity; go_process:
transcription; go_process: regulation of transcription; DNA-dependent
## 6
innexin; go_component: membrane; go_component: gap junction;
go_component: integral to membrane
##
Product Distance_to_TSS
CPG_ISLAND
## 1 ATPase; Ca++ transporting; fast twitch 1 isoform a 291
TRUE
## 2 sarcolemma associated protein 369
TRUE
## 3 mesenchyme homeo box 2 432
TRUE
## 4 homeobox D3 268
FALSE
## 5 zinc finger 398 isoform b 671
TRUE
## 6 pannexin 1 500
TRUE
## CPG_ISLAND_LOCATIONS MIR_CPG_ISLAND MIR_NAMES V34 V35
## 1 16:28797486-28797825 NA NA
## 2 3:57716811-57718675 NA NA
## 3 7:15691512-15693551 NA NA
## 4 NA NA
## 5 7:148453584-148455804 NA NA
## 6 11:93501124-93502564 NA NA

```

```
str(Infinium27)
```

```
## 'data.frame': 27723 obs. of 35 variables:
```

[illegible]

```
## $ Next_Base      : chr  "T" "T" "T" "C" ...
## $ Color_Channel  : chr  "Red" "Red" "Red" "Grn" ...
## $ TSS_Coordinate : int   28797310 57718214 15692819 176737051
148454441 93501742 92883290 11971455 88360731 1135160 ...
## $ Gene_Strand    : chr  "+" "+" "-" "+" ...
## $ Gene_ID        : chr  "GeneID:487" "GeneID:7871"
"GeneID:4223" "GeneID:3232" ...
## $ Symbol         : chr  "ATP2A1" "SLMAP" "MEOX2" "HOXD3" ...
## $ Synonym        : chr  "ATP2A; SERCA1;" "SLAP; KIAA1601;"
"GAX; MOX2;" "HOX4; HOX1D; HOX4A; Hox-4.1; MGC10470;" ...
## $ Accession      : chr  "NM_173201.2" "NM_007159.2"
"NM_005924.3" "NM_006898.4" ...
## $ GID            : chr  "GI:47132613" "GI:56550042"
"GI:55956906" "GI:23510372" ...
## $ Annotation     : chr  "isoform a is encoded by transcript
variant a; sarcoplasmic/endoplasmic reticulum calcium ATPase 1; calcium
pump"| __truncated__ "Sarcolemmal-associated protein; go_component:
integral to plasma membrane; go_component: smooth endoplasmic ret"|
__truncated__ "growth arrest-specific homeo box; go_component: nucleus;
go_function: transcription factor activity; go_process"| __truncated__
"homeobox protein Hox-D3; Hox-4.1; mouse; homolog of; homeo box D3;
go_component: nucleus; go_function: transcri"| __truncated__ ...
## $ Product        : chr  "ATPase; Ca++ transporting; fast
twitch 1 isoform a" "sarcolemma associated protein" "mesenchyme homeo
box 2" "homeobox D3" ...
## $ Distance_to_TSS : int   291 369 432 268 671 500 240 502 57 56
...
## $ CPG_ISLAND     : logi   TRUE TRUE TRUE FALSE TRUE TRUE ...
## $ CPG_ISLAND_LOCATIONS: chr  "16:28797486-28797825" "3:57716811-
57718675" "7:15691512-15693551" "" ...
## $ MIR_CPG_ISLAND  : chr  "" "" "" "" ...
## $ MIR_NAMES       : chr  "" "" "" "" ...
## $ V34             : logi   NA NA NA NA NA NA ...
## $ V35             : logi   NA NA NA NA NA NA ...
```

Let's set stringsAsFactors = TRUE

```
Infinium27 <-
read.table("~/Dropbox/DRD_2025/1/Infinium27.txt", sep="\t", header=T, stringsAsFactors = T)
```

```
dim(Infinium27)
```

```
## [1] 27723    35
```

```
head(Infinium27)
```

##	IlmnID	Name	IlmnStrand	AddressA_ID				
##	1	cg00000292	cg00000292	TOP	990370			
##	2	cg00002426	cg00002426	TOP	6580397			
##	3	cg00003994	cg00003994	TOP	7150184			
##	4	cg00005847	cg00005847	BOT	4850717			
##	5	cg00006414	cg00006414	BOT	6980731			
##	6	cg00007981	cg00007981	TOP	5260689			
##				AlleleA_ProbeSeq	AddressB_ID			
##	1	AAACATTAATTACCAACCACTCTTCCAAAAAACACTTACCATTAAAACCA			6660678			
##	2	AATATAATAACATTACCTTACCCATCTTATAATCAAACCAAACAAAAACA			6100343			
##	3	AATAATAATAATACCCCCTATAATACTAATAACAAACATACCCCTCTTCA			7150392			
##	4	TACTATAATACACCCTATATTTAAAACACTAAACTTACCCCATTTAAAACA			1260113			
##	5	CTCAAAAACCAAACAAAACAAAACCCCAATACTAATCATTAATAAAATCA			4280093			
##	6	CCAAATTCAAAAAACTACAACATAAACTAACTATAAACAAAATAATCACA			6860356			
##				AlleleB_ProbeSeq	GenomeBuild	Chr		
MapInfo								
##	1	AAACATTAATTACCAACCGCTCTTCCAAAAAACACTTACCATTAAAACCG				36	16	
28797601								
##	2	AATATAATAACATTACCTTACCCGTCTTATAATCAAACCAAACGAAAACG				36	3	
57718583								
##	3	AATAATAATAATACCCCCTATAATACTAATAACAAACATACCCCTCTTCG				36	7	
15692387								
##	4	TACTATAATACACCCTATATTTAAAACACTAAACTTACCCCATTTAAAACG				36	2	
176737319								
##	5	CTCGAAAACCGAACAAAACAAAACCCCAATACTAATCGTTAATAAAATCG				36	7	
148453770								
##	6	CCAAATTCAAAAAACTACGACTAAAACATACTATAAACAAAATAATCACG				36	11	
93502242								
##	Ploidy	Species	Source	SourceVersion	SourceStrand			
##	1	diploid Homo sapiens	NCBI:RefSeq	36.1	TOP			
##	2	diploid Homo sapiens	NCBI:RefSeq	36.1	TOP			
##	3	diploid Homo sapiens	NCBI:RefSeq	36.1	BOT			
##	4	diploid Homo sapiens	NCBI:RefSeq	36.1	BOT			
##	5	diploid Homo sapiens	NCBI:RefSeq	36.1	BOT			
##	6	diploid Homo sapiens	NCBI:RefSeq	36.1	BOT			
##				SourceSeq				
##	1	CGGCCTCAATGGTAAGTGTCCCTTGGAAGAGCGGCTGGTAATTAATGCCC						
##	2	CGCTCTCGTCTGGTTTGATCACAAGACGGGCAAGGTAATGTCACCACATT						
##	3	GGTGGTGGTGGTGCCCCCTGTGATGCTGGCTGGCAAACATGCCCTCTTCG						
##	4	TACTGTAATGCACCCTGTATTTAAGGCACTGGGCTTGCCCCATTAAAGCG						
##	5	CTCGGAAACCGAGCAGGGCAAACCCCACTGCTGATCGTTAGTGGGATCG						
##	6	CCAAGTTCAGGGGCTGCGACTGGAGCTGGCTGTGGACAAGATGGTCACG						
##								
TopGenomicSeq								

[illegible]

membrane; go_component: sarcoplasmic reticulum; go_component: smooth endoplasmic reticulum; go_function: ATP binding; go_function: hydrolase activity; go_function: nucleotide binding; go_function: calcium ion binding; go_function: magnesium ion binding; go_function: calcium-transporting ATPase activity; go_function: hydrolase activity; acting on acid anhydrides; catalyzing transmembrane movement of substances; go_process: metabolism; go_process: cation transport; go_process: proton transport; go_process: calcium ion transport; go_process: regulation of striated muscle contraction

2

Sarcolemmal-associated protein; go_component: integral to plasma membrane; go_component: smooth endoplasmic reticulum; go_function: unfolded protein binding; go_process: protein folding; go_process: muscle contraction

3

growth arrest-specific homeo box; go_component: nucleus; go_function: transcription factor activity; go_process: circulation; go_process: development; go_process: regulation of transcription; DNA-dependent

4

homeobox protein Hox-D3; Hox-4.1; mouse; homolog of; homeo box D3; go_component: nucleus; go_function: transcription factor activity; go_process: morphogenesis; go_process: regulation of transcription; DNA-dependent

5

isoform b is encoded by transcript variant 2; zinc finger DNA binding protein ZER6; zinc finger-estrogen receptor interaction; clone 6; zinc finger DNA binding protein p52/p71; go_component: nucleus; go_function: DNA binding; go_function: zinc ion binding; go_function: metal ion binding; go_function: transcriptional activator activity; go_process: transcription; go_process: regulation of transcription; DNA-dependent

6

innexin; go_component: membrane; go_component: gap junction; go_component: integral to membrane

##	Product	Distance_to_TSS
CPG_ISLAND		
## 1	ATPase; Ca++ transporting; fast twitch 1 isoform a	291
TRUE		
## 2	sarcolemma associated protein	369
TRUE		
## 3	mesenchyme homeo box 2	432
TRUE		
## 4	homeobox D3	268
FALSE		
## 5	zinc finger 398 isoform b	671
TRUE		

```
## 6                                pannexin 1                500
TRUE
##   CPG_ISLAND_LOCATIONS MIR_CPG_ISLAND MIR_NAMES V34 V35
## 1 16:28797486-28797825                NA  NA
## 2  3:57716811-57718675                NA  NA
## 3  7:15691512-15693551                NA  NA
## 4                                NA  NA
## 5 7:148453584-148455804                NA  NA
## 6 11:93501124-93502564                NA  NA
```

```
str(Infinium27)
```

```
## 'data.frame':   27723 obs. of  35 variables:
```

```
## $ IlmnID           : Factor w/ 27723 levels
"[Controls]","1090068",...: 146 147 148 149 150 151 152 153 154 155 ...
## $ Name             : Factor w/ 27594 levels "", "Background", ...
4 5 6 7 8 9 10 11 12 13 ...
## $ IlmnStrand       : Factor w/ 19 levels "", "Aqua", "Beige", ...
18 18 18 7 7 18 18 18 7 7 ...
## $ AddressA_ID      : Factor w/ 27723 levels
"", "10008", "10014", ...: 27519 23075 25139 15854 24567 17440 25428 24075
13227 15407 ...
## $ AlleleA_ProbeSeq : Factor w/ 27554 levels
"", "AAAAAAAAAAAAAAAAAACCCACAAAAAATCCCCACTCTAAAACCCACACA", ...: 3752 8322
7998 23233 19861 17649 19785 15295 22091 5098 ...
## $ AddressB_ID      : int  6660678 6100343 7150392 1260113
4280093 6860356 2900438 3780392 6380376 1580192 ...
## $ AlleleB_ProbeSeq : Factor w/ 27556 levels
"", "AAAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAAACCCCGCG", ...: 3343 7801
7495 23235 19165 15929 18859 14256 22063 4784 ...
## $ GenomeBuild      : int  36 36 36 36 36 36 36 36 36 36 ...
## $ Chr              : Factor w/ 25 levels "", "1", "10", "11", ...: 9
17 21 13 21 4 7 11 7 18 ...
## $ MapInfo          : int  28797601 57718583 15692387 176737319
148453770 93502242 92883530 11970953 88360674 1135216 ...
## $ Ploidy           : Factor w/ 2 levels "", "diploid": 2 2 2 2 2
2 2 2 2 2 ...
## $ Species          : Factor w/ 2 levels "", "Homo sapiens": 2 2 2
2 2 2 2 2 2 2 ...
## $ Source           : Factor w/ 2 levels "", "NCBI:RefSeq": 2 2 2
2 2 2 2 2 2 2 ...
## $ SourceVersion    : num  36.1 36.1 36.1 36.1 36.1 36.1 36.1
36.1 36.1 36.1 ...
## $ SourceStrand     : Factor w/ 3 levels "", "BOT", "TOP": 3 3 2 2
2 2 2 2 3 2 ...
```

```

## $ SourceSeq          : Factor w/ 27561 levels
"", "AAAAAACAAATCCAGCGAGGCCAGACAGGATAGGGGCTCCCTGGGGTCCG", ...: 13494 10681
22912 24155 18384 4502 18867 3945 10079 720 ...
## $ TopGenomicSeq      : Factor w/ 27572 levels
"", "AAAAAAAAAAAAAAAAAAAAAAAAAAGGATCAGGCAGCAGACAAGCTAGGAGGCGATG[CG]GCC
ACCGCATTTTAAAGGCCCTCTGGGGACTCCCGCAGAGCTGGCAAAATTTTGTTCAGT", ...: 25638
9256 20453 6684 13942 3513 10524 16966 17248 15372 ...
## $ Next_Base          : Factor w/ 4 levels "", "A", "C", "T": 4 4 4 3
3 4 3 3 3 3 ...
## $ Color_Channel      : Factor w/ 3 levels "", "Grn", "Red": 3 3 3 2
2 3 2 2 2 2 ...
## $ TSS_Coordinate     : int 28797310 57718214 15692819 176737051
148454441 93501742 92883290 11971455 88360731 1135160 ...
## $ Gene_Strand        : Factor w/ 3 levels "", "-", "+": 3 3 2 3 3 3
3 3 3 3 ...
## $ Gene_ID            : Factor w/ 14496 levels "", "GeneID:1", ...:
7222 11562 6842 5728 9572 4071 5932 6149 1492 6638 ...
## $ Symbol             : Factor w/ 14477 levels "", "7A5", "A1BG", ...:
881 11921 7887 6063 14277 9286 2966 6422 13389 4790 ...
## $ Synonym            : Factor w/ 12655 levels "", "(ALPHA)II-
SPECTRIN;", ...: 845 11268 4991 5937 9529 8641 2235 1 1039 1 ...
## $ Accession          : Factor w/ 14529 levels "", "NM_000014.4", ...:
13031 5970 5040 5810 9201 7364 13822 6649 13949 14375 ...
## $ GID                : Factor w/ 14475 levels "", "GI:10047089", ...:
10364 12005 11929 3723 4203 7850 6235 13814 11565 10219 ...
## $ Annotation         : Factor w/ 12894 levels "", "-14 gene;
go_component: cellular component unknown; go_function: molecular
function unknown; go_function: molec"| __truncated__, ...: 7336 10347
4868 5179 7680 5379 1560 5381 7484 1 ...
## $ Product            : Factor w/ 14457 levels "", "1-acylglycerol-
3-phosphate O-acyltransferase 1", ...: 954 11246 7993 4935 14025 9257
2609 6789 12743 5929 ...
## $ Distance_to_TSS    : int 291 369 432 268 671 500 240 502 57 56
...
## $ CPG_ISLAND         : logi TRUE TRUE TRUE FALSE TRUE TRUE ...
## $ CPG_ISLAND_LOCATIONS: Factor w/ 11471 levels "", "1:100087875-
100088785", ...: 3947 8063 9817 1 9776 2230 3410 4999 3387 1 ...
## $ MIR_CPG_ISLAND     : Factor w/ 111 levels "", "1:148473407-
148474609", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ MIR_NAMES          : Factor w/ 75 levels
"", "HS_104;", "HS_107;", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ V34                : logi NA NA NA NA NA NA ...
## $ V35                : logi NA NA NA NA NA NA ...

```

How many probes are there in each chromosome? We can proceed in two ways: 1) We subset the dataframe for each chromosome

```
chr1 <- Infinium27[Infinium27$Chr=="1",]
```

```
dim(chr1)
```

```
## [1] 2904 35
```

```
chr2 <- Infinium27[Infinium27$Chr=="2",]
```

```
dim(chr2)
```

```
## [1] 1712 35
```

2. We use the table function

```
table(Infinium27$Chr)
```

```
##
```

```
##      1  10  11  12  13  14  15  16  17  18  19  2
20  21  22
## 145 2904 1045 1735 1529 493 833 829 1198 1586 395 1905 1712
884 315 645
## 3 4 5 6 7 8 9 X Y
## 1523 1028 1159 1490 1260 942 1076 1085 7
```

Are there missing values?

```
str(Infinium27)
```

```
## 'data.frame': 27723 obs. of 35 variables:
```

```
## $ IlmnID : Factor w/ 27723 levels
"[Controls]","1090068",...: 146 147 148 149 150 151 152 153 154 155 ...
## $ Name : Factor w/ 27594 levels "", "Background", ...:
4 5 6 7 8 9 10 11 12 13 ...
## $ IlmnStrand : Factor w/ 19 levels "", "Aqua", "Beige", ...:
18 18 18 7 7 18 18 18 7 7 ...
## $ AddressA_ID : Factor w/ 27723 levels
"", "10008", "10014", ...: 27519 23075 25139 15854 24567 17440 25428 24075
13227 15407 ...
## $ AlleleA_ProbeSeq : Factor w/ 27554 levels
"", "AAAAAAAAAAAAAAAAAACACAAAAAATCCCCACTCTAAAACCCACACA", ...: 3752 8322
7998 23233 19861 17649 19785 15295 22091 5098 ...
## $ AddressB_ID : int 6660678 6100343 7150392 1260113
4280093 6860356 2900438 3780392 6380376 1580192 ...
## $ AlleleB_ProbeSeq : Factor w/ 27556 levels
"", "AAAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAAACCCCGCG", ...: 3343 7801
7495 23235 19165 15929 18859 14256 22063 4784 ...
```

```

## $ GenomeBuild      : int  36 36 36 36 36 36 36 36 36 36 ...
## $ Chr              : Factor w/ 25 levels "", "1", "10", "11", ...: 9
17 21 13 21 4 7 11 7 18 ...
## $ MapInfo          : int  28797601 57718583 15692387 176737319
148453770 93502242 92883530 11970953 88360674 1135216 ...
## $ Ploidy           : Factor w/ 2 levels "", "diploid": 2 2 2 2 2
2 2 2 2 2 ...
## $ Species          : Factor w/ 2 levels "", "Homo sapiens": 2 2 2
2 2 2 2 2 2 2 ...
## $ Source           : Factor w/ 2 levels "", "NCBI:RefSeq": 2 2 2
2 2 2 2 2 2 2 ...
## $ SourceVersion    : num  36.1 36.1 36.1 36.1 36.1 36.1 36.1
36.1 36.1 36.1 ...
## $ SourceStrand     : Factor w/ 3 levels "", "BOT", "TOP": 3 3 2 2
2 2 2 2 3 2 ...
## $ SourceSeq        : Factor w/ 27561 levels
"", "AAAAAACAAATCCAGCGAGGCCAGACAGGATAGGGGCTCCCTGGGGTCCG", ...: 13494 10681
22912 24155 18384 4502 18867 3945 10079 720 ...
## $ TopGenomicSeq    : Factor w/ 27572 levels
"", "AAAAAAAAAAAAAAAAAAAAAAAAAAGGATCAGGCAGCAGACAAGCTAGGAGGCGATG[CG]GCC
ACCGCATTTAAAGGCCCTCTGGGGACTCCCGCAGAGCTGGCAAATTTTGTTCAGT", ...: 25638
9256 20453 6684 13942 3513 10524 16966 17248 15372 ...
## $ Next_Base       : Factor w/ 4 levels "", "A", "C", "T": 4 4 4 3
3 4 3 3 3 3 ...
## $ Color_Channel    : Factor w/ 3 levels "", "Grn", "Red": 3 3 3 2
2 3 2 2 2 2 ...
## $ TSS_Coordinate   : int  28797310 57718214 15692819 176737051
148454441 93501742 92883290 11971455 88360731 1135160 ...
## $ Gene_Strand      : Factor w/ 3 levels "", "-", "+": 3 3 2 3 3 3
3 3 3 3 ...
## $ Gene_ID          : Factor w/ 14496 levels "", "GeneID:1", ...:
7222 11562 6842 5728 9572 4071 5932 6149 1492 6638 ...
## $ Symbol           : Factor w/ 14477 levels "", "7A5", "A1BG", ...:
881 11921 7887 6063 14277 9286 2966 6422 13389 4790 ...
## $ Synonym          : Factor w/ 12655 levels "", "(ALPHA)II-
SPECTRIN;", ...: 845 11268 4991 5937 9529 8641 2235 1 1039 1 ...
## $ Accession        : Factor w/ 14529 levels "", "NM_000014.4", ...:
13031 5970 5040 5810 9201 7364 13822 6649 13949 14375 ...
## $ GID              : Factor w/ 14475 levels "", "GI:10047089", ...:
10364 12005 11929 3723 4203 7850 6235 13814 11565 10219 ...
## $ Annotation       : Factor w/ 12894 levels "", "-14 gene;
go_component: cellular component unknown; go_function: molecular
function unknown; go_function: molec"| __truncated__, ...: 7336 10347
4868 5179 7680 5379 1560 5381 7484 1 ...

```

```
## $ Product          : Factor w/ 14457 levels "", "1-acylglycerol-
3-phosphate O-acyltransferase 1",...: 954 11246 7993 4935 14025 9257
2609 6789 12743 5929 ...
## $ Distance_to_TSS   : int    291 369 432 268 671 500 240 502 57 56
...
## $ CPG_ISLAND        : logi    TRUE TRUE TRUE FALSE TRUE TRUE ...
## $ CPG_ISLAND_LOCATIONS: Factor w/ 11471 levels "", "1:100087875-
100087875",...: 3947 8063 9817 1 9776 2230 3410 4999 3387 1 ...
## $ MIR_CPG_ISLAND    : Factor w/ 111 levels "", "1:148473407-
148474609",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ MIR_NAMES         : Factor w/ 75 levels
"", "HS_104;", "HS_107;",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ V34               : logi    NA NA NA NA NA NA ...
## $ V35               : logi    NA NA NA NA NA NA ...
```

```
TSS_NA <- Infinium27[is.na(Infinium27$TSS_Coordinate),]
```

```
dim(TSS_NA)
```

```
## [1] 767 35
```

```
head(TSS_NA)
```

##	IlmnID	Name	IlmnStrand	AddressA_ID	
## 68	cg00059930	cg00059930	BOT	1980551	
## 72	cg00061629	cg00061629	TOP	2230014	
## 100	cg00090147	cg00090147	TOP	3800398	
## 127	cg00117172	cg00117172	TOP	2450594	
## 141	cg00129774	cg00129774	TOP	6590626	
## 228	cg00210842	cg00210842	TOP	4890524	
##					AlleleA_ProbeSeq AddressB_ID
## 68	ACAATACAACTACTCCTTATACACAATAAACTTAACAACTAACTTAACA				1050551
## 72	ACAAATTAAAATCAAATAAAAAAACCTCTTTATTAAAATAACAAAACA				5360014
## 100	AATACAAAAAACTACTAAATAACAAAATATTCATACCCTTTCTAACCA				2350040
## 127	CAAAAATAACTTTTAACAAAATTACCCCACTCCACCTAAATTTTAAACA				2370673
## 141	AAACAAATTTAATTCTTAAAACTAAACAAATTTCTTCCCAATAAAACA				4880647
## 228	AAAAATCAACTTACCCAAACCTTTACACCTAAATCTTAATAAAAAAAACA				6980593
##					AlleleB_ProbeSeq GenomeBuild
Chr	MapInfo				
## 68	ACAATACGAACTACTCCTTATACACGATAAACTTAACAACTAACTTAACG				36
13	47792383				
## 72	ACGAATTAAAATCAAATAAAAAAACCGCTCTTTATTAAAATAACAAAACG				36
11	44281834				
## 100	AATACGAAAAAACTACTAAATAACGAACTATTCATACCCTTTCTAACCG				36
8	11599776				
## 127	CAAAAATAACTTTTAACGAAATTACCCCGCTCCACCTAAATTTTAAACG				36
1	25128425				

141 AAACAAATTTAATTCTTAAAACTAAACAAATTTCTTCCCCGATAAAACG 36
12 54646681

228 AAAAATCAACTTACCCAAACCTTTACGCCTAAATCTTAATAAAAAAACG 36
12 97563709

##	Ploidy	Species	Source	SourceVersion	SourceStrand
## 68	diploid	Homo sapiens	NCBI:RefSeq	36.1	TOP
## 72	diploid	Homo sapiens	NCBI:RefSeq	36.1	BOT
## 100	diploid	Homo sapiens	NCBI:RefSeq	36.1	BOT
## 127	diploid	Homo sapiens	NCBI:RefSeq	36.1	TOP
## 141	diploid	Homo sapiens	NCBI:RefSeq	36.1	TOP
## 228	diploid	Homo sapiens	NCBI:RefSeq	36.1	TOP

SourceSeq

68 CGCCAAGCCAGCTGCCAAGCCCACCGTGCACAAGGAGCAGCCCGCATTGC

72 GCGGGTTGGGGTCAGGTGGAGAAGCCGCTCTTTGTTAAGGTGACAGAACG

100 GGTGCGGAGAGGCTGCTGAGTGGCGGAACATATTCATGCCCTTTCTGGCCG

127 CGCCCCAAAACCCAGGTGGAGCGGGGCAACCCCGTTAAAAGTCATTCTCTG

141 CGCCCCACCGGGGAAGAACTTGTCTAGCCCCCAAGAACCAAACCTGCCT

228 CGCCCTCCCCACTAAGACCTAGGCGCAAAGGCTTGGGTAAGTTGACCTCC

##

TopGenomicSeq

68

CAGCGGGGTTGCTCGGGGCACCAACCGATGAGCGCGCTGCACAGAGGGGCCACAGGGCAGA [CG] CCAAGCC
AGCTGCCAAGCCCACCGTGCACAAGGAGCAGCCCGCATTGCCTGCCTGTGGTC

72

GCGCTGGCTGGTGCGGGTTGGGGTCAGGTGGAGAAGCCGCTCTTTGTTAAGGTGACAGAA [CG] TGCTGGG
GGTGGGGGCCGGGGCCAGGGCCGGTGCAACTAGGGGGCCGCTGCCCTTTCTCTG

100

GAGCCGCTCTGAGGTGCGGAGAGGCTGCTGAGTGGCGGAACATATTCATGCCCTTTCTGGC [CG] GCCTCCT
CGCCCTCGGGGCTGGGGTCCAGGGACTGAATGCTCCTCTGGAAGCTCACCACC

127

CGGGTTAGGGGGGCGCAAAACCCCATCCGCCATTTCCGCACCAACGTCTCTACGCAAGG [CG] CCCCAAA
ACCCAGGTGGAGCGGGGCAACCCCGTTAAAAGTCATTCTGCAGGGCGCATCC

141

GACTAGAGACATAGGTAGGAACTTGGACCCAAAGCAGGTACTTGGGAAGAGTGTTTCAGC [CG] CCCCACC
GGGGAAGAACTTGTCTAGCCCCCAAGAACCAAACCTTGCTGGGGCGGGATCA

228

AACTCCTGTTGCCGTTTCCCTCCACCGGCCTGGAGTCTCCCAGTCTTGTCCTGGCAGTGC [CG] CCCTCCC
CACTAAGACCTAGGCGCAAAGGCTTGGGTAAGTTGACCTCCTCGCTTTTCTCC

##	Next_Base	Color_Channel	TSS_Coordinate	Gene_Strand	Gene_ID
----	-----------	---------------	----------------	-------------	---------

## 68	T	Red	NA	+	GeneID:5925
-------	---	-----	----	---	-------------

RB1

## 72	T	Red	NA	-	GeneID:60529
-------	---	-----	----	---	--------------

ALX4

## 100	A	Red	NA	+	GeneID:2626
--------	---	-----	----	---	-------------

GATA4

## 127	C	Grn	NA	-	GeneID:864
RUNX3					
## 141	A	Red	NA	+	GeneID:1017
CDK2					
## 228	A	Red	NA	+	GeneID:317
APAF1					

##		Synonym	Accession	GID
## 68		RB; OSRC;	NT_024524.13	
## 72	FPP; PFM; PFM1; PFM2;	KIAA1788;	NT_009237.17	
## 100		MGC126629;	NT_077531.3	
## 127	AML2; CBFA3; PEBP2aC;	NT_004610.18		
## 141		p33 (CDK2);	NT_029419.11	
## 228		NT_019546.15		

Annotation

68 retinoblastoma-1; retinoblastoma susceptibility protein; retinoblastoma susceptibility protein; go_component: nucleus; go_component: nucleus; go_component: chromatin; go_function: protein binding; go_function: protein binding; go_function: androgen receptor binding; go_function: transcription factor activity; go_function: transcription coactivator activity; go_process: M phase; go_process: G1 phase; go_process: transcription; go_process: cell cycle checkpoint; go_process: negative regulation of cell growth; go_process: androgen receptor signaling pathway; go_process: regulation of transcription; DNA-dependent; go_process: negative regulation of protein kinase activity; go_process: negative regulation of progression through cell cycle; go_process: positive regulation of transcription; DNA-dependent; go_process: negative regulation of transcription from RNA polymerase II promoter

72 homeodomain transcription factor ALX4; parietal foramina 1; parietal foramina 2; go_component: nucleus; go_function: transcription factor activity; go_process: development; go_process: skeletal development; go_process: skeletal development; go_process: regulation of transcription; DNA-dependent

100 GATA-binding protein 4; go_component: nucleus; go_component: nucleus; go_component: nucleus; go_function: DNA binding; go_function: protein binding; go_function: zinc ion binding; go_function: metal ion binding; go_function: sequence-specific DNA binding; go_function: transcription factor activity; go_function: transcription factor activity; go_function: transcriptional activator activity; go_function: RNA polymerase II transcription factor activity; go_process: transcription; go_process: development; go_process: transcription; go_process: heart development; go_process: positive regulation of transcription; go_process: regulation of transcription; DNA-dependent; go_process:

transcription from RNA polymerase II promoter; go_process: positive regulation of transcription; DNA-dependent

127

isoform 2 is encoded by transcript variant 2; core-binding factor; runt domain; alpha subunit 3; encoded by distal 5' portion of the mRNA; go_component: nucleus; go_function: ATP binding; go_function: transcription factor activity; go_process: transcription; go_process: cell proliferation; go_process: regulation of transcription; DNA-dependent; go_process: transcription from RNA polymerase II promoter

141

isoform 1 is encoded by transcript variant 1; cdc2-related protein kinase; cell division kinase 2; p33 protein kinase; go_component: nucleus; go_component: cytoplasm; go_function: ATP binding; go_function: nucleotide binding; go_function: protein binding; go_function: protein binding; go_function: protein binding; go_function: transferase activity; go_function: protein serine/threonine kinase activity; go_function: cyclin-dependent protein kinase activity; go_process: mitosis; go_process: cell cycle; go_process: cell division; go_process: regulation of DNA replication; go_process: protein amino acid phosphorylation; go_process: G2/M transition of mitotic cell cycle; go_process: positive regulation of cell proliferation; go_process: traversing start control point of mitotic cell cycle

228

isoform e is encoded by transcript variant 5; apoptotic protease activating factor; apoptotic protease activating factor 1; go_component: cytosol; go_component: intracellular; go_function: ATP binding; go_function: protein binding; go_function: protein binding; go_function: nucleotide binding; go_function: caspase activator activity; go_process: regulation of apoptosis; go_process: nervous system development; go_process: caspase activation via cytochrome c

Product Distance_to_TSS

CPG_ISLAND

## 68	retinoblastoma 1	NA
TRUE		
## 72	aristaless-like homeobox 4	NA
TRUE		
## 100	GATA binding protein 4	NA
TRUE		
## 127	runt-related transcription factor 3 isoform 2	NA
TRUE		
## 141	cyclin-dependent kinase 2 isoform 1	NA
TRUE		
## 228	apoptotic protease activating factor isoform e	NA
TRUE		
##	CPG_ISLAND_LOCATIONS MIR_CPG_ISLAND MIR_NAMES V34 V35	

```
## 68 13:47792306-47793951 NA NA
## 72 11:44281312-44281846 NA NA
## 100 8:11598711-11600557 NA NA
## 127 1:25127692-25131906 NA NA
## 141 12:54646626-54647300 NA NA
## 228 12:97561659-97563710 NA NA
```

```
TSS_NA_2 <- Infinium27[Infinium27$TSS_Coordinate=="NA",]
```

```
dim(TSS_NA_2)
```

```
## [1] 767 35
```

How many probes are not associated to a gene?

```
Infinium27_No_gene <- Infinium27[Infinium27$Symbol=="",]
```

```
dim(Infinium27_No_gene)
```

```
## [1] 172 35
```

```
str(Infinium27_No_gene)
```

```
## 'data.frame': 172 obs. of 35 variables:
```

```
## $ IlmnID : Factor w/ 27723 levels
"[Controls]","1090068",...: 160 890 1177 1922 7637 8722 8806 9121 10005
10138 ...
## $ Name : Factor w/ 27594 levels "", "Background", ...
18 748 1035 1780 7495 8580 8664 8979 9863 9996 ...
## $ IlmnStrand : Factor w/ 19 levels "", "Aqua", "Beige", ...: 7
7 7 18 18 7 18 18 7 18 ...
## $ AddressA_ID : Factor w/ 27723 levels
"", "10008", "10014", ...: 1871 9689 10214 1522 44 10395 19471 2030 13157
17296 ...
## $ AlleleA_ProbeSeq : Factor w/ 27554 levels
"", "AAAAAAAAAAAAAAAAAACCCACAAAAAATCCCCACTCTAAAACCCACACA", ...: 18049 15152
14421 22125 15351 21894 8387 2999 9046 2763 ...
## $ AddressB_ID : int 670594 3850184 4490653 4880523 6280097
1170152 5870441 580215 870632 6370543 ...
## $ AlleleB_ProbeSeq : Factor w/ 27556 levels
"", "AAAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAAACCCCGCG", ...: 16222 14157
13528 22038 14286 21896 7862 2799 8484 2600 ...
## $ GenomeBuild : int 36 36 36 36 36 36 36 36 36 36 ...
## $ Chr : Factor w/ 25 levels "", "1", "10", "11", ...: 16
13 4 7 21 22 16 16 22 16 ...
## $ MapInfo : int 20710839 88996387 18239903 105211235
38374631 81719991 21316667 21316350 81720205 21303200 ...
```

```

## $ Ploidy : Factor w/ 2 levels "", "diploid": 2 2 2 2 2
2 2 2 2 2 ...
## $ Species : Factor w/ 2 levels "", "Homo sapiens": 2 2 2
2 2 2 2 2 2 2 ...
## $ Source : Factor w/ 2 levels "", "NCBI:RefSeq": 2 2 2
2 2 2 2 2 2 2 ...
## $ SourceVersion : num 36.1 36.1 36.1 36.1 36.1 36.1 36.1
36.1 36.1 36.1 ...
## $ SourceStrand : Factor w/ 3 levels "", "BOT", "TOP": 3 3 2 3
3 2 3 3 2 3 ...
## $ SourceSeq : Factor w/ 27561 levels
"", "AAAAAACAATCCAGCGAGGCCAGACAGGATAGGGGCTCCCTGGGGTCCG", ...: 16142 16165
3375 16608 7026 23944 7000 16403 934 15356 ...
## $ TopGenomicSeq : Factor w/ 27572 levels
"", "AAAAAAAAAAAAAAAAAAAAAAAAAAGGATCAGGCAGCAGACAAGCTAGGAGGCGATG[CG]GCC
ACCGCATTTAAAGGCCCTCTGGGGACTCCCGCAGAGCTGGCAAATTTTGTCACT", ...: 22858
8551 24369 5591 26966 27047 6714 8702 27443 16700 ...
## $ Next_Base : Factor w/ 4 levels "", "A", "C", "T": 3 4 3 4
4 3 4 4 4 3 ...
## $ Color_Channel : Factor w/ 3 levels "", "Grn", "Red": 2 3 2 3
3 2 3 3 3 2 ...
## $ TSS_Coordinate : int 20710659 88996309 18240008 105211138
38374181 81719970 21316556 21316556 81719970 21302536 ...
## $ Gene_Strand : Factor w/ 3 levels "", "-", "+": 3 2 3 2 2 2
2 2 2 3 ...
## $ Gene_ID : Factor w/ 14496 levels "", "GeneID:1", ...:
6081 5242 2081 3053 11006 1906 4299 4299 1906 10044 ...
## $ Symbol : Factor w/ 14477 levels "", "7A5", "A1BG", ...:
1 1 1 1 1 1 1 1 1 1 ...
## $ Synonym : Factor w/ 12655 levels "", "(ALPHA)II-
SPECTRIN;", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Accession : Factor w/ 14529 levels "", "NM_000014.4", ...:
1 1 1 1 1 1 1 1 1 1 ...
## $ GID : Factor w/ 14475 levels "", "GI:10047089", ...:
1 1 1 1 1 1 1 1 1 1 ...
## $ Annotation : Factor w/ 12894 levels "", "-14 gene;
go_component: cellular component unknown; go_function: molecular
function unknown; go_function: molec"| __truncated__, ...: 1 1 1 1 1 1 1
1 1 1 ...
## $ Product : Factor w/ 14457 levels "", "1-acylglycerol-
3-phosphate O-acyltransferase 1", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Distance_to_TSS : int 180 78 105 97 450 21 111 206 235 664
...
## $ CPG_ISLAND : logi FALSE FALSE TRUE FALSE FALSE FALSE
...

```

```
## $ CPG_ISLAND_LOCATIONS: Factor w/ 11471 levels "", "1:100087875-
100088785",...: 1 1 1762 1 1 1 1 1 1 ...
## $ MIR_CPG_ISLAND      : Factor w/ 111 levels "", "1:148473407-
148474609",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ MIR_NAMES           : Factor w/ 75 levels
"", "HS_104;", "HS_107;",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ V34                 : logi  NA NA NA NA NA NA ...
## $ V35                 : logi  NA NA NA NA NA NA ...
```

```
Infinium27_No_gene <- droplevels(Infinium27_No_gene)
```

```
str(Infinium27_No_gene)
```

```
## 'data.frame':    172 obs. of  35 variables:
```

```
## $ IlmnID              : Factor w/ 172 levels
"[Controls]", "1090068",...: 146 147 148 149 150 151 152 153 154 155 ...
## $ Name                : Factor w/ 43 levels "", "Background",...: 4 5
6 7 8 9 10 11 12 13 ...
## $ IlmnStrand          : Factor w/ 19 levels "", "Aqua", "Beige",...: 7
7 7 18 18 7 18 18 7 18 ...
## $ AddressA_ID         : Factor w/ 172 levels
"", "1010193", "1010735",...: 6 11 12 5 2 13 25 7 18 23 ...
## $ AlleleA_ProbeSeq     : Factor w/ 27 levels
"", "AAAAATAATACCATATAACCAATATAAAACAAAACAACTAAACCCAAAACA",...: 20 15 14 23
17 21 7 4 9 3 ...
## $ AddressB_ID         : int   670594 3850184 4490653 4880523 6280097
1170152 5870441 580215 870632 6370543 ...
## $ AlleleB_ProbeSeq     : Factor w/ 27 levels
"", "AAAAATAATACCATATAACCGCGATATAAAACAAAACAACTAAACCCAAAACG",...: 20 15 14 23
17 21 7 4 9 3 ...
## $ GenomeBuild         : int   36 36 36 36 36 36 36 36 36 36 ...
## $ Chr                 : Factor w/ 11 levels "", "11", "14", "15",...: 6
5 2 3 8 9 6 6 9 6 ...
## $ MapInfo             : int   20710839 88996387 18239903 105211235
38374631 81719991 21316667 21316350 81720205 21303200 ...
## $ Ploidy              : Factor w/ 2 levels "", "diploid": 2 2 2 2 2
2 2 2 2 2 ...
## $ Species             : Factor w/ 2 levels "", "Homo sapiens": 2 2 2
2 2 2 2 2 2 2 ...
## $ Source              : Factor w/ 2 levels "", "NCBI:RefSeq": 2 2 2
2 2 2 2 2 2 2 ...
## $ SourceVersion       : num   36.1 36.1 36.1 36.1 36.1 36.1 36.1
36.1 36.1 36.1 ...
## $ SourceStrand        : Factor w/ 3 levels "", "BOT", "TOP": 3 3 2 3
3 2 3 3 2 3 ...
```

```
## $ SourceSeq          : Factor w/ 28 levels
"", "AATTCAGGCAGAGACACACATCCGAACTGTGTCATGTGCATTCCTTTACG", ...: 17 18 6 20
12 23 11 19 2 16 ...
## $ TopGenomicSeq      : Factor w/ 28 levels
"", "AAGACAGCTTGTCAGGGCCCCTGAGGAGCAGCCTCTAGTTCAGCTCTCAGAGATAGGGTC[CG]GGG
CTCCTTTGCTGTGGATCCTCCTGCTCTGGGCTCCAAGTGAGGGGTATAGGGAAACA", ...: 20 11 21
6 26 27 8 12 28 16 ...
## $ Next_Base          : Factor w/ 4 levels "", "A", "C", "T": 3 4 3 4
4 3 4 4 4 3 ...
## $ Color_Channel      : Factor w/ 3 levels "", "Grn", "Red": 2 3 2 3
3 2 3 3 3 2 ...
## $ TSS_Coordinate     : int   20710659 88996309 18240008 105211138
38374181 81719970 21316556 21316556 81719970 21302536 ...
## $ Gene_Strand        : Factor w/ 3 levels "", "-", "+": 3 2 3 2 2 2
2 2 2 3 ...
## $ Gene_ID            : Factor w/ 18 levels "", "GeneID:11191", ...:
11 7 4 5 18 3 6 6 3 15 ...
## $ Symbol             : Factor w/ 1 level "": 1 1 1 1 1 1 1 1 1 1
...
## $ Synonym            : Factor w/ 1 level "": 1 1 1 1 1 1 1 1 1 1
...
## $ Accession          : Factor w/ 1 level "": 1 1 1 1 1 1 1 1 1 1
...
## $ GID                : Factor w/ 1 level "": 1 1 1 1 1 1 1 1 1 1
...
## $ Annotation         : Factor w/ 1 level "": 1 1 1 1 1 1 1 1 1 1
...
## $ Product            : Factor w/ 1 level "": 1 1 1 1 1 1 1 1 1 1
...
## $ Distance_to_TSS    : int   180 78 105 97 450 21 111 206 235 664
...
## $ CPG_ISLAND         : logi   FALSE FALSE TRUE FALSE FALSE FALSE
...
## $ CPG_ISLAND_LOCATIONS: Factor w/ 6 levels "", "11:18239883-
18240212", ...: 1 1 2 1 1 1 1 1 1 1 ...
## $ MIR_CPG_ISLAND     : Factor w/ 1 level "": 1 1 1 1 1 1 1 1 1 1
...
## $ MIR_NAMES          : Factor w/ 1 level "": 1 1 1 1 1 1 1 1 1 1
...
## $ V34                : logi   NA NA NA NA NA NA ...
## $ V35                : logi   NA NA NA NA NA NA ...
```

How many genes there are on each chromosome?

```
str(Infinium27$Gene_ID)
```

```
## Factor w/ 14496 levels "", "GeneID:1", ...: 7222 11562 6842 5728 9572
4071 5932 6149 1492 6638 ...
```

```
Infinium27_gene <- Infinium27[Infinium27$Gene_ID!="",]
```

```
str(Infinium27_gene)
```

```
## 'data.frame': 27578 obs. of 35 variables:
```

```
## $ IlmnID : Factor w/ 27723 levels
"[Controls]", "1090068", ...: 146 147 148 149 150 151 152 153 154 155 ...
## $ Name : Factor w/ 27594 levels "", "Background", ...:
4 5 6 7 8 9 10 11 12 13 ...
## $ IlmnStrand : Factor w/ 19 levels "", "Aqua", "Beige", ...:
18 18 18 7 7 18 18 18 7 7 ...
## $ AddressA_ID : Factor w/ 27723 levels
"", "10008", "10014", ...: 27519 23075 25139 15854 24567 17440 25428 24075
13227 15407 ...
## $ AlleleA_ProbeSeq : Factor w/ 27554 levels
"", "AAAAAAAAAAAAAAAAAACCCACAAAAAATCCCCACTCTAAAACCCCA", ...: 3752 8322
7998 23233 19861 17649 19785 15295 22091 5098 ...
## $ AddressB_ID : int 6660678 6100343 7150392 1260113
4280093 6860356 2900438 3780392 6380376 1580192 ...
## $ AlleleB_ProbeSeq : Factor w/ 27556 levels
"", "AAAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAAACCCCGCG", ...: 3343 7801
7495 23235 19165 15929 18859 14256 22063 4784 ...
## $ GenomeBuild : int 36 36 36 36 36 36 36 36 36 36 ...
## $ Chr : Factor w/ 25 levels "", "1", "10", "11", ...: 9
17 21 13 21 4 7 11 7 18 ...
## $ MapInfo : int 28797601 57718583 15692387 176737319
148453770 93502242 92883530 11970953 88360674 1135216 ...
## $ Ploidy : Factor w/ 2 levels "", "diploid": 2 2 2 2 2
2 2 2 2 2 ...
## $ Species : Factor w/ 2 levels "", "Homo sapiens": 2 2 2
2 2 2 2 2 2 2 ...
## $ Source : Factor w/ 2 levels "", "NCBI:RefSeq": 2 2 2
2 2 2 2 2 2 2 ...
## $ SourceVersion : num 36.1 36.1 36.1 36.1 36.1 36.1 36.1
36.1 36.1 36.1 ...
## $ SourceStrand : Factor w/ 3 levels "", "BOT", "TOP": 3 3 2 2
2 2 2 2 3 2 ...
## $ SourceSeq : Factor w/ 27561 levels
"", "AAAAAACAAATCCAGCGAGGCCAGACAGGATAGGGGCTCCCTGGGGTCCG", ...: 13494 10681
22912 24155 18384 4502 18867 3945 10079 720 ...
## $ TopGenomicSeq : Factor w/ 27572 levels
"", "AAAAAAAAAAAAAAAAAAAAAAAAAAGGATCAGGCAGCAGACAAGCTAGGAGGCGATG[CG]GCC
```

```

ACCGCATTTTAAAAGGCCCTCTGGGGACTCCCGCAGAGCTGGCAAAATTTTGTTCAGT",...: 25638
9256 20453 6684 13942 3513 10524 16966 17248 15372 ...
## $ Next_Base          : Factor w/ 4 levels "", "A", "C", "T": 4 4 4 3
3 4 3 3 3 3 ...
## $ Color_Channel      : Factor w/ 3 levels "", "Grn", "Red": 3 3 3 2
2 3 2 2 2 2 ...
## $ TSS_Coordinate     : int   28797310 57718214 15692819 176737051
148454441 93501742 92883290 11971455 88360731 1135160 ...
## $ Gene_Strand        : Factor w/ 3 levels "", "-", "+": 3 3 2 3 3 3
3 3 3 3 ...
## $ Gene_ID            : Factor w/ 14496 levels "", "GeneID:1",...:
7222 11562 6842 5728 9572 4071 5932 6149 1492 6638 ...
## $ Symbol             : Factor w/ 14477 levels "", "7A5", "A1BG",...:
881 11921 7887 6063 14277 9286 2966 6422 13389 4790 ...
## $ Synonym            : Factor w/ 12655 levels "", "(ALPHA)II-
SPECTRIN;",...: 845 11268 4991 5937 9529 8641 2235 1 1039 1 ...
## $ Accession          : Factor w/ 14529 levels "", "NM_000014.4",...:
13031 5970 5040 5810 9201 7364 13822 6649 13949 14375 ...
## $ GID                : Factor w/ 14475 levels "", "GI:10047089",...:
10364 12005 11929 3723 4203 7850 6235 13814 11565 10219 ...
## $ Annotation         : Factor w/ 12894 levels "", "-14 gene;
go_component: cellular component unknown; go_function: molecular
function unknown; go_function: molec"| __truncated__,...: 7336 10347
4868 5179 7680 5379 1560 5381 7484 1 ...
## $ Product            : Factor w/ 14457 levels "", "1-acylglycerol-
3-phosphate O-acyltransferase 1",...: 954 11246 7993 4935 14025 9257
2609 6789 12743 5929 ...
## $ Distance_to_TSS    : int    291 369 432 268 671 500 240 502 57 56
...
## $ CPG_ISLAND         : logi   TRUE TRUE TRUE FALSE TRUE TRUE ...
## $ CPG_ISLAND_LOCATIONS: Factor w/ 11471 levels "", "1:100087875-
100088785",...: 3947 8063 9817 1 9776 2230 3410 4999 3387 1 ...
## $ MIR_CPG_ISLAND     : Factor w/ 111 levels "", "1:148473407-
148474609",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ MIR_NAMES          : Factor w/ 75 levels
"", "HS_104;", "HS_107;",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ V34                : logi   NA NA NA NA NA NA ...
## $ V35                : logi   NA NA NA NA NA NA ...

```

```
Infinium27_gene <- droplevels(Infinium27_gene)
```

```
dim(Infinium27_gene)
```

```
## [1] 27578    35
```

```
str(Infinium27_gene)
```



```
## 'data.frame':    27578 obs. of  35 variables:
```

```
## $ IlmnID          : Factor w/ 27578 levels  
"cg00000292","cg00002426",...: 1 2 3 4 5 6 7 8 9 10 ...  
## $ Name            : Factor w/ 27578 levels  
"cg00000292","cg00002426",...: 1 2 3 4 5 6 7 8 9 10 ...  
## $ IlmnStrand       : Factor w/ 2 levels "BOT","TOP": 2 2 2 1 1 2  
2 2 1 1 ...  
## $ AddressA_ID      : Factor w/ 27578 levels "10008","10014",...:  
27518 23074 25138 15853 24566 17439 25427 24074 13226 15406 ...  
## $ AlleleA_ProbeSeq : Factor w/ 27553 levels  
"AAAAAAAAAAAAAAAAAACACAAAAAATCCCCACTCTAAAACCCACACA",...: 3751 8321 7997  
23232 19860 17648 19784 15294 22090 5097 ...  
## $ AddressB_ID      : int  6660678 6100343 7150392 1260113  
4280093 6860356 2900438 3780392 6380376 1580192 ...  
## $ AlleleB_ProbeSeq : Factor w/ 27555 levels  
"AAAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAAACCCCGCG",...: 3342 7800 7494  
23234 19164 15928 18858 14255 22062 4783 ...  
## $ GenomeBuild       : int  36 36 36 36 36 36 36 36 36 36 ...  
## $ Chr               : Factor w/ 24 levels "1","10","11",...: 8 16  
20 12 20 3 6 10 6 17 ...  
## $ MapInfo           : int  28797601 57718583 15692387 176737319  
148453770 93502242 92883530 11970953 88360674 1135216 ...  
## $ Ploidy            : Factor w/ 1 level "diploid": 1 1 1 1 1 1 1  
1 1 1 ...  
## $ Species           : Factor w/ 1 level "Homo sapiens": 1 1 1 1 1  
1 1 1 1 1 ...  
## $ Source            : Factor w/ 1 level "NCBI:RefSeq": 1 1 1 1 1  
1 1 1 1 1 ...  
## $ SourceVersion     : num  36.1 36.1 36.1 36.1 36.1 36.1 36.1  
36.1 36.1 36.1 ...  
## $ SourceStrand      : Factor w/ 2 levels "BOT","TOP": 2 2 1 1 1 1  
1 1 2 1 ...  
## $ SourceSeq         : Factor w/ 27560 levels  
"AAAAAACAAATCCAGCGAGGCCAGACAGGATAGGGGCTCCCTGGGGTCCG",...: 13493 10680  
22911 24154 18383 4501 18866 3944 10078 719 ...  
## $ TopGenomicSeq     : Factor w/ 27571 levels  
"AAAAAAAAAAAAAAAAAAAAAAAAAAGGATCAGGCAGCAGACAAGCTAGGAGGCGATG[CG]GCCACC  
GCATTTAAAAGGCCCTCTGGGGACTCCCGCAGAGCTGGCAAAATTTTGTTCAGT",...: 25637 9255  
20452 6683 13941 3512 10523 16965 17247 15371 ...  
## $ Next_Base         : Factor w/ 3 levels "A","C","T": 3 3 3 2 2 3  
2 2 2 2 ...  
## $ Color_Channel     : Factor w/ 2 levels "Grn","Red": 2 2 2 1 1 2  
1 1 1 1 ...  
## $ TSS_Coordinate    : int  28797310 57718214 15692819 176737051  
148454441 93501742 92883290 11971455 88360731 1135160 ...
```

```
## $ Gene_Strand      : Factor w/ 2 levels "-", "+": 2 2 1 2 2 2 2 2
2 2 ...
## $ Gene_ID          : Factor w/ 14495 levels
"GeneID:1","GeneID:10",...: 7221 11561 6841 5727 9571 4070 5931 6148
1491 6637 ...
## $ Symbol           : Factor w/ 14477 levels "", "7A5", "A1BG",...:
881 11921 7887 6063 14277 9286 2966 6422 13389 4790 ...
## $ Synonym          : Factor w/ 12655 levels "", "(ALPHA)II-
SPECTRIN;",...: 845 11268 4991 5937 9529 8641 2235 1 1039 1 ...
## $ Accession        : Factor w/ 14529 levels "", "NM_000014.4",...:
13031 5970 5040 5810 9201 7364 13822 6649 13949 14375 ...
## $ GID              : Factor w/ 14475 levels "", "GI:10047089",...:
10364 12005 11929 3723 4203 7850 6235 13814 11565 10219 ...
## $ Annotation       : Factor w/ 12894 levels "", "-14 gene;
go_component: cellular component unknown; go_function: molecular
function unknown; go_function: molec"| __truncated__,...: 7336 10347
4868 5179 7680 5379 1560 5381 7484 1 ...
## $ Product          : Factor w/ 14457 levels "", "1-acylglycerol-
3-phosphate O-acyltransferase 1",...: 954 11246 7993 4935 14025 9257
2609 6789 12743 5929 ...
## $ Distance_to_TSS   : int   291 369 432 268 671 500 240 502 57 56
...
## $ CPG_ISLAND       : logi   TRUE TRUE TRUE FALSE TRUE TRUE ...
## $ CPG_ISLAND_LOCATIONS: Factor w/ 11471 levels "", "1:100087875-
100088785",...: 3947 8063 9817 1 9776 2230 3410 4999 3387 1 ...
## $ MIR_CPG_ISLAND    : Factor w/ 111 levels "", "1:148473407-
148474609",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ MIR_NAMES         : Factor w/ 75 levels
"", "HS_104;", "HS_107;",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ V34               : logi   NA NA NA NA NA NA ...
## $ V35               : logi   NA NA NA NA NA NA ...
```

```
summary(Infinium27_gene)
```

##	IlmnID	Name	IlmnStrand	AddressA_ID
##	cg00000292:	1	cg00000292:	1
##	cg00002426:	1	cg00002426:	1
##	cg00003994:	1	cg00003994:	1
##	cg00005847:	1	cg00005847:	1
##	cg00006414:	1	cg00006414:	1
##	cg00007981:	1	cg00007981:	1
##	(Other) :27572	(Other) :27572	(Other):27572	
##			AlleleA_ProbeSeq	
##	AddressB_ID			
##	ACAAAAACAACACTAAATTATTAAAAACCTCTATCTAAATATAAACCCA:	3	Min.	:
	10010			

```

## TTACTAATCTCCAAATTATACATCTTAAAATTCTCCTAAACCATCTTACA: 3 1st
Qu.:1850054
## AAAAACTCAACACATAATTACAAAACCTTAATAATCCACAAAATACCACA: 2 Median
:3850062
## AAAATTATTATTACTATTTCTCCAAACCTAACCAATCCCTCTACCAACCA: 2 Mean
:3774257
## AAACAACACCACTCCATAATAACTCTATTCTTTTTTAATAAACTTTAACA: 2 3rd
Qu.:5860010
## AATCCACTTATACAATCATCCTTTTTTATTTAATAACTTAACTAAACAACA: 2 Max.
:7650754
## (Other) :27564
## AlleleB_ProbeSeq
GenomeBuild
## ACAAAAACAACGCTAAATTATTAAAAACCCCTCTATCTAAAATAAAACCCG: 3 Min.
:36
## TTACTAATCTCCGAATTATACATCTTAAAATTCTCCTAAACCATCTTACG: 3 1st
Qu.:36
## AAAAACTCAACACATAATTACAAAACCTTAATAATCCGCAAAATACCGCG: 2 Median
:36
## AAAATTATTATTACTATTTCTCCAAACCTAACCAATCCCTCTACCAACCG: 2 Mean
:36
## AAACAACACCACTCCGTAATAACTCTATTCTTTTTTAATAAACTTTAACG: 2 3rd
Qu.:36
## AATCCACTTATACAATCATCCTTTTTTATTTAATAACTTAACTAAACAACG: 2 Max.
:36
## (Other) :27564
## Chr MapInfo Ploidy Species
## 1 : 2904 Min. : 16170 diploid:27578 Homo
sapiens:27578
## 19 : 1905 1st Qu.: 31106883
## 11 : 1735 Median : 57606732
## 2 : 1712 Mean : 73241132
## 17 : 1586 3rd Qu.:109901061
## 12 : 1529 Max. :247099901
## (Other):16207
## Source SourceVersion SourceStrand
## NCBI:RefSeq:27578 Min. :36.1 BOT:15166
## 1st Qu.:36.1 TOP:12412
## Median :36.1
## Mean :36.1
## 3rd Qu.:36.1
## Max. :36.1
## SourceSeq
## CGGGCCTCACCCCAGACAGAGGGCCCCCAATAATCCAGCGCTGCCTCTGC: 3
## TTGCTGATCTCCGAGTTGTGCATCTTGGGGTTCTCCTGGGCCATCTTGCG: 3

```

```

## ATGAGCCACTGAGACTCTAATAAGATCCAGTGGAAATAACCAGGCTCTCG: 2
## CGAAGTAAATTATGGGATGTCATCCACGCCTGTCCCAAAGGTACCGAATT: 2
## CGCCTGTTAATAAAGATACATGGATATCTTGGGGCTATGAAAGTGGTAAG: 2
## CGCGGCATCCTGCGGATCATCAAGCCCTGCAACCATGTGCTGAGTCTCTC: 2
## (Other) :27564
##
TopGenomicSeq
##
AAGGTTTCCTAGATGAGCCACTGAGACTCTAATAAGATCCAGTGGAAATAACCAGGCTCT [CG] TCGGAAT
ATAAGTCCCAAGGGAAGCTGTGCCAGTCTTGTGGGCGACTGCCTGACTTCTCC: 2
##
AATGAATGAATGGGGCAGCACCCTCCGTGATGGCTCTATTCTTTTTTAATGGACTTTGG [CG] CCTTGGC
CCACTTTCTTCTTTGACAGCACTGGTCTACTCCAGAGCATTCTCTAGCTA: 2
##
CAGGGTAGTACCGTTGCCAGGATGTGACGCCACTGACTTGCGCATTGGAGGTCAGAAGAC [CG] CGAGATT
CTCGCCCTGAGCAACGAGCGACGGCCTGACGTGGCGGAGGGAAGCCGGCCCA: 2
##
GAGGGAAGCAGGCGCAGGCTCCGTGAGGAGGCAAGGTAAGACGCCGAGGGAGGACTGAGG [CG] GGCCTCA
CCCCAGACAGAGGGCCCCCAATAATCCAGCGCTGCCTCTGCTGCCGGGCCTGG: 2
##
GAGGGAAGCCGGCCCCAGGCTCGGTGAGGAGGCAAGGTAAGACGCTGAGGGAGGACTGAGG [CG] GGCCTCA
CCTCAGACAGAGGGCCTCAAATAATCCAGTGCTGCCTCTGCTGCCGGGCCTGG: 2
##
GGATGGCCGGCTTGGTGATGCCCTGGATATTGTGCGCAGTACTTTACGGTGGCGCTTAG [CG] CCGCCTT
TGCCAAGACCCTTCCCGCCTTTGCCGCGGCCAGACATGACGAGCAAGAGGAGT: 2
## (Other)
:27566
## Next_Base Color_Channel TSS_Coordinate Gene_Strand
Gene_ID
## A: 2655 Grn:15500 Min. : 16351 -:13709 GeneID:2778
: 30
## C:15500 Red:12078 1st Qu.: 31231490 +:13869 GeneID:4255
: 26
## T: 9423 Median : 58068957 GeneID:3784
: 23
## Mean : 73600601
GeneID:60529: 23
## 3rd Qu.:110408930 GeneID:5925
: 21
## Max. :247099153
GeneID:55743: 20
## NA's :622 (Other)
:27435
## Symbol
## GNAS : 30
## : 27

```

```

##  MGMT      :    26
##  ALX4      :    23
##  KCNQ1     :    23
##  RB1       :    21
##  (Other):27428
##
Synonym
##
: 3459
##  XL; AHO; GSA; GSP; POH; XL2; GPSA; NESP; GNAS1; PHP1A; PHP1B;
GNASXL; NESP55; C20orf45; MGC33735; XLalphas; dJ309F20.1.1;
dJ806M20.3.3;:    30
##  FPP; PFM; PFM1; PFM2; KIAA1788;
:    23
##  LQT; RWS; WRS; LQT1; ATFB1; KCNA8; KCNA9; Kv1.9; Kv7.1; KVLQT1;
:    23
##  RB; OSRC;
:    21
##  GUD; WAGR; WT33; WIT-2;
:    20
##  (Other)
:24002
##           Accession           GID
##  NT_009237.17:   108           :   649
##  NT_011362.9 :    39   GI:67003556:    12
##  NT_022517.17:   36   GI:16950656:    10
##  NT_007933.14:   33   GI:50726974:    10
##           :    27   GI:29126235:     9
##  NT_026446.13:   26   GI:40217789:     9
##  (Other)      :27309   (Other)      :26879
##
Annotation
##
: 817
##  go_component: integral to membrane
: 151
##  Derived by automated computational analysis using gene prediction
method: GNOMON.           :    92
##  go_function: binding
:    90
##  go_component: nucleus; go_function: zinc ion binding; go_function:
metal ion binding; go_function: nucleic acid binding:    88
##  go_function: protein binding
:    78
##  (Other)
:26262

```

```
##
Product
##
: 105
## guanine nucleotide binding protein; alpha stimulating activity
polypeptide 1 isoform a: 30
## O-6-methylguanine-DNA methyltransferase
: 26
## aristaless-like homeobox 4
: 23
## retinoblastoma 1
: 21
## checkpoint with forkhead and ring finger domains
: 20
## (Other)
:27353
## Distance_to_TSS CPG_ISLAND CPG_ISLAND_LOCATIONS
## Min. : 0.0 Mode :logical : 7572
## 1st Qu.: 120.0 FALSE:7572 16:30490076-30491410 : 12
## Median : 275.0 TRUE :20006 4:129104666-129106616: 11
## Mean : 365.1 12:4252826-4255273 : 10
## 3rd Qu.: 514.0 19:14500498-14502147 : 10
## Max. :1499.0 11:73559258-73560445 : 9
## NA's :622 (Other) :19954
## MIR_CPG_ISLAND MIR_NAMES V34
## :27323 :27323 Mode:logical
## 11:73559258-73560445: 7 HS_250; : 10 NA's:27578
## 2:27659123-27659757 : 7 HSA-MIR-639;: 8
## 16:30490076-30491410: 6 HS_135; : 7
## 4:39204770-39206279 : 6 HS_176; : 7
## 16:66613904-66615043: 5 HS_21; : 7
## (Other) : 224 (Other) : 216
## V35
## Mode:logical
## NA's:27578
##
##
##
##
##
```

```
table(Infinium27_gene$Chr)
```

```
##
```

```
## 1 10 11 12 13 14 15 16 17 18 19 2 20
21 22 3
```

```
## 2904 1045 1735 1529 493 833 829 1198 1586 395 1905 1712 884
315 645 1523
## 4 5 6 7 8 9 X Y
## 1028 1159 1490 1260 942 1076 1085 7
```

```
table(Infinium27_gene$Chr, Infinium27_gene$Gene_Strand)
```

```
##
```

```
##      -      +
## 1 1420 1484
## 10 493 552
## 11 864 871
## 12 794 735
## 13 241 252
## 14 420 413
## 15 422 407
## 16 527 671
## 17 863 723
## 18 187 208
## 19 967 938
## 2 805 907
## 20 406 478
## 21 171 144
## 22 314 331
## 3 777 746
## 4 528 500
## 5 553 606
## 6 760 730
## 7 639 621
## 8 491 451
## 9 567 509
## X 499 586
## Y 1 6
```

```
df_table <- data.frame(table(Infinium27_gene$Chr))
```

```
df_table
```

```
##      Var1 Freq
## 1      1 2904
## 2     10 1045
## 3     11 1735
## 4     12 1529
## 5     13 493
## 6     14 833
```

```
## 7      15  829
## 8      16 1198
## 9      17 1586
## 10     18  395
## 11     19 1905
## 12      2 1712
## 13     20  884
## 14     21  315
## 15     22  645
## 16      3 1523
## 17      4 1028
## 18      5 1159
## 19      6 1490
## 20      7 1260
## 21      8  942
## 22      9 1076
## 23      X 1085
## 24      Y    7
```

```
colnames(df_table) <- c("chr","number of genes")
```

```
df_table
```

```
##      chr number of genes
```

```
## 1      1          2904
## 2     10          1045
## 3     11          1735
## 4     12          1529
## 5     13           493
## 6     14           833
## 7     15           829
## 8     16          1198
## 9     17          1586
## 10    18           395
## 11    19          1905
## 12     2          1712
## 13    20           884
## 14    21           315
## 15    22           645
## 16     3          1523
## 17     4          1028
## 18     5          1159
## 19     6          1490
## 20     7          1260
## 21     8           942
## 22     9          1076
```



```
## 23    X          1085
## 24    Y           7
```

What is the gene with the highest number of probes?

```
gene_table <- table(Infinium27$Symbol)

head(gene_table)
str(gene_table)
gene_table <- data.frame(gene_table)
str(gene_table)
colnames(gene_table) <- c("Gene", "Freq")
head(gene_table)
gene_table <- gene_table[order(gene_table$Freq),]
head(gene_table)
tail(gene_table)
gene_table <- gene_table[order(-gene_table$Freq),]
head(gene_table)
gene_table[gene_table$Freq == max(gene_table$Freq),]
gene_table <- gene_table[!is.na(gene_table$Gene),]
head(gene_table)
gene_table <- gene_table[gene_table$Gene != "",]
head(gene_table)
gene_table <- droplevels(gene_table)
gene_table[gene_table$Freq == max(gene_table$Freq),]
```

CHUNK 9: Exercises:

- What is the chromosome with the highest number of probes?
- What is the chromosome with the highest number of probes with SourceStrand equal to TOP?
- On which chromosomes are the probes not associated to a gene?
- What is the maximum distance of a probe from TSS? And the smallest?
- How many CpG islands are on chromosome 7?

DRD_2025_Lesson_2

First of all, let's clean the workspace and set our the working directory:

```
rm(list=ls())  
setwd("~/Dropbox/DRD_2025/2")
```

CHUNK1: upload the manifest and explore it

The Illumina 450k manifest has been downloaded from this link:

http://support.illumina.com/array/array_kits/infinium_humanmethylation450_beadchip_kit/downloads.html

The csv file is quite large (200 MB). It is a good idea to import in R just the first 20 rows of the file to check that everything is ok. To this aim, we set the nrows argument=20. As this file is a csv, we will use the comma as separator.

```
Illumina450Manifest <- read.table("HumanMethylation450_15017482_v1-  
2.csv", sep=",", header=T, nrows=20)  
  
## Error in scan(file = file, what = what, sep = sep, quote = quote,  
dec = dec, : line 7 did not have 27 elements
```

There are some problems :-)

Let's try to use the readLines() function to check the first lines of the file. You will see that the problem is that the first 7 lines of the file are a sort of header, and that the spreadsheet actually starts from the 8th line. Now we try to import again the file skipping the first 7 lines (argument skip=7). Furthermore, we will set stringsAsFactors = T.

```
readLines("HumanMethylation450_15017482_v1-2.csv", n=20)  
  
## [1] "Illumina, Inc.,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,"  
  
## [2] "[Heading],,,,,,,,,,,,,,,,,,,,,,,,,,,,,,"  
## [3] "Descriptor File Name,BS0010894-  
AQP_content.bpm,,,,,,,,,,,,,,,,,,,,,,,,,,,,,"  
## [4] "Assay Format,Infinium 2,,,,,,,,,,,,,,,,,,,,,,,,,,,,,"  
## [5] "Date Manufactured,6/11/2008,,,,,,,,,,,,,,,,,,,,,,,,,,,,,"  
## [6] "Loci Count ,485553,,,,,,,,,,,,,,,,,,,,,,,,,,,,,"  
## [7] "[Assay],,,,,,,,,,,,,,,,,,,,,,,,,,,,,,"
```

```

## [8]
"IlmnID,Name,AddressA_ID,AlleleA_ProbeSeq,AddressB_ID,AlleleB_ProbeSeq,
Infinium_Design_Type,Next_Base,Color_Channel,Forward_Sequence,Genome_Build,
CHR,MAPINFO,SourceSeq,Chromosome_36,Coordinate_36,Strand,Probe_SNP,
Probe_SNP_10,Random_Loci,Methyl27_Loci,UCSC_RefGene_Name,UCSC_RefGene_Accession,
UCSC_RefGene_Group,UCSC_CpG_Islands_Name,Relation_to_UCSC_CpG_Island,Phantom,DMR,
Enhancer,HMM_Island,Regulatory_Feature_Name,Regulatory_Feature_Group,DHS"
## [9]
"cg00035864,cg00035864,31729416,AAAACACTAACAATCTTATCCACATAAACCCCTTAAATTT
ATCTCAAATTC,, ,II,, ,AATCCAAAGATGATGGAGGAGTGCCCGCTCATGATGTGAAGTACCTGCTCAG
CTGGAAAC[CG]AATTTGAGATAAATTCAAGGGTCTATGTGGACAAGACTGCTAGTGTCTCTCTCTGGATT
G,37,Y,8553009,AGACACTAGCAGTCTTGTCCACATAGACCCTTGAATTTATCTCAAATTCG,Y,861
3009,F,,,,,TTY18,NR_001550,TSS1500,,,,,,,"
## [10]
"cg00050873,cg00050873,32735311,ACAAAAACAACACACAACACTATAATAATTTTAAATA
AATAAACCCCA,31717405,ACGAAAAACAACGCACAACACTATAATAATTTTAAATAAATAAACCCCG
,I,A,Red,TATCTCTGTCTGGCGAGGAGGCAACGCACAACACTGTGGTGGTTTTGGAGTGGGTGGACCC[C
G]GCCAAGACGGCCTGGGCTGACCAGAGACGGGAGGCAGAAAAAGTGGGCAGGTGGTTGCAG,37,Y,936
3356,CGGGGTCCACCCACTCCAAAAACCACCACAGTTGTGCGTTGCCTCCTCGC,Y,9973356,R,,,,
,TSPY4;FAM197Y2,NM_001164471;NR_001553,Body;TSS1500,chrY:9363680-
9363943,N_Shore,,,,Y:9973136-9976273,,,,"
## [11]
"cg00061679,cg00061679,28780415,AAAACATTAAAAAACTAATTCACTACTATTTAATTACTT
TATTTTCCATC,, ,II,, ,TCAACAAATGAGAGACATTGAAGAACTAATTCACTACTATTTGGTTACTTTA
TTTTCCAT[CG]AAGAAAACCTCTTTTTTAAAACTAACACATAAATAAAATGAACGAAGAACAACAACTAAAC
G,37,Y,25314171,CGATGGAAAATAAAGTAACCAATAGTAGTGAATTAGTTCTTCAATGTCT,Y,23
723559,R,,,,,DAZ1;DAZ4;DAZ4,NM_004081;NM_020420;NM_001005375,Body;Body;
Body,,,,,,,"
## [12]
"cg00063477,cg00063477,16712347,TATTCTTCCACACAAAATACTAAACRTATATTTACAAAA
ATACTTCCATC,, ,II,, ,CTCCTGTACTTGTTCATTAAATAATGATTCCTTGATATACCAAGTCTGGAT
AGCGGATT[CG]ATGGAAGCATTTTTGTAAATATACGTTTCAGTATTTTGTGTGGAAGAACAATCTAGCT
G,37,Y,22741795,CGATGGAAGCATTTTTGTAAATATACGTTTCAGTATTTTGTGTGGAAGAAC,Y,21
151183,F,rs9341313,rs13447379,, ,EIF1AY,NM_004681,Body,chrY:22737825-
22738052,S_Shelf,,,,,,,"
## [13]
"cg00121626,cg00121626,19779393,AAAACATAAAAAATAACTTACAAACCAAATACTATACC
CTACAACCTCTC,, ,II,, ,AGGTGAATGAAGAGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCT
ACAACCTCT[CG]CGCTGCCACAAAAAGCCAATTAACGAAGCAAACCTGTTTTATCTTTTCTCTCCCCAGAA
G,37,Y,21664296,AGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACCTCTCG,Y,20
123684,R,,,,,BCORL2,NR_002923,Body,chrY:21664481-
21665063,N_Shore,,,,,,,"
## [14]
"cg00212031,cg00212031,29674443,CCCAATTAACCACAAAACTAAACAAATTATACAATCAA
AAAAACATACA,38703326,CCCAATTAACCGCAAAAACTAAACAAATTATACGATCGAAAAACGTACG

```

, I, T, Red, CCATTGGCCCCCCCCAGTTGGCCGCAGGGACTGAGCAAGTTATGCGGTGCGGAAGACGTG [C
G] TTAAAGGGCTGAAGGGGAGGGACGGAAGTGCAGTCTCTGTGACAGCTCTGAGGTGGGAG, 37, Y, 212
39348, CGCACGTCTTCCCGACCGCATAACTTGCTCAGTCCCTGCGGCCAACTGGG, Y, 19698736, R, ,
,,, TTTY14, NR_001543, TSS200, chrY:21238448-21240005, Island, , , , Y:19697854-
19699393, , , , "

[15]

"cg00213748, cg00213748, 30703409, TTTTAACACCTAACACCATTTTAACAATAAAAAATTCTAC
AAAAAAAACA, 36767301, TTTTAACGCCTAACACCGTTTTAACGATAAAAAATTCTACAAAAAAAACG
, I, A, Red, TCTGTGGGACCATTTTAACGCCTGGCACC GTTTTAACGATGGAGGTCTTGCAGGAGGGGG [C
G] ACCTGGGGTAGGAGGCGTGCTAGTGGTGGATGACATTGTGGCAGAGATGGAGGTGGTGGC, 37, Y, 814
8233, CGCCCCCTCCTGCAGAACCTCCATCGTTAAACGGTGCCAGGCGTTAAAA, Y, 8208233, R, , , ,
,,, chrY:8147877-8148210, S_Shore, , , , Y:8207555-8208234, , , , "

[16]

"cg00214611, cg00214611, 69792329, CTAACCTCCAAACCACACTTTATATACTAAACTACAATA
TAACACAAACA, 46723459, CTAACCTCCGAACCGCGCTTTATATACTAAACTACAATATAACGCGAACG
, I, A, Red, GCGCCGGCAGGACTAGCTTCCGGGCCGCGCTTTGTGTGCTGGGCTGCAGTGTGGCGCGGG [C
G] AGGAAGCTGGTAGGGCGGTTGTGCGCAAGCTCCAGCTGCAGCCTCCGCCTACGTGAGAAGA, 37, Y, 158
15688, CGCCCGCGCCACACTGCAGCCCAGCACACAAAGCGCGGCCCGGAAGCTAG, Y, 14325082, R, ,
,,, TMSB4Y; TMSB4Y, NM_004202; NM_004202, 1stExon; 5'UTR, chrY:15815488-
15815779, Island, , , , Y:14324883-14325218, Y:15815422-
15815706, Promoter_Associated_Cell_type_specific, "

[17]

"cg00223952, cg00223952, 17691457, TTAAC TATACAAAAAATCTCTCCCTTTAAAAAATACAA
CCAAC TTCAAC, , , II, , , TTTCCCGACATTGGCTGTGCAAAAAGGTCTCTCCCTTTAAAGGGTACAGCC
AGCTTCAG [CG] AGGCTGAGGGAACTCTAAGGTTGGGGCGGGTGTGACAAAAGAGCACGTCCCGGACTC
C, 37, Y, 4868996, CGCTGAAGCTGGCTGTACCCTTTAAAGGGAGAGACCTTTTTGCACAGCCA, Y, 492
8996, R, , , , , PCDH11Y, NM_032971, 5'UTR, chrY:4870744-
4871225, N_Shore, , , , Y:4928638-4929278, , , , "

[18]

"cg00243321, cg00243321, 36709370, AATACRAAATCTACRAATCCAACRAAAATCCTCAAAAAA
CCTAAATCCAC, , , II, , , AGACCCGCTCCCTGCAAGACCCACGGGCGTCGCCCTGCTGAGAACCTGGTCC
CACACCTA [CG] TGGACCCAGGTTTCCTGAGGACCTCCGCTGGACCCGCAGATCCCGCACTGGCCAAAGGG
C, 37, Y, 6133740, CGTGGACCCAGGTTTCCTGAGGACCTCCGCTGGACCCGCAGATCCCGCAC, Y, 619
3740, F, , , , , , , , chrY:6133645-6133958, Island, , , , Y:6192225-6194784, , , , "

[19]

"cg00271873, cg00271873, 11808416, CRCCAAATTTATTCAAACCTTCTACAATCTCCCAAATTTT
CAAAC TTATCC, , , II, , , TTCCTGTCCGACGCCAGGTTTGTTTCAGGCTTCTGCAGTCTCCCAAGTTTTC
GGCTTGTC [CG] ACACCTTCTTGACTGAAGAATGATAATGTTTTTTGGTAAGACTGTGCCACTGACAGATGAT
C, 37, Y, 14649908, CGGACAAGCCTGAAAACCTTGGGAGACTGCAGAAGCCTGAACAAACCTGGC, Y, 13
159918, R, , , , , , , , chrY:14649050-14649669, S_Shore, , , , , , , , "

[20]

"cg00272582, cg00272582, 73678371, AAATCTAAAAACCTACAACAATAAAATCTATCRCCTTAA
CTAAAC TAAC, , , II, , , ACTGAGGGCGGGGTCTAAGGGCCTGCAACAATGAGATCTGTGCGCTTGGCT
AGGACTGG [CG] CCGAGAGGCGATAGGTCTCGGGAGAGCCTGGCGCAGGGTGTGGGAGATTAGGAATCCCA
G, 37, Y, 2802965, GGTCTAAGGGCCTGCAACAATGAGATCTGTGCGCTTGGCTAGGACTGGCG, Y, 286

```
2965,R,,,,,ZFY;ZFY;ZFY,NM_003411;NM_001145276;NM_001145275,TSS1500;TSS2
00;TSS1500,chrY:2802964-2804283,Island,,,,,Y:2862947-2864321,,,,"
```

```
Illumina450Manifest <- read.table("HumanMethylation450_15017482_v1-
2.csv",sep="," ,header=T, nrows=20, skip=7, stringsAsFactors = T)
```

```
dim(Illumina450Manifest)
```

```
## [1] 20 33
```

```
head(Illumina450Manifest)
```

##	IlmnID	Name	AddressA_ID
## 1	cg00035864	cg00035864	31729416
## 2	cg00050873	cg00050873	32735311
## 3	cg00061679	cg00061679	28780415
## 4	cg00063477	cg00063477	16712347
## 5	cg00121626	cg00121626	19779393
## 6	cg00212031	cg00212031	29674443

##	AlleleA_ProbeSeq	AddressB_ID
## 1	AAAACACTAACAATCTTATCCACATAAACCCCTTAAATTTATCTCAAATTC	NA
## 2	ACAAAAAACAACACACAACCTATAATAATTTTAAAATAAATAAACCCCA	31717405
## 3	AAAACATTAAAAAACTAATTCACACTACTATTTAATTACTTTATTTTCCATC	NA
## 4	TATTCTTCCACACAAAATACTAAACRTATATTTACAAAAATACTTCCATC	NA
## 5	AAAATAATAAAAAATACTTACAAACCAATACTATACCCTACAACCTCTC	NA
## 6	CCCAATTAACCAACAAAATAACAAATTATACAATCAAAAAACATACA	38703326

##	AlleleB_ProbeSeq
## 1	II
## 2	ACGAAAAAACAACGCACAACCTATAATAATTTTAAAATAAATAAACCCCG
## 3	I
## 4	II
## 5	II
## 6	CCCAATTAACCGCAAAAACTAAACAAATTATACGATCGAAAAACGTACG
##	I

##	Next_Base	Color_Channel
## 1		
## 2	A	Red
## 3		
## 4		
## 5		
## 6	T	Red

```

##
Forward_Sequence
## 1
AATCCAAAGATGATGGAGGAGTGCCCGCTCATGATGTGAAGTACCTGCTCAGCTGGAAAC [CG] AATTTGA
GATAAATTCAAGGGTCTATGTGGACAAGACTGCTAGTGTCTCTCTCTGGATTG
## 2
TATCTCTGTCTGGCGAGGAGGCAACGCACAACACTGTGGTGGTTTTTGGAGTGGGTGGACCC [CG] GCCAAGA
CGGCCTGGGCTGACCAGAGACGGGAGGCAGAAAAAGTGGGCAGGTGGTTGCAG
## 3
TCAACAAATGAGAGACATTGAAGAACTAATTCACACTACTATTTGGTTACTTTATTTTCCAT [CG] AAGAAAA
CCTCTTTTTTAAAACTAACACATAAAATAAAATGAACGAAGAACAACAACTAAACG
## 4
CTCCTGTACTTGTTTCATTAAATAATGATTCCTTGGATATACCAAGTCTGGATAGCGGATT [CG] ATGGAAG
CATTTTTGTAAATATACGTTTCAGTATTTTGTGTGGAAGAACAATCTAGCTG
## 5
AGGTGAATGAAGAGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACCTCT [CG] CGCTGCC
ACAAAAAGCCAATTAACGAAGCAAACCTGTTTTATCTTTTCTCTCCCCAGAAG
## 6
CCATTGGCCCGCCCCAGTTGGCCGCAGGGACTGAGCAAGTTATGCGGTGCGGAAGACGTG [CG] TTAAAGG
GCTGAAGGGGAGGGACGGAACCTGACAGTCTCTGTGACAGCTCTGAGGTGGGAG
## Genome_Build CHR MAPINFO
SourceSeq
## 1 37 Y 8553009
AGACACTAGCAGTCTTGTCCACATAGACCCTTGAATTTATCTCAAATTCG
## 2 37 Y 9363356
CGGGGTCCACCCACTCCAAAAACCACCACAGTTGTGCGTTGCCTCCTCGC
## 3 37 Y 25314171
CGATGGAAAATAAAGTAACCAATAGTAGTGAATTAGTTCTTCAATGTCT
## 4 37 Y 22741795
CGATGGAAGCATTTTTGTAAATATACGTTTCAGTATTTTGTGTGGAAGAAC
## 5 37 Y 21664296
AGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACCTCTCG
## 6 37 Y 21239348
CGCACGTCTTCCCGACCGCATAACTTGCTCAGTCCCTGCGGCCAACTGGG
## Chromosome_36 Coordinate_36 Strand Probe_SNPs Probe_SNPs_10
Random_Loci
## 1 Y 8613009 F
NA
## 2 Y 9973356 R
NA
## 3 Y 23723559 R
NA
## 4 Y 21151183 F rs9341313 rs13447379
NA
## 5 Y 20123684 R
NA

```

```
## 6          Y          19698736          R
NA
##   Methyl27_Loci UCSC_RefGene_Name          UCSC_RefGene_Accession
## 1          NA          TTTY18          NR_001550
## 2          NA   TSPY4;FAM197Y2          NM_001164471;NR_001553
## 3          NA   DAZ1;DAZ4;DAZ4 NM_004081;NM_020420;NM_001005375
## 4          NA          EIF1AY          NM_004681
## 5          NA          BCORL2          NR_002923
## 6          NA          TTTY14          NR_001543
##   UCSC_RefGene_Group UCSC_CpG_Islands_Name
Relation_to_UCSC_CpG_Island Phantom
## 1          TSS1500
NA
## 2          Body;TSS1500   chrY:9363680-9363943
N_Shore          NA
## 3          Body;Body;Body
NA
## 4          Body chrY:22737825-22738052
S_Shelf          NA
## 5          Body chrY:21664481-21665063
N_Shore          NA
## 6          TSS200 chrY:21238448-21240005
Island          NA
##   DMR Enhancer          HMM_Island Regulatory_Feature_Name
## 1 NA          NA          NA
## 2 NA          NA   Y:9973136-9976273          NA
## 3 NA          NA          NA
## 4 NA          NA          NA
## 5 NA          NA          NA
## 6 NA          NA   Y:19697854-19699393          NA
##   Regulatory_Feature_Group DHS
## 1          NA NA
## 2          NA NA
## 3          NA NA
## 4          NA NA
## 5          NA NA
## 6          NA NA
```

```
str(Illumina450Manifest)
```

```
## 'data.frame':   20 obs. of  33 variables:
```

```
## $ IlmnID          : Factor w/ 20 levels
"cg00035864","cg00050873",...: 1 2 3 4 5 6 7 8 9 10 ...
## $ Name            : Factor w/ 20 levels
"cg00035864","cg00050873",...: 1 2 3 4 5 6 7 8 9 10 ...
```

```

## $ AddressA_ID : int 31729416 32735311 28780415
16712347 19779393 29674443 30703409 69792329 36709370 11808416 ...
## $ AlleleA_ProbeSeq : Factor w/ 20 levels
"AAAACACTAACAATCTTATCCACATAAACCCCTTAAATTTATCTCAAATTC",...: 1 10 2 19 4 14
20 17 8 15 ...
## $ AddressB_ID : int NA 31717405 NA NA NA 38703326
36767301 46723459 NA NA ...
## $ AlleleB_ProbeSeq : Factor w/ 6 levels
"", "AACTCTAAACTACCCGACACAAACTCCAAAACTTCTCGAAAAAACTCG",...: 1 3 1 1 1 4
6 5 1 1 ...
## $ Infinium_Design_Type : Factor w/ 2 levels "I","II": 2 1 2 2
2 1 1 1 2 2 ...
## $ Next_Base : Factor w/ 3 levels "", "A", "T": 1 2 1
1 1 3 2 2 1 1 ...
## $ Color_Channel : Factor w/ 2 levels "", "Red": 1 2 1 1
1 2 2 2 1 1 ...
## $ Forward_Sequence : Factor w/ 20 levels
"AAAGTTCTTTTTGCTTCCTGAAATTTCCGGCAGCCAAAAGATCTATAGCGAGTGGCAGTC[CG]ATCCAG
CAAGAGCCAATGAAAGATCCACTCCACAGTGAAGAAGGACATGCAGATGAAGTG",...: 3 15 16 10
7 8 18 14 5 20 ...
## $ Genome_Build : int 37 37 37 37 37 37 37 37 37 37
...
## $ CHR : Factor w/ 1 level "Y": 1 1 1 1 1 1 1
1 1 1 ...
## $ MAPINFO : int 8553009 9363356 25314171
22741795 21664296 21239348 8148233 15815688 6133740 14649908 ...
## $ SourceSeq : Factor w/ 20 levels
"ACCTACCGGGTGCCATAGAGGCTGCGGCAGGGTTCCTGTGGCTTGGGTCG",...: 2 15 5 6 3 8
10 11 17 13 ...
## $ Chromosome_36 : Factor w/ 1 level "Y": 1 1 1 1 1 1 1
1 1 1 ...
## $ Coordinate_36 : int 8613009 9973356 23723559
21151183 20123684 19698736 8208233 14325082 6193740 13159918 ...
## $ Strand : Factor w/ 2 levels "F","R": 1 2 2 1
2 2 2 2 1 2 ...
## $ Probe_SNPs : Factor w/ 2 levels "", "rs9341313": 1
1 1 2 1 1 1 1 1 1 ...
## $ Probe_SNPs_10 : Factor w/ 2 levels "", "rs13447379":
1 1 1 2 1 1 1 1 1 1 ...
## $ Random_Loci : logi NA NA NA NA NA NA ...
## $ Methyl27_Loci : logi NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name : Factor w/ 15 levels
"", "BCORL2", "DAZ1;DAZ4;DAZ4",...: 14 12 3 4 2 13 1 10 1 1 ...
## $ UCSC_RefGene_Accession : Factor w/ 15 levels
"", "NM_001146706;NM_004653;NM_001146705",...: 12 4 6 8 13 10 1 7 1 1 ...

```



```
## $ UCSC_RefGene_Group      : Factor w/ 10 levels
"", "1stExon;5UTR,chrY:15815488-15815779,Island,,,,Y:14324883-
14325218,Y:15815422-15815706,Promoter_Associated_Cell_"|
__truncated__,...: 7 6 5 3 3 10 1 2 1 1 ...
## $ UCSC_CpG_Islands_Name   : Factor w/ 15 levels
"", "chrY:14100002-14100593",...: 1 13 1 6 5 4 11 9 10 3 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 6 levels
"", "Island", "N_Shelf",...: 1 4 1 5 4 2 6 4 2 6 ...
## $ Phantom                 : logi  NA NA NA NA NA NA NA ...
## $ DMR                     : logi  NA NA NA NA NA NA NA ...
## $ Enhancer                : logi  NA NA NA NA NA NA NA ...
## $ HMM_Island              : Factor w/ 11 levels "", "Y:10355560-
10355993",...: 1 10 1 1 1 3 8 6 7 1 ...
## $ Regulatory_Feature_Name  : logi  NA NA NA NA NA NA NA ...
## $ Regulatory_Feature_Group : logi  NA NA NA NA NA NA NA ...
## $ DHS                     : logi  NA NA NA NA NA NA NA ...
```

You can also use the `read.csv` function, which uses the comma as default separator

```
?read.csv
```

```
Illumina450Manifest <- read.csv("HumanMethylation450_15017482_v1-2.csv",header=T,
nrows=20,skip=7,stringsAsFactors = T)
head(Illumina450Manifest)
```

##	IlmnID	Name	AddressA_ID
## 1	cg00035864	cg00035864	31729416
## 2	cg00050873	cg00050873	32735311
## 3	cg00061679	cg00061679	28780415
## 4	cg00063477	cg00063477	16712347
## 5	cg00121626	cg00121626	19779393
## 6	cg00212031	cg00212031	29674443
##		AlleleA_ProbeSeq	AddressB_ID
## 1	AAAACACTAACAATCTTATCCACATAAACCCCTTAAATTTATCTCAAATTC		NA
## 2	ACAAAAAACAACACACAACCTATAATAATTTTAAAATAAATAAACCCCA		31717405
## 3	AAAACATTAAAAAACTAATTCACCTACTATTTAATTACTTTATTTTCCATC		NA
## 4	TATTCTTCCACACAAAATACTAAACRTATATTTACAAAAATACTTCCATC		NA
## 5	AAAACATAATAAAATAACTTACAAACCAATACTATAACCCTACAACCTCTC		NA
## 6	CCCAATTAACCACAAAACTAAACAAATTATACAATCAAAAAACATACA		38703326
##		AlleleB_ProbeSeq	
##		Infinium_Design_Type	
## 1		II	
## 2	ACGAAAAAACAACGCACAACCTATAATAATTTTAAAATAAATAAACCCCG		
I			

```

## 3
II
## 4
II
## 5
II
## 6 CCCAATTAACCGCAAAAACTAAACAAATTATACGATCGAAAAACGTACG
I
##      Next_Base  Color_Channel
## 1
## 2          A          Red
## 3
## 4
## 5
## 6          T          Red
##
Forward_Sequence
## 1
AATCCAAAGATGATGGAGGAGTGCCCGCTCATGATGTGAAGTACCTGCTCAGCTGGAAAC [CG] AATTTGA
GATAAATTCAAGGGTCTATGTGGACAAGACTGCTAGTGTCTCTCTCTGGATTG
## 2
TATCTCTGTCTGGCGAGGAGGCAACGCACAACGTGGTGGTTTTTGGAGTGGGTGGACCC [CG] GCCAAGA
CGGCCTGGGCTGACCAGAGACGGGAGGCAGAAAAAGTGGGCAGGTGGTTGCAG
## 3
TCAACAAATGAGAGACATTGAAGAACTAATTCACTACTATTTGGTTACTTTATTTTCCAT [CG] AAGAAAA
CCTCTTTTTTAAAACTAACACATAAAATAAAATGAACGAAGAACAACCTAAACG
## 4
CTCCTGTACTTGTTTCATTAAATAATGATTCCCTTGATATACCAAGTCTGGATAGCGGATT [CG] ATGGAAG
CATTTTTTGTAATATACGTTTCAGTATTTTGTGTGGAAGAACAATCTAGCTG
## 5
AGGTGAATGAAGAGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACCTCT [CG] CGCTGCC
ACAAAAAGCCAATTAACGAAGCAAACTGTTTTATCTTTTCTCTCCCCAGAAG
## 6
CCATTGGCCCGCCCCAGTTGGCCGCAGGGACTGAGCAAGTTATGCGGTCGGGAAGACGTG [CG] TTAAAGG
GCTGAAGGGGAGGGACGGAAGTACAGTCTCTGTGACAGCTCTGAGGTGGGAG
##      Genome_Build  CHR  MAPINFO
SourceSeq
## 1          37      Y  8553009
AGACACTAGCAGTCTTGTCCACATAGACCCTTGAATTTATCTCAAATTTCG
## 2          37      Y  9363356
CGGGGTCCACCCACTCCAAAAACCACACAGTTGTGCGTTGCCTCCTCGC
## 3          37      Y  25314171
CGATGGAAAATAAAGTAACCAATAGTAGTGAATTAGTTCTTCAATGTCT
## 4          37      Y  22741795
CGATGGAAGCATTTTTGTAAATATACGTTTCAGTATTTTGTGTGGAAGAAC

```

```

## 5          37    Y 21664296
AGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACCTCTCG
## 6          37    Y 21239348
CGCACGTCTTCCCGACCGCATAACTTGCTCAGTCCCTGCGGCCAACTGGG
##   Chromosome_36 Coordinate_36 Strand Probe_SNPs Probe_SNPs_10
Random_Loci
## 1          Y          8613009          F
NA
## 2          Y          9973356          R
NA
## 3          Y          23723559         R
NA
## 4          Y          21151183         F   rs9341313      rs13447379
NA
## 5          Y          20123684         R
NA
## 6          Y          19698736         R
NA
##   Methyl27_Loci UCSC_RefGene_Name          UCSC_RefGene_Accession
## 1          NA          TTTY18                      NR_001550
## 2          NA      TSPY4;FAM197Y2          NM_001164471;NR_001553
## 3          NA      DAZ1;DAZ4;DAZ4 NM_004081;NM_020420;NM_001005375
## 4          NA          EIF1AY                      NM_004681
## 5          NA          BCORL2                      NR_002923
## 6          NA          TTTY14                      NR_001543
##   UCSC_RefGene_Group UCSC_CpG_Islands_Name
Relation_to_UCSC_CpG_Island Phantom
## 1          TSS1500
NA
## 2          Body;TSS1500   chrY:9363680-9363943
N_Shore      NA
## 3          Body;Body;Body
NA
## 4          Body chrY:22737825-22738052
S_Shelf      NA
## 5          Body chrY:21664481-21665063
N_Shore      NA
## 6          TSS200 chrY:21238448-21240005
Island       NA
##   DMR Enhancer          HMM_Island Regulatory_Feature_Name
## 1   NA      NA
## 2   NA      NA   Y:9973136-9976273
## 3   NA      NA
## 4   NA      NA
## 5   NA      NA
## 6   NA      NA   Y:19697854-19699393

```

```
## Regulatory_Feature_Group DHS
## 1 NA
## 2 NA
## 3 NA
## 4 NA
## 5 NA
## 6 NA
```

```
str(Illumina450Manifest)
```

```
## 'data.frame': 20 obs. of 33 variables:
```

```
## $ IlmnID : Factor w/ 20 levels
"cg00035864","cg00050873",...: 1 2 3 4 5 6 7 8 9 10 ...
## $ Name : Factor w/ 20 levels
"cg00035864","cg00050873",...: 1 2 3 4 5 6 7 8 9 10 ...
## $ AddressA_ID : int 31729416 32735311 28780415
16712347 19779393 29674443 30703409 69792329 17691457 36709370 ...
## $ AlleleA_ProbeSeq : Factor w/ 20 levels
"AAACACTAACAACTCTTATCCACATAAACCCCTTAAATTTATCTCAAATTC",...: 1 10 2 18 4 13
20 16 19 8 ...
## $ AddressB_ID : int NA 31717405 NA NA NA 38703326
36767301 46723459 NA NA ...
## $ AlleleB_ProbeSeq : Factor w/ 6 levels
"", "AACTCTAAACTACCCGACACAACTCCAAAACTTCTCGAAAAAACTCG",...: 1 3 1 1 1 4
6 5 1 1 ...
## $ Infinium_Design_Type : Factor w/ 2 levels "I","II": 2 1 2 2
2 1 1 1 2 2 ...
## $ Next_Base : Factor w/ 3 levels "", "A", "T": 1 2 1
1 1 3 2 2 1 1 ...
## $ Color_Channel : Factor w/ 2 levels "", "Red": 1 2 1 1
1 2 2 2 1 1 ...
## $ Forward_Sequence : Factor w/ 20 levels
"AAAGTTCTTTTTGCTTCTGAAATTTCCGGCAGCCAAAAGATCTATAGCGAGTGGCAGTC[CG]ATCCAG
CAAGAGCCAATGAAAGATCCACTCCACAGTGAAGAAGGACATGCAGATGAAGTG",...: 2 14 15 9 6
7 17 13 20 4 ...
## $ Genome_Build : int 37 37 37 37 37 37 37 37 37 37
...
## $ CHR : Factor w/ 1 level "Y": 1 1 1 1 1 1 1
1 1 1 ...
## $ MAPINFO : int 8553009 9363356 25314171
22741795 21664296 21239348 8148233 15815688 4868996 6133740 ...
## $ SourceSeq : Factor w/ 20 levels
"ACCTACCGGGTGCCATAGAGGCTGCGGCAGGGTTCTGTGGCTTGGGTCG",...: 2 15 5 6 3 8
10 11 12 17 ...
## $ Chromosome_36 : Factor w/ 1 level "Y": 1 1 1 1 1 1 1
1 1 1 ...
```

```
## $ Coordinate_36 : int 8613009 9973356 23723559
21151183 20123684 19698736 8208233 14325082 4928996 6193740 ...
## $ Strand : Factor w/ 2 levels "F","R": 1 2 2 1
2 2 2 2 2 1 ...
## $ Probe_SNPs : Factor w/ 2 levels "", "rs9341313": 1
1 1 2 1 1 1 1 1 1 ...
## $ Probe_SNPs_10 : Factor w/ 2 levels "", "rs13447379":
1 1 1 2 1 1 1 1 1 1 ...
## $ Random_Loci : logi NA NA NA NA NA NA ...
## $ Methyl27_Loci : logi NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name : Factor w/ 16 levels
"", "BCORL2", "DAZ1;DAZ4;DAZ4",...: 15 13 3 4 2 14 1 11 9 1 ...
## $ UCSC_RefGene_Accession : Factor w/ 16 levels
"", "NM_001146706;NM_004653;NM_001146705",...: 13 4 6 8 14 11 1 7 9 1 ...
## $ UCSC_RefGene_Group : Factor w/ 11 levels
"", "1stExon;5'UTR",...: 8 7 6 4 4 11 1 2 3 1 ...
## $ UCSC_CpG_Islands_Name : Factor w/ 15 levels
"", "chrY:14649050-14649669",...: 1 13 1 6 5 4 11 3 9 10 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 5 levels
"", "Island", "N_Shore",...: 1 3 1 4 3 2 5 2 3 2 ...
## $ Phantom : logi NA NA NA NA NA NA ...
## $ DMR : logi NA NA NA NA NA NA ...
## $ Enhancer : logi NA NA NA NA NA NA ...
## $ HMM_Island : Factor w/ 12 levels "", "Y:10355560-
10355993",...: 1 11 1 1 1 4 9 3 7 8 ...
## $ Regulatory_Feature_Name : Factor w/ 2 levels "", "Y:15815422-
15815706": 1 1 1 1 1 1 1 2 1 1 ...
## $ Regulatory_Feature_Group : Factor w/ 2 levels
"", "Promoter_Associated_Cell_type_specific": 1 1 1 1 1 1 1 2 1 1 ...
## $ DHS : logi NA NA NA NA NA NA ...
```

The descriptions of the Manifest Column Headings can be found here:

<https://support.illumina.com/content/dam/illumina-support/documents/downloads/productfiles/methylationepic/infinium-methylationepic-manifest-column-headings.pdf>. In summary:

IlmnID: Unique CpG locus identifier from the Illumina CG database

Name: Unique CpG locus identifier from the Illumina CG database

AddressA_ID: Address of probe A

AlleleA_ProbeSeq: Sequence for probe A

AddressB_ID: Address of probe B - Infinium I assays only

AlleleB_ProbeSeq: Sequence for probe B - Infinium I assays only

Infinium_Design_Type: Defines Assay type - Infinium I or Infinium II

Next_Base: Base added at SBE step - Infinium I assays only

Color_Channel: Color of the incorporated base (Red or Green) - Infinium I assays only

Forward_Sequence: Sequence (in 5'-3' orientation) flanking query site

Genome_Build: Genome build on which forward sequence is based

CHR: Chromosome - genome build 37

MAPINFO: Coordinates - genome build 37

SourceSeq: Unconverted design sequence

Chromosome_36: Chromosome - genome build 36

Coordinate_36: Coordinates - genome build 36

Strand: Design strand

Probe_SNPs: Assays with SNPs present within probe >10bp from query site

Probe_SNPs_10: Assays with SNPs present within probe ≤10bp from query site (HM27 carryover or recently discovered)

Random_Loci: Loci which were chosen randomly in the design process

Methyl27_Loci: Present or absent on HumanMethylation27 array

UCSC_RefGene_Name: Gene name (UCSC)

UCSC_RefGene_Accession: Accession number (UCSC)

UCSC_RefGene_Group: Gene region feature category (UCSC)

UCSC_CpG_Islands_Name: CpG island name (UCSC)

Relation_to_UCSC_CpG_Island: Relationship to Canonical CpG Island: Shores - 0-2 kb from CpG island; Shelves - 2-4 kb from CpG island.

Phantom: FANTOM-derived promoter

DMR: Differentially methylated region (experimentally determined)

Enhancer: Enhancer element (informatically-determined)

HMM_Island: Hidden Markov Model Island

Regulatory_Feature_Name: Regulatory feature (informatically determined)

Regulatory_Feature_Group: Regulatory feature category

DHS: DNase hypersensitive site (experimentally determined)

Now we can upload the whole manifest. This step can take some minutes, if you encounter problems you can skip the next two command lines and directly go to “Load the Illumina450Manifest.RData”:

```
Illumina450Manifest <- read.csv("HumanMethylation450_15017482_v1-
2.csv",header=T, skip=7, stringsAsFactors = T)

save(Illumina450Manifest,file="Illumina450Manifest.RData")
```

Remember that if you save your data with save(), it cannot be restored under different name. The original object name will be automatically used when you will load the data.

Load the Illumina450Manifest.RData

```
load("Illumina450Manifest.RData")
```

To know the name of the object that we have just uploaded, we can list the objects in the workspace:

```
ls()

## [1] "Illumina450Manifest"
```

Let’s apply some exploratory functions:

```
dim(Illumina450Manifest)

## [1] 486428      33

head(Illumina450Manifest)

##           IlmnID           Name AddressA_ID
## 1 cg00035864 cg00035864      31729416
## 2 cg00050873 cg00050873      32735311
## 3 cg00061679 cg00061679      28780415
## 4 cg00063477 cg00063477      16712347
## 5 cg00121626 cg00121626      19779393
## 6 cg00212031 cg00212031      29674443
##                                     AlleleA_ProbeSeq AddressB_ID
## 1 AAAACACTAACAATCTTATCCACATAAACCCCTTAAATTTATCTCAAATTC
## 2 ACAAAAAACAACACACAACATAATAATTTTAAAATAAATAAACCCCA      31717405
## 3 AAAACATTAATAAACTAATTCACACTACTATTTAATTACTTTATTTTCCATC
## 4 TATTCTTCCACACAAAATACTAAACRTATATTTACAAAATACTTCCATC
## 5 AAAACTAATAAAAATACTTACAAACCAATACTATACCCTACAACCTCTC
## 6 CCAATTAACCACAAAACTAAACAAATTATACAATCAAAAAACATACA      38703326
##                                     AlleleB_ProbeSeq
## Infinium_Design_Type
```

```

## 1
II
## 2 ACGAAAAACAACGCACAACCTATAATAATTTTAAAATAAATAAACCCCG
I
## 3
II
## 4
II
## 5
II
## 6 CCCAATTAACCGCAAAACTAAACAAATTATACGATCGAAAAACGTACG
I
##   Next_Base   Color_Channel
## 1
## 2           A           Red
## 3
## 4
## 5
## 6           T           Red
##
Forward_Sequence
## 1
AATCCAAAGATGATGGAGGAGTGCCCGCTCATGATGTGAAGTACCTGCTCAGCTGGAAAC [CG] AATTTGA
GATAAATTCAAGGGTCTATGTGGACAAGACTGCTAGTGTCTCTCTCTGGATTG
## 2
TATCTCTGTCTGGCGAGGAGGCAACGCACAACCTGTGGTGGTTTTTGGAGTGGGTGGACCC [CG] GCCAAGA
CGGCCTGGGCTGACCAGAGACGGGAGGCAGAAAAAGTGGGCAGGTGGTTGCAG
## 3
TCAACAAATGAGAGACATTGAAGAACTAATTCACACTACTATTTGGTTACTTTATTTTCCAT [CG] AAGAAAA
CCTCTTTTTTAAAACTAACACATAAATAAAATGAACGAAGAACAACTAAACG
## 4
CTCCTGTACTTGTTTCATTAAATAATGATTCCTTGATATACCAAGTCTGGATAGCGGATT [CG] ATGGAAG
CATTTTTGTAAATATACGTTTCAGTATTTTGTGTGGAAGAACACAATCTAGCTG
## 5
AGGTGAATGAAGAGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACTCT [CG] CGCTGCC
ACAAAAAGCCAATTAACGAAGCAAACTGTTTTATCTTTTCCTCTCCCCAGAAG
## 6
CCATTGGCCCGCCCCAGTTGGCCGCAGGGACTGAGCAAGTTATGCGGTCGGAAGACGTG [CG] TTAAAGG
GCTGAAGGGGAGGGACGGAAGTACAGTCTCTGTGACAGCTCTGAGGTGGGAG
##   Genome_Build   CHR   MAPINFO
SourceSeq
## 1           37     Y   8553009
AGACACTAGCAGTCTTGTCCACATAGACCCTTGAATTTATCTCAAATTTCG
## 2           37     Y   9363356
CGGGGTCCACCCACTCCAAAAACCACCACAGTTGTGCGTTGCCTCCTCGC

```


3 37 Y 25314171
CGATGGAAAATAAAGTAACCAAATAGTAGTGAATTAGTTCTTCAATGTCT

4 37 Y 22741795
CGATGGAAGCATTTTTGTAAATATACGTTTCAGTATTTTGTGTGGAAGAAC

5 37 Y 21664296
AGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACCTCTCG

6 37 Y 21239348
CGCACGTCTTCCCGACCGCATAACTTGCTCAGTCCCTGCGGCCAACTGGG

Chromosome_36 Coordinate_36 Strand Probe_SNPs Probe_SNPs_10
Random_Loci

1 Y 8613009 F

NA

2 Y 9973356 R

NA

3 Y 23723559 R

NA

4 Y 21151183 F rs9341313 rs13447379

NA

5 Y 20123684 R

NA

6 Y 19698736 R

NA

Methyl27_Loci UCSC_RefGene_Name UCSC_RefGene_Accession

1 NA TTTY18 NR_001550

2 NA TSPY4;FAM197Y2 NM_001164471;NR_001553

3 NA DAZ1;DAZ4;DAZ4 NM_004081;NM_020420;NM_001005375

4 NA EIF1AY NM_004681

5 NA BCORL2 NR_002923

6 NA TTTY14 NR_001543

UCSC_RefGene_Group UCSC_CpG_Islands_Name

Relation_to_UCSC_CpG_Island Phantom

1 TSS1500

2 Body;TSS1500 chrY:9363680-9363943

N_Shore

3 Body;Body;Body

4 Body chrY:22737825-22738052

S_Shelf

5 Body chrY:21664481-21665063

N_Shore

6 TSS200 chrY:21238448-21240005

Island

DMR Enhancer HMM_Island Regulatory_Feature_Name

1 NA

2 NA Y:9973136-9976273

3 NA

4 NA

```
## 5          NA
## 6          NA Y:19697854-19699393
##  Regulatory_Feature_Group DHS
## 1          NA
## 2          NA
## 3          NA
## 4          NA
## 5          NA
## 6          NA
```

```
str(Illumina450Manifest)
```

```
## 'data.frame':   486428 obs. of  33 variables:
```

```
##  $ IlmnID                : Factor w/ 486428 levels
"[Controls]", "10627500", ...: 1743 2082 2315 2353 3658 5417 5451 5467
5662 6070 ...
##  $ Name                  : Factor w/ 485593 levels
"", "BISULFITE CONVERSION I", ...: 895 1234 1467 1505 2810 4569 4603 4619
4814 5222 ...
##  $ AddressA_ID           : Factor w/ 485614 levels "", "-
99", "10600322", ...: 163955 171493 142506 49192 74497 146457 155290
448596 56107 200820 ...
##  $ AlleleA_ProbeSeq      : Factor w/ 486428 levels
"", "AAAAAAAAAAAAAAAAAACCAAAAAAAAAATCCCCACTCTAAAACCCACACA", ...: 35018
143551 35762 428168 41002 296102 483734 335382 461498 128017 ...
##  $ AddressB_ID           : Factor w/ 135503 levels
"", "10600313", ...: 1 45525 1 1 1 60142 56512 76868 1 1 ...
##  $ AlleleB_ProbeSeq      : Factor w/ 135502 levels
"", "AAAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAAACCCCGCG", ...: 1 49675 1 1
1 77355 135117 95693 1 1 ...
##  $ Infinium_Design_Type   : Factor w/ 3 levels "", "I", "II": 3 2
3 3 3 2 2 2 3 3 ...
##  $ Next_Base              : Factor w/ 4 levels "", "A", "C", "T": 1
2 1 1 1 4 2 2 1 1 ...
##  $ Color_Channel          : Factor w/ 3 levels "", "Grn", "Red": 1
3 1 1 1 3 3 3 1 1 ...
##  $ Forward_Sequence       : Factor w/ 485513 levels
"", "AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAATATGTCTGCTAAATAATCACAGAGTGAAA[CG]AAG
AGGAAGAACTAAAGTACAGTTTGGAACATAGTACAGTGCCCTGGGTACTGAATCA", ...: 25106
392559 395303 218896 81396 152102 421755 296780 478969 55952 ...
##  $ Genome_Build           : int   37 37 37 37 37 37 37 37 37 37
...
##  $ CHR                    : Factor w/ 25 levels
"", "1", "10", "11", ...: 25 25 25 25 25 25 25 25 25 25 ...
##  $ MAPINFO                : int   8553009 9363356 25314171
22741795 21664296 21239348 8148233 15815688 4868996 6133740 ...
```

```

## $ SourceSeq : Factor w/ 485344 levels
"", "AAAAAAAAAACCAACCCCAAACAAAAACCCCTGTGCCAGAAATTGCATTGC", ...: 28962
285897 116668 116692 29628 123295 148857 152724 189023 323598 ...
## $ Chromosome_36 : Factor w/ 26 levels
"", "1", "10", "11", ...: 26 26 26 26 26 26 26 26 26 ...
## $ Coordinate_36 : Factor w/ 484739 levels
"", "100000057", ...: 451149 483812 205768 185981 178029 173645 442781
102377 331284 376024 ...
## $ Strand : Factor w/ 3 levels "", "F", "R": 2 3 3
2 3 3 3 3 2 ...
## $ Probe_SNPs : Factor w/ 56004 levels
"", "rs10000615", ...: 1 1 1 54643 1 1 1 1 1 1 ...
## $ Probe_SNPs_10 : Factor w/ 35790 levels
"", "rs10000804", ...: 1 1 1 4064 1 1 1 1 1 1 ...
## $ Random_Loci : logi NA NA NA NA NA NA ...
## $ Methyl27_Loci : logi NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name : Factor w/ 42456 levels
"", "A1BG", "A1CF;A1CF;A1CF", ...: 39103 38905 9635 11319 3085 39100 1
37974 27057 1 ...
## $ UCSC_RefGene_Accession : Factor w/ 52670 levels
"", "NM_000014", ...: 48799 14035 19907 20891 49121 48792 1 20105 38174 1
...
## $ UCSC_RefGene_Group : Factor w/ 4165 levels
"", "1stExon", "1stExon;1stExon", ...: 2970 2779 2447 2035 2035 3694 1 201
943 1 ...
## $ UCSC_CpG_Islands_Name : Factor w/ 27177 levels
"", "chr1:10003165-10003585", ...: 1 27170 1 27136 27134 27133 27156 27120
27147 27149 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 6 levels
"", "Island", "N_Shelf", ...: 1 4 1 5 4 2 6 2 4 2 ...
## $ Phantom : Factor w/ 11912 levels "", "high-
CpG:100009860-100010119", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ DMR : Factor w/ 4 levels
"", "CDMR", "DMR", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Enhancer : logi NA NA NA NA NA NA ...
## $ HMM_Island : Factor w/ 63086 levels
"", "1:100088009-100088708", ...: 1 63085 1 1 1 63046 63077 63033 63064
63067 ...
## $ Regulatory_Feature_Name : Factor w/ 47969 levels
"", "1:10001853-10003978", ...: 1 1 1 1 1 1 1 47965 1 1 ...
## $ Regulatory_Feature_Group : Factor w/ 9 levels
"", "Gene_Associated", ...: 1 1 1 1 1 1 1 7 1 1 ...
## $ DHS : logi NA NA NA NA NA NA ...

```

```
colnames(Illumina450Manifest)
```

```
## [1] "IlmnID" "Name"
## [3] "AddressA_ID" "AlleleA_ProbeSeq"
## [5] "AddressB_ID" "AlleleB_ProbeSeq"
## [7] "Infinium_Design_Type" "Next_Base"
## [9] "Color_Channel" "Forward_Sequence"
## [11] "Genome_Build" "CHR"
## [13] "MAPINFO" "SourceSeq"
## [15] "Chromosome_36" "Coordinate_36"
## [17] "Strand" "Probe_SNPs"
## [19] "Probe_SNPs_10" "Random_Loci"
## [21] "Methyl27_Loci" "UCSC_RefGene_Name"
## [23] "UCSC_RefGene_Accession" "UCSC_RefGene_Group"
## [25] "UCSC_CpG_Islands_Name" "Relation_to_UCSC_CpG_Island"
## [27] "Phantom" "DMR"
## [29] "Enhancer" "HMM_Island"
## [31] "Regulatory_Feature_Name" "Regulatory_Feature_Group"
## [33] "DHS"
```

```
head(rownames(Illumina450Manifest))
```

```
## [1] "1" "2" "3" "4" "5" "6"
```

Let's check the distribution of probes across the chromosomes

```
table(Illumina450Manifest$CHR)
```

```
##
##      1      10      11      12      13      14      15      16      17      18
19      2
##  916 46857 24388 28794 24539 12285 15078 15259 21969 27879  5922
25521 34810
##   20   21   22    3    4    5    6    7    8    9    X
Y
## 10379  4243  8552 25159 20464 24327 36611 30017 20950  9861 11232
416
```

Remember that by default, factor levels are ordered according to alphabetical order (1,10,11, etc).

Remember that, to select the column CHR, you can write also:

```
table(Illumina450Manifest[, "CHR"])
```

```
##
##      1      10      11      12      13      14      15      16      17      18
19      2
```

```
##      916 46857 24388 28794 24539 12285 15078 15259 21969 27879 5922
25521 34810
##      20      21      22      3      4      5      6      7      8      9      X
Y
## 10379 4243 8552 25159 20464 24327 36611 30017 20950 9861 11232
416
```

or

```
table(Illumina450Manifest[,]$CHR)
```

```
##
##      1      10      11      12      13      14      15      16      17      18
19      2
##      916 46857 24388 28794 24539 12285 15078 15259 21969 27879 5922
25521 34810
##      20      21      22      3      4      5      6      7      8      9      X
Y
## 10379 4243 8552 25159 20464 24327 36611 30017 20950 9861 11232
416
```

or index it. From the result of the `colnames` function, we know that the column `CHR` is the 12th column, so you can write:

```
table(Illumina450Manifest[,12])
```

```
##
##      1      10      11      12      13      14      15      16      17      18
19      2
##      916 46857 24388 28794 24539 12285 15078 15259 21969 27879 5922
25521 34810
##      20      21      22      3      4      5      6      7      8      9      X
Y
## 10379 4243 8552 25159 20464 24327 36611 30017 20950 9861 11232
416
```

CHUNK 2: create a “clean” version of the manifest

There are 916 probes that do not map in any chromosome. Let’s have a look at them: we will subset the dataframe in order to retain only the rows for which the column `CHR` is empty (nothing between the quotes)

```
notMappedToCHR <- Illumina450Manifest[Illumina450Manifest$CHR=="",]
```

```
dim(notMappedToCHR)
```

```
## [1] 916 33
```

```
str(notMappedToCHR)
```

```
## 'data.frame': 916 obs. of 33 variables:
```

```
## $ IlmnID : Factor w/ 486428 levels
"[Controls]","10627500",...: 486369 486416 486366 486372 486389 486392
486373 486423 486411 486377 ...
## $ Name : Factor w/ 485593 levels
"", "BISULFITE CONVERSION I",...: 485530 485577 485527 485533 485550
485553 485534 485584 485572 485538 ...
## $ AddressA_ID : Factor w/ 485614 levels "", "-
99","10600322",...: 30770 67466 93841 182220 803 274431 365032 290646
200748 103802 ...
## $ AlleleA_ProbeSeq : Factor w/ 486428 levels
"", "AAAAAAAAAAAAAAAAAACCAAAAAAAAAATCCCCACTCTAAACCCCA",...: 398359
467864 246749 456305 176148 464309 133414 330730 329051 296770 ...
## $ AddressB_ID : Factor w/ 135503 levels
"", "10600313",...: 65887 81093 21658 42589 30222 14866 72465 10321 11027
61420 ...
## $ AlleleB_ProbeSeq : Factor w/ 135502 levels
"", "AAAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAACCCCGCG",...: 116081
132181 68540 129864 49227 131450 39524 94505 94051 77480 ...
## $ Infinium_Design_Type : Factor w/ 3 levels "", "I", "II": 2 2
2 2 2 2 2 2 2 2 ...
## $ Next_Base : Factor w/ 4 levels "", "A", "C", "T": 2
3 2 4 4 3 3 4 4 3 ...
## $ Color_Channel : Factor w/ 3 levels "", "Grn", "Red": 3
2 3 3 3 2 2 3 3 2 ...
## $ Forward_Sequence : Factor w/ 485513 levels
"", "AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAATATGTCTGCTAAATAATCACAGAGTGAAA[CG]AAG
AGGAAGAAACTAAAGTACAGTTTGAACATAGTACAGTGCCCCTGGGTACTGAATCA",...: 1 1 1 1
1 1 1 1 1 1 ...
## $ Genome_Build : int NA NA NA NA NA NA NA NA NA NA
...
## $ CHR : Factor w/ 25 levels
"", "1", "10", "11",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ MAPINFO : int NA NA NA NA NA NA NA NA NA NA
...
## $ SourceSeq : Factor w/ 485344 levels
"", "AAAAAAAAAACCAACCCCAACCAAAACCCCTGTGCCAGAAATTGCATTTCG",...: 1 1 1 1 1 1
1 1 1 1 ...
## $ Chromosome_36 : Factor w/ 26 levels
"", "1", "10", "11",...: 1 1 1 1 1 1 1 1 1 1 ...
```

```
## $ Coordinate_36 : Factor w/ 484739 levels
"", "100000057", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Strand : Factor w/ 3 levels "", "F", "R": 1 1 1
1 1 1 1 1 1 1 ...
## $ Probe_SNPs : Factor w/ 56004 levels
"", "rs10000615", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Probe_SNPs_10 : Factor w/ 35790 levels
"", "rs10000804", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Random_Loci : logi NA NA NA NA NA NA ...
## $ Methyl27_Loci : logi NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name : Factor w/ 42456 levels
"", "A1BG", "A1CF;A1CF;A1CF", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ UCSC_RefGene_Accession : Factor w/ 52670 levels
"", "NM_000014", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ UCSC_RefGene_Group : Factor w/ 4165 levels
"", "1stExon", "1stExon;1stExon", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ UCSC_CpG_Islands_Name : Factor w/ 27177 levels
"", "chr1:10003165-10003585", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 6 levels
"", "Island", "N_Shelf", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Phantom : Factor w/ 11912 levels "", "high-
CpG:100009860-100010119", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ DMR : Factor w/ 4 levels
"", "CDMR", "DMR", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Enhancer : logi NA NA NA NA NA NA ...
## $ HMM_Island : Factor w/ 63086 levels
"", "1:100088009-100088708", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Regulatory_Feature_Name : Factor w/ 47969 levels
"", "1:10001853-10003978", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Regulatory_Feature_Group : Factor w/ 9 levels
"", "Gene_Associated", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ DHS : logi NA NA NA NA NA NA ...
```

You will see that R remembers all the factor levels from the original dataframe. Use the `droplevels` function to remove the unused levels!

```
notMappedToCHR <- droplevels(notMappedToCHR)
```

```
dim(notMappedToCHR)
```

```
## [1] 916 33
```

```
str(notMappedToCHR)
```

```
## 'data.frame': 916 obs. of 33 variables:
```

```
## $ IlmnID : Factor w/ 916 levels
"[Controls]", "10627500", ...: 857 904 854 860 877 880 861 911 899 865 ...
```

```

## $ Name : Factor w/ 81 levels "", "BISULFITE
CONVERSION I",...: 18 65 15 21 38 41 22 72 60 26 ...
## $ AddressA_ID : Factor w/ 102 levels "", "-
99", "10622451",...: 10 17 22 37 3 49 60 51 42 26 ...
## $ AlleleA_ProbeSeq : Factor w/ 916 levels
"", "AAAACCTCTCTTTCTATCTCCCACTCTATATCACTCTACACATTTATAA",...: 895 912 45
908 18 909 11 55 54 50 ...
## $ AddressB_ID : Factor w/ 27 levels
"", "14806497",...: 19 23 7 11 9 5 21 2 4 16 ...
## $ AlleleB_ProbeSeq : Factor w/ 26 levels
"", "AACAAACAAAAATACTACTCACAACCTCTACTCATTAAAAAAATCCCTCT",...: 19 25 12 23
9 24 5 16 15 13 ...
## $ Infinium_Design_Type : Factor w/ 3 levels "", "I", "II": 2 2
2 2 2 2 2 2 2 ...
## $ Next_Base : Factor w/ 4 levels "", "A", "C", "T": 2
3 2 4 4 3 3 4 4 3 ...
## $ Color_Channel : Factor w/ 3 levels "", "Grn", "Red": 3
2 3 3 3 2 2 3 3 2 ...
## $ Forward_Sequence : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Genome_Build : int NA NA NA NA NA NA NA NA NA NA
...
## $ CHR : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ MAPINFO : int NA NA NA NA NA NA NA NA NA NA
...
## $ SourceSeq : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Chromosome_36 : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Coordinate_36 : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Strand : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Probe_SNPs : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Probe_SNPs_10 : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Random_Loci : logi NA NA NA NA NA NA ...
## $ Methyl27_Loci : logi NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ UCSC_RefGene_Accession : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ UCSC_RefGene_Group : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...

```



```
## $ UCSC_CpG_Islands_Name      : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Phantom                    : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ DMR                        : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Enhancer                   : logi  NA NA NA NA NA NA ...
## $ HMM_Island                 : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Regulatory_Feature_Name    : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Regulatory_Feature_Group   : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ DHS                        : logi  NA NA NA NA NA NA ...
```

916 lines are not too much...let's try to print them, considering only the first 2 columns:

notMappedToCHR[,c(1,2)]		
##	IlmnID	Name
## 485513	rs10796216	rs10796216
## 485514	rs715359	rs715359
## 485515	rs1040870	rs1040870
## 485516	rs10936224	rs10936224
## 485517	rs213028	rs213028
## 485518	rs2385226	rs2385226
## 485519	rs11034952	rs11034952
## 485520	rs9292570	rs9292570
## 485521	rs654498	rs654498
## 485522	rs1414097	rs1414097
## 485523	rs13369115	rs13369115
## 485524	rs10033147	rs10033147
## 485525	rs3936238	rs3936238
## 485526	rs1520670	rs1520670
## 485527	rs6991394	rs6991394
## 485528	rs951295	rs951295
## 485529	rs10882854	rs10882854
## 485530	rs2208123	rs2208123
## 485531	rs10774834	rs10774834
## 485532	rs10846239	rs10846239
## 485533	rs6471533	rs6471533
## 485534	rs5936512	rs5936512
## 485535	rs6626309	rs6626309
## 485536	rs5926356	rs5926356

## 485537	rs5987737	rs5987737
## 485538	rs1019916	rs1019916
## 485539	rs10457834	rs10457834
## 485540	rs1416770	rs1416770
## 485541	rs1941955	rs1941955
## 485542	rs2125573	rs2125573
## 485543	rs2235751	rs2235751
## 485544	rs2521373	rs2521373
## 485545	rs264581	rs264581
## 485546	rs2804694	rs2804694
## 485547	rs2959823	rs2959823
## 485548	rs5931272	rs5931272
## 485549	rs6546473	rs6546473
## 485550	rs739259	rs739259
## 485551	rs798149	rs798149
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##	485592	13643320	TARGET REMOVAL
##	485593	42790394	TARGET REMOVAL
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##	486087	30675376	NEGATIVE
##	486088	71734366	NEGATIVE

##	486089	22688305	NEGATIVE
##	486090	21705378	NEGATIVE
##	486091	65676384	NEGATIVE
##	486092	48639474	NEGATIVE
##	486093	29617416	NEGATIVE
##	486094	47764352	NEGATIVE
##	486095	50792435	NEGATIVE
##	486096	39742442	NEGATIVE
##	486097	24684310	NEGATIVE
##	486098	70610367	NEGATIVE
##	486099	57726345	NEGATIVE
##	486100	10721502	NEGATIVE
##	486101	52742375	NEGATIVE
##	486102	60726465	NEGATIVE
##	486103	27668419	NEGATIVE
##	486104	73697323	NEGATIVE
##	486105	15719334	NEGATIVE
##	486106	21643393	NEGATIVE
##	486107	37720318	NEGATIVE
##	486108	69752425	NEGATIVE
##	486109	48693431	NEGATIVE
##	486110	64601411	NEGATIVE
##	486111	24705436	NEGATIVE
##	486112	15740334	NEGATIVE
##	486113	24728321	NEGATIVE
##	486114	49680446	NEGATIVE
##	486115	27691388	NEGATIVE
##	486116	29657357	NEGATIVE
##	486117	19612454	NEGATIVE
##	486118	41705442	NEGATIVE
##	486119	31645490	NEGATIVE
##	486120	55673311	NEGATIVE
##	486121	31635492	NEGATIVE
##	486122	64683305	NEGATIVE
##	486123	11748306	NEGATIVE
##	486124	48730356	NEGATIVE
##	486125	11694363	NEGATIVE
##	486126	54665394	NEGATIVE
##	486127	28772416	NEGATIVE
##	486128	40747444	NEGATIVE
##	486129	50809407	NEGATIVE
##	486130	39679476	NEGATIVE
##	486131	74738436	NEGATIVE
##	486132	23734321	NEGATIVE
##	486133	15783320	NEGATIVE
##	486134	66703354	NEGATIVE

##	486135	56793432	NEGATIVE
##	486136	24744393	NEGATIVE
##	486137	14762352	NEGATIVE
##	486138	15794469	NEGATIVE
##	486139	35761501	NEGATIVE
##	486140	52677505	NEGATIVE
##	486141	73601495	NEGATIVE
##	486142	67640436	NEGATIVE
##	486143	51617379	NEGATIVE
##	486144	48683430	NEGATIVE
##	486145	35714458	NEGATIVE
##	486146	69652373	NEGATIVE
##	486147	43724337	NEGATIVE
##	486148	49658386	NEGATIVE
##	486149	70766461	NEGATIVE
##	486150	20661394	NEGATIVE
##	486151	55669483	NEGATIVE
##	486152	34630325	NEGATIVE
##	486153	13710478	NEGATIVE
##	486154	46635484	NEGATIVE
##	486155	34779461	NEGATIVE
##	486156	30762347	NEGATIVE
##	486157	36808443	NEGATIVE
##	486158	45708406	NEGATIVE
##	486159	55670455	NEGATIVE
##	486160	30689326	NEGATIVE
##	486161	42643451	NEGATIVE
##	486162	33730454	NEGATIVE
##	486163	69659385	NEGATIVE
##	486164	59731452	NEGATIVE
##	486165	60632306	NEGATIVE
##	486166	66793462	NEGATIVE
##	486167	56737301	NEGATIVE
##	486168	25621346	NEGATIVE
##	486169	62653340	NEGATIVE
##	486170	56781365	NEGATIVE
##	486171	71762482	NEGATIVE
##	486172	73667338	NEGATIVE
##	486173	15769506	NEGATIVE
##	486174	19775306	NEGATIVE
##	486175	19625434	NEGATIVE
##	486176	71775347	NEGATIVE
##	486177	44618314	NEGATIVE
##	486178	21635460	NEGATIVE
##	486179	59684453	NEGATIVE
##	486180	73789449	NEGATIVE

##	486181	14719400	NEGATIVE
##	486182	26658411	NEGATIVE
##	486183	33765459	NEGATIVE
##	486184	26769463	NEGATIVE
##	486185	42768497	NEGATIVE
##	486186	10744385	NEGATIVE
##	486187	26648509	NEGATIVE
##	486188	17796358	NEGATIVE
##	486189	56626320	NEGATIVE
##	486190	36803355	NEGATIVE
##	486191	68601372	NEGATIVE
##	486192	23631316	NEGATIVE
##	486193	50636432	NEGATIVE
##	486194	20666345	NEGATIVE
##	486195	26669386	NEGATIVE
##	486196	74720399	NEGATIVE
##	486197	33657423	NEGATIVE
##	486198	62672306	NEGATIVE
##	486199	56602420	NEGATIVE
##	486200	72656385	NEGATIVE
##	486201	19658444	NEGATIVE
##	486202	61721300	NEGATIVE
##	486203	26665427	NEGATIVE
##	486204	17799458	NEGATIVE
##	486205	49700377	NEGATIVE
##	486206	60632320	NEGATIVE
##	486207	60755343	NEGATIVE
##	486208	24809342	NEGATIVE
##	486209	22705438	NEGATIVE
##	486210	33651323	NEGATIVE
##	486211	10731326	NEGATIVE
##	486212	50699483	NEGATIVE
##	486213	26770499	NEGATIVE
##	486214	15727485	NEGATIVE
##	486215	36733310	NEGATIVE
##	486216	52665450	NEGATIVE
##	486217	70717333	NEGATIVE
##	486218	62632474	NEGATIVE
##	486219	52672329	NEGATIVE
##	486220	44760439	NEGATIVE
##	486221	16673481	NEGATIVE
##	486222	56788431	NEGATIVE
##	486223	15797317	NEGATIVE
##	486224	23740309	NEGATIVE
##	486225	17641445	NEGATIVE
##	486226	10760363	NEGATIVE

##	486227	21620397	NEGATIVE
##	486228	23786490	NEGATIVE
##	486229	33677481	NEGATIVE
##	486230	14777361	NEGATIVE
##	486231	69729339	NEGATIVE
##	486232	37710397	NEGATIVE
##	486233	60682508	NEGATIVE
##	486234	40713496	NEGATIVE
##	486235	12641437	NEGATIVE
##	486236	48797398	NEGATIVE
##	486237	66764427	NEGATIVE
##	486238	74654383	NEGATIVE
##	486239	45807495	NEGATIVE
##	486240	37634407	NEGATIVE
##	486241	73603488	NEGATIVE
##	486242	41636384	RESTORATION
##	486243	43652393	NORM_A
##	486244	18733503	NORM_G
##	486245	70603486	NORM_A
##	486246	63775337	NORM_G
##	486247	70622474	NORM_A
##	486248	58676498	NORM_G
##	486249	17630478	NORM_A
##	486250	35689446	NORM_G
##	486251	15639322	NORM_A
##	486252	61760309	NORM_G
##	486253	66624473	NORM_A
##	486254	59734347	NORM_G
##	486255	18654457	NORM_A
##	486256	67713480	NORM_G
##	486257	49636425	NORM_A
##	486258	36806404	NORM_G
##	486259	31786303	NORM_A
##	486260	16690480	NORM_G
##	486261	70700334	NORM_A
##	486262	32689421	NORM_G
##	486263	42624324	NORM_A
##	486264	39662476	NORM_G
##	486265	62796402	NORM_A
##	486266	48709470	NORM_G
##	486267	67699483	NORM_A
##	486268	67721390	NORM_G
##	486269	61620502	NORM_A
##	486270	63724326	NORM_G
##	486271	16723352	NORM_A
##	486272	54708463	NORM_G

##	486273	68641317	NORM_A
##	486274	15733328	NORM_G
##	486275	48622429	NORM_A
##	486276	51668449	NORM_G
##	486277	44620377	NORM_A
##	486278	63640430	NORM_G
##	486279	33613483	NORM_A
##	486280	25661400	NORM_G
##	486281	63602392	NORM_A
##	486282	24744355	NORM_G
##	486283	31793441	NORM_A
##	486284	52807455	NORM_G
##	486285	15747446	NORM_A
##	486286	10804473	NORM_G
##	486287	72748406	NORM_A
##	486288	59624459	NORM_G
##	486289	62710448	NORM_A
##	486290	15642403	NORM_G
##	486291	11718309	NORM_A
##	486292	69667417	NORM_G
##	486293	61626504	NORM_A
##	486294	18658347	NORM_G
##	486295	49729303	NORM_A
##	486296	63742354	NORM_G
##	486297	74662463	NORM_A
##	486298	59764324	NORM_G
##	486299	68634356	NORM_A
##	486300	48757493	NORM_G
##	486301	24636465	NORM_A
##	486302	38796356	NORM_G
##	486303	36657375	NORM_A
##	486304	28724405	NORM_G
##	486305	68744490	NORM_A
##	486306	69624309	NORM_G
##	486307	26787359	NORM_C
##	486308	60810444	NORM_T
##	486309	27756327	NORM_C
##	486310	11641439	NORM_T
##	486311	56778357	NORM_C
##	486312	71678368	NORM_T
##	486313	40720405	NORM_C
##	486314	31707322	NORM_T
##	486315	51606350	NORM_C
##	486316	43724438	NORM_T
##	486317	39615318	NORM_C
##	486318	54713306	NORM_T

##	486319	67751351	NORM_C
##	486320	37766401	NORM_T
##	486321	43751315	NORM_C
##	486322	11742370	NORM_T
##	486323	25717484	NORM_C
##	486324	25790300	NORM_T
##	486325	16739368	NORM_C
##	486326	40666333	NORM_T
##	486327	34710461	NORM_C
##	486328	11758414	NORM_T
##	486329	57799313	NORM_C
##	486330	36621472	NORM_T
##	486331	35717396	NORM_C
##	486332	30750303	NORM_T
##	486333	55732329	NORM_C
##	486334	42722426	NORM_T
##	486335	38663386	NORM_C
##	486336	26704364	NORM_T
##	486337	44650346	NORM_C
##	486338	15681446	NORM_T
##	486339	73794434	NORM_C
##	486340	31736353	NORM_T
##	486341	38708375	NORM_C
##	486342	29774306	NORM_T
##	486343	63615474	NORM_C
##	486344	33644415	NORM_T
##	486345	34642451	NORM_C
##	486346	69660330	NORM_T
##	486347	18791508	NORM_C
##	486348	36629493	NORM_T
##	486349	59756399	NORM_C
##	486350	55772319	NORM_T
##	486351	35810456	NORM_C
##	486352	33806355	NORM_T
##	486353	17719492	NORM_C
##	486354	31775443	NORM_T
##	486355	55747501	NORM_C
##	486356	46610354	NORM_T
##	486357	57649399	NORM_C
##	486358	55662399	NORM_T
##	486359	17801437	NORM_C
##	486360	19695407	NORM_T
##	486361	51761481	NORM_C
##	486362	63623369	NORM_T
##	486363	43647400	NORM_C
##	486364	59792509	NORM_T

## 486365	21666326	NORM_C
## 486366	45648306	NORM_T
## 486367	15751492	NORM_C
## 486368	12713402	NORM_T
## 486369	49714326	NORM_C
## 486370	57713321	NORM_T
## 486371	67775306	NORM_C
## 486372	67735372	NORM_T
## 486373	11651448	NORM_C
## 486374	44631336	NORM_T
## 486375	12606314	NORM_C
## 486376	73635489	NORM_T
## 486377	63709386	NORM_C
## 486378	52747473	NORM_T
## 486379	49675379	NORM_C
## 486380	68705481	NORM_T
## 486381	22667482	NORM_C
## 486382	55799383	NORM_T
## 486383	56738460	NORM_C
## 486384	49721486	NORM_T
## 486385	56656442	NORM_C
## 486386	15681421	NORM_T
## 486387	22786434	NORM_C
## 486388	23770483	NORM_T
## 486389	47796462	NORM_C
## 486390	15754447	NORM_T
## 486391	29769366	NORM_C
## 486392	44612371	NORM_T
## 486393	48634496	NORM_C
## 486394	16759387	NORM_T
## 486395	73784382	NORM_C
## 486396	68726417	NORM_T
## 486397	14669466	NORM_C
## 486398	10714330	NORM_T
## 486399	24711322	NORM_C
## 486400	20651487	NORM_T
## 486401	38781340	NORM_C
## 486402	52668393	NORM_T
## 486403	52761413	NORM_C
## 486404	28790326	NORM_T
## 486405	33780372	NORM_C
## 486406	36711421	NORM_T
## 486407	71670310	NORM_C
## 486408	29618498	NORM_T
## 486409	23684492	NORM_C
## 486410	68657384	NORM_T

##	486411	34720366	NORM_C
##	486412	58768373	NORM_T
##	486413	11739368	NORM_C
##	486414	11749491	NORM_T
##	486415	62708477	NORM_C
##	486416	55684415	NORM_T
##	486417	64609487	NORM_C
##	486418	34667361	NORM_T
##	486419	56806464	NORM_C
##	486420	31747339	NORM_T
##	486421	41745372	NORM_C
##	486422	14656336	NORM_T
##	486423	65610474	NORM_C
##	486424	65622344	NORM_T
##	486425	57640346	NORM_C
##	486426	47715450	NORM_T
##	486427	28673402	NORM_C
##	486428	13742412	NORM_T

We can use the table function to have some hints:

```
table(notMappedToCHR$Name) #Try the other possibilities to select the
column "Name"
```

##

##		BISULFITE CONVERSION I	BISULFITE CONVERSION
II			
##		1	12
4			
##	EXTENSION		HYBRIDIZATION
NEGATIVE			
##		4	3
613			
##	NON-POLYMORPHIC		NORM_A
NORM_C			
##		4	32
61			
##	NORM_G		NORM_T
RESTORATION			
##		32	61
1			
##	rs10033147		rs1019916
rs1040870			
##		1	1
1			

##	rs10457834	rs10774834
rs10796216		
##	1	1
1		
##	rs10846239	rs10882854
rs10936224		
##	1	1
1		
##	rs11034952	rs11249206
rs13369115		
##	1	1
1		
##	rs133860	rs1414097
rs1416770		
##	1	1
1		
##	rs1467387	rs1484127
rs1495031		
##	1	1
1		
##	rs1510189	rs1510480
rs1520670		
##	1	1
1		
##	rs1941955	rs1945975
rs2032088		
##	1	1
1		
##	rs2125573	rs213028
rs2208123		
##	1	1
1		
##	rs2235751	rs2385226
rs2468330		
##	1	1
1		
##	rs2521373	rs264581
rs2804694		
##	1	1
1		
##	rs2857639	rs2959823
rs348937		
##	1	1
1		
##	rs3818562	rs3936238
rs4331560		

##	1	1	
1			
##	rs472920	rs4742386	
rs5926356			
##	1	1	
1			
##	rs5931272	rs5936512	
rs5987737			
##	1	1	
1			
##	rs6426327	rs6471533	
rs654498			
##	1	1	
1			
##	rs6546473	rs6626309	
rs6982811			
##	1	1	
1			
##	rs6991394	rs715359	
rs739259			
##	1	1	
1			
##	rs7660805	rs7746156	
rs798149			
##	1	1	
1			
##	rs845016	rs877309	
rs9292570			
##	1	1	
1			
##	rs9363764	rs939290	
rs951295			
##	1	1	
1			
##	rs966367	rs9839873	
SPECIFICITY I			
##	1	1	
12			
##	SPECIFICITY II	STAINING	TARGET
REMOVAL			
##	3	6	
2			

There are 65 probes whose name starts with “rs”. These probes do not measure DNA methylation, but genetic polymorphisms (SNPs). They are included in the 450k design to allow

sample tracking and check for sample mixups. The remaining probes are control probes that we will describe in the next lesson

Now we want to create a new dataframe in which we remove the control and the rs probes. We can perform this task in different ways: - we can create an object that stores the result of the condition:

```
only_in_a_chromosome <- Illumina450Manifest$CHR!=""
```

```
head(only_in_a_chromosome)
```

```
## [1] TRUE TRUE TRUE TRUE TRUE TRUE
```

```
table(only_in_a_chromosome)
```

```
## only_in_a_chromosome
```

```
## FALSE TRUE
```

```
## 916 485512
```

```
Illumina450Manifest_clean <- Illumina450Manifest[only_in_a_chromosome,]
```

```
dim(Illumina450Manifest_clean)
```

```
## [1] 485512 33
```

- or we can apply directly the condition to subset the dataframe

```
Illumina450Manifest_clean <-
```

```
Illumina450Manifest[Illumina450Manifest$CHR!="",]
```

```
dim(Illumina450Manifest_clean)
```

```
## [1] 485512 33
```

3. or we can select the probes which are not included in the notMappedToCHR object

```
Illumina450Manifest_clean <-
```

```
Illumina450Manifest[!Illumina450Manifest$IlmnID %in%  
notMappedToCHR$IlmnID,]
```

```
dim(Illumina450Manifest_clean)
```

```
## [1] 485512 33
```

In any case, always remember to drop the unused levels!

```
str(Illumina450Manifest_clean)
```



```
## 'data.frame':    485512 obs. of  33 variables:
```

```
## $ IlmnID : Factor w/ 486428 levels  
"[Controls]","10627500",...: 1743 2082 2315 2353 3658 5417 5451 5467  
5662 6070 ...  
## $ Name : Factor w/ 485593 levels  
"", "BISULFITE CONVERSION I",...: 895 1234 1467 1505 2810 4569 4603 4619  
4814 5222 ...  
## $ AddressA_ID : Factor w/ 485614 levels "", "-  
99","10600322",...: 163955 171493 142506 49192 74497 146457 155290  
448596 56107 200820 ...  
## $ AlleleA_ProbeSeq : Factor w/ 486428 levels  
"", "AAAAAAAAAAAAAAAAAACCAAAAAAAAAATCCCCACTCTAAAACCCACACA",...: 35018  
143551 35762 428168 41002 296102 483734 335382 461498 128017 ...  
## $ AddressB_ID : Factor w/ 135503 levels  
"", "10600313",...: 1 45525 1 1 1 60142 56512 76868 1 1 ...  
## $ AlleleB_ProbeSeq : Factor w/ 135502 levels  
"", "AAAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAAACCCCGCG",...: 1 49675 1 1  
1 77355 135117 95693 1 1 ...  
## $ Infinium_Design_Type : Factor w/ 3 levels "", "I", "II": 3 2  
3 3 3 2 2 2 3 3 ...  
## $ Next_Base : Factor w/ 4 levels "", "A", "C", "T": 1  
2 1 1 1 4 2 2 1 1 ...  
## $ Color_Channel : Factor w/ 3 levels "", "Grn", "Red": 1  
3 1 1 1 3 3 3 1 1 ...  
## $ Forward_Sequence : Factor w/ 485513 levels  
"", "AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAATATGTCTGCTAAATAATCACAGAGTGAAA[CG]AAG  
AGGAAGAACTAAAGTACAGTTTGGAAACATAGTACAGTGCCCTGGGTACTGAATCA",...: 25106  
392559 395303 218896 81396 152102 421755 296780 478969 55952 ...  
## $ Genome_Build : int 37 37 37 37 37 37 37 37 37 37  
...  
## $ CHR : Factor w/ 25 levels  
"", "1", "10", "11",...: 25 25 25 25 25 25 25 25 25 ...  
## $ MAPINFO : int 8553009 9363356 25314171  
22741795 21664296 21239348 8148233 15815688 4868996 6133740 ...  
## $ SourceSeq : Factor w/ 485344 levels  
"", "AAAAAAAAAACCAACCCCAACAAAAACCCCTGTGCCAGAAATTGCATTTCG",...: 28962  
285897 116668 116692 29628 123295 148857 152724 189023 323598 ...  
## $ Chromosome_36 : Factor w/ 26 levels  
"", "1", "10", "11",...: 26 26 26 26 26 26 26 26 26 ...  
## $ Coordinate_36 : Factor w/ 484739 levels  
"", "100000057",...: 451149 483812 205768 185981 178029 173645 442781  
102377 331284 376024 ...  
## $ Strand : Factor w/ 3 levels "", "F", "R": 2 3 3  
2 3 3 3 3 3 2 ...
```

```
## $ Probe_SNPs : Factor w/ 56004 levels
"", "rs10000615", ...: 1 1 1 54643 1 1 1 1 1 1 ...
## $ Probe_SNPs_10 : Factor w/ 35790 levels
"", "rs10000804", ...: 1 1 1 4064 1 1 1 1 1 1 ...
## $ Random_Loci : logi NA NA NA NA NA NA ...
## $ Methyl27_Loci : logi NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name : Factor w/ 42456 levels
"", "A1BG", "A1CF;A1CF;A1CF", ...: 39103 38905 9635 11319 3085 39100 1
37974 27057 1 ...
## $ UCSC_RefGene_Accession : Factor w/ 52670 levels
"", "NM_000014", ...: 48799 14035 19907 20891 49121 48792 1 20105 38174 1
...
## $ UCSC_RefGene_Group : Factor w/ 4165 levels
"", "1stExon", "1stExon;1stExon", ...: 2970 2779 2447 2035 2035 3694 1 201
943 1 ...
## $ UCSC_CpG_Islands_Name : Factor w/ 27177 levels
"", "chr1:10003165-10003585", ...: 1 27170 1 27136 27134 27133 27156 27120
27147 27149 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 6 levels
"", "Island", "N_Shelf", ...: 1 4 1 5 4 2 6 2 4 2 ...
## $ Phantom : Factor w/ 11912 levels "", "high-
CpG:100009860-100010119", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ DMR : Factor w/ 4 levels
"", "CDMR", "DMR", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Enhancer : logi NA NA NA NA NA NA ...
## $ HMM_Island : Factor w/ 63086 levels
"", "1:100088009-100088708", ...: 1 63085 1 1 1 63046 63077 63033 63064
63067 ...
## $ Regulatory_Feature_Name : Factor w/ 47969 levels
"", "1:10001853-10003978", ...: 1 1 1 1 1 1 1 47965 1 1 ...
## $ Regulatory_Feature_Group : Factor w/ 9 levels
"", "Gene_Associated", ...: 1 1 1 1 1 1 1 7 1 1 ...
## $ DHS : logi NA NA NA NA NA NA ...
```

```

Illumina450Manifest_clean <- droplevels(Illumina450Manifest_clean)

```

```
str(Illumina450Manifest_clean)
```

```
## 'data.frame': 485512 obs. of 33 variables:
```

```
## $ IlmnID : Factor w/ 485512 levels
"cg000000029", "cg000000108", ...: 892 1231 1464 1502 2807 4566 4600 4616
4811 5219 ...
## $ Name : Factor w/ 485512 levels
"cg000000029", "cg000000108", ...: 892 1231 1464 1502 2807 4566 4600 4616
4811 5219 ...
```

```

## $ AddressA_ID : Factor w/ 485512 levels
"10600322","10600328",...: 163924 171461 142476 49179 74479 146426
155259 448531 56092 200778 ...
## $ AlleleA_ProbeSeq : Factor w/ 485512 levels
"AAAAAAAAAAAAAAAAAACACAAAAAATCCCCACTCTAAAACCCACACA",...: 35017 143540
35761 427267 41000 296053 482818 335327 460590 128010 ...
## $ AddressB_ID : Factor w/ 135477 levels
"", "10600313",...: 1 45514 1 1 1 60128 56499 76848 1 1 ...
## $ AlleleB_ProbeSeq : Factor w/ 135477 levels
"", "AAAAAAAAAAAAAAAAAACGCGAAAAAATCCCCGCTCTAAAACCCGCG",...: 1 49667 1 1
1 77344 135092 95678 1 1 ...
## $ Infinium_Design_Type : Factor w/ 2 levels "I","II": 2 1 2 2
2 1 1 1 2 2 ...
## $ Next_Base : Factor w/ 4 levels "", "A", "C", "T": 1
2 1 1 1 4 2 2 1 1 ...
## $ Color_Channel : Factor w/ 3 levels "", "Grn", "Red": 1
3 1 1 1 3 3 3 1 1 ...
## $ Forward_Sequence : Factor w/ 485512 levels
"AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAATATGTCTGCTAAATAATCACAGAGTGAAA[CG]AAGAGG
AAGAAACTAAAGTACAGTTTGGAACATAGTACAGTGCCCCCTGGGTACTGAATCA",...: 25105
392558 395302 218895 81395 152101 421754 296779 478968 55951 ...
## $ Genome_Build : int 37 37 37 37 37 37 37 37 37 37 37
...
## $ CHR : Factor w/ 24 levels
"1","10","11",...: 24 24 24 24 24 24 24 24 24 24 ...
## $ MAPINFO : int 8553009 9363356 25314171
22741795 21664296 21239348 8148233 15815688 4868996 6133740 ...
## $ SourceSeq : Factor w/ 485343 levels
"AAAAAAAAAACCAACCCCAACAAAAACCCCTGTGCCCAGAATTGCATTCG",...: 28961 285896
116667 116691 29627 123294 148856 152723 189022 323597 ...
## $ Chromosome_36 : Factor w/ 25 levels
"1","10","11",...: 25 25 25 25 25 25 25 25 25 25 ...
## $ Coordinate_36 : Factor w/ 484738 levels
"100000057","100000691",...: 451148 483811 205767 185980 178028 173644
442780 102376 331283 376023 ...
## $ Strand : Factor w/ 2 levels "F","R": 1 2 2 1
2 2 2 2 2 1 ...
## $ Probe_SNPs : Factor w/ 56004 levels
"", "rs10000615",...: 1 1 1 54643 1 1 1 1 1 1 ...
## $ Probe_SNPs_10 : Factor w/ 35790 levels
"", "rs10000804",...: 1 1 1 4064 1 1 1 1 1 1 ...
## $ Random_Loci : logi NA NA NA NA NA NA ...
## $ Methyl27_Loci : logi NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name : Factor w/ 42456 levels
"", "A1BG","A1CF;A1CF;A1CF",...: 39103 38905 9635 11319 3085 39100 1
37974 27057 1 ...

```

```
## $ UCSC_RefGene_Accession      : Factor w/ 52670 levels
"", "NM_000014", ...: 48799 14035 19907 20891 49121 48792 1 20105 38174 1
...
## $ UCSC_RefGene_Group          : Factor w/ 4165 levels
"", "1stExon", "1stExon;1stExon", ...: 2970 2779 2447 2035 2035 3694 1 201
943 1 ...
## $ UCSC_CpG_Islands_Name       : Factor w/ 27177 levels
"", "chr1:10003165-10003585", ...: 1 27170 1 27136 27134 27133 27156 27120
27147 27149 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 6 levels
"", "Island", "N_Shelf", ...: 1 4 1 5 4 2 6 2 4 2 ...
## $ Phantom                    : Factor w/ 11912 levels "", "high-
CpG:100009860-100010119", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ DMR                       : Factor w/ 4 levels
"", "CDMR", "DMR", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Enhancer                  : logi  NA NA NA NA NA NA ...
## $ HMM_Island                : Factor w/ 63086 levels
"", "1:100088009-100088708", ...: 1 63085 1 1 1 63046 63077 63033 63064
63067 ...
## $ Regulatory_Feature_Name     : Factor w/ 47969 levels
"", "1:10001853-10003978", ...: 1 1 1 1 1 1 1 47965 1 1 ...
## $ Regulatory_Feature_Group    : Factor w/ 9 levels
"", "Gene_Associated", ...: 1 1 1 1 1 1 1 7 1 1 ...
## $ DHS                      : logi  NA NA NA NA NA NA ...
```

We will save the clean manifest in a new file:

```
save(Illumina450Manifest_clean, file="Illumina450Manifest_clean.RData")
```

CHUNK3: Islands, shores and shelves

Let's use the exploratory function table:

```
table(Illumina450Manifest_clean$Relation_to_UCSC_CpG_Island)
```

```
##
```

```
##           Island N_Shelf N_Shore S_Shelf S_Shore
## 176047 150254 24844 62870 22300 49197
```

How many probes map in Islands?

```
ProbesIslands <-
Illumina450Manifest_clean[Illumina450Manifest_clean$Relation_to_UCSC_Cp
G_Island=="Island",]
```

```
dim(ProbesIslands)
```

```
## [1] 150254      33
```

```
table(ProbesIslands$Relation_to_UCSC_CpG_Island)
```

```
##
```

```
##      Island N_Shelf N_Shore S_Shelf S_Shore
##      0 150254      0      0      0      0
```

```
ProbesIslands <- droplevels(ProbesIslands)
```

```
table(ProbesIslands$Relation_to_UCSC_CpG_Island)
```

```
##
```

```
## Island
## 150254
```

How many probes in Islands map on chromosome 1? I will use a double condition:

```
ProbesIslands_chromosome1 <-
Illumina450Manifest_clean[Illumina450Manifest_clean$Relation_to_UCSC_Cp
G_Island=="Island" & Illumina450Manifest_clean$CHR=="1",]
```

```
dim(ProbesIslands_chromosome1)
```

```
## [1] 14339      33
```

The table function allows also to investigate the combination of 2 columns:

```
table(Illumina450Manifest_clean$Relation_to_UCSC_CpG_Island,Illumina450
Manifest_clean$CHR)
```

```
##
```

```
##      1      10      11      12      13      14      15      16      17
18      19
##      17080 10268 11342  9790  5306  6025  6379  7013  8717
943 3431
##      Island 14339  6885  8355  6862  3179  4472  4510  6966  9301
2769 11780
##      N_Shelf 2324  1124  1458  1223   627   729   683  1386  1706
392  1734
##      N_Shore 6173  2738  3459  3072  1457  1737  1657  2938  3575
777  3907
##      S_Shelf 2097  1074  1287  1074   531   680   632  1143  1495
357  1487
```

```
## S_Shore 4844 2299 2893 2518 1185 1435 1398 2523 3085
684 3182
##
##      2    20    21    22     3     4     5     6     7
8      9
##      14848 1868   995 1553 10595 8480 9694 15235 11584
9111 2722
## Island 9899 4428 1681 3565 7331 5855 7031 9327 8379
5621 3307
## N_Shelf 1435 653 289 610 1043 833 1029 1899 1646
943 457
## N_Shore 4164 1637 631 1263 2957 2593 3409 4766 3926
2491 1896
## S_Shelf 1363 606 248 551 974 837 971 1680 1376
835 433
## S_Shore 3101 1187 399 1010 2259 1866 2193 3704 3106
1949 1046
##
##      X      Y
##      2975   93
## Island 4275 137
## N_Shelf 598 23
## N_Shore 1550 97
## S_Shelf 542 27
## S_Shore 1292 39
```

```
table(ProbesIslands_chromosome1$Relation_to_UCSC_CpG_Island,
ProbesIslands_chromosome1$CHR)
```

```
##
```

```
##      1    10    11    12    13    14    15    16    17
18    19
##      0     0     0     0     0     0     0     0     0
0      0
## Island 14339 0     0     0     0     0     0     0     0
0      0
## N_Shelf 0     0     0     0     0     0     0     0     0
0      0
## N_Shore 0     0     0     0     0     0     0     0     0
0      0
## S_Shelf 0     0     0     0     0     0     0     0     0
0      0
## S_Shore 0     0     0     0     0     0     0     0     0
0      0
##
```

```
##           2    20    21    22    3    4    5    6    7
8      9
##           0    0    0    0    0    0    0    0    0
0      0
##   Island    0    0    0    0    0    0    0    0    0
0      0
##   N_Shelf   0    0    0    0    0    0    0    0    0
0      0
##   N_Shore   0    0    0    0    0    0    0    0    0
0      0
##   S_Shelf   0    0    0    0    0    0    0    0    0
0      0
##   S_Shore   0    0    0    0    0    0    0    0    0
0      0
##
##           X     Y
##           0     0
##   Island    0     0
##   N_Shelf   0     0
##   N_Shore   0     0
##   S_Shelf   0     0
##   S_Shore   0     0
```

```
ProbesIslands_chromosome1 <- droplevels(ProbesIslands_chromosome1)
table(ProbesIslands_chromosome1$Relation_to_UCSC_CpG_Island,ProbesIslands_chromosome1$CHR)
```

```
##
##           1
##   Island 14339
```

How many probes in Islands map also in a gene? Again, I will use a double condition

```
ProbesIslands_Gene <-
Illumina450Manifest_clean[Illumina450Manifest_clean$Relation_to_UCSC_CpG_Island=="Island" & Illumina450Manifest_clean$UCSC_RefGene_Name!="",]
```

```
dim(ProbesIslands_Gene)
```

```
## [1] 127862    33
```

```
ProbesIslands_Gene <- droplevels(ProbesIslands_Gene)
```

```
str(ProbesIslands_Gene)
```

```
## 'data.frame':   127862 obs. of  33 variables:
```

```

## $ IlmnID : Factor w/ 127862 levels
"cg00000622","cg00000658",...: 1197 1208 1506 4827 8244 9147 10723 10769
11264 11367 ...
## $ Name : Factor w/ 127862 levels
"cg00000622","cg00000658",...: 1197 1208 1506 4827 8244 9147 10723 10769
11264 11367 ...
## $ AddressA_ID : Factor w/ 127862 levels
"10600369","10600404",...: 38611 118220 124704 91029 17352 70443 31580
33636 13897 32504 ...
## $ AlleleA_ProbeSeq : Factor w/ 127862 levels
"AAAAAAAAAAAAAAAAAACACAAAAAATCCCCACTCTAAAACCCACACA",...: 82219 93451
25727 87832 91195 44313 47687 57173 51140 4573 ...
## $ AddressB_ID : Factor w/ 67756 levels
"", "10600313",...: 29968 38284 1 1 1 57060 50353 54704 1 1 ...
## $ AlleleB_ProbeSeq : Factor w/ 67756 levels
"", "AAAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAAACCCCGCG",...: 38124 47777
1 1 1 54725 22998 29658 1 1 ...
## $ Infinium_Design_Type : Factor w/ 2 levels "I","II": 1 1 2 2
2 1 1 1 2 2 ...
## $ Next_Base : Factor w/ 4 levels "", "A", "C", "T": 4
2 1 1 1 2 3 2 1 1 ...
## $ Color_Channel : Factor w/ 3 levels "", "Grn", "Red": 3
3 1 1 1 3 2 3 1 1 ...
## $ Forward_Sequence : Factor w/ 127862 levels
"AAAAAAAAAAAAAAAAAACCGGAAATGGGTCTTACCATCTTCTCGGAGCCGGAGTG[CG]AAGAAA
TAAAGAAATAGTGCTTTAAGTCAATGAATTCCTCCTTGGGACCCACTATCGAGA",...: 31336 80035
9453 55821 47431 13681 75354 12396 93153 72112 ...
## $ Genome_Build : int 37 37 37 37 37 37 37 37 37 37
...
## $ CHR : Factor w/ 24 levels
"1","10","11",...: 24 24 24 24 24 24 24 24 24 24 ...
## $ MAPINFO : int 21239348 15815688 2802965
21664717 9234365 6778695 15815552 9194502 2804282 9194524 ...
## $ SourceSeq : Factor w/ 127782 levels
"AAAAAAAAAATCCCCGTTGCTCTTTACACCCCGGAAACGGCAAGGGAACG",...: 27794 37355
108962 99224 78031 38585 64768 112440 22205 99060 ...
## $ Chromosome_36 : Factor w/ 25 levels
"1","10","11",...: 25 25 25 25 25 25 25 25 25 25 ...
## $ Coordinate_36 : Factor w/ 127809 levels
"100009094","100009134",...: 42653 24921 57279 43876 123065 105177 24920
122831 57301 122832 ...
## $ Strand : Factor w/ 2 levels "F","R": 2 2 2 1
2 1 1 1 1 1 ...
## $ Probe_SNPs : Factor w/ 12585 levels
"", "rs10009502",...: 1 1 1 1 1 1 12476 1 1 1 ...

```



```
## $ Probe_SNPs_10 : Factor w/ 7164 levels
"", "rs1000244", ...: 1 1 1 1 1 1 1 1 5857 1 ...
## $ Random_Loci : logi NA NA NA NA NA NA ...
## $ Methyl27_Loci : logi NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name : Factor w/ 25621 levels
"A1BG", "A2BP1;A2BP1", ...: 23305 22633 24691 1909 23185 21765 22633 23185
24691 23185 ...
## $ UCSC_RefGene_Accession : Factor w/ 29080 levels
"NM_000016;NM_000016;NM_001127328;NM_001127328", ...: 27622 10402 9615
27737 6881 22921 10402 6881 9613 6881 ...
## $ UCSC_RefGene_Group : Factor w/ 2668 levels
"1stExon", "1stExon;1stExon", ...: 2316 159 2274 1352 1352 2613 159 1841
946 1841 ...
## $ UCSC_CpG_Islands_Name : Factor w/ 20836 levels
"chr1:10003165-10003585", ...: 20818 20812 20824 20819 20832 20828 20812
20830 20824 20830 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 1 level "Island": 1 1 1 1
1 1 1 1 1 1 ...
## $ Phantom : Factor w/ 8655 levels "", "high-
CpG:100009860-100010119", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ DMR : Factor w/ 4 levels
"", "CDMR", "DMR", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Enhancer : logi NA NA NA NA NA NA ...
## $ HMM_Island : Factor w/ 21551 levels
"", "1:100088009-100088708", ...: 21536 21531 21542 21537 21549 21546
21531 21548 21542 21548 ...
## $ Regulatory_Feature_Name : Factor w/ 15960 levels
"", "1:10001853-10003978", ...: 1 15959 1 1 1 1 15959 1 15960 1 ...
## $ Regulatory_Feature_Group : Factor w/ 9 levels
"", "Gene_Associated", ...: 1 7 1 1 1 1 7 1 7 1 ...
## $ DHS : logi NA NA NA NA NA NA ...
```

Now we want to identify the CpG island (considering also its shores and shelves) with the highest number of probes

```
table(Illumina450Manifest_clean$UCSC_CpG_Islands_Name)
```

```
##
```

```
## chr1:10003165-10003585
```

```
chr1:1002663-1005318
```

```
## 176047 14
```

```
18
```

```
## chr1:100315420-100316009 chr1:100435297-100436070 chr1:100503482-
100504404
```

```
## 18 19
```

```
16
```

##	chr1:10057121-10058108	chr1:100598130-100598754	chr1:100678766-100679098
##	15	16	
7			
##	chr1:100731632-100731999	chr1:100817709-100818899	
	chr1:10092532-10093404		
##	14	14	
13			
##	chr1:101004471-101005885	chr1:1011509-1013402	chr1:101360588-101360832
##	15	9	
12			
##	chr1:101361549-101361808	chr1:101491317-101491680	
	chr1:1016863-1017439		
##	7	15	
8			
##	chr1:101701785-101702068	chr1:101702445-101702745	
	chr1:1018099-1018390		
##	13	8	
3			
##	chr1:1019818-1020026	chr1:1023143-1023347	
	chr1:1024075-1024614		
##	8	11	
5			
##	chr1:1026377-1026625	chr1:10270015-10271306	
	chr1:1027509-1027730		
##	2	14	
6			
##	chr1:10386183-10386390	chr1:104068487-104068913	
	chr1:10458548-10459997		
##	8	13	
18			
##	chr1:1046044-1046340	chr1:10490200-10491072	
	chr1:1050515-1050830		
##	12	19	
8			
##	chr1:1051178-1052445	chr1:10532177-10532800	
	chr1:10534918-10535172		
##	14	13	
11			
##	chr1:106623393-106623854	chr1:10698299-10698910	
	chr1:10699394-10700098		
##	10	8	
4			
##	chr1:10702136-10702340	chr1:10707810-10708140	
	chr1:10713838-10714203		

##	5	16	
11			
##	chr1:10714401-10714708	chr1:10719867-10720581	
	chr1:1072369-1072847		
##	4	10	
16			
##	chr1:10725187-10725617	chr1:10753773-10754176	
	chr1:10755226-10755521		
##	12	6	
3			
##	chr1:10757891-10758252	chr1:107599205-107600410	
	chr1:10764449-10764925		
##	6	15	
7			
##	chr1:107682889-107684463	chr1:108023232-108023442	chr1:108507251-108507841
##	23	8	
13			
##	chr1:10853894-10856964	chr1:108735227-108735513	chr1:108742400-108743201
##	23	6	
15			
##	chr1:10895895-10896117	chr1:109102538-109103212	chr1:109203593-109204378
##	7	9	
11			
##	chr1:109234788-109235161	chr1:109289018-109289822	
	chr1:1093211-1093476		
##	14	11	
8			
##	chr1:109419544-109420038	chr1:109505681-109506214	chr1:109583855-109585032
##	12	13	
19			
##	chr1:109618579-109618832	chr1:109633188-109633632	chr1:109642663-109643384
##	14	16	
14			
##	chr1:109656095-109656611	chr1:109756442-109757098	chr1:109791945-109792723
##	17	12	
13			
##	chr1:109801054-109801700	chr1:1098043-1100584	chr1:109824313-109824526
##	9	46	
5			

## chr1:109825710-109826207	chr1:109940074-109940894	chr1:109968648-109969570
## 8	14	
16		
## chr1:110008962-110010124	chr1:110026630-110026903	chr1:110036545-110036943
## 13	11	
13		
## chr1:110051954-110052505	chr1:110091039-110091590	chr1:110162464-110162716
## 13	18	
20		
## chr1:110185961-110186164	chr1:110198541-110199127	chr1:110210581-110210956
## 7	17	
12		
## chr1:110230238-110230614	chr1:110282351-110283306	chr1:110453143-110453889
## 7	10	
12		
## chr1:110527248-110528026	chr1:110577199-110577942	chr1:110610265-110613303
## 18	17	
22		
## chr1:110626528-110627484	chr1:110672810-110673339	chr1:110692988-110694264
## 11	9	
13		
## chr1:11072319-11073245	chr1:110752256-110754794	chr1:110765665-110766012
## 8	17	
8		
## chr1:110880394-110880624	chr1:110880942-110882800	chr1:1109314-1110145
## 5	16	
19		
## chr1:110949892-110950540	chr1:111098066-111098362	chr1:111148983-111150186
## 17	8	
17		
## chr1:11119030-11120634	chr1:111216244-111217937	chr1:111323223-111323433
## 16	21	
7		
## chr1:111505881-111507007	chr1:1115574-1116006	chr1:11159387-11160136

##	21	24	
19			
##	chr1:111682227-111683303	chr1:111746337-111747303	chr1:111813411-111813853
##	21	21	
9			
##	chr1:111889138-111889539	chr1:111991304-111992202	chr1:112058184-112058590
##	18	17	
8			
##	chr1:112162204-112162837	chr1:112281052-112283277	chr1:112298041-112298992
##	14	19	
20			
##	chr1:112524468-112525385	chr1:112531559-112533773	chr1:112938495-112939113
##	9	24	
11			
##	chr1:113008844-113009154	chr1:113044410-113044992	chr1:113050813-113052301
##	11	6	
17			
##	chr1:113161591-113162571	chr1:113217475-113218097	chr1:11322258-11323036
##	20	15	
16			
##	chr1:113249342-113250198	chr1:113256885-113258149	chr1:113261468-113261765
##	14	15	
10			
##	chr1:113265573-113265787	chr1:1132827-1133218	chr1:113286332-113287172
##	9	8	
14			
##	chr1:11333259-11333547	chr1:113392142-113392807	chr1:113497986-113498206
##	12	10	
10			
##	chr1:113498366-113499312	chr1:113615591-113616260	chr1:1136360-1137156
##	12	14	
9			
##	chr1:113741473-113741745	chr1:113932580-113934070	chr1:1141670-1142150
##	6	13	
15			

## chr1:114301514-114302272	chr1:114354373-114355300	chr1:114447454-114448024
## 16	15	
16		
## chr1:114471965-114472899	chr1:1145568-1145878	
chr1:1146708-1146914		
## 19	8	
2		
## chr1:114695136-114696672	chr1:114696886-114697185	
chr1:1149055-1149524		
## 17	8	
11		
## chr1:115053060-115054264	chr1:115212239-115212985	
chr1:1152522-1153134		
## 13	18	
6		
## chr1:115323075-115323462	chr1:1153784-1154419	
chr1:11538669-11540342		
## 14	9	
13		
## chr1:115397448-115397942	chr1:11561366-11562156	chr1:115631850-115632584
## 13	11	
13		
## chr1:115880167-115881332	chr1:11589838-11590044	
chr1:1159168-1159528		
## 14	6	
10		
## chr1:11596382-11596622	chr1:116184380-116185561	
chr1:1161959-1162281		
## 8	10	
8		
## chr1:1163502-1163895	chr1:116370810-116371303	chr1:116380359-116382364
## 6	13	
21		
## chr1:116518757-116519537	chr1:116694664-116694983	
chr1:1167001-1168985		
## 14	8	
13		
## chr1:116710157-116710546	chr1:116710837-116711260	chr1:116915291-116916813
## 6	7	
16		
## chr1:116961182-116961542	chr1:117077042-117077299	
chr1:11708767-11709277		

##	13	9
8		
##	chr1:11710459-11710788	chr1:117112960-117113883
	chr1:11714217-11715304	
##	2	15
21		
##	chr1:117209824-117210621	chr1:11724113-11724885 chr1:117284383-117284627
##	19	13
11		
##	chr1:1173915-1174263	chr1:11740828-11741717 chr1:117452203-117453452
##	8	16
15		
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##                               7                               9
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## chr12:105501333-105501908 chr12:105629690-105630099 chr12:105724272-
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##                               19                               12
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## chr12:1058481-1059430 chr12:106460673-106460892 chr12:106532106-
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## chr12:106633422-106633642 chr12:106640880-106642374 chr12:106696232-
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##                               11                               11
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## chr12:106751416-106751965 chr12:106974412-106974951 chr12:106975195-
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## chr12:10765955-10766331 chr12:107711603-107714107 chr12:107974842-
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##                               16                               25
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## chr12:108079442-108079893 chr12:108154394-108155137 chr12:108168986-
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##                               14                               13
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## chr12:108237437-108238721 chr12:10826531-10826786 chr12:108297426-
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chr12:111834869-111835085 chr12:111842938-111844392 chr12:111855896-111856750

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## chr12:117476810-117477106 chr12:117484305-117484627 chr12:117536290-
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## chr12:11801853-11802218 chr12:11802752-11802983 chr12:118198767-
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## 9 6
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## chr12:118406590-118407364 chr12:118454391-118454996 chr12:118499103-
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## 14 17
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## chr12:118541054-118542135 chr12:118573730-118574731 chr12:118809324-
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## 11 18
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## chr12:118814014-118814839 chr12:119212110-119212393 chr12:119418591-
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## 12 8
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## chr12:119419105-119419810 chr12:119591856-119592176 chr12:119772353-
119772594
## 9 8
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## chr12:120031494-120033212 chr12:120105502-120106232 chr12:120127973-
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## chr12:120241921-120242233 chr12:120314847-120315169 chr12:120426547-
120428066

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## chr12:120541564-120541788 chr12:120554162-120555224 chr12:120632222-
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##                                8                                10
9
## chr12:120638588-120639116 chr12:120650214-120650528 chr12:120651711-
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##                                15                                6
11
## chr12:120702976-120703541 chr12:120755346-120755672 chr12:120799372-
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##                                11                                9
7
## chr12:120806251-120807026 chr12:120835586-120835927 chr12:120875654-
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##                                10                                7
13
## chr12:120884047-120884703 chr12:120906678-120907894 chr12:120933649-
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##                                17                                14
18
## chr12:120966788-120967038 chr12:120972167-120972447 chr12:121018728-
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##                                16                                10
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## chr12:121022419-121022704 chr12:121078161-121079483 chr12:121124111-
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## chr12:121148239-121148853 chr12:121163472-121163913 chr12:121177115-
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##                                18                                18
13
## chr12:121341414-121342436 chr12:121416485-121416817 chr12:121454153-
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## chr12:121533571-121533923 chr12:121647705-121648216 chr12:121711846-
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chr12:122247781-122248424 chr12:122249843-122250164 chr12:122254907-122255494

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chr12:122261329-122261593 chr12:122265374-122265954 chr12:122277301-122277539

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12 12

12

chr12:123011261-123011765 chr12:123236994-123237664 chr12:123319750-123321043

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##	15	20
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## chr12:123470553-123471102	chr12:123481890-123482117	chr12:123634826-123636176
##	13	9
17		
## chr12:123717367-123718388	chr12:123754049-123754373	chr12:123755246-123756408
##	18	9
20		
## chr12:123848617-123849919	chr12:123868267-123869381	chr12:123873794-123874597
##	9	15
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## chr12:123920582-123921462	chr12:123942639-123943001	chr12:124017593-124018994
##	13	11
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## chr12:124068935-124069699	chr12:124086217-124087015	chr12:124118097-124118479
##	13	16
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## chr12:124155578-124156171	chr12:124196646-124197507	chr12:12419679-12420541
##	11	17
18		
## chr12:124246524-124247254	chr12:124272128-124272585	chr12:124421581-124422163
##	13	9
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## chr12:124798796-124799224	chr12:124808972-124809176	chr12:124809316-124810144
##	7	7
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## chr12:12502942-12503465 chr12:125039119-125039344 chr12:125050796-
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## chr12:127212279-127212529	chr12:127630511-127631199	chr12:12764751-12765205
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##	10	13
14		
## chr12:129429603-129429843	chr12:12966123-12966349	chr12:129787735-129788160
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## 4 8
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## 9 8
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19 22

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##	15	11
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##	16	11
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##	12	8
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19		
## chr12:133073397-133073632	chr12:133102168-133102604	chr12:133125659-133126011
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## chr12:133131896-133132157	chr12:133135293-133135722	chr12:133136392-133137480
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##	8	16
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##	10	8
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chr12:133168946-133169191 chr12:133177826-133178055 chr12:133178608-133178857

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chr12:133303808-133304234 chr12:133308488-133310334 chr12:133335487-133335851

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2		
##	chrX:49643335-49643654	chrX:49643906-49644501
	chrX:49687027-49687993	
##	9	4
20		
##	chrX:50024336-50024556	chrX:50212429-50212666
	chrX:50213658-50213983	
##	8	5
8		
##	chrX:50214459-50214677	chrX:50556224-50557179
	chrX:51075272-51076331	
##	7	14
18		
##	chrX:51138953-51139255	chrX:51149661-51150911
	chrX:51238706-51240104	
##	10	14
8		
##	chrX:51360851-51361777	chrX:51424890-51425822
	chrX:51453549-51454481	
##	6	1
1		
##	chrX:51486414-51487010	chrX:51637830-51638076
	chrX:52004915-52005456	
##	9	16
8		
##	chrX:52925885-52926421	chrX:52949907-52950633
	chrX:52963993-52964738	
##	1	6
7		
##	chrX:53024599-53024966	chrX:53029708-53030093
	chrX:53111281-53112016	
##	6	7
14		
##	chrX:53117470-53118033	chrX:53123311-53123777
	chrX:53253783-53254826	

##	8	8
13		
##	chrX:53283829-53284038	chrX:53349666-53350613
	chrX:53448968-53449772	
##	6	15
15		
##	chrX:53461178-53461463	chrX:53468578-53469023
	chrX:53710495-53711457	
##	13	8
6		
##	chrX:53713068-53713794	chrX:54070312-54071311
	chrX:54208939-54209723	
##	13	12
11		
##	chrX:54383872-54385477	chrX:54521445-54522749
	chrX:54556428-54556959	
##	9	13
15		
##	chrX:54665781-54666274	chrX:54834761-54835010
	chrX:54835220-54835525	
##	8	12
7		
##	chrX:55026359-55026985	chrX:55187321-55188007
	chrX:55515067-55515544	
##	12	15
13		
##	chrX:56258951-56259159	chrX:56589945-56590514
	chrX:57021591-57022044	
##	17	14
10		
##	chrX:57147679-57147997	chrX:57163646-57163868
	chrX:57618402-57619328	
##	10	7
10		
##	chrX:57935678-57936921	chrX:5811007-5811304
	chrX:5821102-5821699	
##	10	6
6		
##	chrX:6144120-6145906	chrX:62570952-62571426
	chrX:62974713-62975218	
##	16	8
13		
##	chrX:63263904-63264129	chrX:63424198-63426190
	chrX:63614957-63615423	
##	6	16
14		

##	chrX:64254408-64255291	chrX:64626743-64628045
	chrX:64754316-64754805	
##	10	10
15		
##	chrX:64771406-64772193	chrX:64887549-64887837
	chrX:65041896-65042304	
##	8	14
6		
##	chrX:66763684-66764077	chrX:66766037-66766279
	chrX:67352650-67352923	
##	15	16
6		
##	chrX:67652946-67653907	chrX:67718427-67719071
	chrX:67906111-67906475	
##	12	14
7		
##	chrX:67913299-67914134	chrX:68003253-68003613
	chrX:68048659-68050252	
##	14	6
33		
##	chrX:68060178-68060558	chrX:68114440-68114710
	chrX:68348399-68348775	
##	10	8
9		
##	chrX:68369818-68370085	chrX:68384913-68385404
	chrX:68441385-68441694	
##	6	11
7		
##	chrX:68525550-68525891	chrX:68723731-68726051
	chrX:68758191-68758647	
##	5	21
7		
##	chrX:68835648-68837063	chrX:69282384-69283285
	chrX:69353104-69353904	
##	20	16
13		
##	chrX:69509644-69509962	chrX:69653707-69654661
	chrX:69665102-69665355	
##	15	10
16		
##	chrX:69674799-69675305	chrX:70150675-70151002
	chrX:70273009-70273255	
##	17	12
8		
##	chrX:70287890-70288359	chrX:70315496-70315853
	chrX:70316349-70316671	

##	13	6
8		
##	chrX:70338280-70338866	chrX:70389535-70389800
	chrX:70401862-70402213	
##	10	5
7		
##	chrX:70444046-70444270	chrX:70473812-70474287
	chrX:70503261-70503963	
##	13	22
11		
##	chrX:70585715-70587063	chrX:7065693-7066488
	chrX:70712049-70712821	
##	11	10
14		
##	chrX:70752638-70752965	chrX:70798022-70798353
	chrX:70823513-70824103	
##	14	10
7		
##	chrX:70887645-70887941	chrX:70981816-70983829
	chrX:71092472-71093196	
##	8	1
7		
##	chrX:71130679-71131820	chrX:71238328-71238675
	chrX:71350941-71352137	
##	9	7
14		
##	chrX:71458560-71459202	chrX:71496700-71497227
	chrX:71524832-71527299	
##	14	13
31		
##	chrX:71900454-71900727	chrX:71933681-71934265
	chrX:72223598-72224080	
##	2	10
2		
##	chrX:72298626-72299108	chrX:72667157-72667590
	chrX:72782945-72783165	
##	2	14
13		
##	chrX:73164003-73164247	chrX:73325164-73325522
	chrX:73512193-73512569	
##	13	6
11		
##	chrX:73523919-73524125	chrX:73639898-73640778
	chrX:73641647-73642011	
##	10	6
6		

##	chrX:73642494-73642766	chrX:73755645-73756784
	chrX:73833999-73834597	
##	6	10
13		
##	chrX:74143852-74145176	chrX:74742765-74743117
	chrX:75249328-75249628	
##	14	13
5		
##	chrX:75369195-75369865	chrX:75648708-75649533
	chrX:77041003-77041725	
##	1	15
14		
##	chrX:77150837-77151247	chrX:77165924-77166295
	chrX:77359303-77359979	
##	22	5
17		
##	chrX:77394831-77395292	chrX:7895077-7895608
	chrX:79590720-79591007	
##	9	18
12		
##	chrX:79828770-79829059	chrX:79830094-79830539
	chrX:80064923-80065136	
##	5	5
11		
##	chrX:80259089-80259493	chrX:82763571-82764462
	chrX:82765442-82765776	
##	6	13
5		
##	chrX:83004829-83005105	chrX:83442396-83443335
	chrX:84189065-84189591	
##	3	12
8		
##	chrX:84498772-84499884	chrX:85402943-85403877
	chrX:8698857-8700500	
##	12	15
19		
##	chrX:8751285-8751608	chrX:8769177-8769559
	chrX:90689602-90690119	
##	8	16
6		
##	chrX:91033993-91034623	chrX:91036684-91037269
	chrX:92928546-92928786	
##	4	4
14		
##	chrX:9432830-9434520	chrX:95939630-95940366
	chrX:9733154-9734064	

##	20	12
11		
##	chrX:9753566-9755101	chrX:9858608-9858930
	chrX:9863527-9863822	
##	11	8
8		
##	chrX:9912585-9912968	chrX:99194334-99194598
	chrX:9960205-9960408	
##	8	9
6		
##	chrX:99661653-99665358	chrX:99666131-99666381
	chrX:99666900-99667224	
##	11	8
6		
##	chrX:9982513-9984583	chrX:99891299-99891794
	chrX:99986423-99987185	
##	8	13
12		
##	chrY:10032933-10034686	chrY:10036737-10037179
	chrY:13328678-13329051	
##	3	5
1		
##	chrY:13462591-13463914	chrY:13470631-13471733
	chrY:13477252-13478448	
##	1	1
2		
##	chrY:13585173-13586108	chrY:13910043-13910469
	chrY:14074239-14075008	
##	1	2
5		
##	chrY:14075675-14076160	chrY:14077836-14078044
	chrY:14100002-14100593	
##	2	3
6		
##	chrY:14107137-14107480	chrY:14532115-14533600
	chrY:14649050-14649669	
##	3	5
8		
##	chrY:15016544-15017298	chrY:15591259-15591720
	chrY:15815488-15815779	
##	13	6
8		
##	chrY:15863345-15864834	chrY:16635739-16636047
	chrY:16636290-16636991	
##	6	12
4		

##	chrY:16941822-16942188	chrY:17567579-17568617
	chrY:19671324-19671587	
##	5	8
1		
##	chrY:19679723-19679935	chrY:19691214-19691762
	chrY:20488010-20488557	
##	2	2
3		
##	chrY:20499841-20500053	chrY:20508190-20508452
	chrY:21154603-21155040	
##	1	2
1		
##	chrY:21238448-21240005	chrY:21664481-21665063
	chrY:21729143-21729556	
##	9	7
12		
##	chrY:22737825-22738052	chrY:23566571-23566864
	chrY:23696580-23696878	
##	17	5
1		
##	chrY:24040559-24040857	chrY:24329015-24329313
	chrY:24454804-24455083	
##	2	1
5		
##	chrY:24549398-24549695	chrY:2657802-2658099
	chrY:2802964-2804283	
##	2	9
17		
##	chrY:28555535-28555932	chrY:4867956-4868586
	chrY:4870744-4871225	
##	3	5
3		
##	chrY:6114064-6114584	chrY:6133645-6133958
	chrY:6134498-6134748	
##	5	5
2		
##	chrY:6173546-6173839	chrY:6388383-6388652
	chrY:6778574-6780028	
##	3	1
10		
##	chrY:7141521-7143224	chrY:7428179-7428424
	chrY:8147877-8148210	
##	10	4
6		
##	chrY:9174129-9174443	chrY:9174873-9175189
	chrY:9193597-9194776	

```
##                                5                                1
5
##      chrY:9195206-9195726      chrY:9215484-9216005
chrY:9234221-9235400
##                                2                                5
1
##      chrY:9235830-9236350      chrY:9302757-9303020
chrY:9303622-9303934
##                                1                                1
3
##      chrY:9304474-9305072      chrY:9323067-9324246
chrY:9343351-9343614
##                                2                                1
3
##      chrY:9344959-9345275      chrY:9363680-9363943
chrY:9364545-9364859
##                                1                                2
2
##      chrY:9365289-9365809      chrY:9385471-9385777
chrY:9478175-9478445
##                                2                                7
2
##      chrY:9706548-9706841      chrY:9745559-9745993
chrY:9930123-9930391
##                                2                                7
2
```

Uh, that was a bad idea, it printed a lot of things in the Console!

```
nlevels(Illumina450Manifest_clean$UCSC_CpG_Islands_Name)
```

```
## [1] 27177
```

```
head(table(Illumina450Manifest_clean$UCSC_CpG_Islands_Name))
```

```
##
```

```
##                                chr1:10003165-10003585      chr1:1002663-
1005318
```

```
##                                176047                                14
18
```

```
## chr1:100315420-100316009 chr1:100435297-100436070 chr1:100503482-
100504404
```

```
##                                18                                19
16
```

I will convert the output of the table function in a data.frame

```
island_table <-  
data.frame(table(Illumina450Manifest_clean$UCSC_CpG_Islands_Name))
```

```
head(island_table)
```

##	Var1	Freq
## 1		176047
## 2	chr1:10003165-10003585	14
## 3	chr1:1002663-1005318	18
## 4	chr1:100315420-100316009	18
## 5	chr1:100435297-100436070	19
## 6	chr1:100503482-100504404	16

I will order the rows of the dataframe using the order function as follows:

```
island_table <- island_table[order(island_table[,2]),]
```

or, in alternative, I can write:

```
island_table <- island_table[order(island_table$Freq),]
```

```
head(island_table)
```

##	Var1	Freq
## 261	chr1:135124-135563	1
## 284	chr1:143647186-143647525	1
## 286	chr1:143687039-143687337	1
## 292	chr1:144491618-144492007	1
## 322	chr1:146076519-146077373	1
## 323	chr1:146081810-146082820	1

```
tail(island_table)
```

##	Var1	Freq
## 22083	chr6:30710307-30712440	110
## 22154	chr6:32939275-32940954	112
## 22169	chr6:33266302-33267582	115
## 22124	chr6:31830299-31830948	117
## 22130	chr6:31939730-31940559	117
## 1		176047

I can also order from the largest to the smallest:

```
island_table <- island_table[order(-island_table[,2]),]
```

```
head(island_table)
```

##	Var1	Freq
## 1		176047
## 22124	chr6:31830299-31830948	117
## 22130	chr6:31939730-31940559	117
## 22169	chr6:33266302-33267582	115
## 22154	chr6:32939275-32940954	112
## 22083	chr6:30710307-30712440	110

I will select two CpG islands with 117 probes each

```
highest_probes <-
Illumina450Manifest_clean[Illumina450Manifest_clean$UCSC_CpG_Islands_Name=="chr6:31830299-31830948"|Illumina450Manifest_clean$UCSC_CpG_Islands_Name=="chr6:31939730-31940559",]
```

```
dim(highest_probes)
```

```
## [1] 234 33
```

or, in alternative, I can write:

```
highest_probes <-
Illumina450Manifest_clean[Illumina450Manifest_clean$UCSC_CpG_Islands_Name %in% c("chr6:31830299-31830948","chr6:31939730-31940559"),]
```

```
dim(highest_probes)
```

```
## [1] 234 33
```

```
highest_probes <- droplevels(highest_probes)
```

```
str(highest_probes)
```

```
## 'data.frame': 234 obs. of 33 variables:
```

```
## $ IlmnID : Factor w/ 234 levels
"cg00123181","cg00179217",...: 1 2 3 4 5 6 7 8 9 10 ...
## $ Name : Factor w/ 234 levels
"cg00123181","cg00179217",...: 1 2 3 4 5 6 7 8 9 10 ...
## $ AddressA_ID : Factor w/ 234 levels
"11636360","11714435",...: 15 133 179 95 19 223 26 103 75 141 ...
## $ AlleleA_ProbeSeq : Factor w/ 234 levels
"AAAAAAAAATCAACAAAACAACTTATCTTAAAAATTTTAAAAACCCAC",...: 185 177 8 182
43 188 231 26 95 94 ...
## $ AddressB_ID : Factor w/ 67 levels
"", "10610406",...: 1 1 1 1 1 21 42 1 29 1 ...
```

```

## $ AlleleB_ProbeSeq          : Factor w/ 67 levels
"", "AAAAAACGTCAACTTATCGTCCCTCAAACCCGTCAATACTAAAAAACG", ...: 1 1 1 1 1
52 66 1 25 1 ...
## $ Infinium_Design_Type      : Factor w/ 2 levels "I","II": 2 2 2 2
2 1 1 2 1 2 ...
## $ Next_Base                 : Factor w/ 4 levels "", "A", "C", "T": 1
1 1 1 1 2 4 1 2 1 ...
## $ Color_Channel             : Factor w/ 3 levels "", "Grn", "Red": 1
1 1 1 1 3 3 1 3 1 ...
## $ Forward_Sequence          : Factor w/ 234 levels
"AAAAAATTGGAGGAAGAAACCCAGAGTGAGCACTCTGCAGGTACCCTTTCTACCACTTCC[CG]TTAATT
TCCACCTTCTGCTAGGGACCTCAGGCCTTCCGATGGTCCCAGGGTGCAATCCA", ...: 183 136 189
73 229 137 44 161 224 72 ...
## $ Genome_Build              : int   37 37 37 37 37 37 37 37 37 37
...
## $ CHR                       : Factor w/ 1 level "6": 1 1 1 1 1 1 1
1 1 1 ...
## $ MAPINFO                   : int   31832796 31831340 31830812
31831510 31937355 31939797 31939865 31829960 31938176 31939758 ...
## $ SourceSeq                 : Factor w/ 233 levels
"AAATATTTCCCCCTAACTATACATACACCCTATGTATATGTGGCAAACG", ...: 194 84 105 13
189 225 232 124 30 67 ...
## $ Chromosome_36             : Factor w/ 1 level "6": 1 1 1 1 1 1 1
1 1 1 ...
## $ Coordinate_36             : Factor w/ 234 levels
"31934923", "31935089", ...: 94 67 61 73 140 185 189 43 154 183 ...
## $ Strand                    : Factor w/ 2 levels "F", "R": 1 2 1 2
2 2 1 1 1 1 ...
## $ Probe_SNPs                : Factor w/ 29 levels
"", "rs11538808", ...: 1 1 1 1 1 1 15 1 19 1 ...
## $ Probe_SNPs_10             : Factor w/ 17 levels
"", "rs11538809", ...: 1 8 1 1 1 1 1 1 1 1 ...
## $ Random_Loci               : logi   NA NA NA NA NA NA ...
## $ Methyl27_Loci             : logi   NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name          : Factor w/ 16 levels
"DOM3Z;STK19;DOM3Z;STK19;STK19", ...: 9 10 5 7 8 1 1 5 16 3 ...
## $ UCSC_RefGene_Accession     : Factor w/ 18 levels
"NM_000434", "NM_000434;NM_000434", ...: 12 13 1 3 11 8 8 1 18 7 ...
## $ UCSC_RefGene_Group         : Factor w/ 18 levels
"1stExon", "3'UTR", ...: 9 3 17 14 9 5 5 9 16 6 ...
## $ UCSC_CpG_Islands_Name      : Factor w/ 2 levels "chr6:31830299-
31830948", ...: 1 1 1 1 2 2 2 1 2 2 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 5 levels
"Island", "N_Shelf", ...: 5 5 1 5 2 1 1 3 3 1 ...
## $ Phantom                   : Factor w/ 7 levels "", "high-
CpG:31938530-31938659", ...: 6 1 1 1 1 3 3 1 1 3 ...

```



```
## $ DMR : Factor w/ 1 level "": 1 1 1 1 1 1 1
1 1 1 ...
## $ Enhancer : logi NA NA NA NA NA NA ...
## $ HMM_Island : Factor w/ 6 levels "","6:31937833-
31938927",...: 1 1 2 3 1 5 5 2 1 5 ...
## $ Regulatory_Feature_Name : Factor w/ 5 levels "","6:31829892-
31831092",...: 1 3 2 3 1 5 5 2 1 5 ...
## $ Regulatory_Feature_Group : Factor w/ 3 levels
"","Promoter_Associated",...: 1 2 2 2 1 2 2 2 1 2 ...
## $ DHS : logi NA NA NA NA NA NA ...
```

Exercise 1: Try to have a look at the probe having the IlmnID “cg19560547”, which is in the promoter of WRN gene...does it map to a CpG island or not according to the Illumina Infinium450k manifest?

Exercise 2: How many probes in the array are in the open sea?

Exercise 3: How many probes map in genic sequences but not in CpG island, shores and shelves?

Exercise 4: What is the coordinate (defined according to the MAPINFO column) of the 10th probe on chromosome 7?

CHUNK4: Genes

```
levels(Illumina450Manifest_clean$UCSC_RefGene_Name)[1:20]
```

```
## [1] ""
## [2] "A1BG"
## [3] "A1CF;A1CF;A1CF"
## [4] "A1CF;A1CF;A1CF;A1CF;A1CF;A1CF"
## [5] "A2BP1;A2BP1"
## [6] "A2BP1;A2BP1;A2BP1"
## [7] "A2BP1;A2BP1;A2BP1;A2BP1"
## [8] "A2BP1;A2BP1;A2BP1;A2BP1;A2BP1;A2BP1"
## [9] "A2BP1;A2BP1;A2BP1;A2BP1;A2BP1;A2BP1;A2BP1;A2BP1;A2BP1"
## [10] "A2LD1"
## [11] "A2LD1;A2LD1"
## [12] "A2M"
## [13] "A2ML1"
## [14] "A2ML1;A2ML1"
## [15] "A4GALT"
## [16] "A4GNT"
## [17] "A4GNT;A4GNT"
## [18] "AAA1;NPSR1;NPSR1"
## [19] "AAAS"
```

```
## [20] "AAAS;AAAS"
```

Have a look at this:

```
Illumina450Manifest_clean[Illumina450Manifest_clean$IlmnID=="cg03817621",]
```

##	IlmnID	Name	AddressA_ID
##	264192	cg03817621	cg03817621
##			31624459
##			AlleleA_ProbeSeq
##			AddressB_ID
##	264192	TTCAAAAATCCCCACCCRAAAAAAATTTTCCTTAATTATAACRAATTCAAC	
##		AlleleB_ProbeSeq	Infinium_Design_Type
##			Next_Base
##			Color_Channel
##	264192		II
##			Forward_Sequence
##	264192	ACCTGAGTAATTTTCAGAGATCCCCACCCGAAAAGGTTTCCTTGATTATGGCGAATTCAA[CG]TGTCAC	
##		CGGGTTATGCTCTAATCGTCATATCAAACCTCAAAGGTTTCAGTGGCTTTAGGT	
##		Genome_Build	CHR
##			MAPINFO
##	264192	37	10
##		52645399	
##			SourceSeq
##			Chromosome_36
##	264192	TCAGAGATCCCCACCCGAAAAGGTTTCCTTGATTATGGCGAATTCAACG	
##		10	
##		Coordinate_36	Strand
##			Probe_SNPs
##			Probe_SNPs_10
##			Random_Loci
##			Methyl27_Loci
##	264192	52315405	R
##			NA
##			TRUE
##			UCSC_RefGene_Name
##	264192	A1CF;A1CF;A1CF;A1CF;A1CF;A1CF	
##			UCSC_RefGene_Accession
##	264192	NM_138933;NM_138932;NM_014576;NM_138932;NM_014576;NM_138933	
##			UCSC_RefGene_Group
##			UCSC_CpG_Islands_Name
##	264192	5'UTR;1stExon;5'UTR;5'UTR;1stExon;1stExon	
##		Relation_to_UCSC_CpG_Island	Phantom
##			DMR
##			Enhancer
##	264192		low-CpG:52315394-52315446
##			NA
##		HMM_Island	Regulatory_Feature_Name
##			Regulatory_Feature_Group
##			DHS
##	264192		
##			NA

http://genome-euro.ucsc.edu/cgi-bin/hgTracks?db=hg19&lastVirtModeType=default&lastVirtModeExtraState=&virtModeType=default&virtMode=0&nonVirtPosition=&position=chr10%3A52472902-52731702&hgside=213618992_FWAq8vB3Gv9iJTJYY7xyDqECVmFA

Have a look at this:

Illumina450Manifest_clean[Illumina450Manifest_clean\$Name=="cg00048759",]									
##	IlmnID	Name	AddressA_ID						
##	199932	cg00048759	cg00048759	38618418					
##					AlleleA_ProbeSeq				
					AddressB_ID				
##	199932	CAAAATTTAAAAATATACAAACTAATACACACCACATTTAAATCACACCA							
		26745306							
##					AlleleB_ProbeSeq				
					Infinium_Design_Type				
##	199932	CGAAATTTAAAAATATACGAACTAATACGCGCCACGTTTAAATCACGCCG							
		I							
##		Next_Base	Color_Channel						
##	199932	C	Grn						
##									
					Forward_Sequence				
##	199932								
		CCCAAGCCCACGTGAGAGGGCAGGACGCCTGAGAGCTTGAGGCCACACGAGGCTGTGGAG[CG]GCGTGAC							
		TCAAACGTGGCGCGCATCAGCTCGCACACTTCCAAACCTCGCGATAGCTACTG							
##		Genome_Build	CHR	MAPINFO					
##	199932		37	7	99775422				
##						SourceSeq			
					Chromosome_36				
##	199932	CGGCGTGACTCAAACGTGGCGCGCATCAGCTCGCACACTTCCAAACCTCG							
		7							
##		Coordinate_36	Strand	Probe_SNPs	Probe_SNPs_10	Random_Loci			
		Methyl27_Loci							
##	199932	99613358	F			NA			
##									
##		UCSC_RefGene_Name	UCSC_RefGene_Accession	UCSC_RefGene_Group					
##	199932	STAG3;GPC2	NM_012447;NM_152742	TSS200;TSS1500					
##		UCSC_CpG_Islands_Name	Relation_to_UCSC_CpG_Island						
##	199932	chr7:99774733-99775583		Island					
##			Phantom	DMR	Enhancer	HMM_Island			
##	199932	high-CpG:99613314-99613542		NA	7:99612670-99613854				
##									
##		Regulatory_Feature_Name	Regulatory_Feature_Group	DHS					
##	199932	7:99774737-99775599		Promoter_Associated	NA				

http://genome-euro.ucsc.edu/cgi-bin/hgTracks?db=hg19&lastVirtModeType=default&lastVirtModeExtraState=&virtModeType=default&virtMode=0&nonVirtPosition=&position=chr7%3A99773884-99776433&hgid=228211450_jK6pt7GD6xw8kdT4sfPebiyQil2f

CHUNK 5: Infinium I and Infinium II probes

Let's investigate Infinium I and Infinium II probes in the manifest

```
str(Illumina450Manifest_clean)
```

```
## 'data.frame': 485512 obs. of 33 variables:
```

```
## $ IlmnID : Factor w/ 485512 levels
"cg00000029","cg00000108",...: 892 1231 1464 1502 2807 4566 4600 4616
4811 5219 ...
## $ Name : Factor w/ 485512 levels
"cg00000029","cg00000108",...: 892 1231 1464 1502 2807 4566 4600 4616
4811 5219 ...
## $ AddressA_ID : Factor w/ 485512 levels
"10600322","10600328",...: 163924 171461 142476 49179 74479 146426
155259 448531 56092 200778 ...
## $ AlleleA_ProbeSeq : Factor w/ 485512 levels
"AAAAAAAAAAAAAAAAAACACAAAAAATCCCCACTCTAAAACCCACACA",...: 35017 143540
35761 427267 41000 296053 482818 335327 460590 128010 ...
## $ AddressB_ID : Factor w/ 135477 levels
"", "10600313",...: 1 45514 1 1 1 60128 56499 76848 1 1 ...
## $ AlleleB_ProbeSeq : Factor w/ 135477 levels
"", "AAAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAAACCCCGCG",...: 1 49667 1 1
1 77344 135092 95678 1 1 ...
## $ Infinium_Design_Type : Factor w/ 2 levels "I","II": 2 1 2 2
2 1 1 1 2 2 ...
## $ Next_Base : Factor w/ 4 levels "", "A", "C", "T": 1
2 1 1 1 4 2 2 1 1 ...
## $ Color_Channel : Factor w/ 3 levels "", "Grn", "Red": 1
3 1 1 1 3 3 3 1 1 ...
## $ Forward_Sequence : Factor w/ 485512 levels
"AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAATATGTCTGCTAAATAATCACAGAGTGAAA[CG]AAGAGG
AAGAACTAAAGTACAGTTTGGAACATAGTACAGTGCCCCCTGGGTACTGAATCA",...: 25105
392558 395302 218895 81395 152101 421754 296779 478968 55951 ...
## $ Genome_Build : int 37 37 37 37 37 37 37 37 37 37 37
...
## $ CHR : Factor w/ 24 levels
"1","10","11",...: 24 24 24 24 24 24 24 24 24 24 ...
```

```

## $ MAPINFO : int 8553009 9363356 25314171
22741795 21664296 21239348 8148233 15815688 4868996 6133740 ...
## $ SourceSeq : Factor w/ 485343 levels
"AAAAAAAAAACCAACCCCAACAAAAACCCCTGTGCCAGAAATTCATTCG",...: 28961 285896
116667 116691 29627 123294 148856 152723 189022 323597 ...
## $ Chromosome_36 : Factor w/ 25 levels
"1","10","11",...: 25 25 25 25 25 25 25 25 25 25 ...
## $ Coordinate_36 : Factor w/ 484738 levels
"100000057","100000691",...: 451148 483811 205767 185980 178028 173644
442780 102376 331283 376023 ...
## $ Strand : Factor w/ 2 levels "F","R": 1 2 2 1
2 2 2 2 2 1 ...
## $ Probe_SNPs : Factor w/ 56004 levels
"", "rs10000615",...: 1 1 1 54643 1 1 1 1 1 1 ...
## $ Probe_SNPs_10 : Factor w/ 35790 levels
"", "rs10000804",...: 1 1 1 4064 1 1 1 1 1 1 ...
## $ Random_Loci : logi NA NA NA NA NA NA ...
## $ Methyl27_Loci : logi NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name : Factor w/ 42456 levels
"", "A1BG", "A1CF;A1CF;A1CF",...: 39103 38905 9635 11319 3085 39100 1
37974 27057 1 ...
## $ UCSC_RefGene_Accession : Factor w/ 52670 levels
"", "NM_000014",...: 48799 14035 19907 20891 49121 48792 1 20105 38174 1
...
## $ UCSC_RefGene_Group : Factor w/ 4165 levels
"", "1stExon", "1stExon;1stExon",...: 2970 2779 2447 2035 2035 3694 1 201
943 1 ...
## $ UCSC_CpG_Islands_Name : Factor w/ 27177 levels
"", "chr1:10003165-10003585",...: 1 27170 1 27136 27134 27133 27156 27120
27147 27149 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 6 levels
"", "Island", "N_Shelf",...: 1 4 1 5 4 2 6 2 4 2 ...
## $ Phantom : Factor w/ 11912 levels "", "high-
CpG:100009860-100010119",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ DMR : Factor w/ 4 levels
"", "CDMR", "DMR",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Enhancer : logi NA NA NA NA NA NA ...
## $ HMM_Island : Factor w/ 63086 levels
"", "1:100088009-100088708",...: 1 63085 1 1 1 63046 63077 63033 63064
63067 ...
## $ Regulatory_Feature_Name : Factor w/ 47969 levels
"", "1:10001853-10003978",...: 1 1 1 1 1 1 1 47965 1 1 ...
## $ Regulatory_Feature_Group : Factor w/ 9 levels
"", "Gene_Associated",...: 1 1 1 1 1 1 1 7 1 1 ...
## $ DHS : logi NA NA NA NA NA NA ...

```

```
table(Illumina450Manifest_clean$Infinium_Design_Type)
```

```
##
```

```
##      I      II
```

```
## 135476 350036
```

```
table(Illumina450Manifest_clean$Color_Channel)
```

```
##
```

```
##      Grn      Red
```

```
## 350036 46289 89187
```

What is the number of Infinium I and Infinium II probes in Islands, shores, shelves and not CpG rich regions??

```
TypeI <-  
Illumina450Manifest_clean[Illumina450Manifest_clean$Infinium_Design_Type=="I",]
```

```
dim(TypeI)
```

```
## [1] 135476      33
```

```
TypeI <- droplevels(TypeI)
```

```
TypeII <-  
Illumina450Manifest_clean[Illumina450Manifest_clean$Infinium_Design_Type=="II",]  
dim(TypeII)
```

```
## [1] 350036      33
```

```
TypeII <- droplevels(TypeII)
```

```
table(TypeI$Relation_to_UCSC_CpG_Island)
```

```
##
```

```
##      Island N_Shelf N_Shore S_Shelf S_Shore
```

```
## 28518 77674 3666 12064 3247 10307
```

```
table(TypeII$Relation_to_UCSC_CpG_Island)
```

```
##
```

```
##      Island N_Shelf N_Shore S_Shelf S_Shore
```

```
## 147529 72580 21178 50806 19053 38890
```

Alternatively (and much faster):

```
table(Illumina450Manifest_clean$Relation_to_UCSC_CpG_Island,
Illumina450Manifest_clean$Infinium_Design_Type)
```

```
##
```

```
##           I      II
##      28518 147529
##   Island  77674  72580
##   N_Shelf   3666  21178
##   N_Shore  12064  50806
##   S_Shelf   3247  19053
##   S_Shore  10307  38890
```

```
table(Illumina450Manifest_clean$Infinium_Design_Type,Illumina450Manifes
t_clean$Relation_to_UCSC_CpG_Island)
```

```
##
```

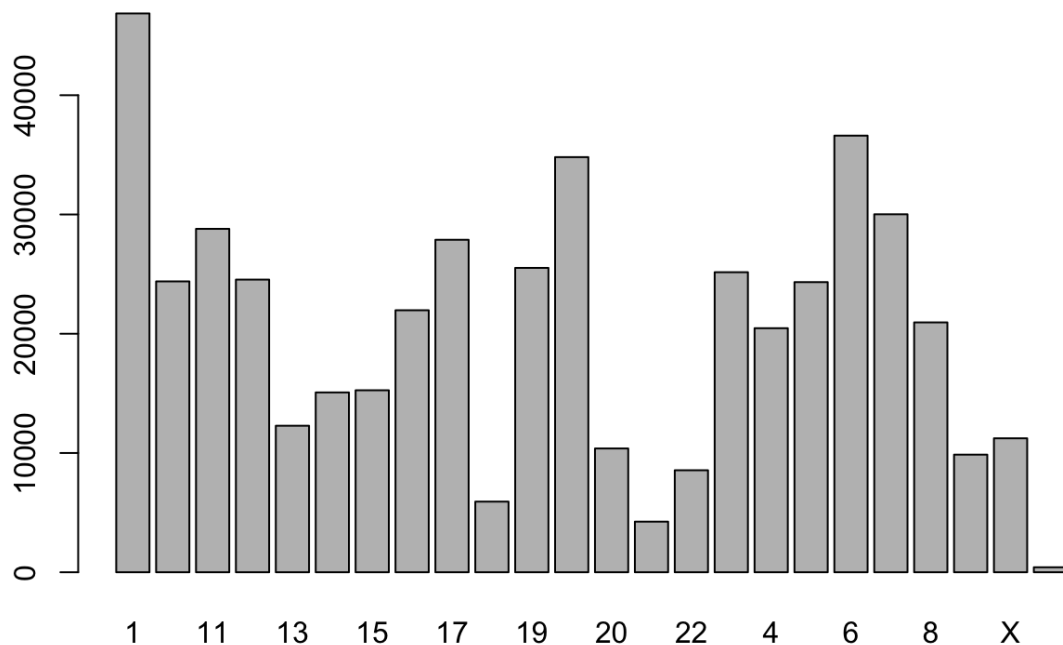
```
##           Island N_Shelf N_Shore S_Shelf S_Shore
##   I      28518  77674   3666  12064   3247   10307
##   II 147529  72580   21178  50806  19053   38890
```

CHUNK 6: Barplots

A nice and short tutorial can be found here: <http://www.statmethods.net/graphs/bar.html> I want to generate a barplot of the probes mapping in each chromosome:

```
counts <- table(Illumina450Manifest_clean$CHR)
```

```
barplot(counts)
```



```
levels(Illumina450Manifest_clean$CHR)
```

```
## [1] "1" "10" "11" "12" "13" "14" "15" "16" "17" "18" "19" "2"
"20" "21" "22"
```

```
## [16] "3" "4" "5" "6" "7" "8" "9" "X" "Y"
```

R uses alphabetical order by default. I can reorder the levels of the chromosomes:

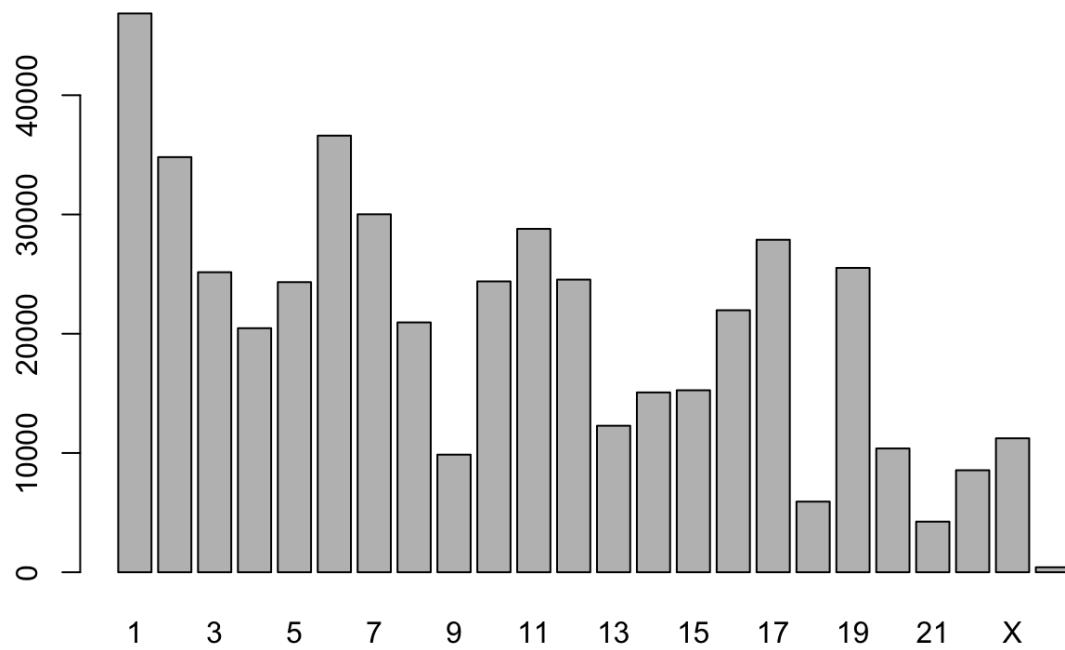
```
Illumina450Manifest_clean$CHR <-
factor(Illumina450Manifest_clean$CHR, levels=c("1", "2", "3", "4", "5", "6", "
7", "8", "9", "10", "11", "12", "13", "14", "15", "16", "17", "18", "19", "20", "21",
"22", "X", "Y"))
```

```
levels(Illumina450Manifest_clean$CHR)
```

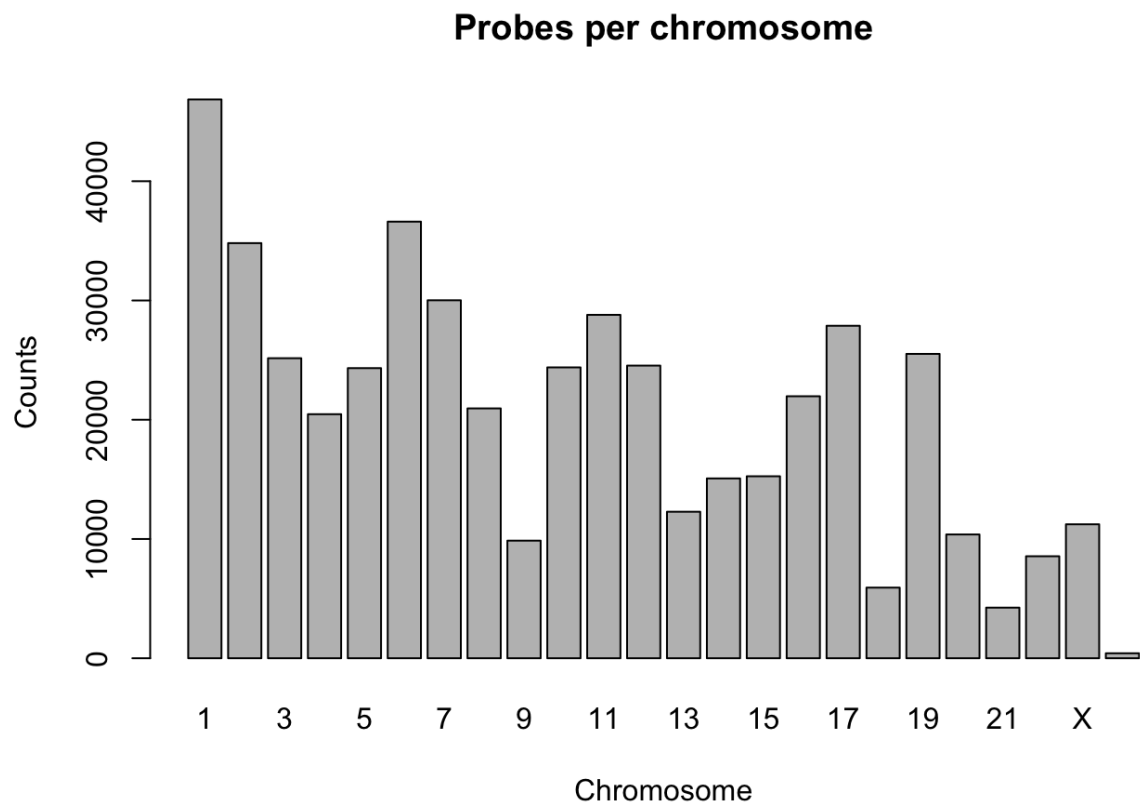
```
## [1] "1" "2" "3" "4" "5" "6" "7" "8" "9" "10" "11" "12"
"13" "14" "15"
```

```
## [16] "16" "17" "18" "19" "20" "21" "22" "X" "Y"
```

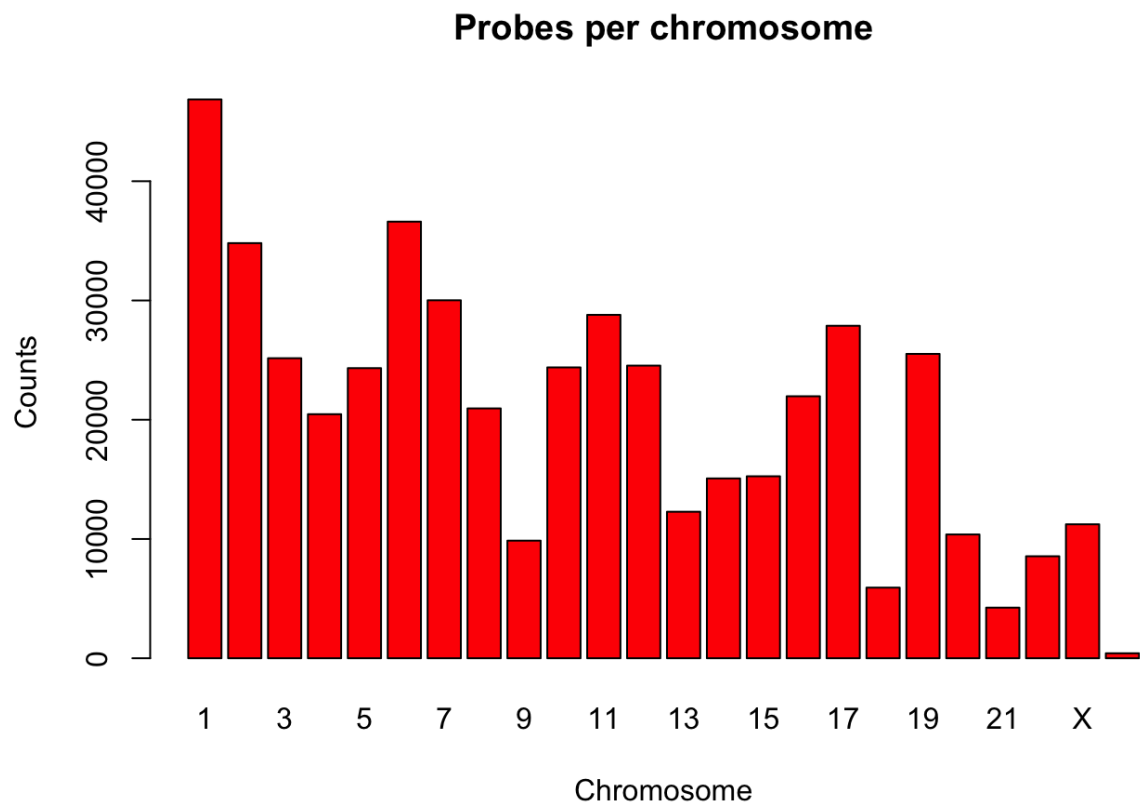
```
barplot(table(Illumina450Manifest_clean$CHR))
```

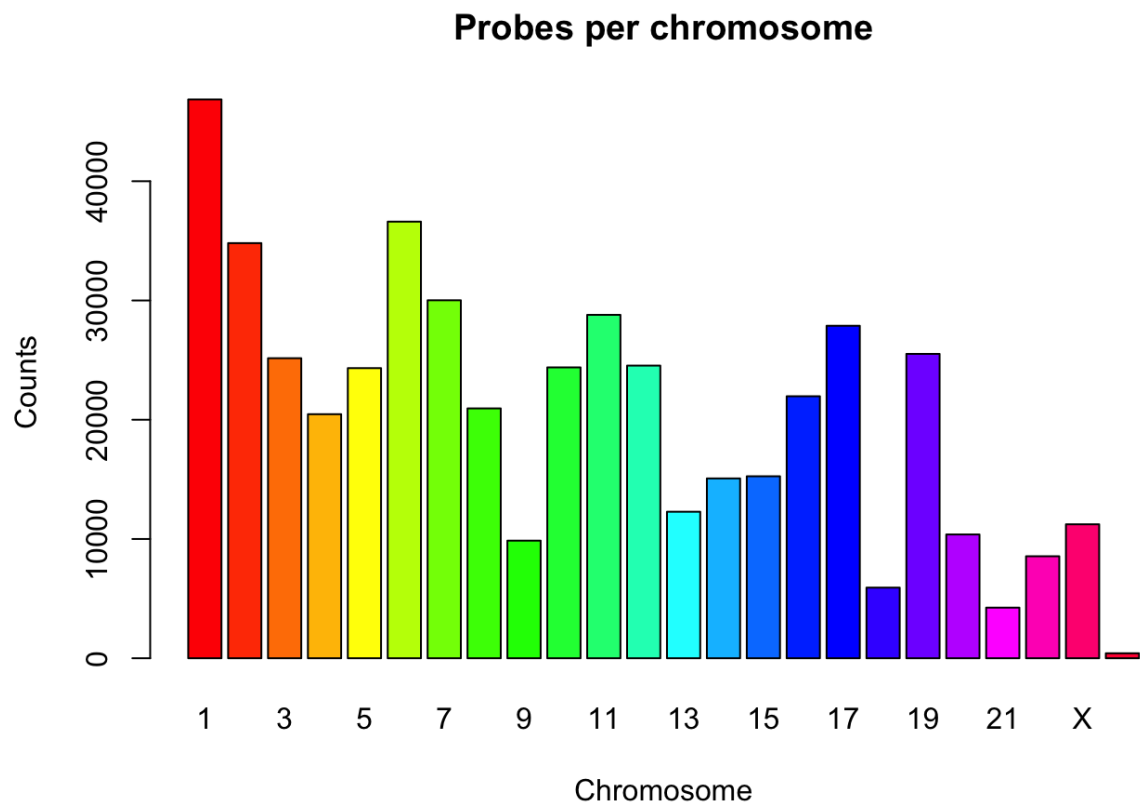
```
barplot(table(Illumina450Manifest_clean$CHR),main="Probes per  
chromosome",xlab="Chromosome",ylab="Counts")
```



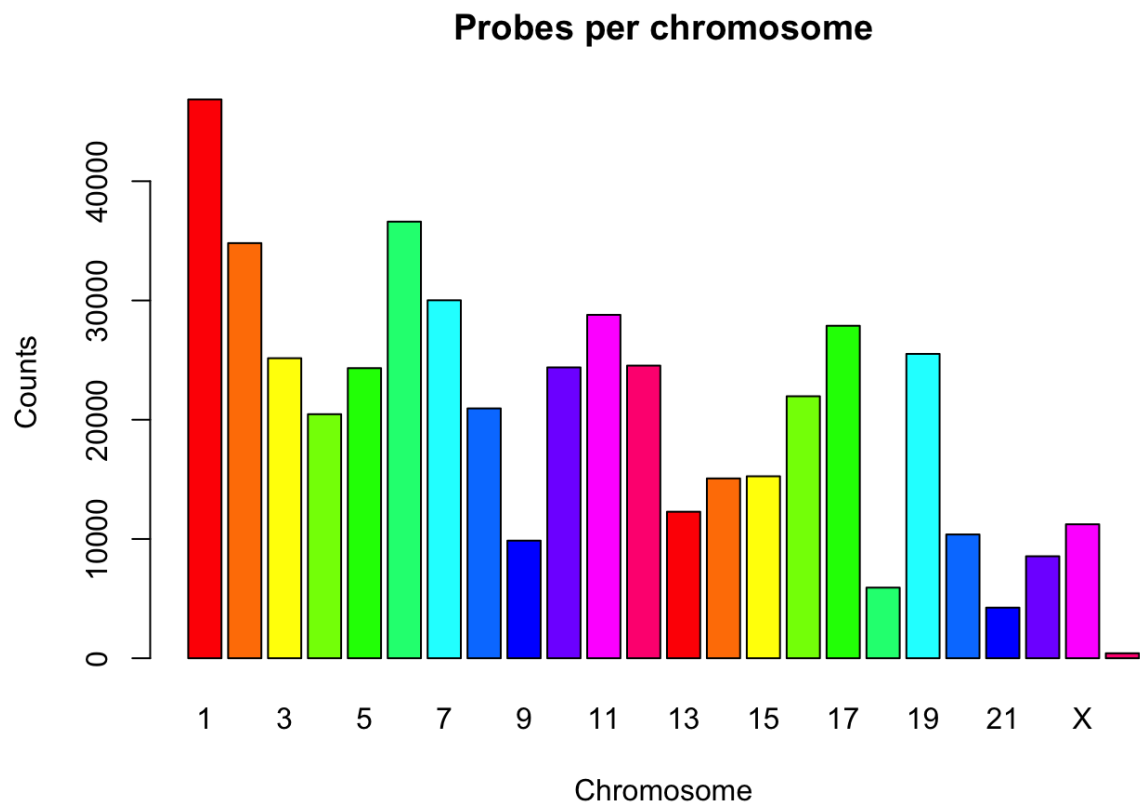
```
barplot(table(Illumina450Manifest_clean$CHR),main="Probes per  
chromosome",xlab="Chromosome",ylab="Counts",col="red")
```



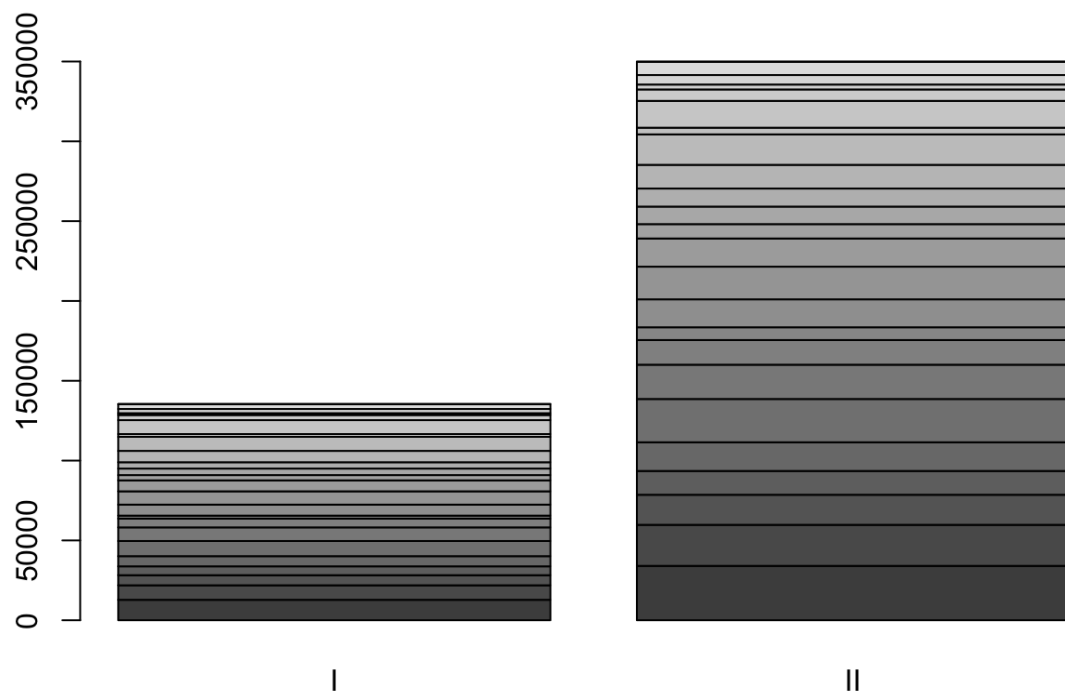
```
barplot(table(Illumina450Manifest_clean$CHR),main="Probes per  
chromosome",xlab="Chromosome",ylab="Counts",col=rainbow(24))
```



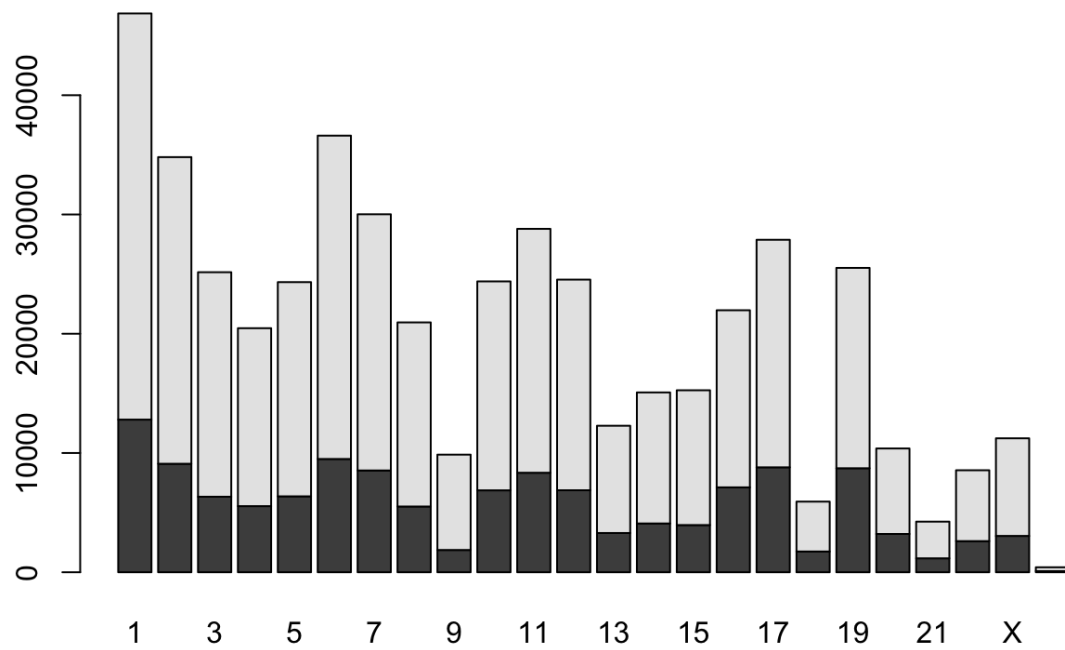
```
barplot(table(Illumina450Manifest_clean$CHR),main="Probes per  
chromosome",xlab="Chromosome",ylab="Counts",col=rainbow(12))
```



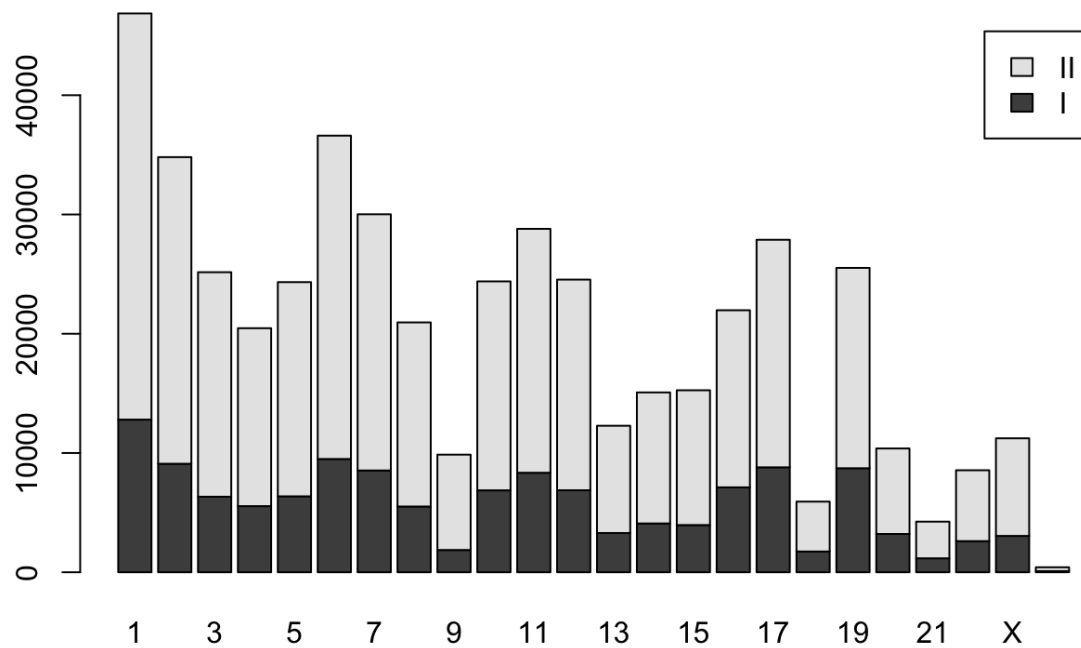
```
barplot(table(Illumina450Manifest_clean$CHR,  
Illumina450Manifest_clean$Infinium_Design_Type))
```



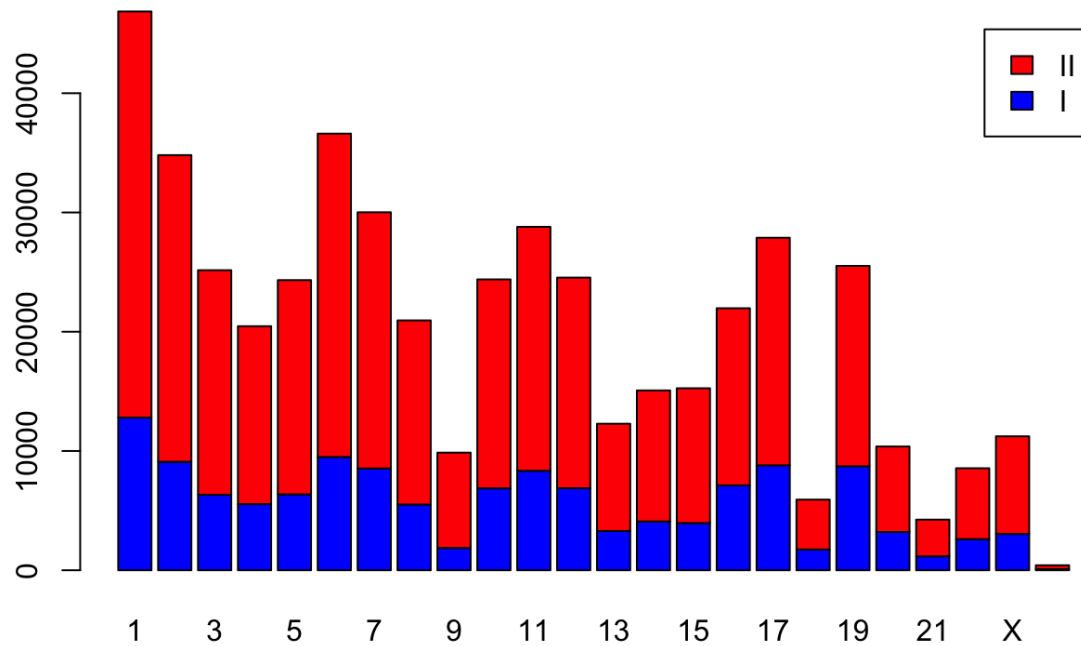
```
barplot(table(Illumina450Manifest_clean$Infinium_Design_Type,  
Illumina450Manifest_clean$CHR))
```



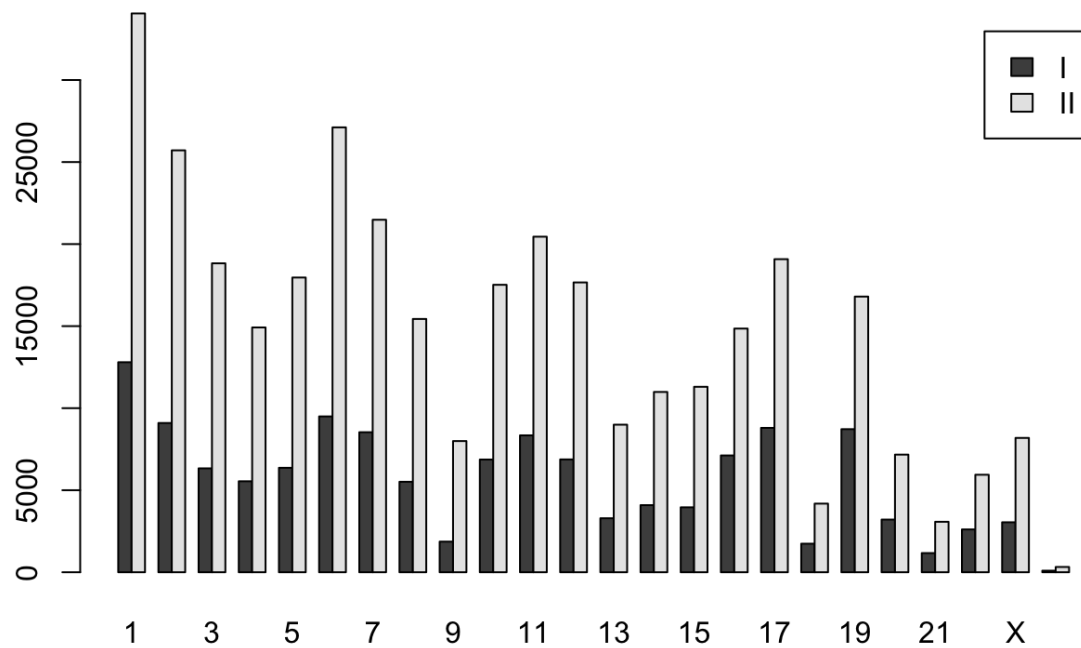
```
barplot(table(Illumina450Manifest_clean$Infinium_Design_Type,
Illumina450Manifest_clean$CHR),
legend=levels(Illumina450Manifest_clean$Infinium_Design_Type))
```



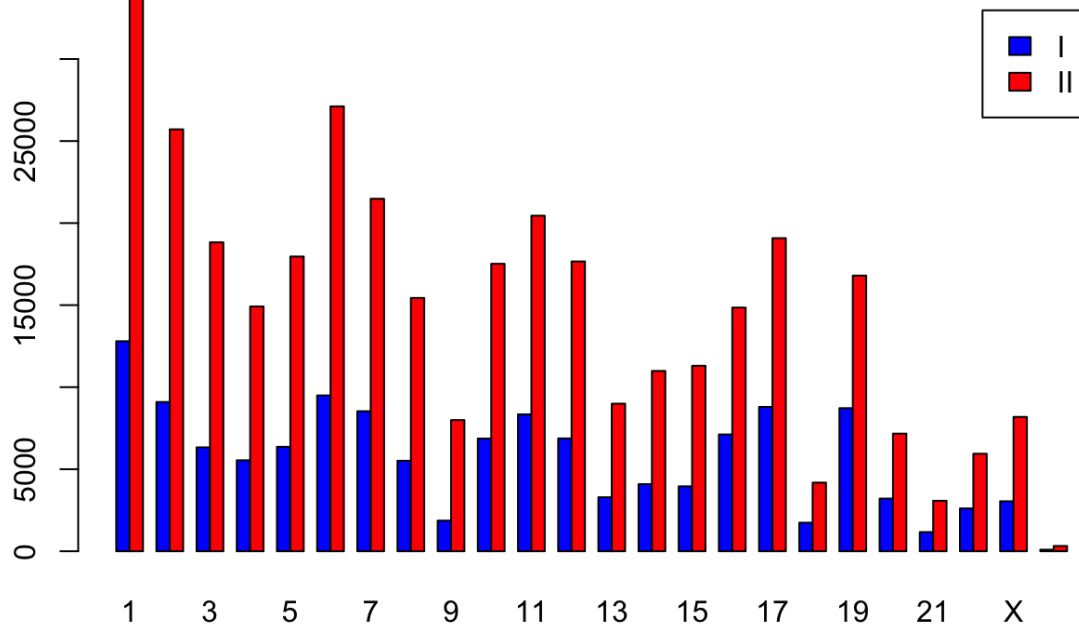
```
barplot(table(Illumina450Manifest_clean$Infinium_Design_Type,
Illumina450Manifest_clean$CHR),
legend=levels(Illumina450Manifest_clean$Infinium_Design_Type),col=c("blue", "red"))
```

```
barplot(table(Illumina450Manifest_clean$Infinium_Design_Type,
Illumina450Manifest_clean$CHR),beside=T,legend=levels(Illumina450Manife
st_clean$Infinium_Design_Type))
```



```
barplot(table(Illumina450Manifest_clean$Infinium_Design_Type,
Illumina450Manifest_clean$CHR),beside=T,legend=levels(Illumina450Manife
st_clean$Infinium_Design_Type),col=c("blue","red"))
```



Exercise 5: Create a barplot representing the number of type I and II probes in each region (islands, shores, shelves and open sea)

Exercise 6: Load the Infinium27k manifest (the file Infinium27.txt): how many probes are in common between 27k and 450k? And where do the common probes map (Island, shore, shelves, genic or not genic)

Supplementary

Is the ratio between typel and typell constant between the chromosomes?

```
table(Illumina450Manifest_clean$Infinium_Design_Type,
      Illumina450Manifest_clean$CHR)
```

```
##
```

```
##      1      2      3      4      5      6      7      8      9     10
11     12
##  I 12800  9096  6331  5544  6365  9497  8534  5513  1866  6869
8343  6876
##  II 34057 25714 18828 14920 17962 27114 21483 15437  7995 17519
20451 17663
```

```
##
##          13      14      15      16      17      18      19      20      21      22
X      Y
##   I    3291   4090   3957   7118   8798   1740   8720   3210   1168   2610
3045      95
##   II   8994  10988  11302  14851  19081   4182  16801   7169   3075   5942
8187    321
```

```
df_table <-
data.frame(table(Illumina450Manifest_clean$Infinium_Design_Type,
Illumina450Manifest_clean$CHR))
```

```
df_table
```

##	Var1	Var2	Freq
## 1	I	1	12800
## 2	II	1	34057
## 3	I	2	9096
## 4	II	2	25714
## 5	I	3	6331
## 6	II	3	18828
## 7	I	4	5544
## 8	II	4	14920
## 9	I	5	6365
## 10	II	5	17962
## 11	I	6	9497
## 12	II	6	27114
## 13	I	7	8534
## 14	II	7	21483
## 15	I	8	5513
## 16	II	8	15437
## 17	I	9	1866
## 18	II	9	7995
## 19	I	10	6869
## 20	II	10	17519
## 21	I	11	8343
## 22	II	11	20451
## 23	I	12	6876
## 24	II	12	17663
## 25	I	13	3291
## 26	II	13	8994
## 27	I	14	4090
## 28	II	14	10988
## 29	I	15	3957
## 30	II	15	11302
## 31	I	16	7118

```
## 32    II    16 14851
## 33     I    17  8798
## 34    II    17 19081
## 35     I    18  1740
## 36    II    18  4182
## 37     I    19  8720
## 38    II    19 16801
## 39     I    20  3210
## 40    II    20  7169
## 41     I    21  1168
## 42    II    21  3075
## 43     I    22  2610
## 44    II    22  5942
## 45     I     X  3045
## 46    II     X  8187
## 47     I     Y   95
## 48    II     Y  321
```

```
df_table_I <- df_table[df_table$Var1=="I",]
```

```
df_table_I <- droplevels(df_table_I)
df_table_II <- df_table[df_table$Var1=="II",]
df_table_II <- droplevels(df_table_II)
head(df_table_I)
```

##	Var1	Var2	Freq
## 1	I	1	12800
## 3	I	2	9096
## 5	I	3	6331
## 7	I	4	5544
## 9	I	5	6365
## 11	I	6	9497

```
head(df_table_II)
```

##	Var1	Var2	Freq
## 2	II	1	34057
## 4	II	2	25714
## 6	II	3	18828
## 8	II	4	14920
## 10	II	5	17962
## 12	II	6	27114

Let's check that the order of the chromosomes is the same in df_table_I and df_table_II

```
df_table_I[,2]==df_table_II[,2]
```

```
## [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

```
## [16] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

```
table(df_table_I[,2]==df_table_II[,2])
```

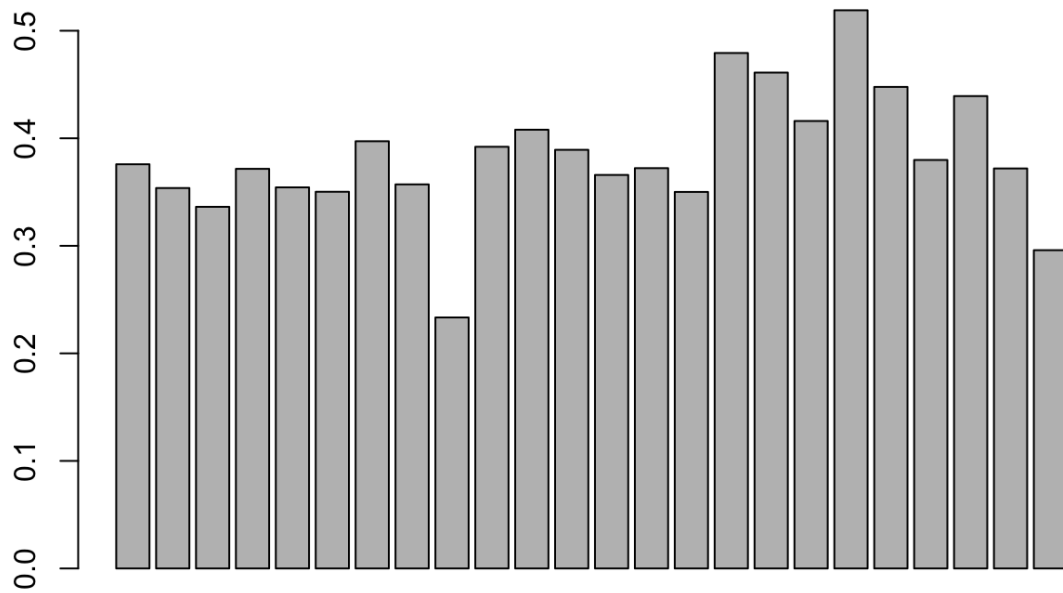
```
##
```

```
## TRUE
```

```
## 24
```

```
ratio_I_II <- df_table_I[,3]/df_table_II[,3]
```

```
barplot(ratio_I_II)
```



```
ratio_I_II
```

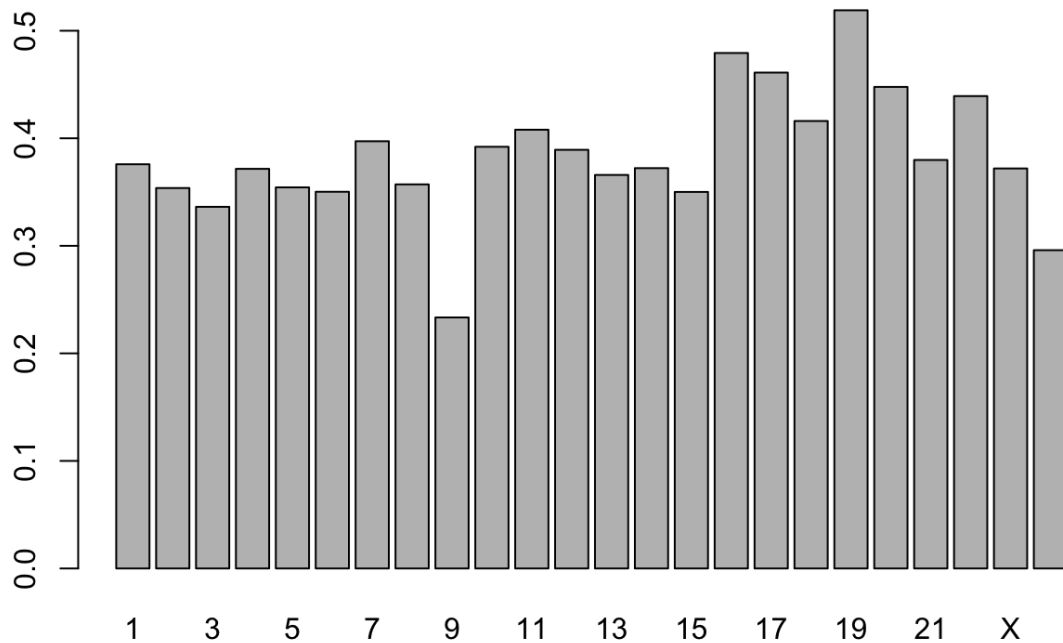
```
## [1] 0.3758405 0.3537373 0.3362545 0.3715818 0.3543592 0.3502619  
0.3972443
```

```
## [8] 0.3571290 0.2333959 0.3920886 0.4079507 0.3892883 0.3659106  
0.3722242
```

```
## [15] 0.3501150 0.4792943 0.4610869 0.4160689 0.5190167 0.4477612
0.3798374
## [22] 0.4392460 0.3719311 0.2959502
```

ratio_I_II is just a vector, without names. If I want to add the chromosome names to the plots

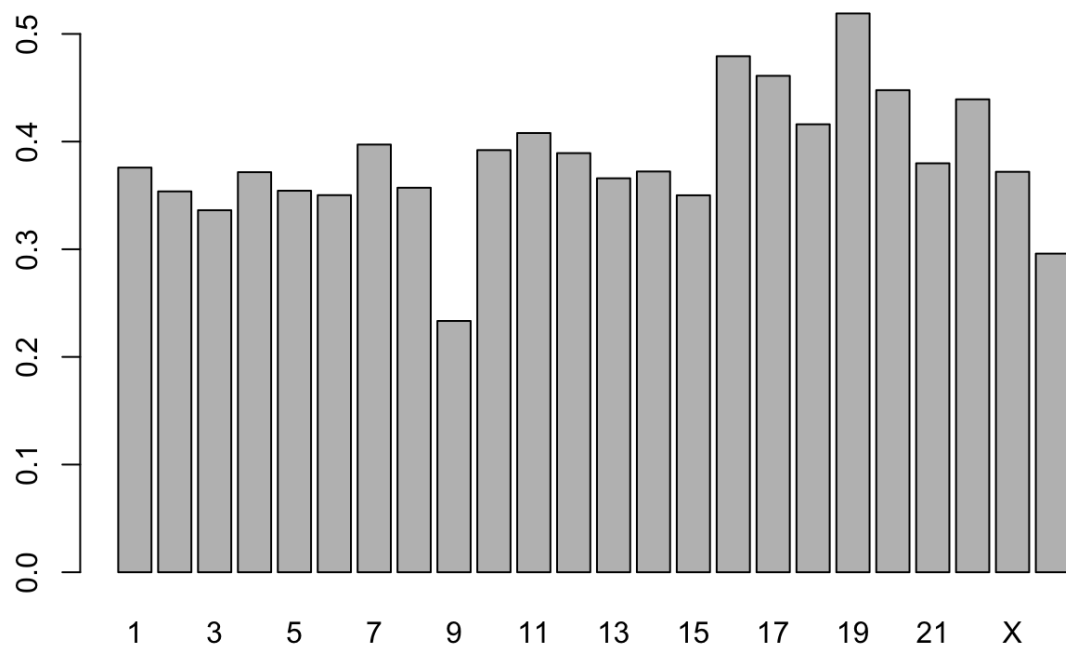
```
barplot(ratio_I_II, names.arg=levels(df_table_I$Var2))
```



or, in alternative:

```
names(ratio_I_II) <- levels(df_table_I$Var2)
```

```
barplot(ratio_I_II)
```



DRD_2025_Lesson_3

```
rm(list=ls())
```

```
setwd("~/Dropbox/DRD_2025/3")  
suppressMessages(library(minfi))
```

```
## Warning: package 'GenomicRanges' was built under R version 4.4.1  
## Warning: package 'S4Vectors' was built under R version 4.4.1  
## Warning: package 'IRanges' was built under R version 4.4.1  
## Warning: package 'matrixStats' was built under R version 4.4.1  
## Warning: package 'locfit' was built under R version 4.4.1
```

```
suppressMessages(library(minfiData))
```

```
vignette("minfi")
```

```
## starting httpd help server ... done
```

CHUNK 1: Import raw data

List the files included in the folder ~/Dropbox/DRD_2025/3/Input_Data

```
list.files("~/Dropbox/DRD_2025/3/Input_Data")
```

```
## [1] "9344737127_R01C02_Grn.idat" "9344737127_R01C02_Red.idat"  
  
## [3] "9344737127_R02C02_Grn.idat" "9344737127_R02C02_Red.idat"  
## [5] "9344737127_R03C02_Grn.idat" "9344737127_R03C02_Red.idat"  
## [7] "9344737127_R06C01_Grn.idat" "9344737127_R06C01_Red.idat"  
## [9] "9376538140_R01C01_Grn.idat" "9376538140_R01C01_Red.idat"  
## [11] "9376538140_R02C01_Grn.idat" "9376538140_R02C01_Red.idat"  
## [13] "9376538140_R04C01_Grn.idat" "9376538140_R04C01_Red.idat"  
## [15] "9376538140_R05C01_Grn.idat" "9376538140_R05C01_Red.idat"  
## [17] "SampleSheet.csv"
```

Let's have a look at the csv

```
SampleSheet <-  
read.table("~/Dropbox/DRD_2025/3/Input_Data/SampleSheet.csv", sep="," , header=T)
```

```
SampleSheet
```

##	Sample_ID	Sample_Well	Array	Slide	Group	Gender	BMI
## 1	21050	G10	R01C02	9344737127	A	F	21.64
## 2	21060	A11	R03C02	9344737127	A	F	20.82
## 3	22041	F10	R06C01	9344737127	B	F	26.57
## 4	22051	H10	R02C02	9344737127	B	F	26.56
## 5	21090	F11	R02C01	9376538140	A	F	17.51
## 6	21100	A12	R05C01	9376538140	A	F	24.56
## 7	23081	E11	R01C01	9376538140	B	F	22.15
## 8	23091	H11	R04C01	9376538140	B	M	23.53

or

```
SampleSheet <-
read.csv("~/Dropbox/DRD_2025/3/Input_Data/SampleSheet.csv",header=T)
```

SampleSheet

##	Sample_ID	Sample_Well	Array	Slide	Group	Gender	BMI
## 1	21050	G10	R01C02	9344737127	A	F	21.64
## 2	21060	A11	R03C02	9344737127	A	F	20.82
## 3	22041	F10	R06C01	9344737127	B	F	26.57
## 4	22051	H10	R02C02	9344737127	B	F	26.56
## 5	21090	F11	R02C01	9376538140	A	F	17.51
## 6	21100	A12	R05C01	9376538140	A	F	24.56
## 7	23081	E11	R01C01	9376538140	B	F	22.15
## 8	23091	H11	R04C01	9376538140	B	M	23.53

Set the directory in which the raw data are stored and load the samplesheet using the function `read.metharray.sheet`

```
baseDir <- ("~/Dropbox/DRD_2025/3/Input_data")
```

```
targets <- read.metharray.sheet(baseDir)
```

```
## [read.metharray.sheet] Found the following CSV files:
```

```
## [1] "/Users/francesco/Dropbox/DRD_2025/3/Input_data/SampleSheet.csv"
```

targets

##	Sample_ID	Sample_Well	Array	Slide	Group	Gender	BMI
## 1	21050	G10	R01C02	9344737127	A	F	21.64
## 2	21060	A11	R03C02	9344737127	A	F	20.82
## 3	22041	F10	R06C01	9344737127	B	F	26.57
## 4	22051	H10	R02C02	9344737127	B	F	26.56
## 5	21090	F11	R02C01	9376538140	A	F	17.51

```
## 6      21100      A12 R05C01 9376538140      A      F 24.56
## 7      23081      E11 R01C01 9376538140      B      F 22.15
## 8      23091      H11 R04C01 9376538140      B      M 23.53
##
##                               Basename
## 1 /Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R01C02
## 2 /Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R03C02
## 3 /Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R06C01
## 4 /Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R02C02
## 5 /Users/francesco/Dropbox/DRD_2025/3/Input_data/9376538140_R02C01
## 6 /Users/francesco/Dropbox/DRD_2025/3/Input_data/9376538140_R05C01
## 7 /Users/francesco/Dropbox/DRD_2025/3/Input_data/9376538140_R01C01
## 8 /Users/francesco/Dropbox/DRD_2025/3/Input_data/9376538140_R04C01
```

Create an object of class `RGChannelSet` using the function `read.metharray.exp`

```
?read.metharray.exp
```

```
RGset <- read.metharray.exp(targets = targets)
save(RGset,file="RGset.RData")
```

Let's explore the `RGset` object:

```
RGset
```

```
## class: RGChannelSet
## dim: 622399 8
## metadata(0):
## assays(2): Green Red
## rownames(622399): 10600313 10600322 ... 74810490 74810492
## rowData names(0):
## colnames(8): 9344737127_R01C02 9344737127_R03C02 ...
9376538140_R01C01
## 9376538140_R04C01
## colData names(9): Sample_ID Sample_Well ... Basename filenames
## Annotation
## array: IlluminaHumanMethylation450k
## annotation: ilmn12.hg19
```

```
str(RGset)
```

```
## Formal class 'RGChannelSet' [package "minfi"] with 6 slots
## ..@ annotation      : Named chr [1:2]
"IlluminaHumanMethylation450k" "ilmn12.hg19"
## .. ..- attr(*, "names")= chr [1:2] "array" "annotation"
## ..@ colData          :Formal class 'DFrame' [package "S4Vectors"]
with 6 slots
```

```

## .. .. .@ rownames      : chr [1:8] "9344737127_R01C02"
"9344737127_R03C02" "9344737127_R06C01" "9344737127_R02C02" ...
## .. .. .@ nrows         : int 8
## .. .. .@ elementType    : chr "ANY"
## .. .. .@ elementMetadata: NULL
## .. .. .@ metadata       : list()
## .. .. .@ listData       :List of 9
## .. .. . $ Sample_ID    : int [1:8] 21050 21060 22041 22051 21090
21100 23081 23091
## .. .. . $ Sample_Well: chr [1:8] "G10" "A11" "F10" "H10" ...
## .. .. . $ Array        : chr [1:8] "R01C02" "R03C02" "R06C01"
"R02C02" ...
## .. .. . $ Slide        : chr [1:8] "9344737127" "9344737127"
"9344737127" "9344737127" ...
## .. .. . $ Group        : chr [1:8] "A" "A" "B" "B" ...
## .. .. . $ Gender       : chr [1:8] "F" "F" "F" "F" ...
## .. .. . $ BMI          : num [1:8] 21.6 20.8 26.6 26.6 17.5 ...
## .. .. . $ Basename     : chr [1:8]
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R01C02"
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R03C02"
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R06C01"
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R02C02" ...
## .. .. . $ filenames    : chr [1:8]
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R01C02"
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R03C02"
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R06C01"
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R02C02" ...
## ..@ assays             :Formal class 'SimpleAssays' [package
"SummarizedExperiment"] with 1 slot
## .. .. .@ data:Formal class 'SimpleList' [package "S4Vectors"]
with 4 slots
## .. .. . . .@ listData      :List of 2
## .. .. . . . $ Green: int [1:622399, 1:8] 274 6329 1548 2000
1815 3046 1090 759 3482 209 ...
## .. .. . . . . .- attr(*, "dimnames")=List of 2
## .. .. . . . . . $ : chr [1:622399] "10600313" "10600322"
"10600328" "10600336" ...
## .. .. . . . . . $ : chr [1:8] "9344737127_R01C02"
"9344737127_R03C02" "9344737127_R06C01" "9344737127_R02C02" ...
## .. .. . . . . . $ Red : int [1:622399, 1:8] 526 1807 1189 14881
6328 1070 351 12889 610 328 ...
## .. .. . . . . .- attr(*, "dimnames")=List of 2
## .. .. . . . . . $ : chr [1:622399] "10600313" "10600322"
"10600328" "10600336" ...
## .. .. . . . . . $ : chr [1:8] "9344737127_R01C02"
"9344737127_R03C02" "9344737127_R06C01" "9344737127_R02C02" ...

```

```
## ..@ elementType      : chr "ANY"
## ..@ elementMetadata: NULL
## ..@ metadata         : list()
## ..@ NAMES            : chr [1:622399] "10600313" "10600322"
"10600328" "10600336" ...
## ..@ elementMetadata:Formal class 'DFrame' [package "S4Vectors"]
with 6 slots
## ..@ rownames         : NULL
## ..@ nrows            : int 622399
## ..@ elementType      : chr "ANY"
## ..@ elementMetadata: NULL
## ..@ metadata         : list()
## ..@ listData         : Named list()
## ..@ metadata         : list()
```

```
?RGChannelSet
```

We extract the Green and Red Channels using the functions `getGreen` and `getRed`

```
Red <- data.frame(getRed(RGset))
```

```
dim(Red)
```

```
## [1] 622399      8
```

```
head(Red)
```

```
##           X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
## 10600313           526           556           660
## 10600322           1807          1457          1369
## 10600328           1189          1279          1120
## 10600336          14881          14211          14680
## 10600345           6328          4476          4010
## 10600353           1070          1005           982
##           X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## 10600313           641           483           652
## 10600322           1914          1649          2325
## 10600328           1129          1763          1621
## 10600336          15868          15747          15581
## 10600345           6040          6398          5747
## 10600353           1196          1139           954
##           X9376538140_R01C01 X9376538140_R04C01
## 10600313           618           661
## 10600322           1840          1599
## 10600328           1796          1706
## 10600336          14770          15405
```

```
## 10600345          4541          4433
## 10600353          911          1044
```

```
Green <- data.frame(getGreen(RGset))
```

```
dim(Green)
```

```
## [1] 622399      8
```

```
head(Green)
```

```
##          X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
```

```
## 10600313          274          313          328
```

```
## 10600322          6329          6949          7470
```

```
## 10600328          1548          1528          1828
```

```
## 10600336          2000          1512          1618
```

```
## 10600345          1815          3251          3459
```

```
## 10600353          3046          3615          3565
```

```
##          X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
```

```
## 10600313          292          358          395
```

```
## 10600322          6947          8002          8322
```

```
## 10600328          1192          2060          2169
```

```
## 10600336          1715          1744          2134
```

```
## 10600345          2066          888          2655
```

```
## 10600353          3437          4211          3803
```

```
##          X9376538140_R01C01 X9376538140_R04C01
```

```
## 10600313          268          358
```

```
## 10600322          6719          8093
```

```
## 10600328          2327          2361
```

```
## 10600336          1634          1740
```

```
## 10600345          2618          3594
```

```
## 10600353          3498          4152
```

CHUNK 2: Addresses, probes and out of band probes

```
load('~/.Dropbox/DRD_2025/2/Illumina450Manifest_clean.RData')
```

```
ls()
```

```
## [1] "baseDir"          "Green"
```

```
## [3] "Illumina450Manifest_clean" "Red"
```

```
## [5] "RGset"            "SampleSheet"
```

```
## [7] "targets"
```

```
head(Illumina450Manifest_clean)
```

##	IlmnID	Name	AddressA_ID
----	--------	------	-------------

## 1	cg00035864	cg00035864	31729416
## 2	cg00050873	cg00050873	32735311
## 3	cg00061679	cg00061679	28780415
## 4	cg00063477	cg00063477	16712347
## 5	cg00121626	cg00121626	19779393
## 6	cg00212031	cg00212031	29674443

##		AlleleA_ProbeSeq	AddressB_ID
----	--	------------------	-------------

## 1	AAAACACTAACAATCTTATCCACATAAACCCCTTAAATTTATCTCAAATTC		
## 2	ACAAAAAAACAACACACAACCTATAATAATTTTTTAAAATAAATAAACCCCA		31717405
## 3	AAAACATTAAAAAACTAATTCACTACTATTTAATTACTTTATTTTCCATC		
## 4	TATTCTTCCACACAAAATACTAAACRTATATTTACAAAAATACTTCCATC		
## 5	AAAATAATAAAAAATACTTACAAACCAATACTATACCCTACAACCTCTC		
## 6	CCCAATTAACCACAAAACTAAACAAATTATACAATCAAAAAACATACA		38703326

##		AlleleB_ProbeSeq
----	--	------------------

Infinium_Design_Type

1

II

2 ACGAAAAAACAACGCACAACCTATAATAATTTTTTAAAATAAATAAACCCCG

I

3

II

4

II

5

II

6 CCCAATTAACCGCAAAAACTAAACAAATTATACGATCGAAAAACGTACG

I

##	Next_Base	Color_Channel
----	-----------	---------------

1

## 2	A	Red
------	---	-----

3

4

5

## 6	T	Red
------	---	-----

##

Forward_Sequence

1

AATCCAAAGATGATGGAGGAGTGCCCGCTCATGATGTGAAGTACCTGCTCAGCTGGAAAC [CG] AATTTGA

GATAAATTCAAGGGTCTATGTGGACAAGACTGCTAGTGTCTCTCTCTGGATTG

2

TATCTCTGTCTGGCGAGGAGGCAACGCACAACCTGTGGTGGTTTTTGGAGTGGGTGGACCC [CG] GCCAAGA

CGGCCTGGGCTGACCAGAGACGGGAGGCAGAAAAAGTGGGCAGGTGGTTGCAG

```

## 3
TCAACAAATGAGAGACATTGAAGAACTAATTCACACTACTATTTGGTTACTTTATTTTCCAT [CG] AAGAAAA
CCTCTTTTTTAAAACTAACACATAAAATAAAATGAACGAAGAACAACCTAAACG
## 4
CTCCTGTACTTGTTTCATTAAATAATGATTCCCTTGGATATACCAAGTCTGGATAGCGGATT [CG] ATGGAAG
CATTTTTTGTAATATACGTTTCAGTATTTTGTGTGGAAGAACAATCTAGCTG
## 5
AGGTGAATGAAGAGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACCTCT [CG] CGCTGCC
ACAAAAAGCCAATTAACGAAGCAAACCTGTTTTATCTTTTCTCTCCCCAGAAG
## 6
CCATTGGCCCGCCCCAGTTGGCCGCAGGGACTGAGCAAGTTATGCGGTCGGAAGACGTG [CG] TTAAAGG
GCTGAAGGGGAGGGACGGAACCTGACAGTCTCTGTGACAGCTCTGAGGTGGGAG
##   Genome_Build CHR   MAPINFO
SourceSeq
## 1           37   Y   8553009
AGACACTAGCAGTCTTGTCCACATAGACCCTTGAATTTATCTCAAATTCG
## 2           37   Y   9363356
CGGGGTCCACCCACTCCAAAAACCACCACAGTTGTGCGTTGCCTCCTCGC
## 3           37   Y  25314171
CGATGGAAAATAAAGTAACCAAATAGTAGTGAATTAGTTCTTCAATGTCT
## 4           37   Y  22741795
CGATGGAAGCATTTTTGTAAATATACGTTTCAGTATTTTGTGTGGAAGAAC
## 5           37   Y  21664296
AGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACCTCTCG
## 6           37   Y  21239348
CGCACGTCTTCCCGACCGCATAAATTGCTCAGTCCCTGCGGCCAACTGGG
##   Chromosome_36 Coordinate_36 Strand Probe_SNPs Probe_SNPs_10
Random_Loci
## 1           Y           8613009           F
NA
## 2           Y           9973356           R
NA
## 3           Y           23723559          R
NA
## 4           Y           21151183          F   rs9341313       rs13447379
NA
## 5           Y           20123684          R
NA
## 6           Y           19698736          R
NA
##   Methyl27_Loci UCSC_RefGene_Name          UCSC_RefGene_Accession
## 1           NA           TTTY18              NR_001550
## 2           NA           TSPY4;FAM197Y2        NM_001164471;NR_001553
## 3           NA           DAZ1;DAZ4;DAZ4  NM_004081;NM_020420;NM_001005375
## 4           NA           EIF1AY              NM_004681
## 5           NA           BCORL2              NR_002923

```



```
## 6          NA          TTTY14          NR_001543
## UCSC_RefGene_Group UCSC_CpG_Islands_Name
Relation_to_UCSC_CpG_Island Phantom
## 1          TSS1500
## 2      Body;TSS1500      chrY:9363680-9363943
N_Shore
## 3      Body;Body;Body
## 4          Body chrY:22737825-22738052
S_Shelf
## 5          Body chrY:21664481-21665063
N_Shore
## 6          TSS200 chrY:21238448-21240005
Island
## DMR Enhancer          HMM_Island Regulatory_Feature_Name
## 1          NA
## 2          NA      Y:9973136-9976273
## 3          NA
## 4          NA
## 5          NA
## 6          NA      Y:19697854-19699393
## Regulatory_Feature_Group DHS
## 1          NA
## 2          NA
## 3          NA
## 4          NA
## 5          NA
## 6          NA
```

From Addresses to probes:

I want to check the probe having address 10600313 (The first rowname in Red and Green objects)

```
illumina450Manifest_clean[illumina450Manifest_clean$AddressA_ID=="10600313",]
```

##	[1] IlmnID	Name
##	[3] AddressA_ID	AlleleA_ProbeSeq
##	[5] AddressB_ID	AlleleB_ProbeSeq
##	[7] Infinium_Design_Type	Next_Base
##	[9] Color_Channel	Forward_Sequence
##	[11] Genome_Build	CHR
##	[13] MAPINFO	SourceSeq
##	[15] Chromosome_36	Coordinate_36
##	[17] Strand	Probe_SNPs
##	[19] Probe_SNPs_10	Random_Loci

```
## [21] Methyl27_Loci UCSC_RefGene_Name
## [23] UCSC_RefGene_Accession UCSC_RefGene_Group
## [25] UCSC_CpG_Islands_Name Relation_to_UCSC_CpG_Island
## [27] Phantom DMR
## [29] Enhancer HMM_Island
## [31] Regulatory_Feature_Name Regulatory_Feature_Group
## [33] DHS
## <0 rows> (or 0-length row.names)
```

there is not an AddressA_ID with code 10600313...

```

Illumina450Manifest_clean[Illumina450Manifest_clean$AddressB_ID=="10600
313",]
```

##	IlmnID	Name	AddressA_ID
##	311232	cg25192902	cg25192902
##			10679328
##			AlleleA_ProbeSeq
##			AddressB_ID
##	311232	TAAACAACCAACAAACAAACCAAAAACTAAAACAACCAACCAATCAACA	
##			10600313
##			AlleleB_ProbeSeq
##			Infinium_Design_Type
##	311232	TAAACAACCAACAAACGAACCGAAAACTAAAACGACCAACCAATCAACG	
##			I
##		Next_Base	Color_Channel
##	311232	A	Red
##			
##			Forward_Sequence
##	311232		
##			CGGAGAGCTGCGCGCAGGCCCTTCGCGCGATCTGCCGTCATATCAGCTTATGGCGCCGGT[CG]CTGATTG
##			GCTGGCCGCTCCAGCCTCCCGGCCCGCTTGCTGGCTGCCCAGCTGCTAGGACA
##		Genome_Build	CHR MAPINFO
##	311232	37 11	121163419
##			SourceSeq
##			Chromosome_36
##	311232		CGCTGATTGGCTGGCCGCTCCAGCCTCCCGGCCCGCTTGCTGGCTGCCCCA
##			11
##		Coordinate_36	Strand Probe_SNPs Probe_SNPs_10 Random_Loci
##			Methyl27_Loci
##	311232	120668629	F NA
##			NA
##		UCSC_RefGene_Name	UCSC_RefGene_Accession
##			UCSC_RefGene_Group
##	311232	SC5DL;SC5DL;SC5DL	NM_006918;NM_001024956;NM_006918
##			5'UTR;TSS200;1stExon
##		UCSC_CpG_Islands_Name	Relation_to_UCSC_CpG_Island

```
## 311232 chr11:121163244-121163952 Island
## Phantom DMR Enhancer
HMM_Island
## 311232 high-CpG:120668534-120668733 NA 11:120668464-
120669162
## Regulatory_Feature_Name Regulatory_Feature_Group DHS
## 311232 11:121162646-121164137 Promoter_Associated NA
```

...but there is an AddressB_ID with code 10600313, and it is associated to the probe cg25192902, type I, Red; the Address_A of this probe is 10679328

I want to check the probe having address 10600322 (The second rowname in Red and Green objects)

```
illumina450Manifest_clean[illumina450Manifest_clean$AddressA_ID=="10600322",]
```

```
## IlmnID Name AddressA_ID
## 430973 cg00226849 cg00226849 10600322
## AlleleA_ProbeSeq
AddressB_ID
## 430973 TAAAAACCCCAAAAAATAACATTAAATATCTCTCTATACCCTAACCCAC
## AlleleB_ProbeSeq Infinium_Design_Type Next_Base Color_Channel
## 430973 II
## Forward_Sequence
## 430973
TTTGTCTCTCCAGATGTGAATAGCTCCACTATAACCAGCCTCGTCTTCCTTCCGGGGGACAA[CG]TGGGTCA
GGGCACAGAGAGATATTTAATGTCACCCTCTTGGGGCTTTCATGGGACTCCCT
## Genome_Build CHR MAPINFO
## 430973 37 18 72201785
## SourceSeq
Chromosome_36
## 430973 CGTGGGTCAGGGCACAGAGAGATATTTAATGTCACCCTCTTGGGGCTTTC
18
## Coordinate_36 Strand Probe_SNPs Probe_SNPs_10 Random_Loci
Methyl27_Loci
## 430973 70352765 F NA
NA
## UCSC_RefGene_Name UCSC_RefGene_Accession UCSC_RefGene_Group
## 430973 CNDP1;CNDP1 NM_032649;NM_032649 1stExon;5'UTR
## UCSC_CpG_Islands_Name Relation_to_UCSC_CpG_Island
## 430973
## Phantom DMR Enhancer HMM_Island
## 430973 low-CpG:70352660-70352818 NA
## Regulatory_Feature_Name Regulatory_Feature_Group DHS
```

430973

NA

There is an AddressA_ID with code 10600322, and it is associated to the probe cg00226849, type II. Note that the column Color_Channel is empty, as all the Type II probes will emit in Red or Green depending on the methylation status of the target CpG. In addition, also Address_B is empty, as type II probes never have an Address_B. Indeed, if you digit

```
Illumina450Manifest_clean[Illumina450Manifest_clean$AddressB_ID=="10600322",]
```

```
## [1] IlmnID Name
## [3] AddressA_ID AlleleA_ProbeSeq
## [5] AddressB_ID AlleleB_ProbeSeq
## [7] Infinium_Design_Type Next_Base
## [9] Color_Channel Forward_Sequence
## [11] Genome_Build CHR
## [13] MAPINFO SourceSeq
## [15] Chromosome_36 Coordinate_36
## [17] Strand Probe_SNPs
## [19] Probe_SNPs_10 Random_Loci
## [21] Methyl27_Loci UCSC_RefGene_Name
## [23] UCSC_RefGene_Accession UCSC_RefGene_Group
## [25] UCSC_CpG_Islands_Name Relation_to_UCSC_CpG_Island
## [27] Phantom DMR
## [29] Enhancer HMM_Island
## [31] Regulatory_Feature_Name Regulatory_Feature_Group
## [33] DHS
## <0 rows> (or 0-length row.names)
```

the resulting dataframe will be empty.

From probes to Addresses

```
head(Illumina450Manifest_clean)
```

```
## IlmnID Name AddressA_ID
## 1 cg00035864 cg00035864 31729416
## 2 cg00050873 cg00050873 32735311
## 3 cg00061679 cg00061679 28780415
## 4 cg00063477 cg00063477 16712347
## 5 cg00121626 cg00121626 19779393
## 6 cg00212031 cg00212031 29674443
## AlleleA_ProbeSeq AddressB_ID
## 1 AAAACACTAACAATCTTATCCACATAAACCCCTTAAATTTATCTCAAATTC
## 2 ACAAAAAACAACACACAACATAATAATTTTAAAAATAAATAAACCCCA 31717405
## 3 AAAACATTAAAAAACTAATTCACACTACTATTTAATTACTTTATTTTCCATC
## 4 TATTCTTCCACACAAAATACTAAACRTATATTTACAAAAATACTTCCATC
```

```

## 5 AAAACTAATAAAAAATAACTTACAAACCAAATACTATACCCTACAACCTCTC
## 6 CCCAATTAACCACAAAACTAAACAAATTATACAATCAAAAAACATACA      38703326
##                               AlleleB_ProbeSeq
Infinium_Design_Type
## 1
II
## 2 ACGAAAAACAACGCACAACCTATAATAATTTTAAAATAAATAAACCCCG
I
## 3
II
## 4
II
## 5
II
## 6 CCCAATTAACCGCAAAAACTAAACAAATTATACGATCGAAAAACGTACG
I
##   Next_Base   Color_Channel
## 1
## 2           A           Red
## 3
## 4
## 5
## 6           T           Red
##
Forward_Sequence
## 1
AATCCAAAGATGATGGAGGAGTGCCCGCTCATGATGTGAAGTACCTGCTCAGCTGGAAAC [CG] AATTTGA
GATAAATTCAAGGGTCTATGTGGACAAGACTGCTAGTGTCTCTCTCTGGATTG
## 2
TATCTCTGTCTGGCGAGGAGGCAACGCACAACCTGTGGTGGTTTTTGGAGTGGGTGGACCC [CG] GCCAAGA
CGGCCTGGGCTGACCAGAGACGGGAGGCAGAAAAAGTGGGCAGGTGGTTGCAG
## 3
TCAACAAATGAGAGACATTGAAGAACTAATTCACTACTATTTGGTTACTTTATTTTCCAT [CG] AAGAAAA
CCTCTTTTTTAAAACTAACACATAAATAAAATGAACGAAGAACAACAACTAAACG
## 4
CTCCTGTACTTGTTTCATTAAATAATGATTCCTTGGATATACCAAGTCTGGATAGCGGATT [CG] ATGGAAG
CATTTTTGTAAATATACGTTCAAGTATTTTGTGTGGAAGAACAATCTAGCTG
## 5
AGGTGAATGAAGAGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACCTCT [CG] CGCTGCC
ACAAAAAGCCAATTAACGAAGCAAACCTGTTTTATCTTTTCTCTCCCCAGAAG
## 6
CCATTGGCCCGCCCCAGTTGGCCGCAGGGACTGAGCAAGTTATGCGGTGCGGAAGACGTG [CG] TTAAAGG
GCTGAAGGGGAGGGACGGAACCTGACAGTCTCTGTGACAGCTCTGAGGTGGGAG
##   Genome_Build CHR   MAPINFO
SourceSeq

```

```

## 1          37   Y   8553009
AGACACTAGCAGTCTTGTCCACATAGACCCTTGAATTTATCTCAAATTTCG
## 2          37   Y   9363356
CGGGGTCCACCCACTCCAAAAACCACCACAGTTGTGCGTTGCCTCCTCGC
## 3          37   Y  25314171
CGATGGAAAATAAAGTAACCAAATAGTAGTGAATTAGTTCTTCAATGTCT
## 4          37   Y  22741795
CGATGGAAGCATTTTTGTAAATATACGTTTCAGTATTTTGTGTGGAAGAAC
## 5          37   Y  21664296
AGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACCTCTCG
## 6          37   Y  21239348
CGCACGTCTTCCCGACCGCATAACTTGCTCAGTCCCTGCGGCCAACTGGG
##   Chromosome_36 Coordinate_36 Strand Probe_SNPs Probe_SNPs_10
Random_Loci
## 1          Y          8613009          F
NA
## 2          Y          9973356          R
NA
## 3          Y          23723559         R
NA
## 4          Y          21151183         F   rs9341313       rs13447379
NA
## 5          Y          20123684         R
NA
## 6          Y          19698736         R
NA
##   Methyl27_Loci UCSC_RefGene_Name          UCSC_RefGene_Accession
## 1          NA          TTTY18              NR_001550
## 2          NA   TSPY4;FAM197Y2          NM_001164471;NR_001553
## 3          NA   DAZ1;DAZ4;DAZ4 NM_004081;NM_020420;NM_001005375
## 4          NA          EIF1AY              NM_004681
## 5          NA          BCORL2              NR_002923
## 6          NA          TTTY14              NR_001543
##   UCSC_RefGene_Group UCSC_CpG_Islands_Name
Relation_to_UCSC_CpG_Island Phantom
## 1          TSS1500
## 2          Body;TSS1500   chrY:9363680-9363943
N_Shore
## 3          Body;Body;Body
## 4          Body chrY:22737825-22738052
S_Shelf
## 5          Body chrY:21664481-21665063
N_Shore
## 6          TSS200 chrY:21238448-21240005
Island
##   DMR Enhancer          HMM_Island Regulatory_Feature_Name

```

```
## 1 NA
## 2 NA Y:9973136-9976273
## 3 NA
## 4 NA
## 5 NA
## 6 NA Y:19697854-19699393
## Regulatory_Feature_Group DHS
## 1 NA
## 2 NA
## 3 NA
## 4 NA
## 5 NA
## 6 NA
```

cg00035864, type II (the first probe in the Illumina450Manifest_clean object)

```
Red[rownames (Red) == "31729416", ]
```

##	X9344737127_R01C02	X9344737127_R03C02	X9344737127_R06C01
##	31729416	236	234
##	X9344737127_R02C02	X9376538140_R02C01	X9376538140_R05C01
##	31729416	327	299
##	X9376538140_R01C01	X9376538140_R04C01	
##	31729416	284	5058

```
Green[rownames(Green)=="31729416",]
```

##	X9344737127_R01C02	X9344737127_R03C02	X9344737127_R06C01
##	31729416	208	215
##	X9344737127_R02C02	X9376538140_R02C01	X9376538140_R05C01
##	31729416	204	229
##	X9376538140_R01C01	X9376538140_R04C01	
##	31729416	182	1351

cg00050873, type I Red (the second probe in the Illumina450Manifest_clean object)

```

Illumina450Manifest_clean[Illumina450Manifest_clean$IlmnID=="cg00050873
"]

```

##	IlmnID	Name	AddressA	ID
----	--------	------	----------	----

```
## 2 cg00050873 cg00050873 32735311
## AlleleA_ProbeSeq AddressB_ID
## 2 ACAAAAAACAACACACAACACTATAATAATTTTTAAATAAAATAAACCCCA 31717405
## AlleleB_ProbeSeq
Infinium Design Type
```

```
## 2 ACGAAAAACAACGCACAACCTATAATAATTTTAAAATAAATAAACCCCG
I
##   Next_Base Color_Channel
## 2      A              Red
##
Forward_Sequence
## 2
TATCTCTGTCTGGCGAGGAGGCAACGCACAACCTGTGGTGGTTTTTGGAGTGGGTGGACCC[CG]GCCAAGA
CGGCCTGGGCTGACCAGAGACGGGAGGCAGAAAAAGTGGGCAGGTGGTTGCAG
##   Genome_Build CHR MAPINFO
SourceSeq
## 2      37      Y 9363356
CGGGGTCCACCCACTCCAAAAACCACCACAGTTGTGCGTTGCCTCCTCGC
##   Chromosome_36 Coordinate_36 Strand Probe_SNPs Probe_SNPs_10
Random_Loci
## 2      Y      9973356      R
NA
##   Methyl27_Loci UCSC_RefGene_Name UCSC_RefGene_Accession
UCSC_RefGene_Group
## 2      NA      TSPY4;FAM197Y2 NM_001164471;NR_001553
Body;TSS1500
##   UCSC_CpG_Islands_Name Relation_to_UCSC_CpG_Island Phantom DMR
Enhancer
## 2 chrY:9363680-9363943      N_Shore
NA
##           HMM_Island Regulatory_Feature_Name Regulatory_Feature_Group
DHS
## 2 Y:9973136-9976273
NA
```

```
Red[rownames(Red)=="32735311",]
```

```
##      X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
## 32735311      337      376      343
##      X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## 32735311      389      300      303
##      X9376538140_R01C01 X9376538140_R04C01
## 32735311      249      1001
```

```
Red[rownames(Red)=="31717405",]
```

```
##      X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
## 31717405      338      397      362
##      X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## 31717405      399      374      271
##      X9376538140_R01C01 X9376538140_R04C01
```



```
## 31717405
```

```
268
```

```
9478
```

But the two addresses are present also when I look at the Green object: these are out of band signals!

```
Green[rownames(Green)=="32735311",]
```

```
##          X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
## 32735311          196          195          230
##          X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## 32735311          179          188          190
##          X9376538140_R01C01 X9376538140_R04C01
## 32735311          195          332
```

```
Green[rownames(Green)=="31717405",]
```

```
##          X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
## 31717405          228          268          201
##          X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## 31717405          225          221          260
##          X9376538140_R01C01 X9376538140_R04C01
## 31717405          172          930
```

Exercise 1: Let's consider the probe cg03360716. Is it a type I or type II probe? What is/are its addresses? What are the associated red and green fluorescences?

CHUNK 3: Class “IlluminaMethylationManifest”

The RGChannelSet stores also a manifest object that contains the probe design information of the array (IlluminaMethylationManifest).

```
?IlluminaMethylationManifest
```

From the help page, you see that getManifest and getManifestInfo are two accessor functions for this class. Let's check their results:

```
getManifest(RGset)
```

```
## IlluminaMethylationManifest object
## Annotation
## array: IlluminaHumanMethylation450k
## Number of type I probes: 135476
## Number of type II probes: 350036
```

```
## Number of control probes: 850
## Number of SNP type I probes: 25
## Number of SNP type II probes: 40
```

```
getManifestInfo(RGset)
```

```
## [1] 485512
```

getManifestInfo does not seem really useful! :-D

What about the getProbeInfo() function? It returns a data.frame giving the type I, type II or control probes

```
getProbeInfo(RGset)
```

```
## DataFrame with 135476 rows and 8 columns
```

```
##           Name      AddressA      AddressB      Color
NextBase
##      <character> <character> <character> <character>
<DNAStrngSet>
## 1      cg00050873      32735311      31717405      Red
A
## 2      cg00212031      29674443      38703326      Red
T
## 3      cg00213748      30703409      36767301      Red
A
## 4      cg00214611      69792329      46723459      Red
A
## 5      cg00455876      27653438      69732350      Red
A
## ...           ...           ...           ...           ...
...
## 135472 cg27620176      45755371      14722402      Red
T
## 135473 cg27645955      16743317      63806469      Grn
C
## 135474 cg27647370      19658466      66783465      Red
A
## 135475 cg27652464      42785497      14603505      Red
A
## 135476 cg27653384      52770334      15676404      Grn
C
##           ProbeSeqA           ProbeSeqB      nCpG
##           <DNAStrngSet>           <DNAStrngSet> <integer>
## 1      ACAAAAAAAC...ATAAACCCCA ACGAAAAAAC...ATAAACCCCG      2
## 2      CCCAATTAAC...AAACATACA  CCCAATTAAC...AAACGTACG      4
```

```
## 3      TTTTAACACC...AAAAAAAACA TTTTAACGCC...AAAAAAAACG      3
## 4      CTAACCTCCA...AACACAAACA CTAACCTCCG...AACGCGAACG      5
## 5      AACTCTAAAC...AAAAAACTCA AACTCTAAAC...AAAAAACTCG      2
## ...      ...      ...      ...
## 135472 AACATCCCAA...CACCTAACCA AACATCCCAA...CACCTAACCG      4
## 135473 AACACAAAT...AAACCCTACA AACACGAAT...GAACCCTACG      5
## 135474 TATACAAACA...TCCACAACCA TATACAAACG...TCCACAACCG      3
## 135475 AATAAAAAAA...AATCCTAACA AATAAAAAAA...GATCCTAACG      2
## 135476 CACCAAAATA...ATAAAAATCA CGCCAAAATA...GTAAAAATCG      2
```

Only the first and the last rows of the object are printed. We will create an object df containing the result of this function:

```
df <- data.frame(getProbeInfo(RGset))
```

```
dim(df)
```

```
## [1] 135476      8
```

Strange...this df has 135476 rows, not 485512...why?

```
head(getProbeInfo(RGset, type = "II"))
```

```
## DataFrame with 6 rows and 4 columns
```

```
##      Name      AddressA      ProbeSeqA      nCpG
##      <character> <character>      <DNASetString> <integer>
## 1  cg00035864    31729416 AAAACACTAA...TCTCAAATTC      0
## 2  cg00061679    28780415 AAAACATTAA...ATTTTCCATC      0
## 3  cg00063477    16712347 TATTCTTCCA...TACTTCCATC      1
## 4  cg00121626    19779393 AAAACTAATA...TACAACTCTC      0
## 5  cg00223952    17691457 TTAACCTATAC...CAACTTCAAC      0
## 6  cg00243321    36709370 AATACRAAAT...CTAAATCCAC      3
```

```
df_II <- data.frame(getProbeInfo(RGset, type = "II"))
```

```
dim(df_II)
```

```
## [1] 350036      4
```

```
350036+135476 #485512
```

```
## [1] 485512
```

This means that in the default settings, the getProbeInfo function returns only Type I probes. However, this is not really specified in the help page. Take home message: when you work with a package, try before you trust!

CHUNK 4: Extract methylated and unmethylated signals

We will use the function `MSet.raw`

```
MSet.raw <- preprocessRaw(RGset)
```

```
MSet.raw
```

```
## class: MethylSet
## dim: 485512 8
## metadata(0):
## assays(2): Meth Unmeth
## rownames(485512): cg00050873 cg00212031 ... ch.22.47579720R
##   ch.22.48274842R
## rowData names(0):
## colnames(8): 9344737127_R01C02 9344737127_R03C02 ...
9376538140_R01C01
##   9376538140_R04C01
## colData names(9): Sample_ID Sample_Well ... Basename filenames
## Annotation
##   array: IlluminaHumanMethylation450k
##   annotation: ilmn12.hg19
## Preprocessing
##   Method: Raw (no normalization or bg correction)
##   minfi version: 1.50.0
##   Manifest version: 0.4.0
```

```
?MethylSet
```

Note that now the number of rows is 485512, exactly the number of probes according to the Manifest!

```
save(MSet.raw, file="MSet_raw.RData")
```

```
Meth <- getMeth(MSet.raw)
str(Meth)
```

```
##   int [1:485512, 1:8] 338 319 318 294 315 579 220 316 184 192 ...
##   - attr(*, "dimnames")=List of 2
##     ..$ : chr [1:485512] "cg00050873" "cg00212031" "cg00213748"
"cg00214611" ...
```

```
##   ..$ : chr [1:8] "9344737127_R01C02" "9344737127_R03C02"
"9344737127_R06C01" "9344737127_R02C02" ...
```

```
head(Meth)
```

##	9344737127_R01C02	9344737127_R03C02	9344737127_R06C01
## cg00050873	338	397	362
## cg00212031	319	285	227
## cg00213748	318	356	279
## cg00214611	294	276	308
## cg00455876	315	306	417
## cg01707559	579	420	378
##	9344737127_R02C02	9376538140_R02C01	9376538140_R05C01
## cg00050873	399	374	271
## cg00212031	290	305	261
## cg00213748	272	260	288
## cg00214611	347	315	288
## cg00455876	300	348	369
## cg01707559	568	812	531
##	9376538140_R01C01	9376538140_R04C01	
## cg00050873	268	9478	
## cg00212031	272	465	
## cg00213748	324	1674	
## cg00214611	261	240	
## cg00455876	242	3722	
## cg01707559	440	594	

```
Unmeth <- getUnmeth(MSet.raw)
```

```
str(Unmeth)
```

```
##   int [1:485512, 1:8] 337 300 332 210 345 458 273 272 211 313 ...
##   - attr(*, "dimnames")=List of 2
##   ..$ : chr [1:485512] "cg00050873" "cg00212031" "cg00213748"
"cg00214611" ...
##   ..$ : chr [1:8] "9344737127_R01C02" "9344737127_R03C02"
"9344737127_R06C01" "9344737127_R02C02" ...
```

```
head(Unmeth)
```

##	9344737127_R01C02	9344737127_R03C02	9344737127_R06C01
## cg00050873	337	376	343
## cg00212031	300	299	339
## cg00213748	332	265	320
## cg00214611	210	275	328

```
## cg00455876          345          637          532
## cg01707559          458          467          643
##          9344737127_R02C02 9376538140_R02C01 9376538140_R05C01
## cg00050873          389          300          303
## cg00212031          296          347          261
## cg00213748          328          348          264
## cg00214611          261          319          247
## cg00455876          430          437          452
## cg01707559          510          795          741
##          9376538140_R01C01 9376538140_R04C01
## cg00050873          249          1001
## cg00212031          297          6506
## cg00213748          309          430
## cg00214611          169          5991
## cg00455876          572          1148
## cg01707559          637          8257
```

Let's check what happens to the probes that we considered before when we move from RGset to MethySet cg00035864, type II (the first probe in the Illumina450Manifest_clean object)

```
Red[rownames(Red)=="31729416",]
```

```
##          X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
## 31729416          236          234          267
##          X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## 31729416          327          299          326
##          X9376538140_R01C01 X9376538140_R04C01
## 31729416          284          5058
```

```
Green[rownames(Green)=="31729416",]
```

```
##          X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
## 31729416          208          215          213
##          X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## 31729416          204          229          245
##          X9376538140_R01C01 X9376538140_R04C01
## 31729416          182          1351
```

```
Unmeth[rownames(Unmeth)=="cg00035864",]
```

```
## 9344737127_R01C02 9344737127_R03C02 9344737127_R06C01
9344737127_R02C02
##          236          234          267
327
## 9376538140_R02C01 9376538140_R05C01 9376538140_R01C01
9376538140_R04C01
```

```
##                299                326                284
5058
```

```
Meth[rownames(Meth)=="cg00035864",]
```

```
## 9344737127_R01C02 9344737127_R03C02 9344737127_R06C01
9344737127_R02C02
```

```
##                208                215                213
204
```

```
## 9376538140_R02C01 9376538140_R05C01 9376538140_R01C01
9376538140_R04C01
```

```
##                229                245                182
1351
```

cg00050873, type I Red (the second probe in the Illumina450Manifest_clean object)

```
Illumina450Manifest_clean[Illumina450Manifest_clean$IlmnID=="cg00050873",]
```

```
##      IlmnID      Name AddressA_ID
```

```
## 2 cg00050873 cg00050873      32735311
```

```
##                                     AlleleA_ProbeSeq AddressB_ID
```

```
## 2 ACAAAAAACAACACACAACACTATAATAATTTTAAAAATAAATAAACCCCA      31717405
```

```
##                                     AlleleB_ProbeSeq
```

```
Infinium_Design_Type
```

```
## 2 ACGAAAAACAACGCACAACACTATAATAATTTTAAAAATAAATAAACCCCG
```

```
I
```

```
##      Next_Base Color_Channel
```

```
## 2      A      Red
```

```
##
```

```
Forward_Sequence
```

```
## 2
```

```
TATCTCTGTCTGGCGAGGAGGCAACGCACAACACTGTGGTGGTTTTTGGAGTGGGTGGACCC[CG]GCCAAGA
CGGCCTGGGCTGACCAGAGACGGGAGGCAGAAAAAGTGGGCAGGTGGTTGCAG
```

```
##      Genome_Build CHR MAPINFO
```

```
SourceSeq
```

```
## 2      37      Y 9363356
```

```
CGGGGTCCACCACTCCAAAAACCACACAGTTGTGCGTTGCCTCCTCGC
```

```
##      Chromosome_36 Coordinate_36 Strand Probe_SNPs Probe_SNPs_10
```

```
Random_Loci
```

```
## 2      Y      9973356      R
```

```
NA
```

```
##      Methyl27_Loci UCSC_RefGene_Name UCSC_RefGene_Accession
```

```
UCSC_RefGene_Group
```

```
## 2      NA      TSPY4;FAM197Y2 NM_001164471;NR_001553
```

```
Body;TSS1500
```

```
## UCSC_CpG_Islands_Name Relation_to_UCSC_CpG_Island Phantom DMR
Enhancer
## 2 chrY:9363680-9363943 N_Shore
NA
## HMM_Island Regulatory_Feature_Name Regulatory_Feature_Group
DHS
## 2 Y:9973136-9976273
NA
```

```
Red[rownames(Red)=="32735311",]
```

```
## X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
## 32735311 337 376 343
## X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## 32735311 389 300 303
## X9376538140_R01C01 X9376538140_R04C01
## 32735311 249 1001
```

```
Red[rownames(Red)=="31717405",]
```

```
## X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
## 31717405 338 397 362
## X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## 31717405 399 374 271
## X9376538140_R01C01 X9376538140_R04C01
## 31717405 268 9478
```

But the two addresses are present also when I look at the Green object: these are out of band signals!

```
Green[rownames(Green)=="32735311",]
```

```
## X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
## 32735311 196 195 230
## X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## 32735311 179 188 190
## X9376538140_R01C01 X9376538140_R04C01
## 32735311 195 332
```

```
Green[rownames(Green)=="31717405",]
```

```
## X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
## 31717405 228 268 201
## X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## 31717405 225 221 260
## X9376538140_R01C01 X9376538140_R04C01
```



```
## 31717405          172          930
```

```
Unmeth[rownames(Unmeth)=="cg00050873",]
```

```
## 9344737127_R01C02 9344737127_R03C02 9344737127_R06C01  
9344737127_R02C02
```

```
##          337          376          343  
389
```

```
## 9376538140_R02C01 9376538140_R05C01 9376538140_R01C01  
9376538140_R04C01
```

```
##          300          303          249  
1001
```

```
Meth[rownames(Meth)=="cg00050873",]
```

```
## 9344737127_R01C02 9344737127_R03C02 9344737127_R06C01  
9344737127_R02C02
```

```
##          338          397          362  
399
```

```
## 9376538140_R02C01 9376538140_R05C01 9376538140_R01C01  
9376538140_R04C01
```

```
##          374          271          268  
9478
```

CHUNK 5: QC plot and control probes

First of all, we consider the median of methylation and unmethylation channels for each sample, using the function `getQC`

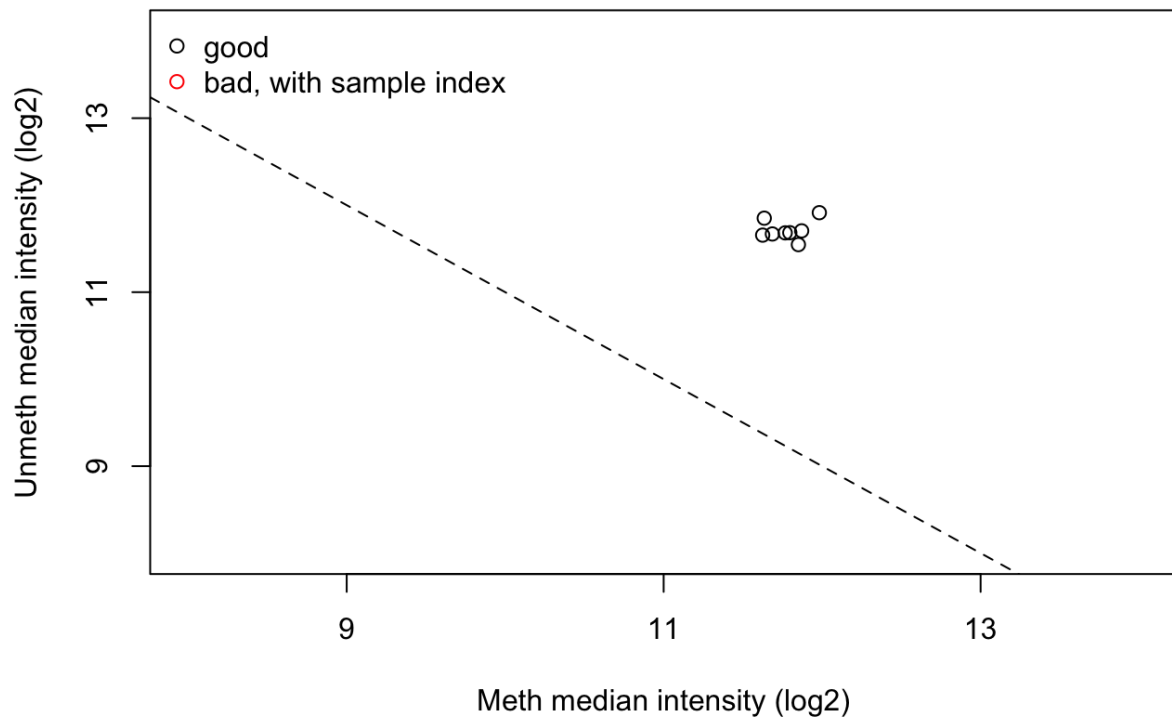
```
qc <- getQC(MSet.raw)
```

```
qc
```

```
## DataFrame with 8 rows and 2 columns
```

```
##          mMed      uMed  
##          <numeric> <numeric>  
## 9344737127_R01C02  11.6861  11.6689  
## 9344737127_R03C02  11.7960  11.6834  
## 9344737127_R06C01  11.8502  11.5464  
## 9344737127_R02C02  11.7669  11.6817  
## 9376538140_R02C01  11.6348  11.8504  
## 9376538140_R05C01  11.9823  11.9136  
## 9376538140_R01C01  11.6248  11.6555  
## 9376538140_R04C01  11.8708  11.7048
```

```
plotQC(qc)
```



All the samples have high median methylation and unmethylation signals, i.e. they have good quality!

Then, we check the control probes. The Illumina guide ([GenomeStudio_control_probes.pdf](#)) tells us the expected intensities for each type of control probes.

To know the names and the numbers of the different types of control probes, we apply the `getProbeInfo` function to the `RGset` object.

```
getProbeInfo(RGset, type = "Control")
```

```
## DataFrame with 848 rows and 4 columns
```

##	Address	Type	Color	ExtendedType
##	<character>	<character>	<character>	<character>
## 1	27630314	STAINING	Red	DNP (High)
## 2	43603326	STAINING	Purple	DNP (Bkg)
## 3	41666334	STAINING	Green	Biotin (High)
## 4	34648333	STAINING	Blue	Biotin (Bkg)
## 5	63642461	EXTENSION	Red	Extension (A)
## ...	...	...	...	...

```
## 844      65622344      NORM_T      Purple      Norm_T59
## 845      57640346      NORM_C      Green      Norm_C60
## 846      47715450      NORM_T      Purple      Norm_T60
## 847      28673402      NORM_C      Green      Norm_C61
## 848      13742412      NORM_T      Purple      Norm_T61
```

```
df_TypeControl <- data.frame(getProbeInfo(RGset, type = "Control"))
str(df_TypeControl)
```

```
## 'data.frame':      848 obs. of  4 variables:
## $ Address      : chr  "27630314" "43603326" "41666334" "34648333"
## ...
## $ Type         : chr  "STAINING" "STAINING" "STAINING" "STAINING"
## ...
## $ Color        : chr  "Red" "Purple" "Green" "Blue" ...
## $ ExtendedType: chr  "DNP (High)" "DNP (Bkg)" "Biotin (High)"
## "Biotin (Bkg)" ...
```

```
table(df_TypeControl$Type)
```

```
##
## BISULFITE CONVERSION I BISULFITE CONVERSION II
EXTENSION
##              12              4
4
##          HYBRIDIZATION          NEGATIVE          NON-
POLYMORPHIC
##              3              613
4
##              NORM_A              NORM_C
NORM_G
##              32              61
32
##              NORM_T              RESTORATION
SPECIFICITY I
##              61              1
12
##          SPECIFICITY II          STAINING          TARGET
REMOVAL
##              3              4
2
```

Then, we can use the controlSripPlot function to plot the intensity values of each type of controls probes in our samples

```
controlStripPlot(RGset, controls="NEGATIVE")
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be  
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be  
specified by
```

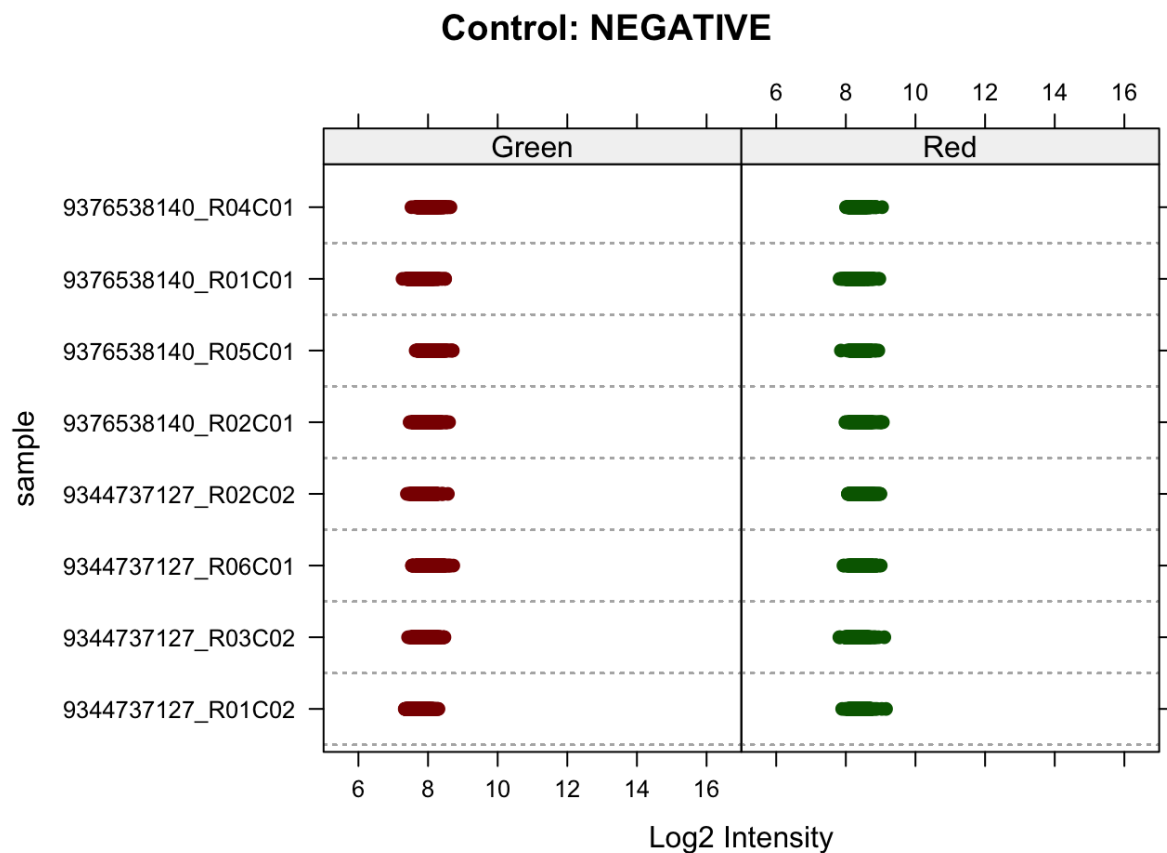
```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be  
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be  
specified by
```

```
## the caller; using TRUE
```



Negative controls are all fine, as all below 1000 ($\log_2(1000)=10$) Note that there is a minor error in the package...green and red labels are swapped (only in R4.0.0 or higher). You can compare the results of these controls with the plots in Lesson 3 slides...everything looks fine!

```
controlStripPlot(RGset, controls="EXTENSION")
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

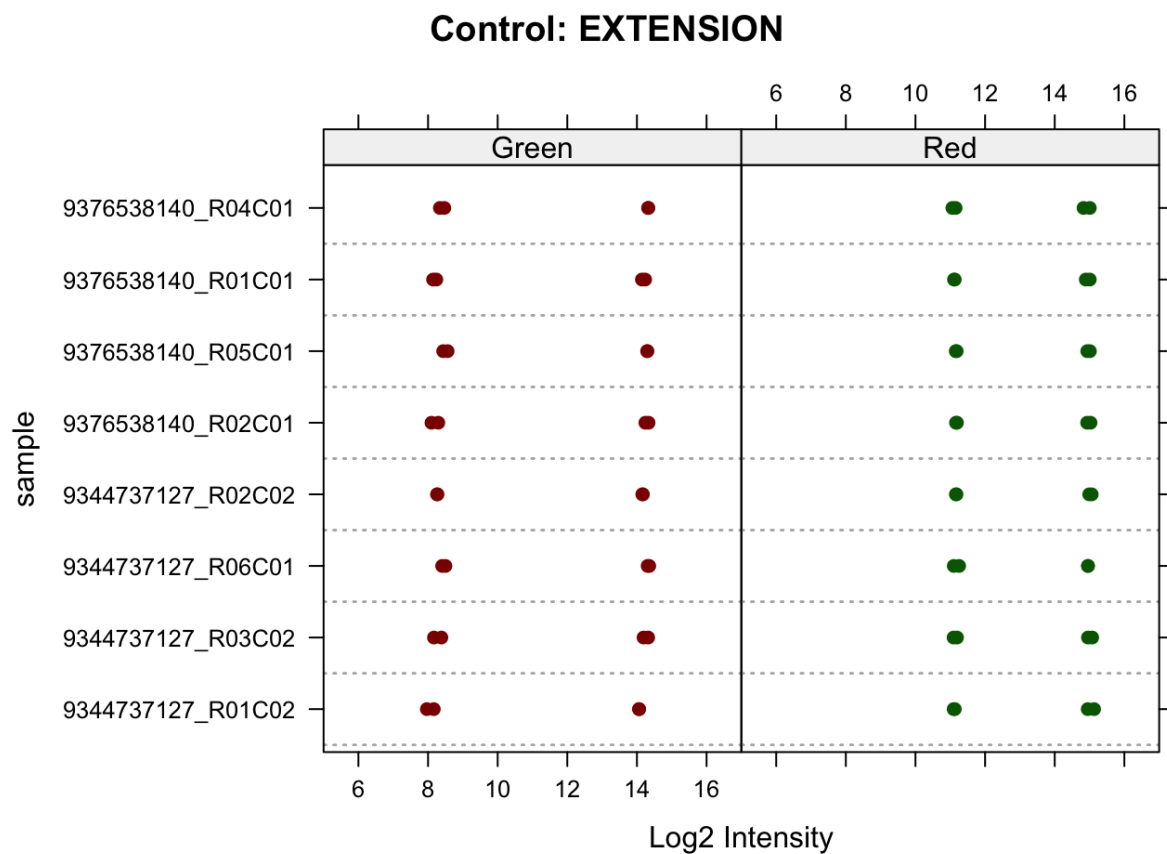
```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```



```
controlStripPlot(RGset, controls="STAINING")
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

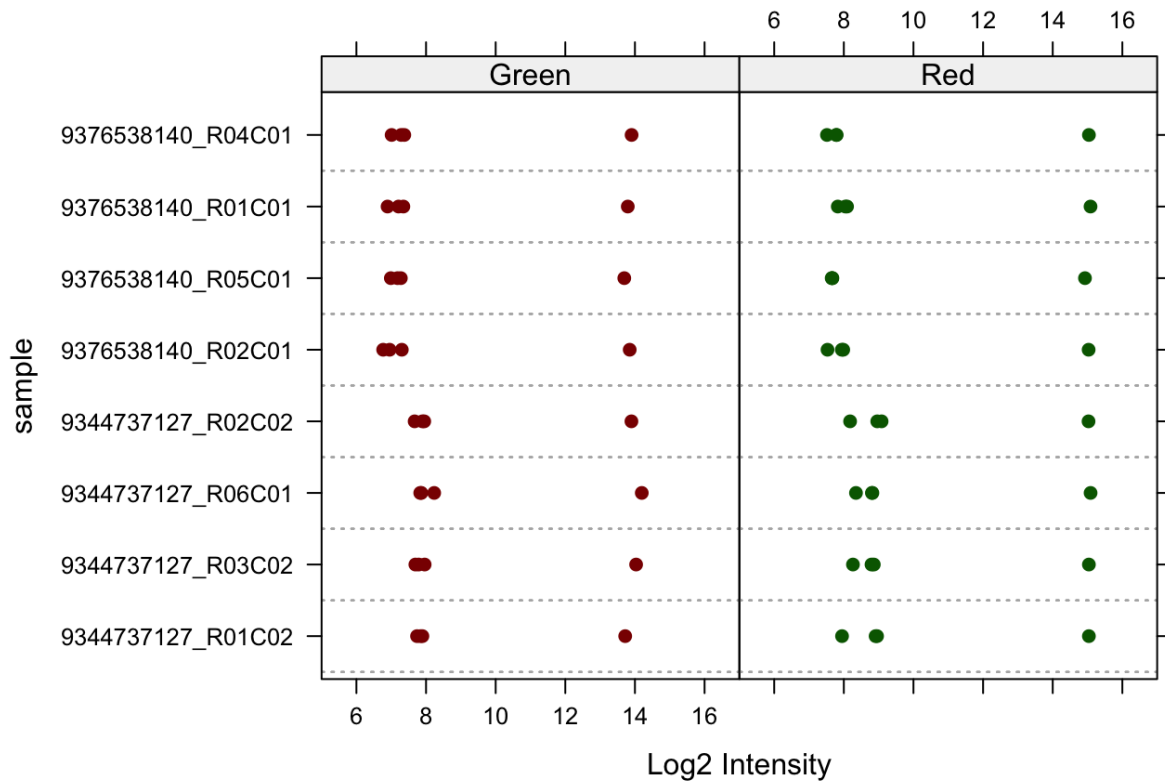
```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
## the caller; using TRUE
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
## the caller; using TRUE
```

Control: STAINING



```
controlStripPlot(RGset, controls="BISULFITE CONVERSION I")
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

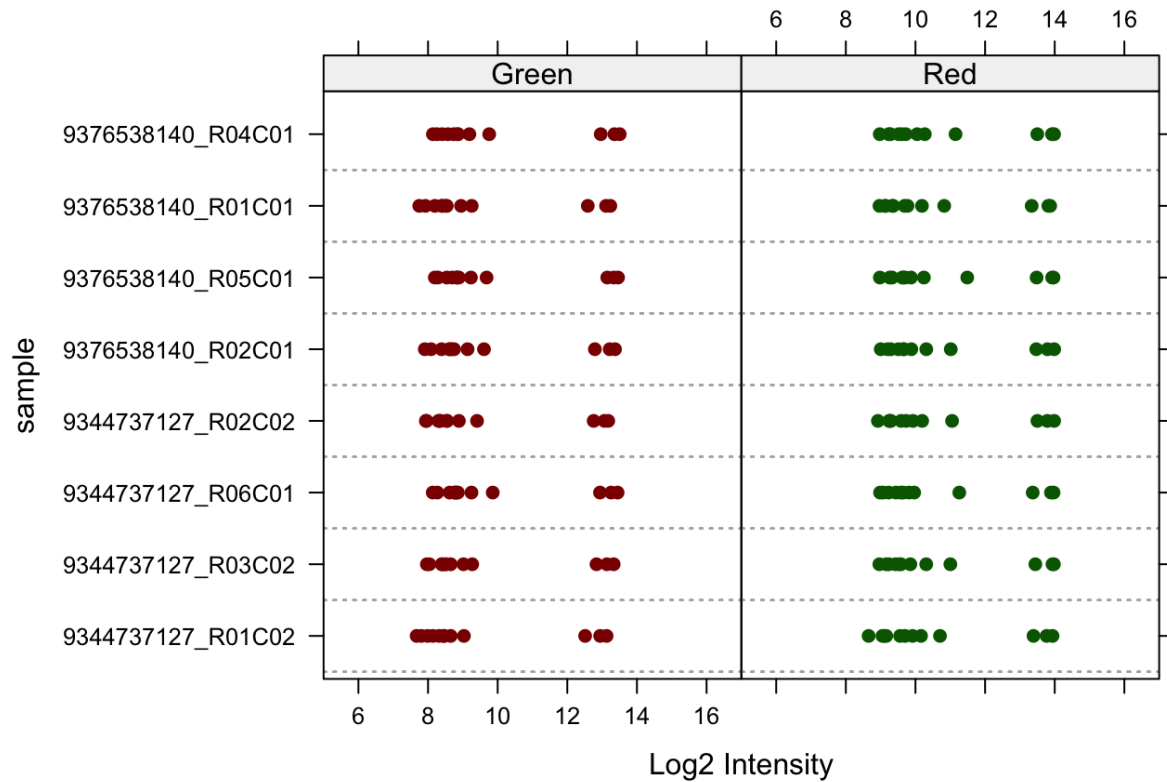
```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

Control: BISULFITE CONVERSION I



```
controlStripPlot(RGset, controls="BISULFITE CONVERSION II")
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

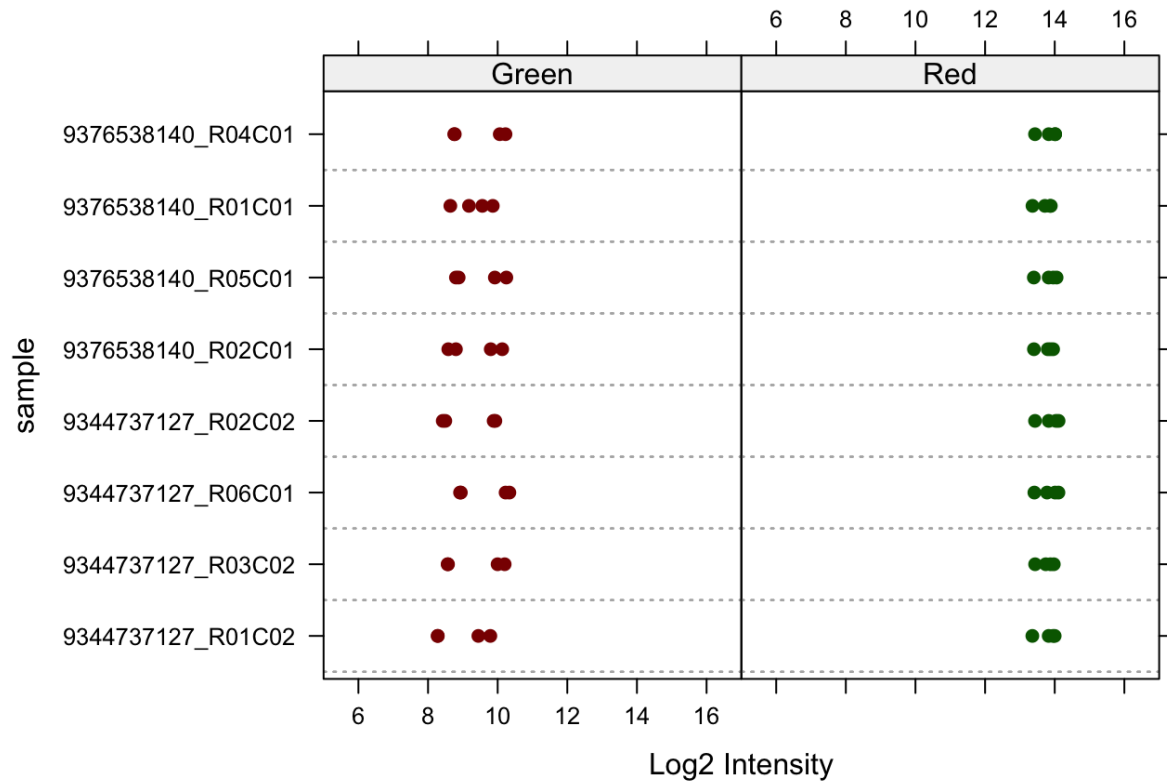
```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

Control: BISULFITE CONVERSION II



```
controlStripPlot(RGset, controls="SPECIFICITY I")
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

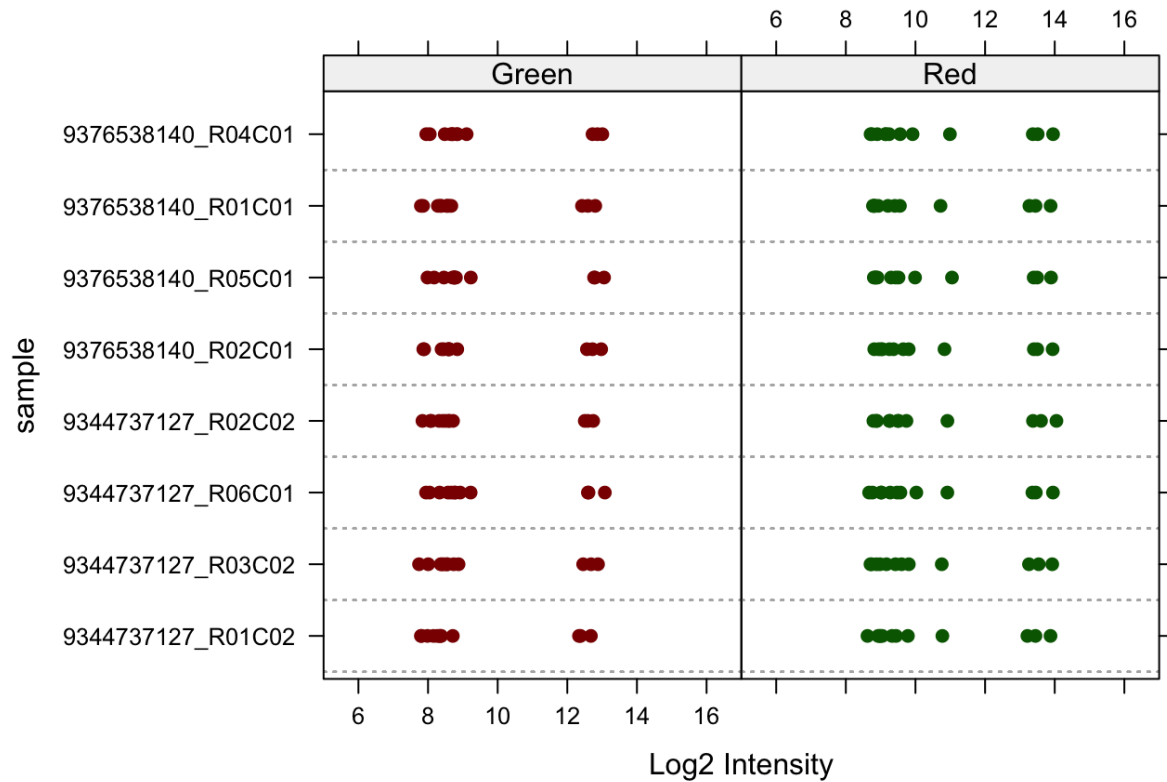
```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```


Control: SPECIFICITY I



```
controlStripPlot(RGset, controls="SPECIFICITY II")
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

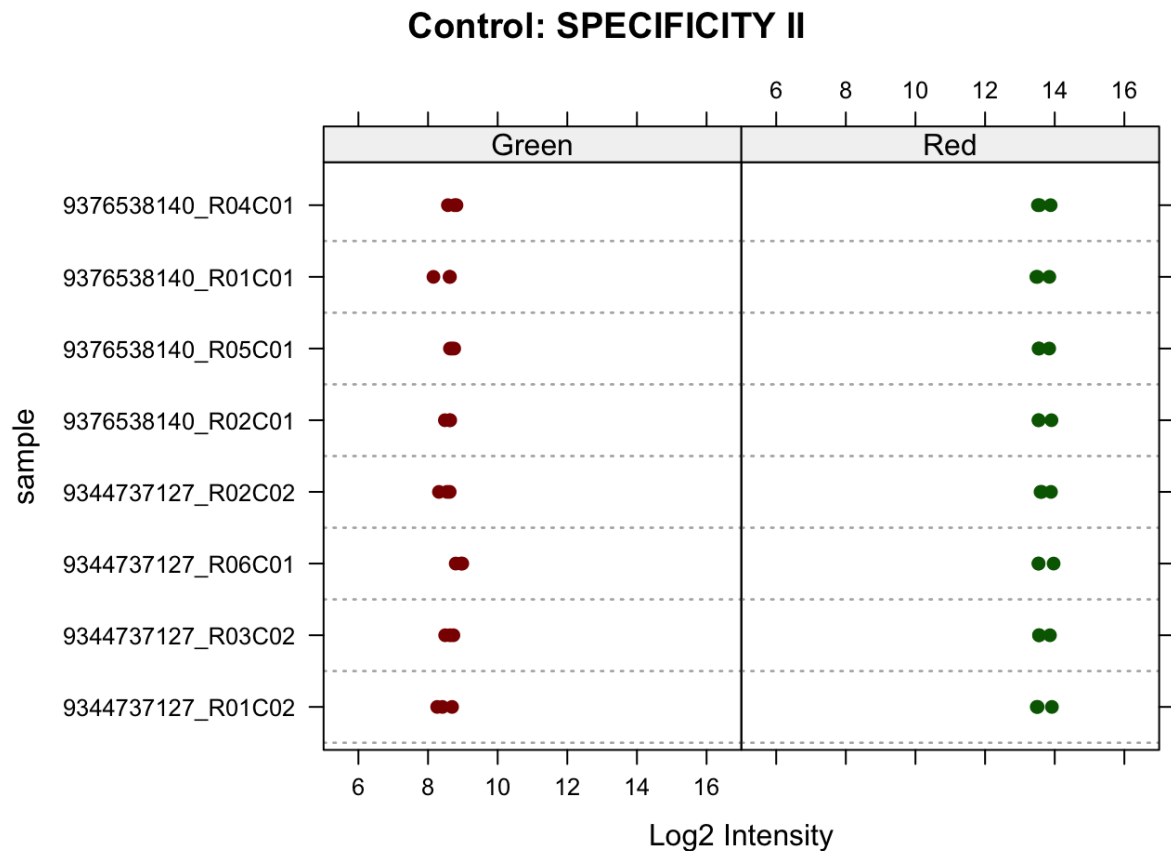
```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```

```
## Warning in type.convert.default(X[[i]], ...): 'as.is' should be
specified by
```

```
## the caller; using TRUE
```



CHUNK 6: Detection p-value and filtering

```
?detectionP
```

The detectionP() functions takes a bit. You can try to jump forward by directly loading the provided detP.RData object

```
detP <- detectionP(RGset)
save(detP,file="detP.RData")
load("detP.RData") # or detP <- readRDS("detP.rds")
str(detP)
```

```
##  num [1:485512, 1:8] 0.402 0.716 0.547 0.988 0.488 ...
##  - attr(*, "dimnames")=List of 2
##    ..$ : chr [1:485512] "cg00050873" "cg00212031" "cg00213748"
##        "cg00214611" ...
```

```
##    ..$ : chr [1:8] "9344737127_R01C02" "9344737127_R03C02"
"9344737127_R06C01" "9344737127_R02C02" ...
```

dim(detP)				
##	[1]	485512	8	
head(detP)				
##		9344737127_R01C02	9344737127_R03C02	9344737127_R06C01
##	cg00050873	4.015766e-01	2.073492e-02	4.330467e-01
##	cg00212031	7.162885e-01	8.046760e-01	9.489066e-01
##	cg00213748	5.466899e-01	6.145728e-01	8.861492e-01
##	cg00214611	9.880296e-01	9.137651e-01	7.661954e-01
##	cg00455876	4.883024e-01	1.702124e-06	4.287404e-04
##	cg01707559	1.370547e-08	7.643943e-05	9.885952e-06
##		9344737127_R02C02	9376538140_R02C01	9376538140_R05C01
##	cg00050873	5.664387e-02	5.609165e-01	9.507839e-01
##	cg00212031	9.159458e-01	6.881236e-01	9.942861e-01
##	cg00213748	8.796076e-01	8.779965e-01	9.784878e-01
##	cg00214611	8.544530e-01	7.783007e-01	9.895595e-01
##	cg00455876	2.317359e-01	6.077951e-02	5.994328e-03
##	cg01707559	2.676472e-09	9.474742e-46	2.313609e-24
##		9376538140_R01C01	9376538140_R04C01	
##	cg00050873	9.599809e-01	0.000000e+00	
##	cg00212031	8.200040e-01	0.000000e+00	
##	cg00213748	4.552471e-01	9.05742e-149	
##	cg00214611	9.991770e-01	0.000000e+00	
##	cg00455876	1.267429e-03	0.000000e+00	
##	cg01707559	2.198449e-13	0.000000e+00	

Note that in the detP object, rownames are the CpG probes names

Now we want to consider a detectionP threshold of 0.05

```
failed <- detP>0.05
```

```
head(failed)
```

##	9344737127_R01C02 9344737127_R03C02 9344737127_R06C01			
## cg00050873	TRUE	FALSE	TRUE	
## cg00212031	TRUE	TRUE	TRUE	
## cg00213748	TRUE	TRUE	TRUE	
## cg00214611	TRUE	TRUE	TRUE	
## cg00455876	TRUE	FALSE	FALSE	
## cg01707559	FALSE	FALSE	FALSE	
##	9344737127_R02C02	9376538140_R02C01	9376538140_R05C01	

```
## cg00050873      TRUE      TRUE      TRUE
## cg00212031      TRUE      TRUE      TRUE
## cg00213748      TRUE      TRUE      TRUE
## cg00214611      TRUE      TRUE      TRUE
## cg00455876      TRUE      TRUE     FALSE
## cg01707559     FALSE     FALSE     FALSE
##          9376538140_R01C01 9376538140_R04C01
## cg00050873      TRUE      FALSE
## cg00212031      TRUE      FALSE
## cg00213748      TRUE      FALSE
## cg00214611      TRUE      FALSE
## cg00455876     FALSE     FALSE
## cg01707559     FALSE     FALSE
```

```
dim(failed)
```

```
## [1] 485512      8
```

```
table(failed)
```

```
## failed
```

```
## FALSE TRUE
```

```
## 3880745 3351
```

`summary(failed)` is particularly useful, as it returns the number of failed (TRUE) and not failed (FALSE) positions for each sample

```
summary(failed)
```

```
## 9344737127_R01C02 9344737127_R03C02 9344737127_R06C01
9344737127_R02C02
```

```
## Mode :logical      Mode :logical      Mode :logical      Mode :logical
```

```
## FALSE:485013      FALSE:485084      FALSE:485032      FALSE:485049
```

```
## TRUE :499         TRUE :428        TRUE :480         TRUE :463
```

```
## 9376538140_R02C01 9376538140_R05C01 9376538140_R01C01
```

```
9376538140_R04C01
```

```
## Mode :logical      Mode :logical      Mode :logical      Mode :logical
```

```
## FALSE:485071      FALSE:485092      FALSE:484989      FALSE:485415
```

```
## TRUE :441         TRUE :420        TRUE :523         TRUE :97
```

Unfortunately, `minfi` does not provide any function that performs filtering of samples/probes on the basis of detection p-value! Other packages, like `watermelon`, allow to perform this action using convenient functions.

In any case, we can manually operate on the `detP` object to filter out failed samples and probes

First of all, we want to calculate the percentage of failed positions in each sample, and filter out those samples that have a % of failed positions higher than a certain threshold, for example 5%.

To calculate the fraction of failed postions per sample, we can use the function `colMeans()` and apply it to the “failed” object that we have previously created

```
?colMeans
```

```
## Help on topic 'colMeans' was found in the following packages:
```

```
##
##   Package              Library
##   DelayedArray
##   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library
##   Matrix
##   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library
##   base
##   /Library/Frameworks/R.framework/Resources/library
##   MatrixGenerics
##   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library
##   SparseArray
##   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library
##
##
## Using the first match ...
```

```
means_of_columns <- colMeans(failed)
```

```
head(means_of_columns)
```

```
## 9344737127_R01C02 9344737127_R03C02 9344737127_R06C01
9344737127_R02C02
##      0.0010277810      0.0008815436      0.0009886470
0.0009536325
## 9376538140_R02C01 9376538140_R05C01
##      0.0009083195      0.0008650662
```

Each value corresponds to the ratio (number of TRUE)/(number of TRUE+FALSE) for each column (that is, for each sample)

For example, for sample 9344737127_R01C02 we had 499 TRUE and 485013 FALSE → $499/(499+485013)= 0.001027781$

We can decide, for example, to retain only the samples in which <5 % of sites has a bad detection p-value (greater than 0.05, the threshold that we used to calculate the object “failed”)

```
samples_to_be_retained <- means_of_columns<0.05
```

```
samples_to_be_retained
```

```
## 9344737127_R01C02 9344737127_R03C02 9344737127_R06C01
9344737127_R02C02
```

```
##          TRUE          TRUE          TRUE
TRUE
## 9376538140_R02C01 9376538140_R05C01 9376538140_R01C01
9376538140_R04C01
##          TRUE          TRUE          TRUE
TRUE
```

Then I subset the RGset object in order to retain only the selected samples:

```
RGset <- RGset[samples_to_be_retained]
```

RGset

```
## class: RGChannelSet
## dim: 622399 8
## metadata(0):
## assays(2): Green Red
## rownames(622399): 10600313 10600322 ... 74810490 74810492
## rowData names(0):
## colnames(8): 9344737127_R01C02 9344737127_R03C02 ...
9376538140_R01C01
## 9376538140_R04C01
## colData names(9): Sample_ID Sample_Well ... Basename filenames
## Annotation
## array: IlluminaHumanMethylation450k
## annotation: ilmn12.hg19
```

In this example, all the samples have < 5% of probes with a bad detection pValue, and therefore I can retain all the samples

The next step is to identify the probes that have a bad detectionP. To calculate the fraction of failed samples per probes, we can use the function rowMeans() and apply it to the “failed” object that we have previously created

```
means_of_rows <- rowMeans(failed)
```

head(means_of_rows)

```
## cg00050873 cg00212031 cg00213748 cg00214611 cg00455876 cg01707559
##          0.750          0.875          0.875          0.875          0.375          0.000
```

Each value corresponds to the ratio (number of TRUE)/(number of TRUE+FALSE) for each row (that is, for each probe)

head(failed)				
##	9344737127_R01C02	9344737127_R03C02	9344737127_R06C01	
## cg00050873	TRUE	FALSE	TRUE	
## cg00212031	TRUE	TRUE	TRUE	
## cg00213748	TRUE	TRUE	TRUE	
## cg00214611	TRUE	TRUE	TRUE	
## cg00455876	TRUE	FALSE	FALSE	
## cg01707559	FALSE	FALSE	FALSE	
##	9344737127_R02C02	9376538140_R02C01	9376538140_R05C01	
## cg00050873	TRUE	TRUE	TRUE	
## cg00212031	TRUE	TRUE	TRUE	
## cg00213748	TRUE	TRUE	TRUE	
## cg00214611	TRUE	TRUE	TRUE	
## cg00455876	TRUE	TRUE	FALSE	
## cg01707559	FALSE	FALSE	FALSE	
##	9376538140_R01C01	9376538140_R04C01		
## cg00050873	TRUE	FALSE		
## cg00212031	TRUE	FALSE		
## cg00213748	TRUE	FALSE		
## cg00214611	TRUE	FALSE		
## cg00455876	FALSE	FALSE		
## cg01707559	FALSE	FALSE		

For example, for the probe cg00050873 we had 6 TRUE and 2 FALSE → $6/(6+2) = 0.750$

We can decide, for example, to retain only the probes in which <1 % of samples has a bad detection p-value (greater than 0.05, the threshold that we used to calculate the object “failed”)

probes_to_be_retained <- means_of_rows<0.01						
head(probes_to_be_retained)						
##	cg00050873	cg00212031	cg00213748	cg00214611	cg00455876	cg01707559
##	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE
str(probes_to_be_retained)						
##	Named logi [1:485512] FALSE FALSE FALSE FALSE FALSE TRUE ...					
##	- attr(*, "names")= chr [1:485512] "cg00050873" "cg00212031" "cg00213748" "cg00214611" ...					

Note that the “probes_to_be_retained” object is a vector of logical vectors with names (that is, each element of the vector has a name). We can access to these names using the function `names()`

```
names(probes_to_be_retained[1:10])

## [1] "cg00050873" "cg00212031" "cg00213748" "cg00214611"
##      "cg00455876"

## [6] "cg01707559" "cg02004872" "cg02011394" "cg02050847"
##      "cg02233190"
```

returns the names of the first 10 probes

```
table(probes_to_be_retained)

## probes_to_be_retained

## FALSE  TRUE
##   1121 484391
```

counts the number of TRUE and FALSE in the vector.

Now I want to select the probes that I want to remove: these are the probes with “FALSE” in the probes_to_be_retained vector

```
probes_to_be_removed <-
probes_to_be_retained[probes_to_be_retained==FALSE]

length(probes_to_be_removed)
```

```
## [1] 1121

names_probes_to_be_removed <- names(probes_to_be_removed)

head(names_probes_to_be_removed)
```

```
## [1] "cg00050873" "cg00212031" "cg00213748" "cg00214611" "cg00455876"
## [6] "cg02004872"
```

Unfortunately, as we will see in the next lesson, most of the normalization procedures implemented in minfi work on the RGset object and do not like missing probes. If the number of bad probes is low, they will not affect the normalization procedures. Therefore, we can retain the bad probes at this step, but save them in order to remove them after the normalization step.

```
save(names_probes_to_be_removed, file="names_probes_to_be_removed.RData"
)
```

Exercise 2: load the RGset_CRC.RData or RGset_CRC.rds object (this is from an IlluminaHumanMethylationEPIC experiment and you will need to install the IlluminaHumanMethylationEPICmanifest from Bioconductor). How many samples does it

contain? Use plotQC to check the quality of the samples and check NEGATIVE and EXTENSION controls...do you see anything wrong with these data?

DRD_2025_Lesson_4

```
rm(list=ls())
```

```
setwd("~/Dropbox/DRD_2025/4")  
suppressMessages(library(minfi))
```

```
## Warning: package 'GenomicRanges' was built under R version 4.4.1  
## Warning: package 'S4Vectors' was built under R version 4.4.1  
## Warning: package 'IRanges' was built under R version 4.4.1  
## Warning: package 'matrixStats' was built under R version 4.4.1  
## Warning: package 'locfit' was built under R version 4.4.1
```

CHUNK 1: Beta and M values

We load the MSet object, which contains the methylation and unmethylation signals

```
load("~/Dropbox/DRD_2025/3/MSet_raw.RData")
```

```
ls()
```

```
## [1] "MSet.raw"
```

```
MSet.raw
```

```
## class: MethySet
```

```
## dim: 485512 8  
## metadata(0):  
## assays(2): Meth Unmeth  
## rownames(485512): cg00050873 cg00212031 ... ch.22.47579720R  
## ch.22.48274842R  
## rowData names(0):  
## colnames(8): 9344737127_R01C02 9344737127_R03C02 ...  
9376538140_R01C01  
## 9376538140_R04C01  
## colData names(9): Sample_ID Sample_Well ... Basename filenames  
## Annotation  
## array: IlluminaHumanMethylation450k  
## annotation: ilmn12.hg19  
## Preprocessing  
## Method: Raw (no normalization or bg correction)
```

```
## minfi version: 1.50.0
## Manifest version: 0.4.0
```

Remember that it is a `MethylSet` class object. If we want to know the accessor functions for this class of objects, we can go to the help page:

```
?MethylSet
```

The functions `getBeta` and `getM` allow to retrieve beta and M values matrices:

```
beta <- getBeta(MSet.raw)
```

```
class(beta)
```

```
## [1] "matrix" "array"
```

```
dim(beta)
```

```
## [1] 485512      8
```

```
head(beta)
```

```
##           9344737127_R01C02 9344737127_R03C02 9344737127_R06C01
## cg00050873      0.5007407      0.5135834      0.5134752
## cg00212031      0.5153473      0.4880137      0.4010601
## cg00213748      0.4892308      0.5732689      0.4657763
## cg00214611      0.5833333      0.5009074      0.4842767
## cg00455876      0.4772727      0.3244963      0.4394099
## cg01707559      0.5583414      0.4735062      0.3702253
##           9344737127_R02C02 9376538140_R02C01 9376538140_R05C01
## cg00050873      0.5063452      0.5548961      0.4721254
## cg00212031      0.4948805      0.4677914      0.5000000
## cg00213748      0.4533333      0.4276316      0.5217391
## cg00214611      0.5707237      0.4968454      0.5383178
## cg00455876      0.4109589      0.4433121      0.4494519
## cg01707559      0.5269017      0.5052894      0.4174528
##           9376538140_R01C01 9376538140_R04C01
## cg00050873      0.5183752      0.90447562
## cg00212031      0.4780316      0.06670492
## cg00213748      0.5118483      0.79562738
## cg00214611      0.6069767      0.03851709
## cg00455876      0.2972973      0.76427105
## cg01707559      0.4085422      0.06711106
```

```
summary(beta)
```

```
## 9344737127_R01C02 9344737127_R03C02 9344737127_R06C01
9344737127_R02C02
## Min. :0.0000 Min. :0.0000 Min. :0.0000 Min.
:0.0000
## 1st Qu.:0.1234 1st Qu.:0.1003 1st Qu.:0.1028 1st
Qu.:0.1052
## Median :0.5496 Median :0.5916 Median :0.6426 Median
:0.5848
## Mean :0.4879 Mean :0.4952 Mean :0.5116 Mean
:0.4928
## 3rd Qu.:0.7949 3rd Qu.:0.8178 3rd Qu.:0.8408 3rd
Qu.:0.8082
## Max. :1.0000 Max. :1.0000 Max. :1.0000 Max.
:1.0000
## NA's :13 NA's :18 NA's :14 NA's :14
## 9376538140_R02C01 9376538140_R05C01 9376538140_R01C01
9376538140_R04C01
## Min. :0.01148 Min. :0.0000 Min. :0.009437 Min.
:0.00000
## 1st Qu.:0.10393 1st Qu.:0.1254 1st Qu.:0.093736 1st
Qu.:0.09656
## Median :0.49236 Median :0.5564 Median :0.572396 Median
:0.61699
## Mean :0.47257 Mean :0.4954 Mean :0.485536 Mean
:0.49915
## 3rd Qu.:0.81101 3rd Qu.:0.8127 3rd Qu.:0.807982 3rd
Qu.:0.82909
## Max. :0.98989 Max. :0.9915 Max. :1.000000 Max.
:0.99289
## NA's :1 NA's :2 NA's :3 NA's :3
```

In our dataset, minimum for beta values is 0, maximum is 1. We have also some NAs: these are the positions for which both the Methylation and Unmethylation values in the MSet.raw are equal to 0 (by default, in minfi the offset is set to 0)

```
M <- getM(MSet.raw)
```

```
dim(M)
```

```
## [1] 485512      8
```

```
head(M)
```

```
##          9344737127_R01C02 9344737127_R03C02 9344737127_R06C01
## cg00050873      0.004274655      0.078406345      0.07778112
## cg00212031      0.088593923     -0.069183565     -0.57859298
```

```
## cg00213748      -0.062156476      0.425884882      -0.19780678
## cg00214611      0.485426827      0.005236648      -0.09076546
## cg00455876     -0.131244533     -1.057761720     -0.35137886
## cg01707559      0.338215750     -0.153033222     -0.76643250
##                9344737127_R02C02 9376538140_R02C01 9376538140_R05C01
## cg00050873      0.03661859      0.31807577      -0.1610249
## cg00212031     -0.02954428     -0.18612642      0.0000000
## cg00213748     -0.27008916     -0.42057568      0.1255309
## cg00214611      0.41088586     -0.01820460      0.2215578
## cg00455876     -0.51937416     -0.32854597     -0.2927020
## cg01707559      0.15539368      0.03052487     -0.4807617
##                9376538140_R01C01 9376538140_R04C01
## cg00050873      0.10608726      3.243141
## cg00212031     -0.12685628     -3.806468
## cg00213748      0.06838697      1.960891
## cg00214611      0.62702656     -4.641691
## cg00455876     -1.24100810      1.696955
## cg01707559     -0.53378985     -3.797083
```

summary(M)

```
## 9344737127_R01C02 9344737127_R03C02 9344737127_R06C01
9344737127_R02C02
```

```
## Min.      : -Inf   Min.      : -Inf   Min.      : -Inf   Min.      : -
Inf
## 1st Qu.: -2.8282   1st Qu.: -3.1659   1st Qu.: -3.1261   1st Qu.: -
3.088
## Median : 0.2873    Median : 0.5349    Median : 0.8462    Median :
0.494
## Mean    :      NaN   Mean    :      NaN   Mean    :      NaN   Mean    :
NaN
## 3rd Qu.: 1.9545    3rd Qu.: 2.1662    3rd Qu.: 2.4012    3rd Qu.:
2.075
## Max.      :      Inf   Max.      :      Inf   Max.      :      Inf   Max.      :
Inf
## NA's      :13       NA's      :18       NA's      :14       NA's      :14
## 9376538140_R02C01 9376538140_R05C01 9376538140_R01C01
9376538140_R04C01
## Min.      : -6.4279   Min.      : -Inf   Min.      : -6.7137   Min.      : -
Inf
## 1st Qu.: -3.1080    1st Qu.: -2.8022    1st Qu.: -3.2733    1st Qu.: -
3.2259
## Median : -0.0441    Median : 0.3269    Median : 0.4207    Median :
0.6879
## Mean      : -0.3761   Mean      : -Inf   Mean      :      Inf   Mean      : -
Inf
```

```
## 3rd Qu.: 2.1014    3rd Qu.: 2.1171    3rd Qu.: 2.0731    3rd Qu.:
2.2783
## Max.      : 6.6138    Max.      : 6.8600    Max.      :    Inf    Max.      :
7.1254
## NA's      :1          NA's      :2          NA's      :3          NA's      :3
```

In our dataset, minimum for M values is -Inf (Methylation value=0, Unmethylation value>0), maximum is +Inf (Methylation value>0, Unmethylation value=0). We have also some NAs: these are the positions for which both the Methylation and Unmethylation values in the MSet.raw are equal to 0 (by default, in minfi the offset is set to 0)

Note! The output of getBeta and getM functions is a matrix. We can also convert the output of these functions to a dataframe (as we have done with the output of getRed and getGreen functions in the last lesson):

```
beta_df <- data.frame(getBeta(MSet.raw))
```

```
class(beta_df)
```

```
## [1] "data.frame"
```

Do you note some differences between beta and beta_df?

```
dim(beta)
```

```
## [1] 485512      8
```

```
dim(beta_df)
```

```
## [1] 485512      8
```

```
head(beta)
```

```
##           9344737127_R01C02 9344737127_R03C02 9344737127_R06C01
```

```
## cg00050873      0.5007407      0.5135834      0.5134752
```

```
## cg00212031      0.5153473      0.4880137      0.4010601
```

```
## cg00213748      0.4892308      0.5732689      0.4657763
```

```
## cg00214611      0.5833333      0.5009074      0.4842767
```

```
## cg00455876      0.4772727      0.3244963      0.4394099
```

```
## cg01707559      0.5583414      0.4735062      0.3702253
```

```
##           9344737127_R02C02 9376538140_R02C01 9376538140_R05C01
```

```
## cg00050873      0.5063452      0.5548961      0.4721254
```

```
## cg00212031      0.4948805      0.4677914      0.5000000
```

```
## cg00213748      0.4533333      0.4276316      0.5217391
```

```
## cg00214611      0.5707237      0.4968454      0.5383178
```

```
## cg00455876      0.4109589      0.4433121      0.4494519
```

```
## cg01707559      0.5269017      0.5052894      0.4174528
```

```
##           9376538140_R01C01 9376538140_R04C01
```

```
## cg00050873      0.5183752      0.90447562
## cg00212031      0.4780316      0.06670492
## cg00213748      0.5118483      0.79562738
## cg00214611      0.6069767      0.03851709
## cg00455876      0.2972973      0.76427105
## cg01707559      0.4085422      0.06711106
```

head(beta_df)			
##	X9344737127_R01C02	X9344737127_R03C02	X9344737127_R06C01
## cg00050873	0.5007407	0.5135834	0.5134752
## cg00212031	0.5153473	0.4880137	0.4010601
## cg00213748	0.4892308	0.5732689	0.4657763
## cg00214611	0.5833333	0.5009074	0.4842767
## cg00455876	0.4772727	0.3244963	0.4394099
## cg01707559	0.5583414	0.4735062	0.3702253
##	X9344737127_R02C02	X9376538140_R02C01	X9376538140_R05C01
## cg00050873	0.5063452	0.5548961	0.4721254
## cg00212031	0.4948805	0.4677914	0.5000000
## cg00213748	0.4533333	0.4276316	0.5217391
## cg00214611	0.5707237	0.4968454	0.5383178
## cg00455876	0.4109589	0.4433121	0.4494519
## cg01707559	0.5269017	0.5052894	0.4174528
##	X9376538140_R01C01	X9376538140_R04C01	
## cg00050873	0.5183752	0.90447562	
## cg00212031	0.4780316	0.06670492	
## cg00213748	0.5118483	0.79562738	
## cg00214611	0.6069767	0.03851709	
## cg00455876	0.2972973	0.76427105	
## cg01707559	0.4085422	0.06711106	

Now we want to plot the density distribution of mean (across the 8 samples) beta and M values for each CpG

First of all, we have to calculate the mean of each row of the matrix across the 8 samples.

To calculate the mean, we will use the `mean()` function. As we know that in our input there are some missing values, we will specify that NA values should be stripped (see this page <https://www.statmethods.net/input/missingdata.html>) using the `na.rm=T` argument.

To calculate the mean of each row of the matrix, we will use the `apply()` function, which takes as input a dataframe or matrix and returns a vector (see here for example: <http://petewerner.blogspot.it/2012/12/using-apply-sapply-lapply-in-r.html>). Note that the "MARGIN" argument in the function allows to specify that the function `mean()` should be applied to each row (set "MARGIN" to 1) or to each column (set "MARGIN" to 2) of the matrix/dataframe.

```
mean_of_beta <- apply(beta,1,mean,na.rm=T)
```

```
head(beta)
```

```
##           9344737127_R01C02 9344737127_R03C02 9344737127_R06C01
## cg00050873      0.5007407      0.5135834      0.5134752
## cg00212031      0.5153473      0.4880137      0.4010601
## cg00213748      0.4892308      0.5732689      0.4657763
## cg00214611      0.5833333      0.5009074      0.4842767
## cg00455876      0.4772727      0.3244963      0.4394099
## cg01707559      0.5583414      0.4735062      0.3702253
##           9344737127_R02C02 9376538140_R02C01 9376538140_R05C01
## cg00050873      0.5063452      0.5548961      0.4721254
## cg00212031      0.4948805      0.4677914      0.5000000
## cg00213748      0.4533333      0.4276316      0.5217391
## cg00214611      0.5707237      0.4968454      0.5383178
## cg00455876      0.4109589      0.4433121      0.4494519
## cg01707559      0.5269017      0.5052894      0.4174528
##           9376538140_R01C01 9376538140_R04C01
## cg00050873      0.5183752      0.90447562
## cg00212031      0.4780316      0.06670492
## cg00213748      0.5118483      0.79562738
## cg00214611      0.6069767      0.03851709
## cg00455876      0.2972973      0.76427105
## cg01707559      0.4085422      0.06711106
```

```
head(mean_of_beta)
```

```
## cg00050873 cg00212031 cg00213748 cg00214611 cg00455876 cg01707559
## 0.5605021 0.4264787 0.5298070 0.4774873 0.4508088 0.4159213
```

```
dim(beta)
```

```
## [1] 485512      8
```

```
length(mean_of_beta)
```

```
## [1] 485512
```

Now we can calculate the density distribution

```
?density
```

```
## Help on topic 'density' was found in the following packages:
```

```
##
##   Package              Library
```



```
## stats
/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library
## BiocGenerics
/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library
##
##
## Using the first match ...
```

```
d_mean_of_beta <- density(mean_of_beta)
```

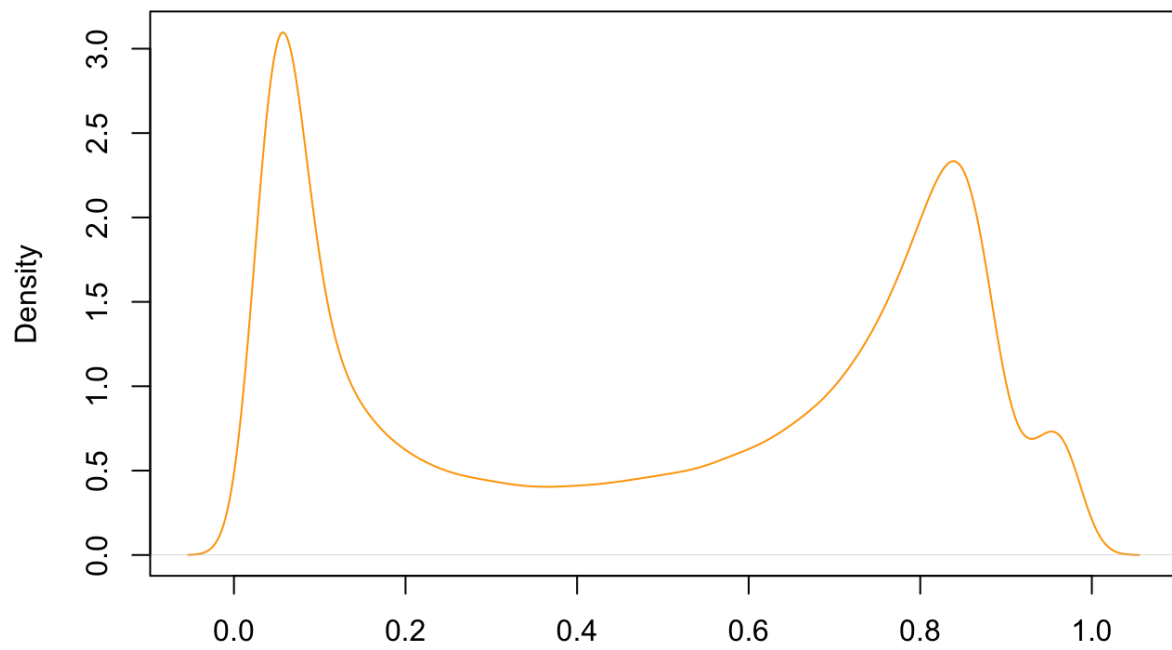
```
d_mean_of_beta
```

```
##
```

```
## Call:
## density.default(x = mean_of_beta)
##
## Data: mean_of_beta (485512 obs.); Bandwidth 'bw' = 0.02181
##
##      x              y
## Min.   :-0.05329   Min.   :0.0004777
## 1st Qu.: 0.22380   1st Qu.:0.4386833
## Median : 0.50089   Median :0.6414534
## Mean    : 0.50089   Mean    :0.9004647
## 3rd Qu.: 0.77798   3rd Qu.:1.1724923
## Max.    : 1.05507   Max.    :3.0967845
```

```
plot(d_mean_of_beta,main="Density of Beta Values",col="orange")
```

Density of Beta Values

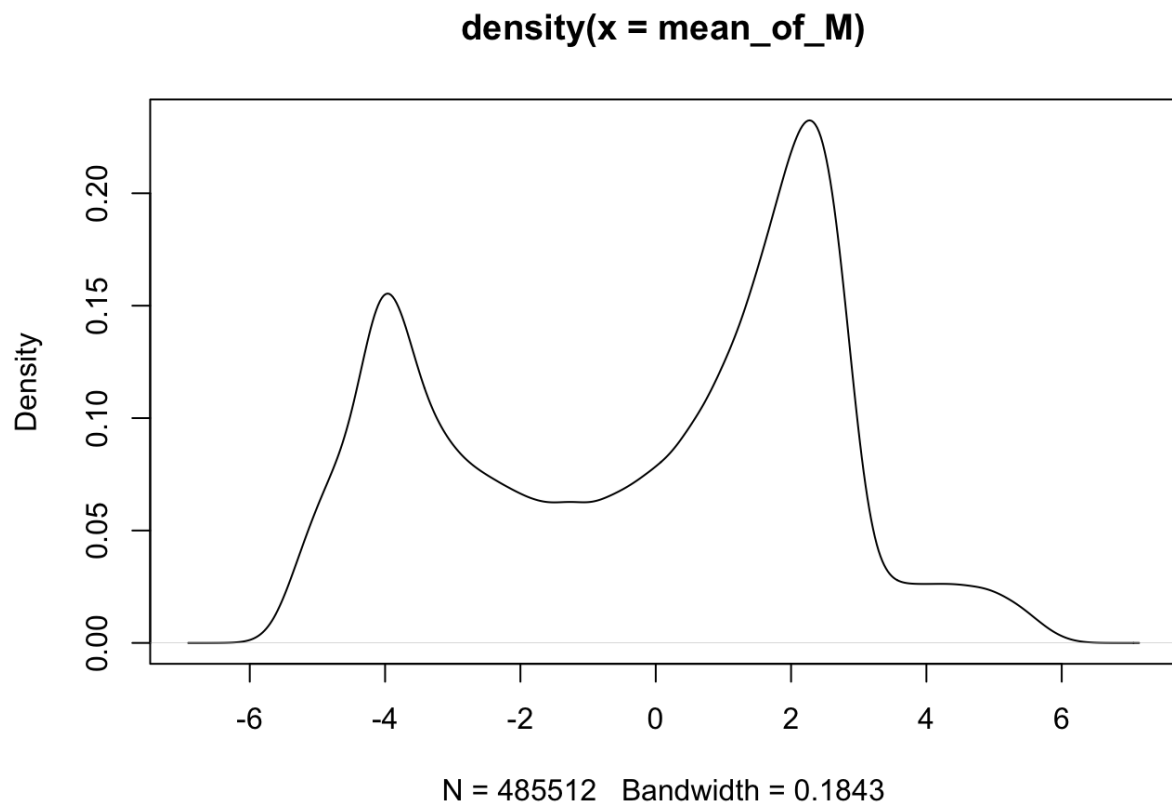


N = 485512 Bandwidth = 0.02181

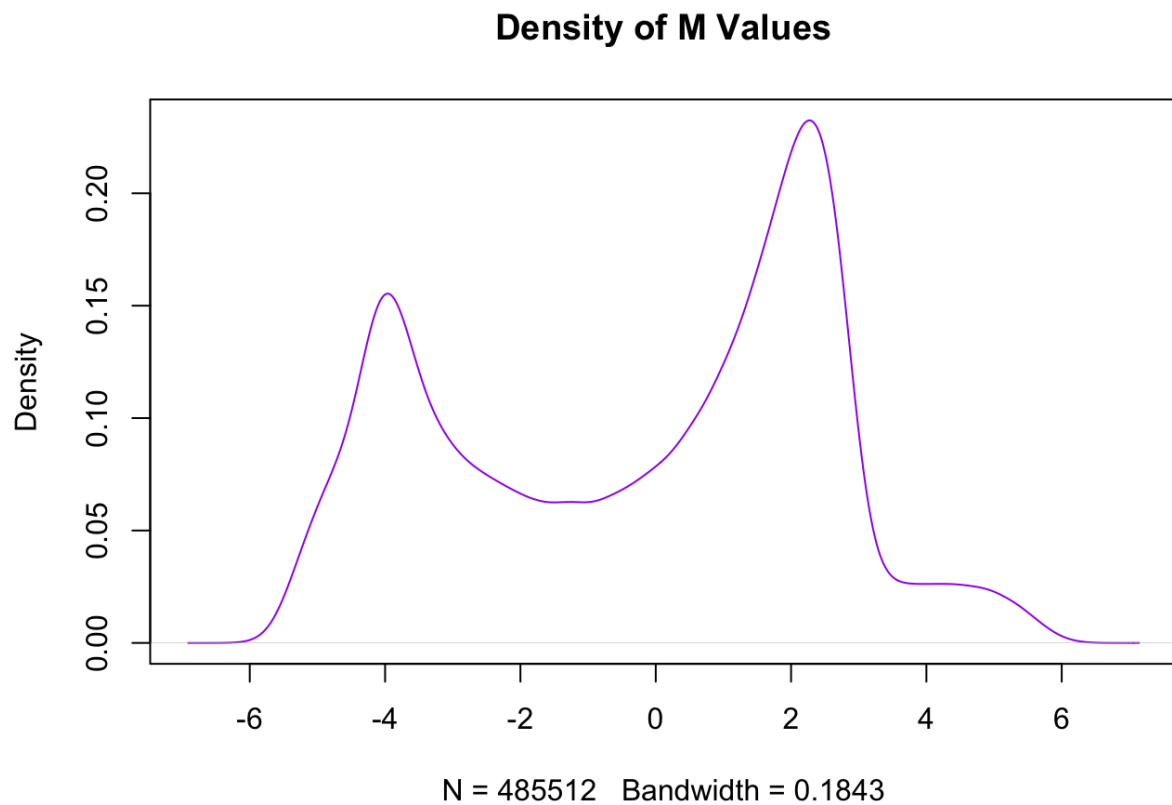
We apply the same steps on M values:

```
mean_of_M <- apply(M, 1, mean, na.rm=T)
```

```
d_mean_of_M <- density(mean_of_M)  
plot(d_mean_of_M)
```



```
plot(d_mean_of_M,main="Density of M Values",col="purple")
```

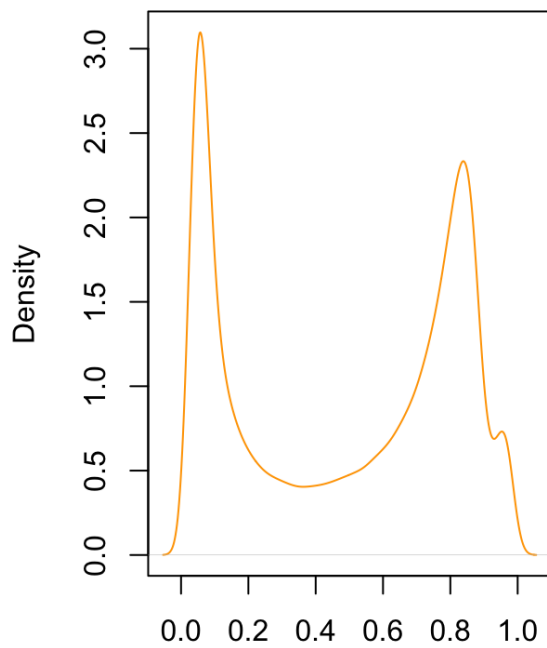


We can put several plots in the same panel by setting the option “mfrow” in the `par()` function (<http://www.statmethods.net/advgraphs/layout.html>). You can find a quick guide to plots here: <http://www.statmethods.net/graphs/index.html>

```
par(mfrow=c(1,2))
```

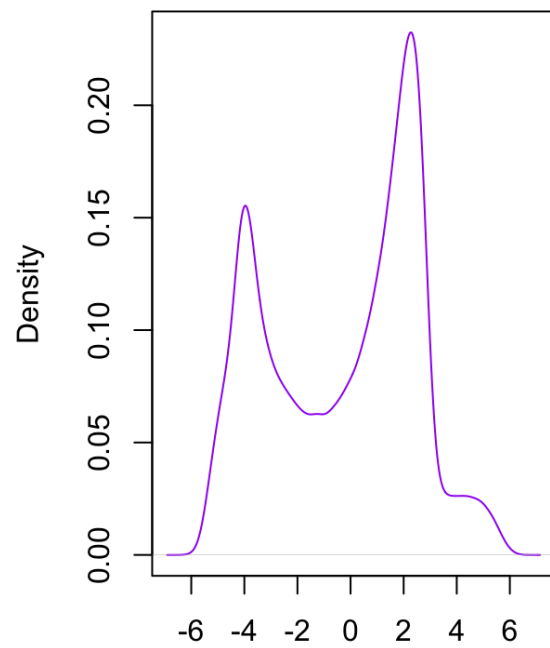
```
plot(d_mean_of_beta,main="Density of Beta Values",col="orange")  
plot(d_mean_of_M,main="Density of M Values",col="purple")
```

Density of Beta Values



N = 485512 Bandwidth = 0.02181

Density of M Values



N = 485512 Bandwidth = 0.1843

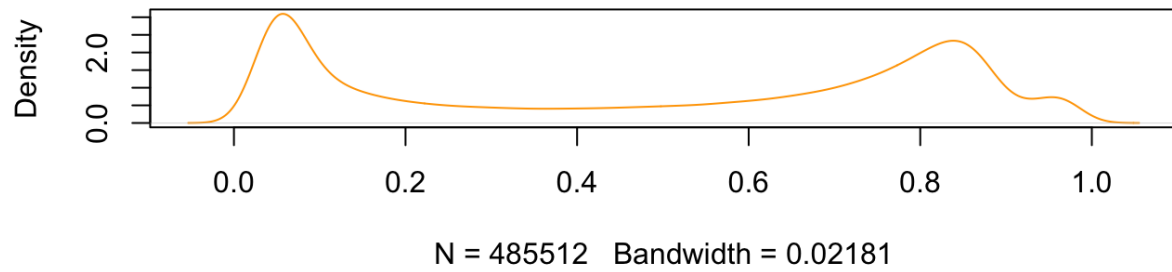
or

```
par(mfrow=c(2,1))
```

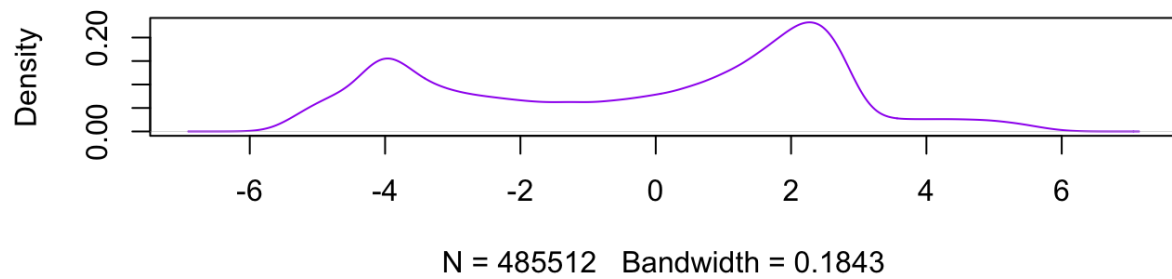
```
plot(d_mean_of_beta,main="Density of Beta Values",col="orange")
```

```
plot(d_mean_of_M,main="Density of M Values",col="purple")
```

Density of Beta Values

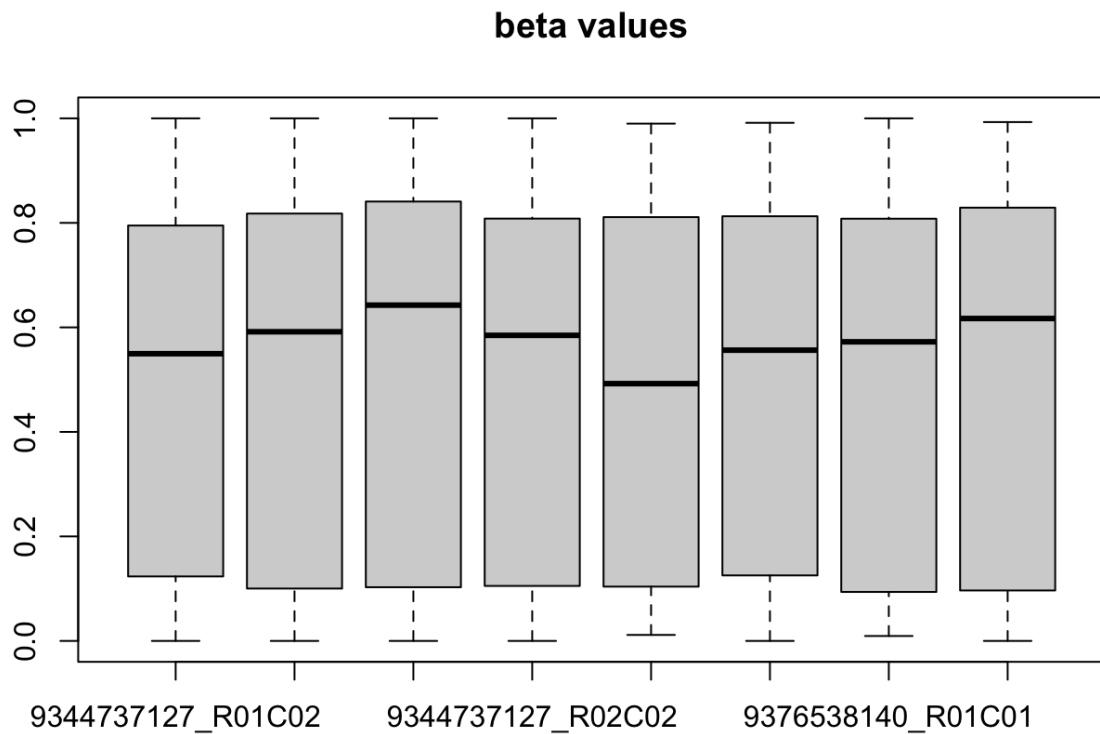


Density of M Values



Another useful plot to represent our data is boxplots:

```
boxplot(beta, main="beta values")
```



CHUNK 2: Save plots

You can save the graph in a variety of formats from the menu File -> Save As. You can also save the graph via code using one of the following functions:

Function Output to

`pdf("mygraph.pdf")` pdf file

`win.metafile("mygraph.wmf")` windows metafile

`png("mygraph.png")` png file

`jpeg("mygraph.jpg")` jpeg file

`bmp("mygraph.bmp")` bmp file

`postscript("mygraph.ps")` postscript file

For example (the file will be saved in the working directory):

```
pdf("Picture1.pdf",width=10,height=5)
```

```
par(mfrow=c(1,2))
```

```
plot(d_mean_of_beta,main="Density of Beta Values",col="blue")
boxplot(beta)
dev.off() #shuts down the graphical device
```

```
## quartz_off_screen
```

```
## 2
```

CHUNK 3: plot the density distribution of beta values for Infinium I and II probes

We want to subset the dataframe of beta values according to Type I and II First of all, we need to know what are the Type I and Type II probes. To this aim, we need our manifest file.

```
load('~/.Dropbox/DRD_2025/2/Illumina450Manifest_clean.RData')
```

```
ls()
```

```
## [1] "beta" "beta_df"
## [3] "d_mean_of_beta" "d_mean_of_M"
## [5] "Illumina450Manifest_clean" "M"
## [7] "mean_of_beta" "mean_of_M"
## [9] "MSet.raw"
```

```
str(Illumina450Manifest_clean)
```

```
## 'data.frame': 485512 obs. of 33 variables:
```

```
## $ IlmnID : Factor w/ 485512 levels
"cg000000029","cg000000108",...: 892 1231 1464 1502 2807 4566 4600 4616
4811 5219 ...
## $ Name : Factor w/ 485512 levels
"cg000000029","cg000000108",...: 892 1231 1464 1502 2807 4566 4600 4616
4811 5219 ...
## $ AddressA_ID : Factor w/ 485512 levels
"10600322","10600328",...: 163924 171461 142476 49179 74479 146426
155259 448531 56092 200778 ...
## $ AlleleA_ProbeSeq : Factor w/ 485512 levels
"AAAAAAAAAAAAAAAAAACACAAAAAATCCCCACTCTAAAACCCACACA",...: 35017 143540
35761 427267 41000 296053 482818 335327 460590 128010 ...
## $ AddressB_ID : Factor w/ 135477 levels
"", "10600313",...: 1 45514 1 1 1 60128 56499 76848 1 1 ...
```



```

## $ AlleleB_ProbeSeq          : Factor w/ 135477 levels
"", "AAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAACCCCGCG", ...: 1 49667 1 1
1 77344 135092 95678 1 1 ...
## $ Infinium_Design_Type      : Factor w/ 2 levels "I","II": 2 1 2 2
2 1 1 1 2 2 ...
## $ Next_Base                 : Factor w/ 4 levels "", "A", "C", "T": 1
2 1 1 1 4 2 2 1 1 ...
## $ Color_Channel             : Factor w/ 3 levels "", "Grn", "Red": 1
3 1 1 1 3 3 3 1 1 ...
## $ Forward_Sequence          : Factor w/ 485512 levels
"AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAATATGTCTGCTAAATAATCACAGAGTGAAA[CG]AAGAGG
AAGAAACTAAAGTACAGTTTGGAACATAGTACAGTGCCCCTGGGTACTGAATCA", ...: 25105
392558 395302 218895 81395 152101 421754 296779 478968 55951 ...
## $ Genome_Build              : int  37 37 37 37 37 37 37 37 37 37
...
## $ CHR                       : Factor w/ 24 levels
"1","10","11", ...: 24 24 24 24 24 24 24 24 24 24 ...
## $ MAPINFO                   : int  8553009 9363356 25314171
22741795 21664296 21239348 8148233 15815688 4868996 6133740 ...
## $ SourceSeq                 : Factor w/ 485343 levels
"AAAAAAAAAACCAACCCCAAACAAAACCCCTGTGCCCAGAATTGCATTCG", ...: 28961 285896
116667 116691 29627 123294 148856 152723 189022 323597 ...
## $ Chromosome_36             : Factor w/ 25 levels
"1","10","11", ...: 25 25 25 25 25 25 25 25 25 25 ...
## $ Coordinate_36             : Factor w/ 484738 levels
"100000057","100000691", ...: 451148 483811 205767 185980 178028 173644
442780 102376 331283 376023 ...
## $ Strand                    : Factor w/ 2 levels "F","R": 1 2 2 1
2 2 2 2 2 1 ...
## $ Probe_SNPs                : Factor w/ 56004 levels
"", "rs10000615", ...: 1 1 1 54643 1 1 1 1 1 1 ...
## $ Probe_SNPs_10             : Factor w/ 35790 levels
"", "rs10000804", ...: 1 1 1 4064 1 1 1 1 1 1 ...
## $ Random_Loci               : logi  NA NA NA NA NA NA ...
## $ Methyl27_Loci             : logi  NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name         : Factor w/ 42456 levels
"", "A1BG", "A1CF;A1CF;A1CF", ...: 39103 38905 9635 11319 3085 39100 1
37974 27057 1 ...
## $ UCSC_RefGene_Accession    : Factor w/ 52670 levels
"", "NM_000014", ...: 48799 14035 19907 20891 49121 48792 1 20105 38174 1
...
## $ UCSC_RefGene_Group        : Factor w/ 4165 levels
"", "1stExon", "1stExon;1stExon", ...: 2970 2779 2447 2035 2035 3694 1 201
943 1 ...

```

```
## $ UCSC_CpG_Islands_Name      : Factor w/ 27177 levels
"", "chr1:10003165-10003585", ...: 1 27170 1 27136 27134 27133 27156 27120
27147 27149 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 6 levels
"", "Island", "N_Shelf", ...: 1 4 1 5 4 2 6 2 4 2 ...
## $ Phantom                    : Factor w/ 11912 levels "", "high-
CpG:100009860-100010119", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ DMR                        : Factor w/ 4 levels
"", "CDMR", "DMR", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Enhancer                   : logi  NA NA NA NA NA NA ...
## $ HMM_Island                 : Factor w/ 63086 levels
"", "1:100088009-100088708", ...: 1 63085 1 1 1 63046 63077 63033 63064
63067 ...
## $ Regulatory_Feature_Name     : Factor w/ 47969 levels
"", "1:10001853-10003978", ...: 1 1 1 1 1 1 1 47965 1 1 ...
## $ Regulatory_Feature_Group    : Factor w/ 9 levels
"", "Gene_Associated", ...: 1 1 1 1 1 1 1 7 1 1 ...
## $ DHS                        : logi  NA NA NA NA NA NA ...
```

We subset the `Illumina450Manifest_clean` in two dataframes, containing only type I (dfI) or type II (dfII) probes

```
dfI <-
Illumina450Manifest_clean[Illumina450Manifest_clean$Infinium_Design_Type=="I",]
```

```
dfI <- droplevels(dfI)
dim(dfI)
```

```
## [1] 135476      33
```

```
str(dfI)
```

```
## 'data.frame':   135476 obs. of  33 variables:
```

```
## $ IlmnID                      : Factor w/ 135476 levels
"cg00000622", "cg00000769", ...: 311 1089 1099 1106 2280 8496 10019 10061
10250 11112 ...
## $ Name                        : Factor w/ 135476 levels
"cg00000622", "cg00000769", ...: 311 1089 1099 1106 2280 8496 10019 10061
10250 11112 ...
## $ AddressA_ID                 : Factor w/ 135476 levels
"10600328", "10600369", ...: 47762 40844 43302 125252 36295 74624 33361
35577 70485 133756 ...
## $ AlleleA_ProbeSeq            : Factor w/ 135476 levels
"AAAAAAAAAAAAAAAAAACCAAAAAAAAAATCCCCACTCTAAAACCCACACA", ...: 46999 97650
135088 103784 39883 50177 55080 65466 80679 129398 ...
```

```

## $ AddressB_ID : Factor w/ 135476 levels
"10600313","10600353",...: 45513 60127 56498 76847 124346 114306 100943
109732 132012 11815 ...
## $ AlleleB_ProbeSeq : Factor w/ 135476 levels
"AAAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAAACCCCGCG",...: 49666 77343
135091 95677 35570 107971 45139 58259 69459 129398 ...
## $ Infinium_Design_Type : Factor w/ 1 level "I": 1 1 1 1 1 1 1
1 1 1 ...
## $ Next_Base : Factor w/ 3 levels "A","C","T": 1 3
1 1 1 1 2 1 2 2 ...
## $ Color_Channel : Factor w/ 2 levels "Grn","Red": 2 2
2 2 2 2 1 2 1 1 ...
## $ Forward_Sequence : Factor w/ 135476 levels
"AAAAAAAAAAAAAAAAAAAAAAAAAAGCTTGCCCCAACATCGGTATATTCCAAATTCG[CG]CGTGCT
TCCGAAGGCTCATATTCTCGGCCCTTTCAGACGCTCTTAAAATTTTCTCCTAG",...: 113476
36406 121253 84324 53100 15696 80195 14484 40097 106165 ...
## $ Genome_Build : int 37 37 37 37 37 37 37 37 37 37
...
## $ CHR : Factor w/ 24 levels
"1","10","11",...: 24 24 24 24 24 24 24 24 24 24 ...
## $ MAPINFO : int 9363356 21239348 8148233
15815688 9385539 6778695 15815552 9194502 22918038 22737591 ...
## $ SourceSeq : Factor w/ 135431 levels
"AAAAAAAAAATCCCCGTTGCTCTTTACACCCCGGAAACGGCAAGGGAACG",...: 83347 29309
36834 39256 98271 40812 71752 120603 16821 126495 ...
## $ Chromosome_36 : Factor w/ 25 levels
"1","10","11",...: 25 25 25 25 25 25 25 25 25 25 ...
## $ Coordinate_36 : Factor w/ 135401 levels
"100002815","100004665",...: 135150 47475 123882 27977 135363 111177
27976 130482 51025 50776 ...
## $ Strand : Factor w/ 2 levels "F","R": 2 2 2 2
2 1 1 1 1 1 ...
## $ Probe_SNPs : Factor w/ 16579 levels
"", "rs10000615",...: 1 1 1 1 1 1 16416 1 1 1 ...
## $ Probe_SNPs_10 : Factor w/ 9588 levels
"", "rs10000804",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Random_Loci : logi NA NA NA NA NA NA ...
## $ Methyl27_Loci : logi NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name : Factor w/ 27920 levels
"", "A1BG", "A2BP1;A2BP1",...: 25400 25525 1 24787 1 23876 24787 25399
20786 7435 ...
## $ UCSC_RefGene_Accession : Factor w/ 31786 levels
"", "NM_000014",...: 7604 29856 1 11389 1 24652 11389 7602 2965 11954 ...
## $ UCSC_RefGene_Group : Factor w/ 2621 levels
"", "1stExon", "1stExon;1stExon",...: 1685 2267 1 132 1 2571 132 1798 132
2267 ...

```

```
## $ UCSC_CpG_Islands_Name      : Factor w/ 25488 levels
"", "chr1:10003165-10003585", ...: 25484 25462 25477 25455 25486 25474
25455 25479 1 25465 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 6 levels
"", "Island", "N_Shelf", ...: 4 2 6 2 2 2 2 1 4 ...
## $ Phantom                    : Factor w/ 9197 levels "", "high-
CpG:100009860-100010119", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ DMR                        : Factor w/ 4 levels
"", "CDMR", "DMR", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Enhancer                   : logi  NA NA NA NA NA NA ...
## $ HMM_Island                 : Factor w/ 46581 levels
"", "1:100088009-100088708", ...: 46580 46557 46576 46551 46581 46573
46551 46577 46564 46563 ...
## $ Regulatory_Feature_Name    : Factor w/ 22970 levels
"", "1:10001853-10003978", ...: 1 1 1 22968 1 1 22968 1 1 1 ...
## $ Regulatory_Feature_Group   : Factor w/ 9 levels
"", "Gene_Associated", ...: 1 1 1 7 1 1 7 1 1 1 ...
## $ DHS                        : logi  NA NA NA NA NA NA ...
```

```
dfII <-
Illumina450Manifest_clean[Illumina450Manifest_clean$Infinium_Design_Type=="II",]
```

```
dfII <- droplevels(dfII)
```

Remember that in the beta matrix, which contains the beta values, the names of the probes are stored in the rownames of the matrix:

```
head(beta)
```

##	9344737127_R01C02	9344737127_R03C02	9344737127_R06C01
## cg00050873	0.5007407	0.5135834	0.5134752
## cg00212031	0.5153473	0.4880137	0.4010601
## cg00213748	0.4892308	0.5732689	0.4657763
## cg00214611	0.5833333	0.5009074	0.4842767
## cg00455876	0.4772727	0.3244963	0.4394099
## cg01707559	0.5583414	0.4735062	0.3702253
##	9344737127_R02C02	9376538140_R02C01	9376538140_R05C01
## cg00050873	0.5063452	0.5548961	0.4721254
## cg00212031	0.4948805	0.4677914	0.5000000
## cg00213748	0.4533333	0.4276316	0.5217391
## cg00214611	0.5707237	0.4968454	0.5383178
## cg00455876	0.4109589	0.4433121	0.4494519
## cg01707559	0.5269017	0.5052894	0.4174528
##	9376538140_R01C01	9376538140_R04C01	
## cg00050873	0.5183752	0.90447562	

```
## cg00212031      0.4780316      0.06670492
## cg00213748      0.5118483      0.79562738
## cg00214611      0.6069767      0.03851709
## cg00455876      0.2972973      0.76427105
## cg01707559      0.4085422      0.06711106
```

Now we subset the beta matrix in order to retain only the rows whose name is in the first column of dfl...

```
beta_I <- beta[rownames(beta) %in% dfI$ilmnID,]
dim(beta_I)
```

```
## [1] 135476      8
```

... or in the first column of dfII

```
beta_II <- beta[rownames(beta) %in% dfII$ilmnID,]
dim(beta_II)
```

```
## [1] 350036      8
```

For each probe in the mean_of_beta_I and mean_of_beta_II matrices, we calculate the mean of beta values across the 8 samples...

```
mean_of_beta_I <- apply(beta_I,1,mean)
mean_of_beta_II <- apply(beta_II,1,mean)
```

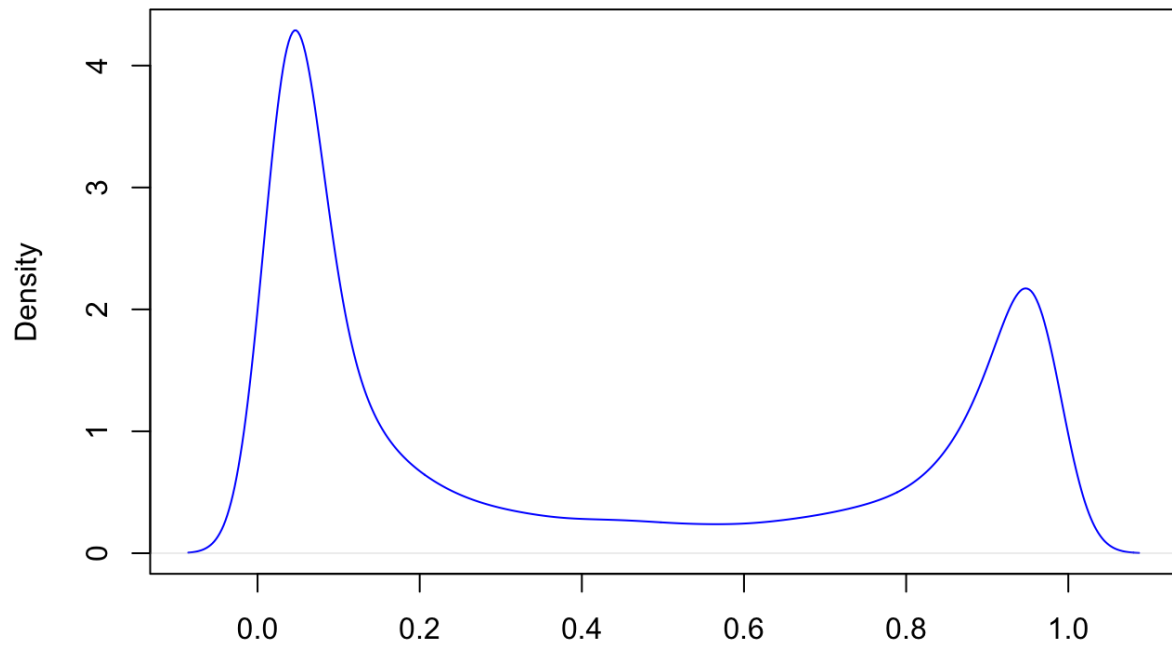
... and then we calculate the density distribution of the 2 vectors of mean values:

```
d_mean_of_beta_I <- density(mean_of_beta_I,na.rm=T)
d_mean_of_beta_II <- density(mean_of_beta_II,na.rm=T)
```

Finally, we can plot the densities.

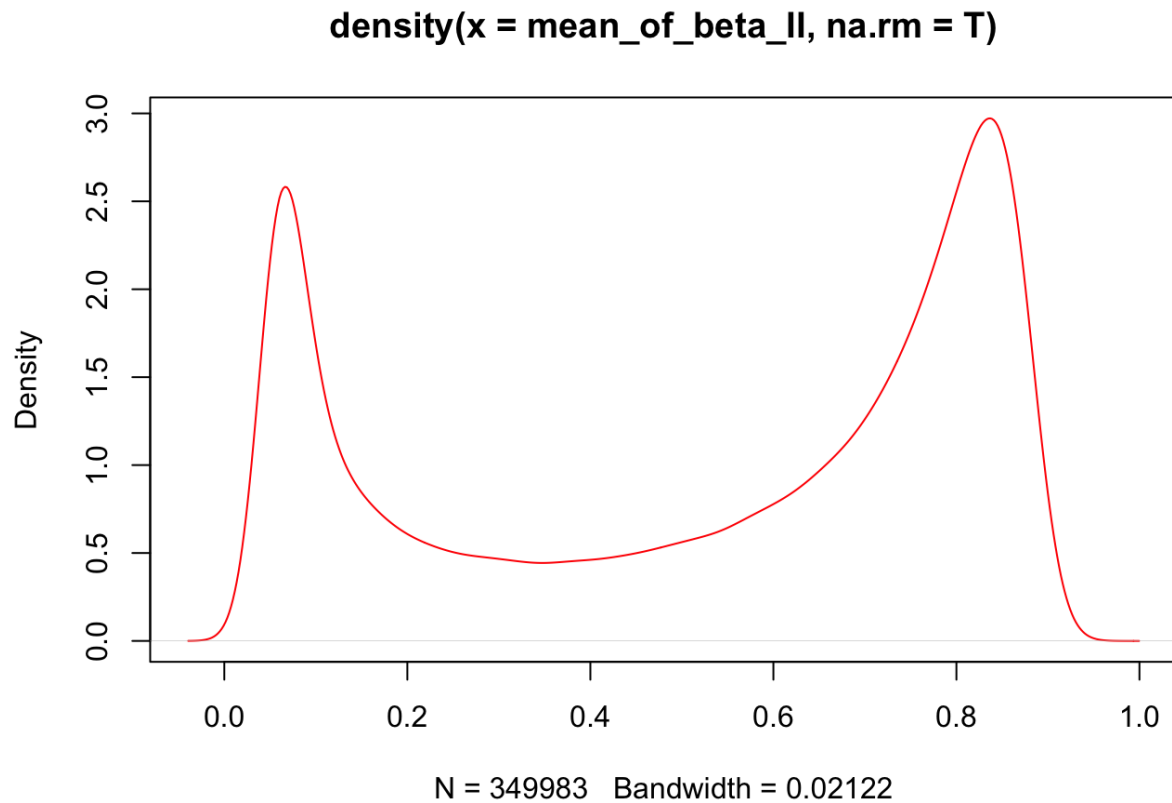
```
plot(d_mean_of_beta_I,col="blue")
```

density(x = mean_of_beta_I, na.rm = T)



N = 135476 Bandwidth = 0.03255

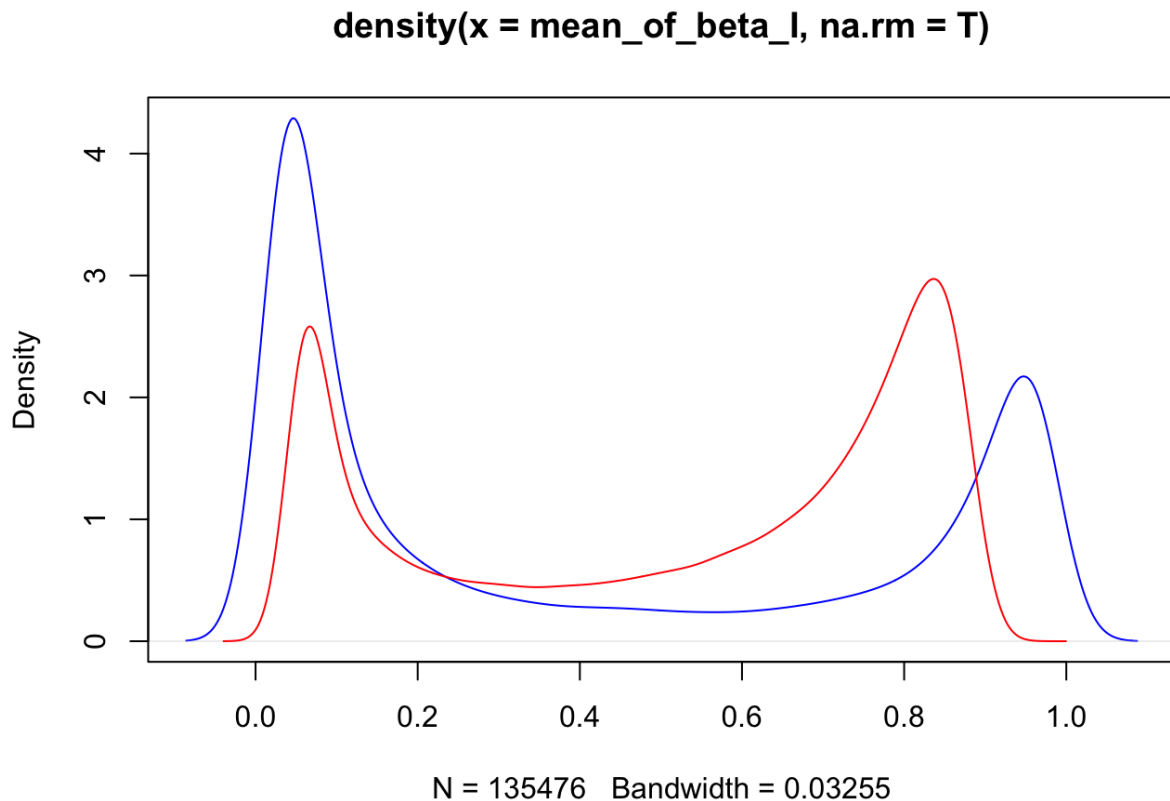
```
plot(d_mean_of_beta_II,col="red")
```



But in this case it is more meaningful to overlay the two density distributions. To this aim, we first open a plot using the `plot()` function, and then we add a new line, coloured with a different color, using the `lines()` function

```
plot(d_mean_of_beta_I,col="blue")
```

```
lines(d_mean_of_beta_II,col="red")
```



Note that the density distribution of beta values from Infinium type II probes is shifted towards the centre. Moreover, type I probes have higher density at low methylation levels.

CHUNK 4: Normalization

We will apply the `preprocessQuantile` function. To evaluate the effect of the normalization, we will compare raw and normalized data according to 3 types of plots:

- densities of mean beta values for type I and type II probes
- densities of standard deviation of beta values for type I and type II probes
- boxplots of beta values

For Raw data, we already have the matrix of beta values, the `d_mean_of_beta_I` and `d_mean_of_beta_II` objects; we need to calculate the densities of the standard deviations, which can be calculated using the function `sd()`:

```
sd_of_beta_I <- apply(beta_I, 1, sd, na.rm=T)
```

```
sd_of_beta_II <- apply(beta_II, 1, sd, na.rm=T)
```

```
d_sd_of_beta_I <- density(sd_of_beta_I,)
```

```
d_sd_of_beta_II <- density(sd_of_beta_II)
```


Now we can perform the normalization using the preprocessQuantile function.

```
?preprocessQuantile
```

According to the help page, the input can be an RGset or MehyISet object. Let's load the RGset object:

```
load("~/Dropbox/DRD_2025/3/RGset.RData")
```

RGset

```
## class: RGChannelSet
## dim: 622399 8
## metadata(0):
## assays(2): Green Red
## rownames(622399): 10600313 10600322 ... 74810490 74810492
## rowData names(0):
## colnames(8): 9344737127_R01C02 9344737127_R03C02 ...
9376538140_R01C01
## 9376538140_R04C01
## colData names(9): Sample_ID Sample_Well ... Basename filenames
## Annotation
## array: IlluminaHumanMethylation450k
## annotation: ilmn12.hg19
```

```
preprocessQuantile_results <- preprocessQuantile(RGset)

## [preprocessQuantile] Mapping to genome.

## Loading required package: IlluminaHumanMethylation450kmanifest

## Loading required package:
IlluminaHumanMethylation450kanno.ilmn12.hg19

## [preprocessQuantile] Fixing outliers.

## [preprocessQuantile] Quantile normalizing.
```

You can have the following error: "there is no package called 'IlluminaHumanMethylation450kanno.ilmn12.hg19'; in this case you have to install the missing package (it will take a bit):

if (!requireNamespace("BiocManager", quietly = TRUE))

```
install.packages("BiocManager")
```

BiocManager::install("IlluminaHumanMethylation450kanno.ilmn12.hg19")

```

preprocessQuantile_results <- preprocessQuantile(RGset)

## [preprocessQuantile] Mapping to genome.
## [preprocessQuantile] Fixing outliers.
## [preprocessQuantile] Quantile normalizing.

str(preprocessQuantile_results)

## Formal class 'GenomicRatioSet' [package "minfi"] with 8 slots

## ..@ preprocessMethod: Named chr [1:4] "preprocessQuantile" "Raw
(no normalization or bg correction)" "1.50.0" "0.4.0"
## .. ..- attr(*, "names")= chr [1:4] "mu.norm" "rg.norm" "minfi"
"manifest"
## ..@ annotation : Named chr [1:2]
"IlluminaHumanMethylation450k" "ilmn12.hg19"
## .. ..- attr(*, "names")= chr [1:2] "array" "annotation"
## ..@ rowRanges : Formal class 'GRanges' [package
"GenomicRanges"] with 7 slots
## .. .. ..@ seqnames : Formal class 'Rle' [package "S4Vectors"]
with 4 slots
## .. .. .. ..@ values : Factor w/ 24 levels
"chr1","chr2",...: 1 2 3 4 5 6 7 8 9 10 ...
## .. .. .. ..@ lengths : int [1:24] 46857 34810 25159
20464 24327 36611 30017 20950 9861 24388 ...
## .. .. .. ..@ elementMetadata: NULL
## .. .. .. ..@ metadata : list()
## .. .. ..@ ranges : Formal class 'IRanges' [package
"IRanges"] with 6 slots
## .. .. .. ..@ start : int [1:485512] 15865 18827 29407
29425 29435 68849 69591 91550 135252 449076 ...
## .. .. .. ..@ width : int [1:485512] 1 1 1 1 1 1 1 1 1
1 ...
## .. .. .. ..@ NAMES : chr [1:485512] "cg13869341"
"cg14008030" "cg12045430" "cg20826792" ...
## .. .. .. ..@ elementType : chr "ANY"
## .. .. .. ..@ elementMetadata: NULL
## .. .. .. ..@ metadata : list()
## .. .. ..@ strand : Formal class 'Rle' [package "S4Vectors"]
with 4 slots
## .. .. .. ..@ values : Factor w/ 3 levels "+","-","*": 3
## .. .. .. ..@ lengths : int 485512
## .. .. .. ..@ elementMetadata: NULL
## .. .. .. ..@ metadata : list()
## .. .. ..@ seqinfo : Formal class 'Seqinfo' [package
"GenomeInfoDb"] with 4 slots

```

```

##    .. .. .. .. ..@ seqnames      : chr [1:24] "chr1" "chr2" "chr3"
"chr4" ...
##    .. .. .. .. ..@ seqlengths    : int [1:24] NA NA NA NA NA NA NA NA NA
NA ...
##    .. .. .. .. ..@ is_circular: logi [1:24] NA NA NA NA NA NA ...
##    .. .. .. .. ..@ genome        : chr [1:24] "hg19" "hg19" "hg19"
"hg19" ...
##    .. .. ..@ elementMetadata:Formal class 'DFrame' [package
"S4Vectors"] with 6 slots
##    .. .. .. .. ..@ rownames      : NULL
##    .. .. .. .. ..@ nrows         : int 485512
##    .. .. .. .. ..@ elementType   : chr "ANY"
##    .. .. .. .. ..@ elementMetadata: NULL
##    .. .. .. .. ..@ metadata      : list()
##    .. .. .. .. ..@ listData      : Named list()
##    .. .. ..@ elementType        : chr "ANY"
##    .. .. ..@ metadata           : list()
##    ..@ colData                  :Formal class 'DFrame' [package "S4Vectors"]
with 6 slots
##    .. .. ..@ rownames            : chr [1:8] "9344737127_R01C02"
"9344737127_R03C02" "9344737127_R06C01" "9344737127_R02C02" ...
##    .. .. ..@ nrows              : int 8
##    .. .. ..@ elementType        : chr "ANY"
##    .. .. ..@ elementMetadata: NULL
##    .. .. ..@ metadata           : list()
##    .. .. ..@ listData           :List of 12
##    .. .. .. ..$ Sample_ID       : int [1:8] 21050 21060 22041 22051 21090
21100 23081 23091
##    .. .. .. ..$ Sample_Well    : chr [1:8] "G10" "A11" "F10" "H10" ...
##    .. .. .. ..$ Array         : chr [1:8] "R01C02" "R03C02" "R06C01"
"R02C02" ...
##    .. .. .. ..$ Slide          : chr [1:8] "9344737127" "9344737127"
"9344737127" "9344737127" ...
##    .. .. .. ..$ Group          : chr [1:8] "A" "A" "B" "B" ...
##    .. .. .. ..$ Gender         : chr [1:8] "F" "F" "F" "F" ...
##    .. .. .. ..$ BMI            : num [1:8] 21.6 20.8 26.6 26.6 17.5 ...
##    .. .. .. ..$ Basename       : chr [1:8]
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R01C02"
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R03C02"
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R06C01"
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R02C02" ...
##    .. .. .. ..$ filenames     : chr [1:8]
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R01C02"
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R03C02"
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R06C01"
"/Users/francesco/Dropbox/DRD_2025/3/Input_data/9344737127_R02C02" ...

```

```

##    .. .. .. ..$ xMed          : Named num [1:8] 13.1 13.2 13.3 13.2
13.2 ...
##    .. .. .. ..- attr(*, "names")= chr [1:8] "9344737127_R01C02"
"9344737127_R03C02" "9344737127_R06C01" "9344737127_R02C02" ...
##    .. .. .. ..$ yMed          : Named num [1:8] 9.11 9.19 9.27 9.22
9.21 ...
##    .. .. .. ..- attr(*, "names")= chr [1:8] "9344737127_R01C02"
"9344737127_R03C02" "9344737127_R06C01" "9344737127_R02C02" ...
##    .. .. .. ..$ predictedSex: Named chr [1:8] "F" "F" "F" "F" ...
##    .. .. .. ..- attr(*, "names")= chr [1:8] "9344737127_R01C02"
"9344737127_R03C02" "9344737127_R06C01" "9344737127_R02C02" ...
##    ..@ assays                  :Formal class 'SimpleAssays' [package
"SummarizedExperiment"] with 1 slot
##    .. .. ..@ data:Formal class 'SimpleList' [package "S4Vectors"]
with 4 slots
##    .. .. .. ..@ listData       :List of 2
##    .. .. .. .. ..$ M : num [1:485512, 1:8] 1.5 1.17 -2.68 -2.28 -
4.01 ...
##    .. .. .. .. ..- attr(*, "dimnames")=List of 2
##    .. .. .. .. ..$ : chr [1:485512] "cg13869341"
"cg14008030" "cg12045430" "cg20826792" ...
##    .. .. .. .. ..$ : chr [1:8] "9344737127_R01C02"
"9344737127_R03C02" "9344737127_R06C01" "9344737127_R02C02" ...
##    .. .. .. .. ..$ CN: num [1:485512, 1:8] 13.8 13.6 13.3 13.8
13.4 ...
##    .. .. .. .. ..- attr(*, "dimnames")=List of 2
##    .. .. .. .. ..$ : chr [1:485512] "cg13869341"
"cg14008030" "cg12045430" "cg20826792" ...
##    .. .. .. .. ..$ : chr [1:8] "9344737127_R01C02"
"9344737127_R03C02" "9344737127_R06C01" "9344737127_R02C02" ...
##    .. .. .. .. ..@ elementType   : chr "ANY"
##    .. .. .. .. ..@ elementMetadata: NULL
##    .. .. .. .. ..@ metadata      : list()
##    ..@ NAMES                     : NULL
##    ..@ elementMetadata :Formal class 'DFrame' [package "S4Vectors"]
with 6 slots
##    .. .. ..@ rownames           : NULL
##    .. .. ..@ nrows              : int 485512
##    .. .. ..@ elementType        : chr "ANY"
##    .. .. ..@ elementMetadata: NULL
##    .. .. ..@ metadata           : list()
##    .. .. ..@ listData           : Named list()
##    ..@ metadata                 : list()

```

```
class(preprocessQuantile_results)
```

```
## [1] "GenomicRatioSet"
```

```
## attr(,"package")
```

```
## [1] "minfi"
```

```
preprocessQuantile_results
```

```
## class: GenomicRatioSet
```

```
## dim: 485512 8
```

```
## metadata(0):
```

```
## assays(2): M CN
```

```
## rownames(485512): cg13869341 cg14008030 ... cg08265308 cg14273923
```

```
## rowData names(0):
```

```
## colnames(8): 9344737127_R01C02 9344737127_R03C02 ...
```

```
9376538140_R01C01
```

```
## 9376538140_R04C01
```

```
## colData names(12): Sample_ID Sample_Well ... yMed predictedSex
```

```
## Annotation
```

```
## array: IlluminaHumanMethylation450k
```

```
## annotation: ilmn12.hg19
```

```
## Preprocessing
```

```
## Method: Raw (no normalization or bg correction)
```

```
## minfi version: 1.50.0
```

```
## Manifest version: 0.4.0
```

```
?GenomicRatioSet
```

You see that `getBeta` is among the accessor functions of a `GenomicRatioSet` class object

```
beta_preprocessQuantile <- getBeta(preprocessQuantile_results)
```

```
head(beta_preprocessQuantile)
```

```
##           9344737127_R01C02 9344737127_R03C02 9344737127_R06C01
```

```
## cg13869341      0.73914779      0.76950459      0.6995062
```

```
## cg14008030      0.69185593      0.64592974      0.6885158
```

```
## cg12045430      0.13474718      0.11699326      0.1004484
```

```
## cg20826792      0.17078769      0.28879485      0.2947075
```

```
## cg00381604      0.05846309      0.09592991      0.1045477
```

```
## cg20253340      0.42958351      0.51681548      0.4396129
```

```
##           9344737127_R02C02 9376538140_R02C01 9376538140_R05C01
```

```
## cg13869341      0.7184184      0.78299289      0.77009158
```

```
## cg14008030      0.6031361      0.63207688      0.56279678
```

```
## cg12045430      0.1218535      0.14349114      0.12472814
```

```
## cg20826792      0.2819566      0.22464011      0.20651747
```

```
## cg00381604      0.1000693      0.09964109      0.08688974
```

```
## cg20253340          0.4760213          0.46439970          0.37458736
##                9376538140_R01C01 9376538140_R04C01
## cg13869341          0.7751171          0.8177119
## cg14008030          0.6861439          0.6460894
## cg12045430          0.4254814          0.1188433
## cg20826792          0.3439327          0.2847663
## cg00381604          0.3128018          0.1086204
## cg20253340          0.4502786          0.6004477
```

Now, as we have done before for raw data, we divide the `beta_preprocessQuantile` matrix according to type I and type II probes, calculate the mean and the standard deviation for each probe across the 8 samples and calculate the density distributions

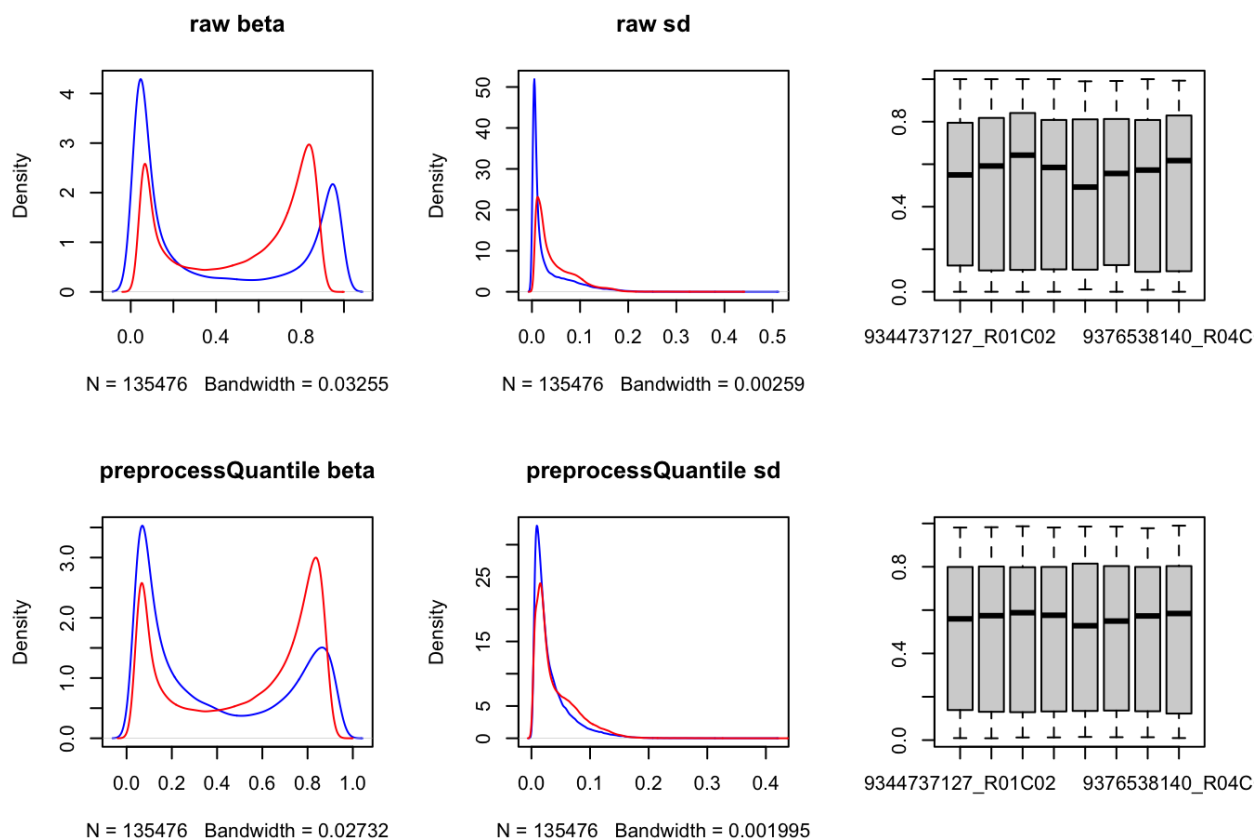
```
beta_preprocessQuantile_I <-
beta_preprocessQuantile[rownames(beta_preprocessQuantile) %in%
dfI$IlmnID,]

beta_preprocessQuantile_II <-
beta_preprocessQuantile[rownames(beta_preprocessQuantile) %in% dfII$IlmnID,]
mean_of_beta_preprocessQuantile_I <- apply(beta_preprocessQuantile_I,1,mean)
mean_of_beta_preprocessQuantile_II <- apply(beta_preprocessQuantile_II,1,mean)
d_mean_of_beta_preprocessQuantile_I <-
density(mean_of_beta_preprocessQuantile_I,na.rm=T)
d_mean_of_beta_preprocessQuantile_II <-
density(mean_of_beta_preprocessQuantile_II,na.rm=T)
sd_of_beta_preprocessQuantile_I <- apply(beta_preprocessQuantile_I,1,sd)
sd_of_beta_preprocessQuantile_II <- apply(beta_preprocessQuantile_II,1,sd)
d_sd_of_beta_preprocessQuantile_I <- density(sd_of_beta_preprocessQuantile_I,na.rm=T)
d_sd_of_beta_preprocessQuantile_II <-
density(sd_of_beta_preprocessQuantile_II,na.rm=T)
```

And here it is our beautiful plot!

```
par(mfrow=c(2,3))

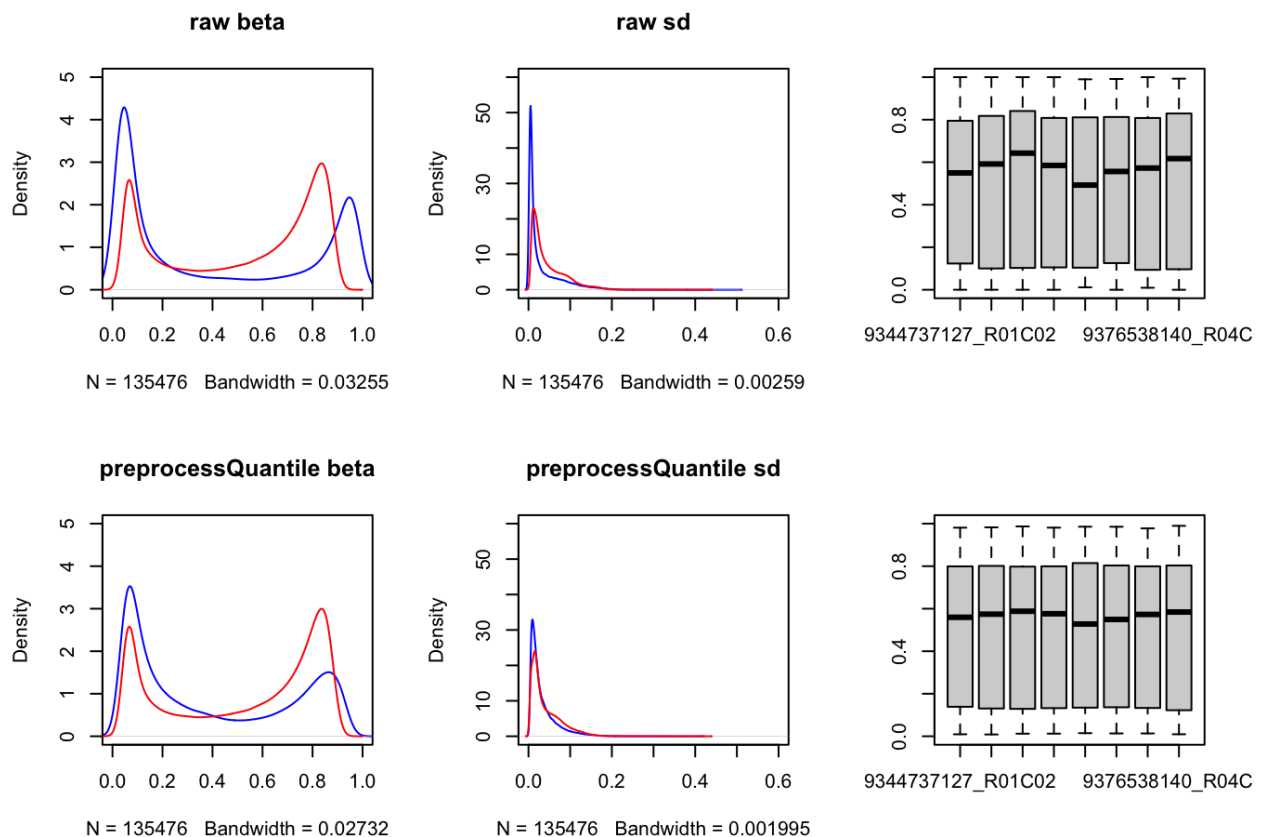
plot(d_mean_of_beta_I,col="blue",main="raw beta")
lines(d_mean_of_beta_II,col="red")
plot(d_sd_of_beta_I,col="blue",main="raw sd")
lines(d_sd_of_beta_II,col="red")
boxplot(beta)
plot(d_mean_of_beta_preprocessQuantile_I,col="blue",main="preprocessQuantile beta")
lines(d_mean_of_beta_preprocessQuantile_II,col="red")
plot(d_sd_of_beta_preprocessQuantile_I,col="blue",main="preprocessQuantile sd")
lines(d_sd_of_beta_preprocessQuantile_II,col="red")
boxplot(beta_preprocessQuantile)
```



Note that it is easier to compare the plots if we have the same scales on x and y axes; to this aim, we can specify the xlim and ylim arguments:

```
par(mfrow=c(2,3))

plot(d_mean_of_beta_I,col="blue",main="raw beta",xlim=c(0,1),ylim=c(0,5))
lines(d_mean_of_beta_II,col="red")
plot(d_sd_of_beta_I,col="blue",main="raw sd",xlim=c(0,0.6),ylim=c(0,60))
lines(d_sd_of_beta_II,col="red")
boxplot(beta,ylim=c(0,1))
plot(d_mean_of_beta_preprocessQuantile_I,col="blue",main="preprocessQuantile
beta",xlim=c(0,1),ylim=c(0,5))
lines(d_mean_of_beta_preprocessQuantile_II,col="red")
plot(d_sd_of_beta_preprocessQuantile_I,col="blue",main="preprocessQuantile
sd",xlim=c(0,0.6),ylim=c(0,60))
lines(d_sd_of_beta_preprocessQuantile_II,col="red")
boxplot(beta_preprocessQuantile,ylim=c(0,1))
```



We can save this plot in a pdf

```
pdf("Plot_comparison_raw_preprocessQuantile.pdf",height=7,width=15)
```

```
par(mfrow=c(2,3))
plot(d_mean_of_beta_I,col="blue",main="raw beta",xlim=c(0,1),ylim=c(0,5))
lines(d_mean_of_beta_II,col="red")
plot(d_sd_of_beta_I,col="blue",main="raw sd",xlim=c(0,0.6),ylim=c(0,60))
lines(d_sd_of_beta_II,col="red")
boxplot(beta,ylim=c(0,1))
plot(d_mean_of_beta_preprocessQuantile_I,col="blue",main="preprocessQuantile
beta",xlim=c(0,1),ylim=c(0,5))
lines(d_mean_of_beta_preprocessQuantile_II,col="red")
plot(d_sd_of_beta_preprocessQuantile_I,col="blue",main="preprocessQuantile
sd",xlim=c(0,0.6),ylim=c(0,60))
lines(d_sd_of_beta_preprocessQuantile_II,col="red")
boxplot(beta_preprocessQuantile,ylim=c(0,1))
dev.off()
```

```
## quartz_off_screen
```

```
##
```

```
2
```


Try by yourself preprocessFunnorm, preprocessNoob and preprocessSWAN

DRD_2025_Lesson_4.1

CHUNK 1: PCA. PCA can be used as a diagnostic plot for the detection of outliers and batch effects.

We will use the function **prcomp()** to calculate the PCA on our matrix of normalized beta values.

Let's load our quantile-normalized DNA methylation data (Lesson 4) and the samplesheet with phenotypic information (Lesson 3):

```
load("~/Dropbox/DRD_2025/4/beta_preprocessQuantile.RData")
```

```
str(beta_preprocessQuantile)
```

```
##  num [1:485512, 1:8] 0.7391 0.6919 0.1347 0.1708 0.0585 ...
##  - attr(*, "dimnames")=List of 2
##    ..$ : chr [1:485512] "cg13869341" "cg14008030" "cg12045430"
##        "cg20826792" ...
##    ..$ : chr [1:8] "9344737127_R01C02" "9344737127_R03C02"
##        "9344737127_R06C01" "9344737127_R02C02" ...
```

```
pheno <-
read.csv("~/Dropbox/DRD_2025/3/Input_data/SampleSheet.csv",header=T,
stringsAsFactors=T)
```

```
str(pheno)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ Sample_ID   : int  21050 21060 22041 22051 21090 21100 23081 23091
##  $ Sample_Well : Factor w/ 8 levels "A11","A12","E11",...: 6 1 4 7 5 2
##  $ Array       : Factor w/ 8 levels "R01C01","R01C02",...: 2 5 8 4 3 7
##  $ Slide       : num  9.34e+09 9.34e+09 9.34e+09 9.34e+09 9.38e+09
##  ...
##  $ Group       : Factor w/ 2 levels "A","B": 1 1 2 2 1 1 2 2
##  $ Gender      : Factor w/ 2 levels "F","M": 1 1 1 1 1 1 1 2
##  $ BMI         : num  21.6 20.8 26.6 26.6 17.5 ...
```

Our beta matrix has samples in columns and CpG probes in rows: the **prcomp()** function should be applied to the transposed matrix, which is achieved using the **t()** function.

```
?prcomp
```

```
pca_results <- prcomp(t(beta_preprocessQuantile),scale=T)
```

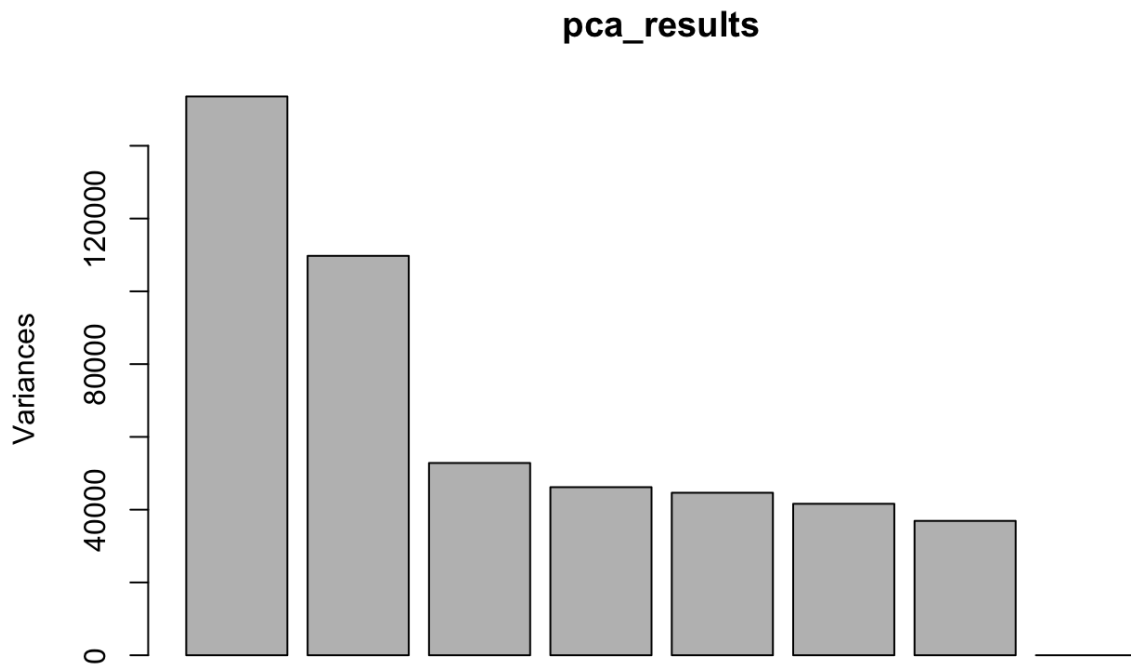
We can print and plot the variance accounted for each component

```
print(summary(pca_results))
```

```
## Importance of components:
```

##		PC1	PC2	PC3	PC4
PC5	PC6				
## Standard deviation		391.8592	331.2883	229.8494	214.90614
		211.30538	204.00466		
## Proportion of Variance		0.3163	0.2261	0.1088	0.09513
		0.09196	0.08572		
## Cumulative Proportion		0.3163	0.5423	0.6511	0.74627
		0.83823	0.92395		
##		PC7	PC8		
## Standard deviation		192.15406	3.88e-12		
## Proportion of Variance		0.07605	0.00e+00		
## Cumulative Proportion		1.00000	1.00e+00		

```
plot(pca_results)
```



We ask the str() of pca_object

```
str(pca_results)
```

```
## List of 5
```

```
## $ sdev      : num [1:8] 392 331 230 215 211 ...
## $ rotation: num [1:485512, 1:8] -0.000628 0.000745 0.000112
0.001532 0.000512 ...
## ..- attr(*, "dimnames")=List of 2
## .. ..$ : chr [1:485512] "cg13869341" "cg14008030" "cg12045430"
"cg20826792" ...
## .. ..$ : chr [1:8] "PC1" "PC2" "PC3" "PC4" ...
## $ center   : Named num [1:485512] 0.759 0.645 0.161 0.262 0.121 ...
## ..- attr(*, "names")= chr [1:485512] "cg13869341" "cg14008030"
"cg12045430" "cg20826792" ...
## $ scale    : Named num [1:485512] 0.038 0.0454 0.1077 0.0563 0.0791
...
## ..- attr(*, "names")= chr [1:485512] "cg13869341" "cg14008030"
"cg12045430" "cg20826792" ...
## $ x        : num [1:8, 1:8] -158.6 190.4 424.1 83.7 -837 ...
## ..- attr(*, "dimnames")=List of 2
## .. ..$ : chr [1:8] "9344737127_R01C02" "9344737127_R03C02"
"9344737127_R06C01" "9344737127_R02C02" ...
## .. ..$ : chr [1:8] "PC1" "PC2" "PC3" "PC4" ...
## - attr(*, "class")= chr "prcomp"
```

The principal components of interest are stored in the element named “x” of the list.

```
pca_results$x
```

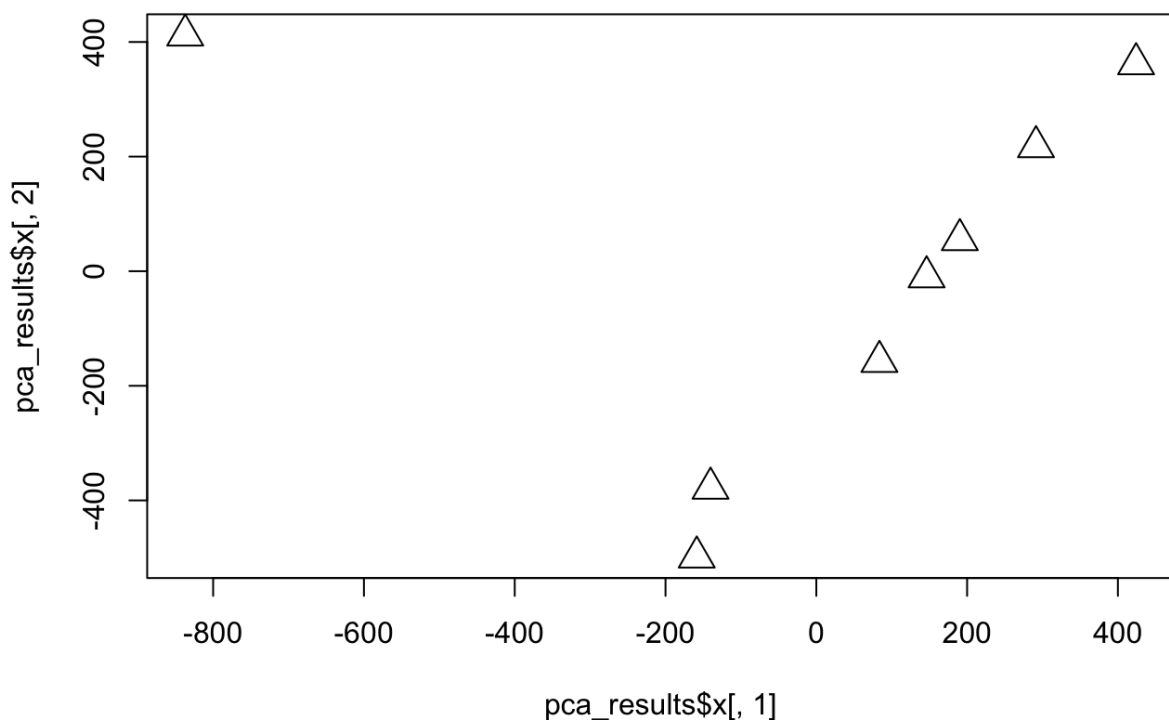
```
##          PC1          PC2          PC3          PC4
PC5
```

```
## 9344737127_R01C02 -158.58603 -499.02472 -275.41715532 122.618476
63.103916
## 9344737127_R03C02 190.36771 54.85588 -10.44934909 -422.344011
286.008070
## 9344737127_R06C01 424.12532 361.80718 -190.88012776 -38.629569 -
340.971394
## 9344737127_R02C02 83.72623 -157.69358 -165.95380244 121.382496
25.566631
## 9376538140_R02C01 -837.02417 411.90073 -27.57991961 6.732864
1.214454
## 9376538140_R05C01 -140.28255 -378.89468 382.74022937 -124.204288 -
259.654230
## 9376538140_R01C01 146.29094 -10.05157 0.09490818 21.863162
19.690139
## 9376538140_R04C01 291.38255 217.10075 287.44521666 312.580870
205.042413
```

##		PC6	PC7	PC8
##	9344737127_R01C02	195.6548455	-184.495680	2.740577e-12
##	9344737127_R03C02	77.9751406	26.604105	2.441508e-12
##	9344737127_R06C01	127.9446991	-39.219304	-5.711227e-12
##	9344737127_R02C02	-148.9712326	406.012545	7.395274e-13
##	9376538140_R02C01	0.3277534	6.308337	-2.850194e-12
##	9376538140_R05C01	32.9674550	41.149318	3.550466e-12
##	9376538140_R01C01	-431.4333949	-234.806100	1.403493e-12
##	9376538140_R04C01	145.5347339	-21.553222	5.454472e-12

We can plot PC1 and PC2 and check if samples cluster according to some variable; the argument “cex” defines the size of the dots, the argument “pch” the dot type:

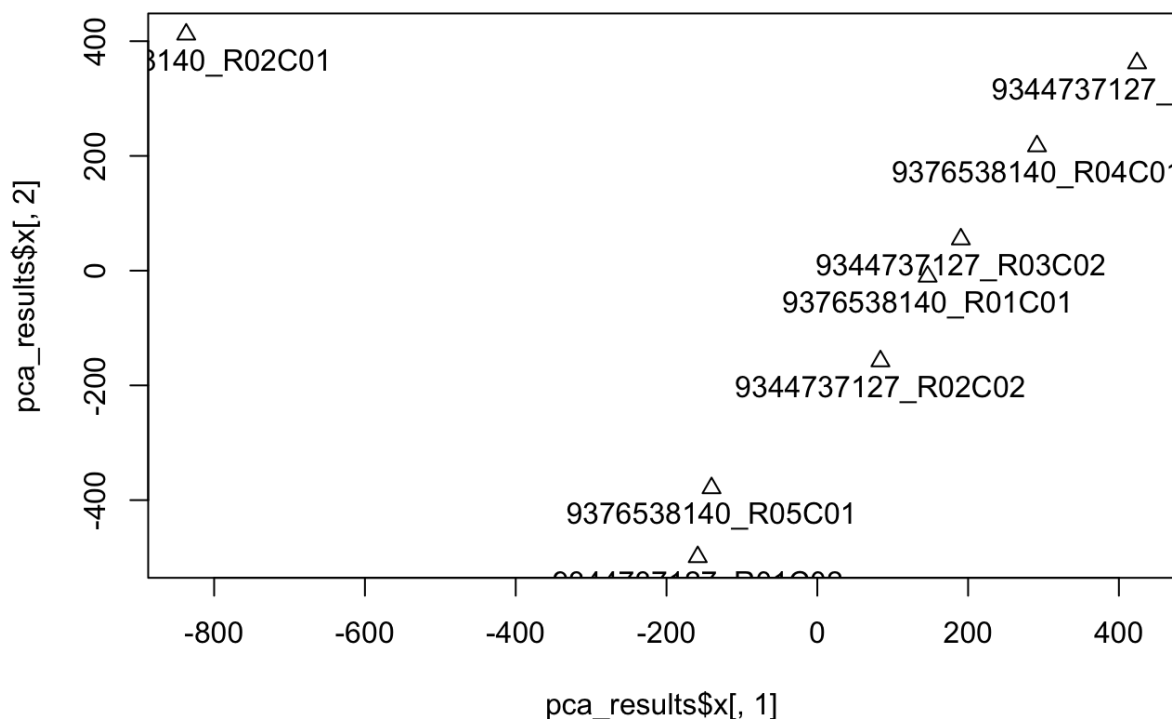
```
plot(pca_results$x[,1], pca_results$x[,2], cex=2, pch=2)
```



We can label the dots using the text() function and setting the “labels” argument to the rownames of pca_results\$x; the argument “pos” defines the position of the text:

```
?text
```

```
plot(pca_results$x[,1], pca_results$x[,2], cex=1, pch=2)
text(pca_results$x[,1], pca_results$x[,2], labels=rownames(pca_results$x), pos=1)
```



Sample 9376538140_R02C01 seems to be an outlier...

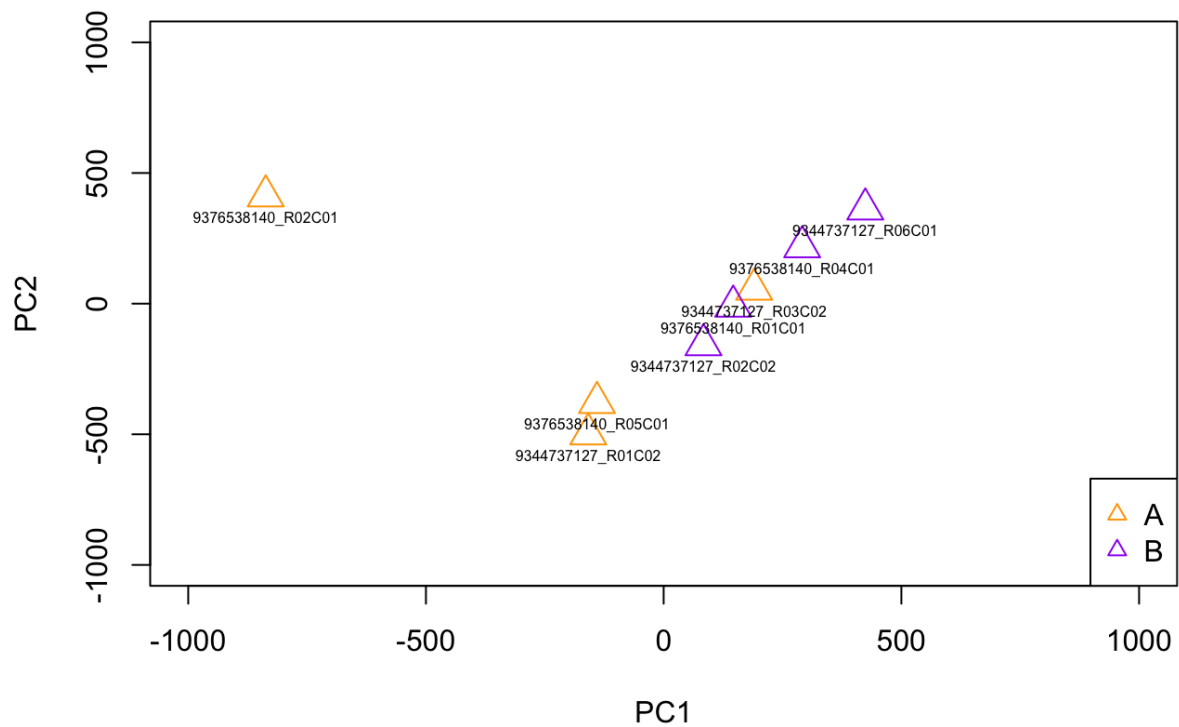
In this dataset we have few samples, so it is ok to plot their names. When you work with larger datasets, on the contrary, it is easier to colour the dots according to some column of the pheno file of interest, for example Group:

```
pheno$Group
## [1] A A B B A A B B
## Levels: A B
```

We define a palette of colours; each colour will be assigned to each levels of the factor according to the order of levels. For example, in this case Group A will be in orange, Group B in purple. We will also adjust the margins and add a legend to the plot.

```
levels(pheno$Group)
## [1] "A" "B"
palette(c("orange", "purple"))
```

```
plot(pca_results$x[,1],
pca_results$x[,2],cex=2,pch=2,col=pheno$Group,xlab="PC1",ylab="PC2",xlim=c(-
1000,1000),ylim=c(-1000,1000))
text(pca_results$x[,1],pca_results$x[,2],labels=rownames(pca_results$x),cex=0.5,pos=1)
legend("bottomright",legend=levels(pheno$Group),col=c(1:nlevels(pheno$Group)),pch=2)
```



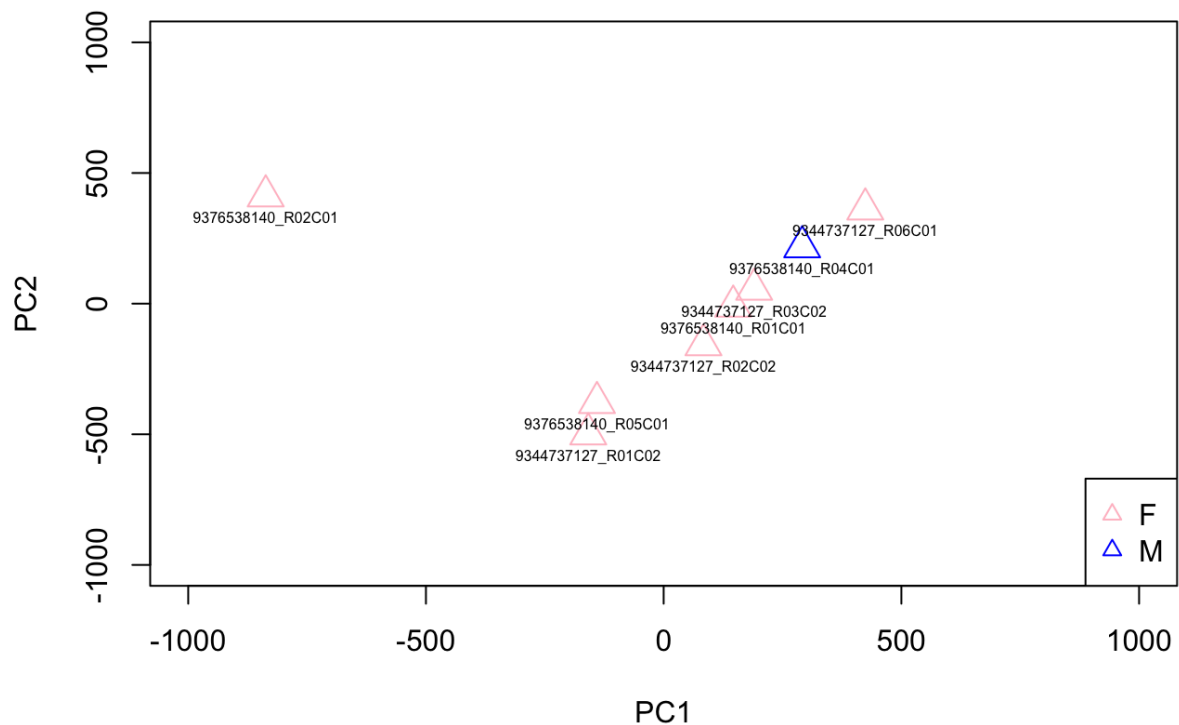
Another example coloring according to the sex of the subjects.

```
levels(pheno$Gender)
```

```
## [1] "F" "M"
```

```
palette(c("pink","blue"))
```

```
plot(pca_results$x[,1],
pca_results$x[,2],cex=2,pch=2,col=pheno$Gender,xlab="PC1",ylab="PC2",xlim=c(-
1000,1000),ylim=c(-1000,1000))
text(pca_results$x[,1],pca_results$x[,2],labels=rownames(pca_results$x),cex=0.5,pos=1)
legend("bottomright",legend=levels(pheno$Gender),col=c(1:nlevels(pheno$Group)),pch=2)
```



CHUNK 2: Batch Effect Removal. Omics experiments are sometimes affected by batch effects, that is unwanted variations in the data that arise due to external factors like experimental conditions or equipment, rather than biological differences.

Let's preview our data to check for batch effects:

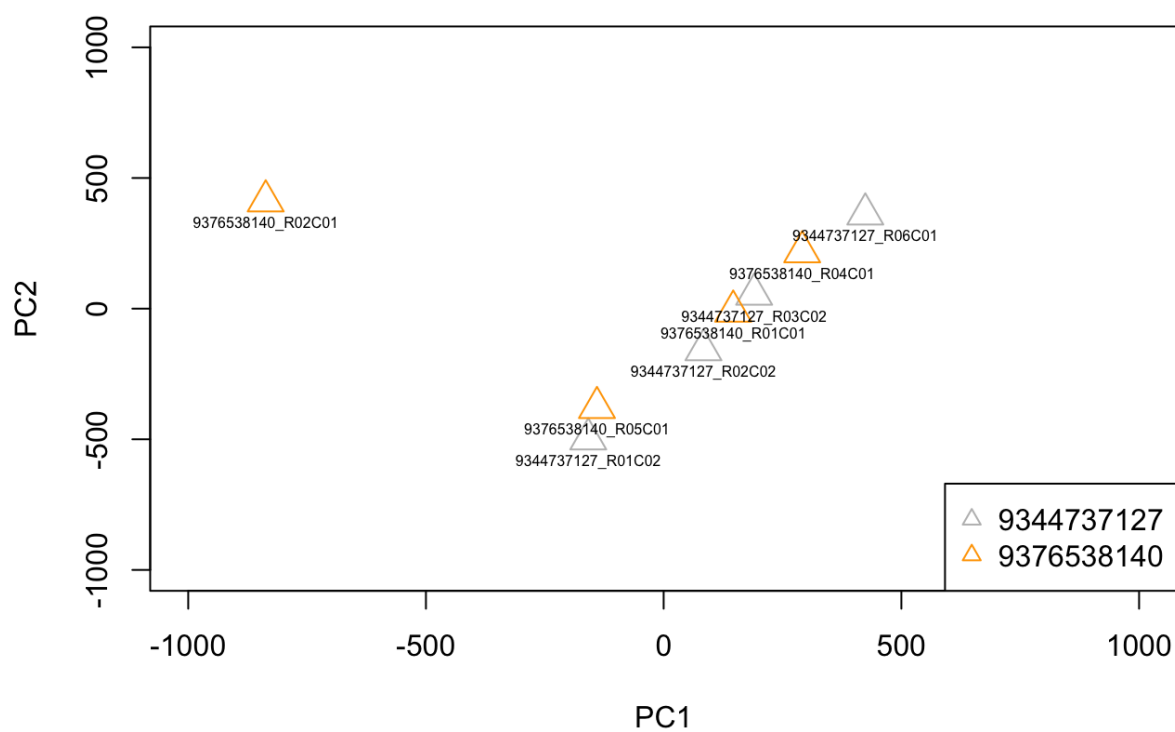
```
levels(pheno$Slide)
```

```
## NULL
```

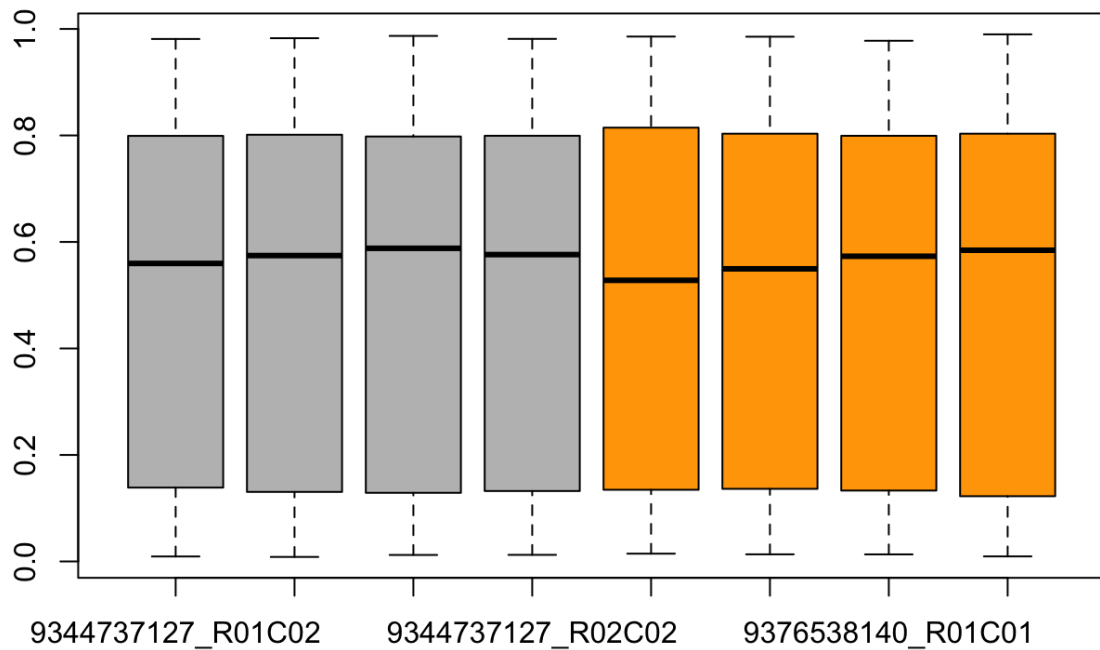
pheno\$Slide has no levels because it is read as a numeric vector. Convert it into a factor for correct pca plot coloring.

```
pheno$Slide <- factor(pheno$Slide)
```

```
palette(c("grey","orange"))
plot(pca_results$x[,1],
pca_results$x[,2],cex=2,pch=2,col=pheno$Slide,xlab="PC1",ylab="PC2",xlim=c(-
1000,1000),ylim=c(-1000,1000))
text(pca_results$x[,1],pca_results$x[,2],labels=rownames(pca_results$x),cex=0.5,pos=1)
legend("bottomright",legend=levels(pheno$Slide),col=c(1:nlevels(pheno$Group)),pch=2)
```

```
boxplot(beta_preprocessQuantile, col=pheno$Slide)
```



We do not see a clear batch effect in our dataset, good!

Although our data seems to be devoid of batch effects let's check what happens if we apply batch effect correction.

Batch effect correction on DNA methylation data is performed using the **ComBat()** function, which is part of the **sva** package.

ComBat() adjusts for known batches using an empirical Bayesian framework, helping to remove these batch effects and make the data more reliable for downstream analysis.

Sva package sva can be downloaded and installed by running:

```
BiocManager::install("sva")
```

ComBat() works on a matrix with features (in this case, methylation values) in the rows and samples in the columns. We will create a first model, in which we will include the variable in which we are interested in (in this case, Sample_Group) and an object named batch in which we will include the known source of batch (in this case, the column Slide).

```
library(sva)

## Loading required package: mgcv

## Loading required package: nlme
```

```
## This is mgcv 1.9-1. For overview type 'help("mgcv-package")'.
## Loading required package: genefilter
## Loading required package: BiocParallel

mod.null <- model.matrix(~as.factor(Group), data = pheno)

batch <- as.factor(pheno$Slide)
combat_edata <- ComBat(dat = beta_preprocessQuantile, batch = batch, mod = mod.null)
```

```
## Found2batches
## Adjusting for1covariate(s) or covariate level(s)
## Standardizing Data across genes
## Fitting L/S model and finding priors
## Finding parametric adjustments
## Adjusting the Data

str(combat_edata)

##  num [1:485512, 1:8] 0.7548 0.681 0.1618 0.18 0.0774 ...
## - attr(*, "dimnames")=List of 2
##  ..$ : chr [1:485512] "cg13869341" "cg14008030" "cg12045430"
##      "cg20826792" ...
##  ..$ : chr [1:8] "9344737127_R01C02" "9344737127_R03C02"
##      "9344737127_R06C01" "9344737127_R02C02" ...
```

```
Quantile_Combat_beta <- as.data.frame(combat_edata)
```

And now let's see how ComBat has modified our data! We will calculate again PCA and plot PC1 and PC2 according to Group, Gender and Slide.

```
Tbeta_ComBat <- t(Quantile_Combat_beta)

pca_result_ComBat <- prcomp(Tbeta_ComBat, scale = TRUE)

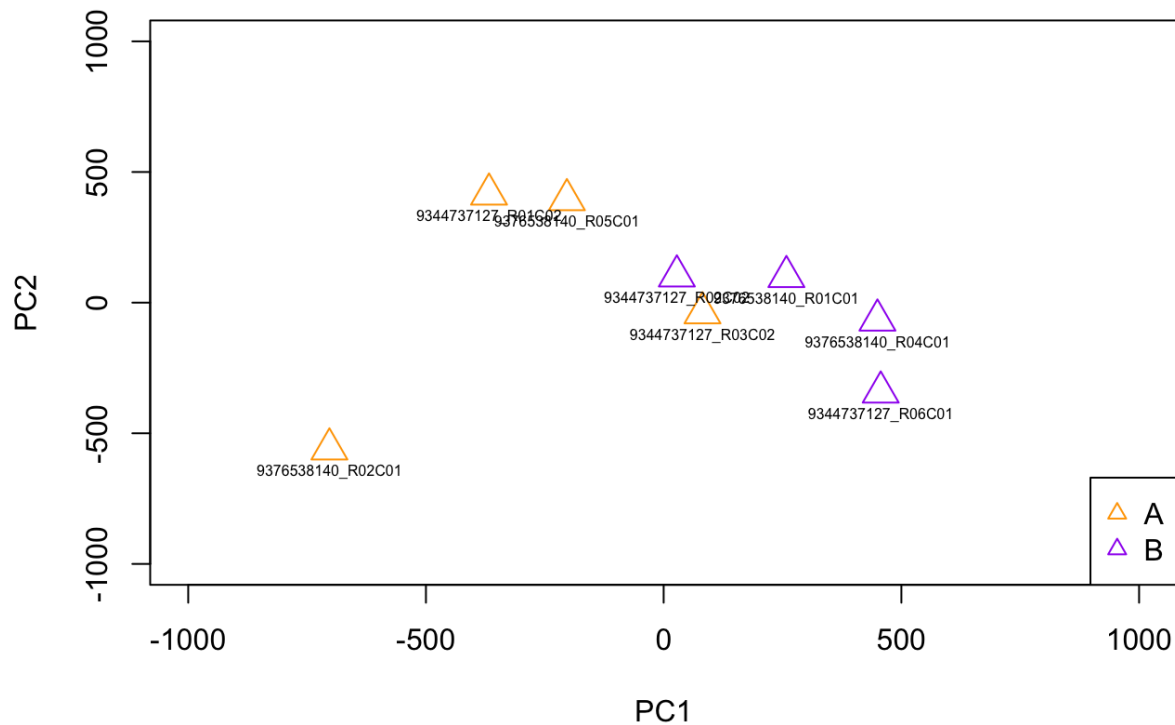
levels(pheno$Group)
```

```
## [1] "A" "B"

palette(c("orange", "purple"))

plot(pca_result_ComBat$x[,1],
pca_result_ComBat$x[,2], cex=2, pch=2, col=pheno$Group, xlab="PC1", ylab="PC2", xlim=c(-1000,1000), ylim=c(-1000,1000))
```

```
text(pca_result_ComBat$x[,1],
pca_result_ComBat$x[,2],labels=rownames(pca_result_ComBat$x),cex=0.5,pos=1)
legend("bottomright",legend=levels(pheno$Group),col=c(1:nlevels(pheno$Group)),pch=2)
```

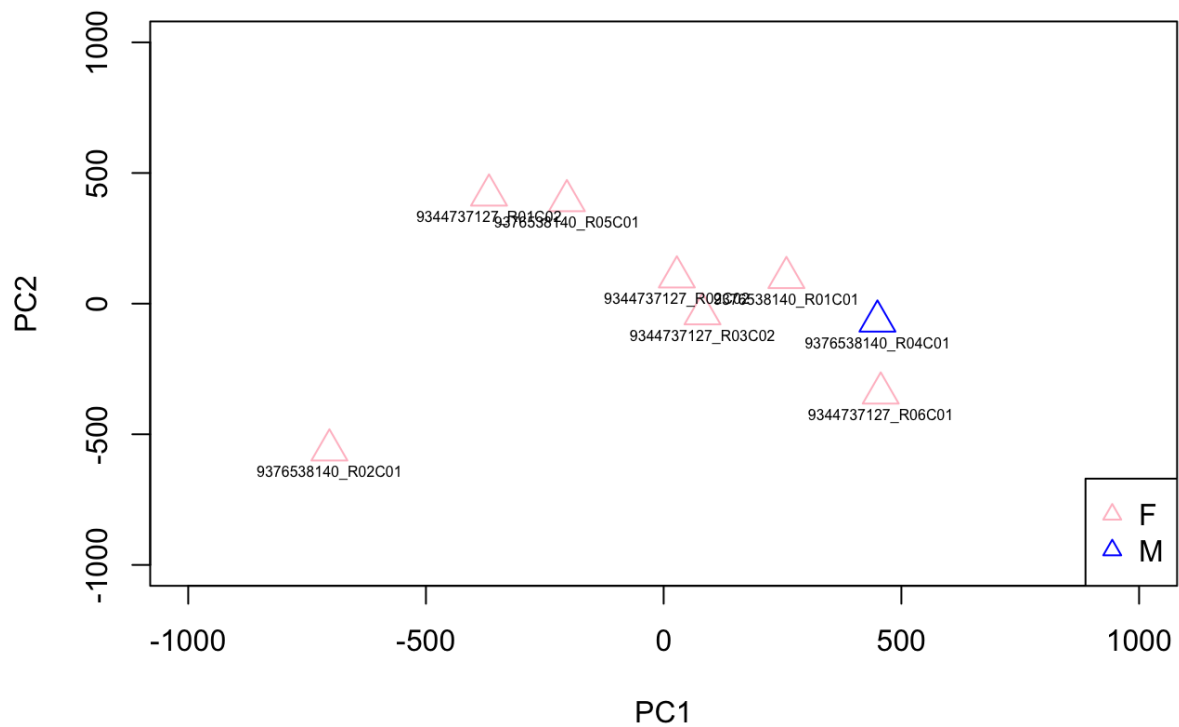


```
levels(pheno$Gender)
```

```
## [1] "F" "M"
```

```
palette(c("pink", "blue"))
```

```
plot(pca_result_ComBat$x[,1],
pca_result_ComBat$x[,2],cex=2,pch=2,col=pheno$Gender,xlab="PC1",ylab="PC2",xlim=c(-
1000,1000),ylim=c(-1000,1000))
text(pca_result_ComBat$x[,1],
pca_result_ComBat$x[,2],labels=rownames(pca_result_ComBat$x),cex=0.5,pos=1)
legend("bottomright",legend=levels(pheno$Gender),col=c(1:nlevels(pheno$Group)),pch=2)
```

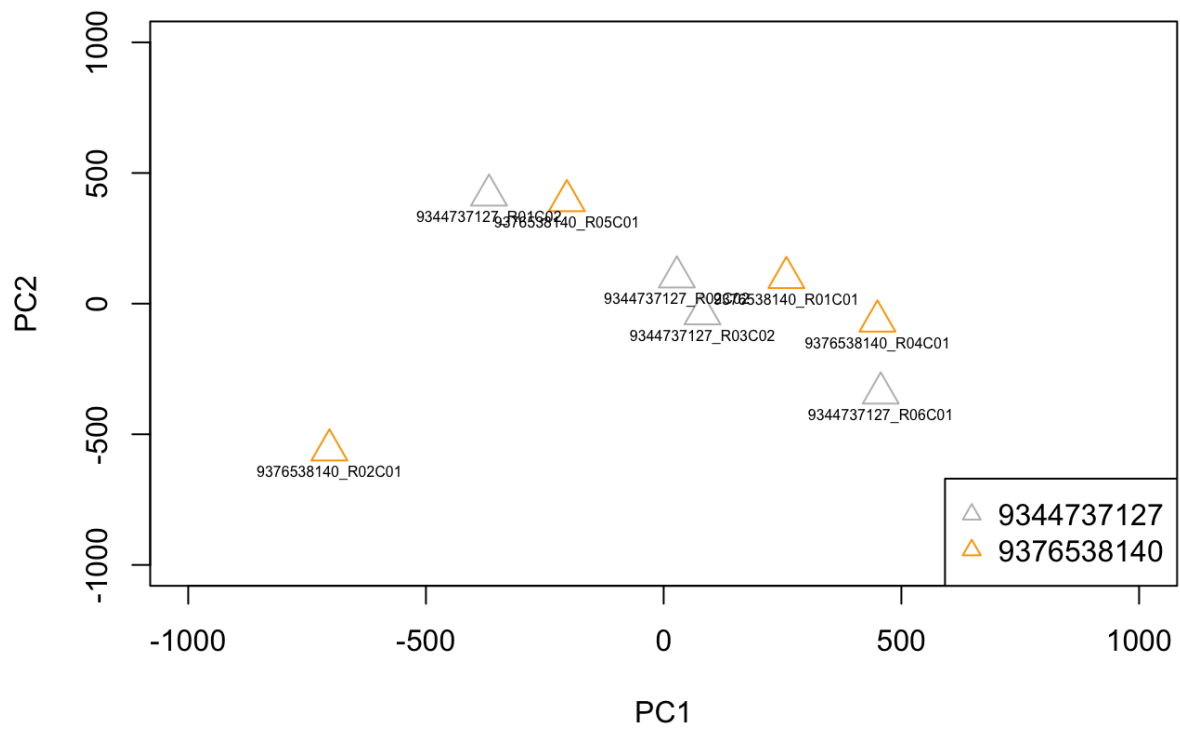


```
levels(pheno$Slide)
```

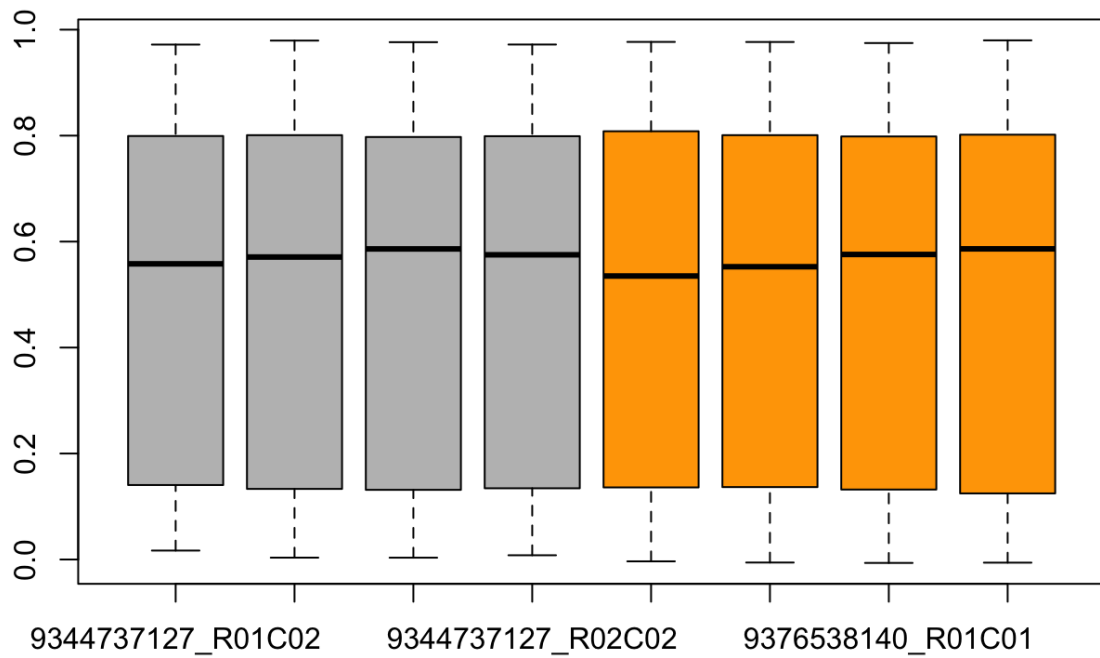
```
## [1] "9344737127" "9376538140"
```

```
palette(c("grey", "orange"))
```

```
plot(pca_result_ComBat$x[,1],
pca_result_ComBat$x[,2],cex=2,pch=2,col=pheno$Slide,xlab="PC1",ylab="PC2",xlim=c(-
1000,1000),ylim=c(-1000,1000))
text(pca_result_ComBat$x[,1],
pca_result_ComBat$x[,2],labels=rownames(pca_result_ComBat$x),cex=0.5,pos=1)
legend("bottomright",legend=levels(pheno$Slide),col=c(1:nlevels(pheno$Group)),pch=2)
```



```
boxplot(Quantile_Combat_beta,col=pheno$Slide)
```



Batch has a minimal effect on data variability. Batch effect correction did not result in large changes in beta value distribution nor in data variability.

Exercise 1: Load `beta_CRC.RData` and `Samplesheet_CRC.txt`. Plot beta value distribution (using boxplot) and Principal Components (using PCA plot) and check for batch effects. Do you notice a batch effect on data? Remove batch effect from data using `ComBat()` function. Check the dataset again using the plotting functions above. Are there any differences compared to uncorrected data?

DRD_2025_Lesson_5

In this lesson we will use preprocessQuantile as normalization method for our data. Let's recalculate beta values after preprocessQuantile normalization and read the file containing the samplesheet: We will recalculate the preprocessQuantile_results object

```
rm(list=ls())
```

```
setwd("~/Dropbox/DRD_2025/5")
```

```
suppressMessages(library(minfi))
```

```
## Warning: package 'GenomicRanges' was built under R version 4.4.1
```

```
## Warning: package 'S4Vectors' was built under R version 4.4.1
```

```
## Warning: package 'IRanges' was built under R version 4.4.1
```

```
## Warning: package 'matrixStats' was built under R version 4.4.1
```

```
## Warning: package 'locfit' was built under R version 4.4.1
```

```
load("~/Dropbox/DRD_2025/3/RGset.RData")
```

```
preprocessQuantile_results <- preprocessQuantile(RGset)
```

```
## [preprocessQuantile] Mapping to genome.
```

```
## Loading required package: IlluminaHumanMethylation450kmanifest
```

```
## Loading required package:
```

```
IlluminaHumanMethylation450kanno.ilmn12.hg19
```

```
## [preprocessQuantile] Fixing outliers.
```

```
## [preprocessQuantile] Quantile normalizing.
```

CHUNK 1: homoscedasticity and heteroscedasticity

We want to check for homoscedasticity and heteroscedasticity in beta and M values

```
beta_preprocessQuantile <- getBeta(preprocessQuantile_results)
```

```
M_preprocessQuantile <- getM(preprocessQuantile_results)
```

```
beta_preprocessQuantile_mean <- apply(beta_preprocessQuantile,1, mean,na.rm=T)
```

```
beta_preprocessQuantile_stddev <- apply(beta_preprocessQuantile,1, sd,na.rm=T)
```

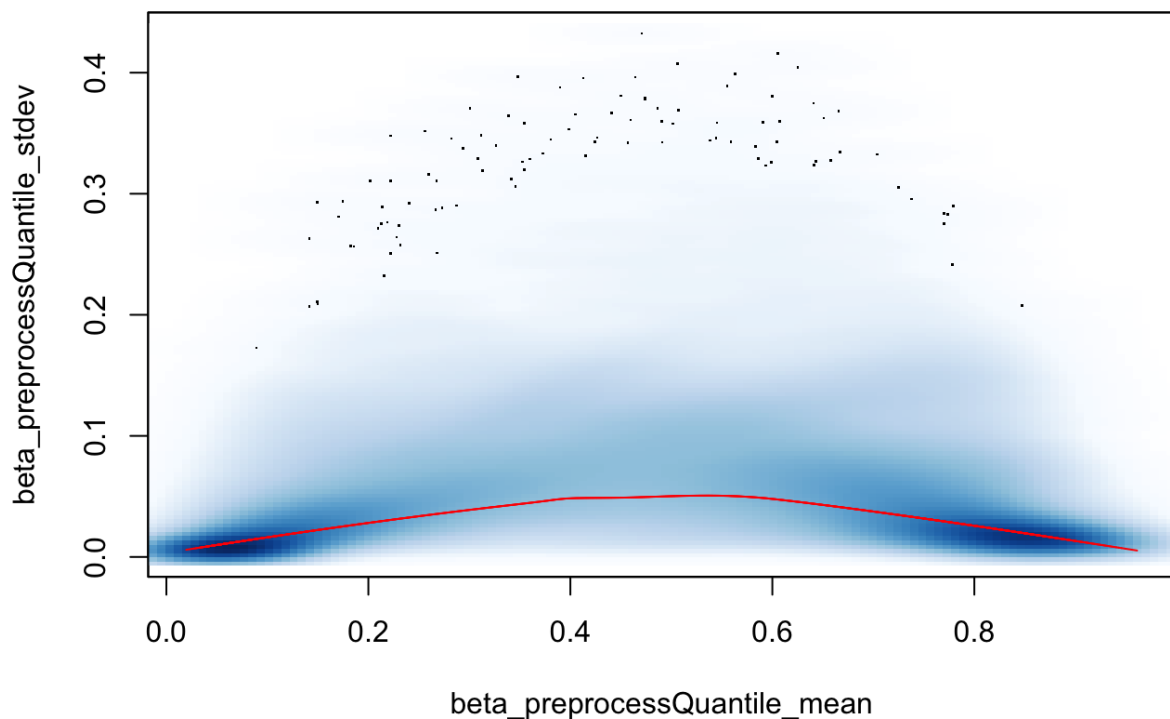


```
M_preprocessQuantile_mean <- apply(M_preprocessQuantile,1, mean,na.rm=T)
M_preprocessQuantile_stdev <- apply(M_preprocessQuantile,1, sd,na.rm=T)
```

Now we want to plot, for each probe, sd against mean. The plot will contain 485512 dots, they are really a lot. For this reason in this case it is more convenient to use the function `smoothScatter()` instead of `plot()`. The lowess function carries out a locally weighted regression of y on x

```
?smoothScatter
```

```
smoothScatter(beta_preprocessQuantile_mean, beta_preprocessQuantile_stdev)
?lowess
lines(lowess(beta_preprocessQuantile_mean, beta_preprocessQuantile_stdev), col="red")
#lowess carries out a locally weighted regression of y on x
```



```
pdf("Beta_M_heteoschedasticity_omoscedasticity.pdf")
```

```
par(mfrow=c(1,2))
smoothScatter(beta_preprocessQuantile_mean, beta_preprocessQuantile_stdev)
lines(lowess(beta_preprocessQuantile_mean, beta_preprocessQuantile_stdev), col="red")
#lowess carries out a locally weighted regression of y on x
```

```
smoothScatter(M_preprocessQuantile_mean, M_preprocessQuantile_stdev)
lines(lowess(M_preprocessQuantile_mean, M_preprocessQuantile_stdev), col="red")
dev.off()
```

```
## quartz_off_screen
```

```
## 2
```

For this dataset, we do not observe large differences between Beta and M values!

CHUNK 2: Identification of Differentially methylated positions

We will perform the differential methylation analysis on beta values from data normalized using preprocessQuantile. The analytical steps performed in this tutorial can be applied to any normalization method and to both beta values and M values.

Let's read the file containing the samplesheet:

```
pheno <-
read.csv("~/Dropbox/DRD_2025/3/Input_data/SampleSheet.csv", header=T,
stringsAsFactors=T)
```

```
str(pheno)
```

```
## 'data.frame': 8 obs. of 7 variables:
```

```
## $ Sample_ID : int 21050 21060 22041 22051 21090 21100 23081 23091
## $ Sample_Well: Factor w/ 8 levels "A11","A12","E11",...: 6 1 4 7 5 2
3 8
## $ Array      : Factor w/ 8 levels "R01C01","R01C02",...: 2 5 8 4 3 7
1 6
## $ Slide      : num 9.34e+09 9.34e+09 9.34e+09 9.34e+09 9.38e+09
...
## $ Group      : Factor w/ 2 levels "A","B": 1 1 2 2 1 1 2 2
## $ Gender     : Factor w/ 2 levels "F","M": 1 1 1 1 1 1 1 2
## $ BMI        : num 21.6 20.8 26.6 26.6 17.5 ...
```

Our aim is to identify the CpG probes that are differentially methylated between samples belonging to groups A and B. We will use a parametric test (t-test) and a non parametric test (Mann-Whitney test). Please note that these statistical approaches are very basic, and usually researchers adopt much more sophisticated statistical approaches, which allows also to take into account the effect of potential confounding factors. However, in our examples we will keep the things simple, and we will learn how to extract the info in which we are interested in (in this case, the p-value) from the output of a function.

Let's use the t-test to compare the first CpG of beta_preprocessQuantile between groups A and B:

```
? t.test
```

```
t_test <- t.test(beta_preprocessQuantile[1,] ~ pheno$Group)
t_test
```

```
##
```

```
## Welch Two Sample t-test
##
## data: beta_preprocessQuantile[1, ] by pheno$Group
## t = 0.44666, df = 3.7029, p-value = 0.68
## alternative hypothesis: true difference in means between group A and
group B is not equal to 0
## 95 percent confidence interval:
## -0.06905351 0.09454516
## sample estimates:
## mean in group A mean in group B
## 0.7654342 0.7526884
```

Let's use the Mann-Whitney test to compare the first CpG of beta_preprocessQuantile between groups A and B:

```
wilcox <- wilcox.test(beta_preprocessQuantile[1,]~ pheno$Group)
```

```
wilcox
```

```
##
```

```
## Wilcoxon rank sum exact test
##
## data: beta_preprocessQuantile[1, ] by pheno$Group
## W = 9, p-value = 0.8857
## alternative hypothesis: true location shift is not equal to 0
```

Note that both the t.test and the wicox.test functions return a list:

```
str(t_test)
```

```
## List of 10
```

```
## $ statistic : Named num 0.447
## ..- attr(*, "names")= chr "t"
## $ parameter : Named num 3.7
## ..- attr(*, "names")= chr "df"
## $ p.value : num 0.68
## $ conf.int : num [1:2] -0.0691 0.0945
```

```
##   ..- attr(*, "conf.level")= num 0.95
##   $ estimate   : Named num [1:2] 0.765 0.753
##   ..- attr(*, "names")= chr [1:2] "mean in group A" "mean in group
B"
##   $ null.value : Named num 0
##   ..- attr(*, "names")= chr "difference in means between group A and
group B"
##   $ stderr     : num 0.0285
##   $ alternative: chr "two.sided"
##   $ method     : chr "Welch Two Sample t-test"
##   $ data.name  : chr "beta_preprocessQuantile[1, ] by pheno$Group"
##   - attr(*, "class")= chr "htest"
```

```
str(wilcox)
```

```
## List of 7
```

```
##   $ statistic  : Named num 9
##   ..- attr(*, "names")= chr "W"
##   $ parameter  : NULL
##   $ p.value    : num 0.886
##   $ null.value : Named num 0
##   ..- attr(*, "names")= chr "location shift"
##   $ alternative: chr "two.sided"
##   $ method     : chr "Wilcoxon rank sum exact test"
##   $ data.name  : chr "beta_preprocessQuantile[1, ] by pheno$Group"
##   - attr(*, "class")= chr "htest"
```

We are interested in the “p.value” element of these lists.

```
t_test$p.value
```

```
## [1] 0.6799849
```

```
wilcox$p.value
```

```
## [1] 0.8857143
```

For 1 row (1 CpG probe) it is easy, but...how can I apply the 2 tests to each row of the dataframe (that is, to each probe of the microarray) and extract the p-value? We will use the `apply()` function, that we have already met in Lesson 4 to calculate the mean of each row of the matrix. However, unlike the `mean()` function, `t.test()` and `wilcox.test()` functions do not return a value, but a list; therefore, we have to create an ad hoc function:

```
My_ttest_function <- function(x) {
```

```
  t_test <- t.test(x~ pheno$Group)
  return(t_test$p.value)
```

```
}
```

Let's apply the function to few rows of beta_preprocessQuantile

```
firstbeta_preprocessQuantile <- beta_preprocessQuantile[1:10,]
```

```
pValues_ttest <- apply(firstbeta_preprocessQuantile,1, My_ttest_function)
pValues_ttest
```

```
## cg13869341 cg14008030 cg12045430 cg20826792 cg00381604 cg20253340
cg21870274
```

```
## 0.67998491 0.52253961 0.48786355 0.04189629 0.26590952 0.38041135
0.12745604
```

```
## cg03130891 cg24335620 cg16162899
```

```
## 0.64948694 0.26401633 0.37096580
```

Great, everything works fine!

Let's do the same with the non-parametric test

```
My_mannwhitney_function <- function(x) {
```

```
  wilcox <- wilcox.test(x~ pheno$Group)
```

```
  return(wilcox$p.value)
```

```
}
```

```
pValues_wilcox <- apply(firstbeta_preprocessQuantile,1, My_mannwhitney_function)
```

```
pValues_wilcox
```

```
## cg13869341 cg14008030 cg12045430 cg20826792 cg00381604 cg20253340
cg21870274
```

```
## 0.88571429 0.68571429 0.68571429 0.11428571 0.02857143 0.48571429
0.11428571
```

```
## cg03130891 cg24335620 cg16162899
```

```
## 0.68571429 0.68571429 0.48571429
```

You can apply your brand new functions to all the rows of the beta_preprocessQuantile matrix.

As this step can take several minutes, in this tutorial we will apply the parametric and non parametric functions only to the first 20000 probes

```
first20k_beta_preprocessQuantile <- beta_preprocessQuantile[1:20000,]
```

```
pValues_ttest_first20k <- apply(first20k_beta_preprocessQuantile,1, My_ttest_function)
length(pValues_ttest_first20k)
```

```
## [1] 20000
```

```
pValues_wilcox_first20k <- apply(first20k_beta_preprocessQuantile,1,
My_mannwhitney_function)
```

```
length(pValues_wilcox_first20k)
```

```
## [1] 20000
```

We can create a data.frame with all the beta values and the pValue column

```
final_ttest_first20k <- data.frame(first20k_beta_preprocessQuantile,
pValues_ttest_first20k)
```

```
head(final_ttest_first20k)
```

##	X9344737127_R01C02	X9344737127_R03C02	X9344737127_R06C01
## cg13869341	0.73914779	0.76950459	0.6995062
## cg14008030	0.69185593	0.64592974	0.6885158
## cg12045430	0.13474718	0.11699326	0.1004484
## cg20826792	0.17078769	0.28879485	0.2947075
## cg00381604	0.05846309	0.09592991	0.1045477
## cg20253340	0.42958351	0.51681548	0.4396129
##	X9344737127_R02C02	X9376538140_R02C01	X9376538140_R05C01
## cg13869341	0.7184184	0.78299289	0.77009158
## cg14008030	0.6031361	0.63207688	0.56279678
## cg12045430	0.1218535	0.14349114	0.12472814
## cg20826792	0.2819566	0.22464011	0.20651747
## cg00381604	0.1000693	0.09964109	0.08688974
## cg20253340	0.4760213	0.46439970	0.37458736
##	X9376538140_R01C01	X9376538140_R04C01	
pValues_ttest_first20k			
## cg13869341	0.7751171	0.8177119	
0.67998491			
## cg14008030	0.6861439	0.6460894	
0.52253961			
## cg12045430	0.4254814	0.1188433	
0.48786355			
## cg20826792	0.3439327	0.2847663	
0.04189629			
## cg00381604	0.3128018	0.1086204	
0.26590952			
## cg20253340	0.4502786	0.6004477	
0.38041135			

```
dim(final_ttest_first20k)
```

```
## [1] 20000 9
```

```
final_wilcox_first20k <- data.frame(first20k_beta_preprocessQuantile,
pValues_wilcox_first20k)
```

```
head(final_wilcox_first20k)
```

##	X9344737127_R01C02	X9344737127_R03C02	X9344737127_R06C01
## cg13869341	0.73914779	0.76950459	0.6995062
## cg14008030	0.69185593	0.64592974	0.6885158
## cg12045430	0.13474718	0.11699326	0.1004484
## cg20826792	0.17078769	0.28879485	0.2947075
## cg00381604	0.05846309	0.09592991	0.1045477
## cg20253340	0.42958351	0.51681548	0.4396129
##	X9344737127_R02C02	X9376538140_R02C01	X9376538140_R05C01
## cg13869341	0.7184184	0.78299289	0.77009158
## cg14008030	0.6031361	0.63207688	0.56279678
## cg12045430	0.1218535	0.14349114	0.12472814
## cg20826792	0.2819566	0.22464011	0.20651747
## cg00381604	0.1000693	0.09964109	0.08688974
## cg20253340	0.4760213	0.46439970	0.37458736
##	X9376538140_R01C01	X9376538140_R04C01	
pValues_wilcox_first20k			
## cg13869341	0.7751171	0.8177119	
0.88571429			
## cg14008030	0.6861439	0.6460894	
0.68571429			
## cg12045430	0.4254814	0.1188433	
0.68571429			
## cg20826792	0.3439327	0.2847663	
0.11428571			
## cg00381604	0.3128018	0.1086204	
0.02857143			
## cg20253340	0.4502786	0.6004477	
0.48571429			

```
dim(final_wilcox_first20k)
```

```
## [1] 20000 9
```

We can order the probes on the basis of the pValues column (from the smallest to the largest value)

```
final_ttest_first20k <-
final_ttest_first20k[order(final_ttest_first20k$pValues_ttest_first20k)
,]
```

```
head(final_ttest_first20k)
```

##	X9344737127_R01C02	X9344737127_R03C02	X9344737127_R06C01
## cg00147160	0.40907489	0.40395268	0.34677567
## cg14914422	0.32145688	0.30750562	0.24472666
## cg09473867	0.04168016	0.04317311	0.05235453
## cg25978218	0.20257688	0.25247419	0.45067145
## cg18154117	0.64307543	0.58601626	0.36963643
## cg11799650	0.20758273	0.22121794	0.34532280
##	X9344737127_R02C02	X9376538140_R02C01	X9376538140_R05C01
## cg00147160	0.36517947	0.42370284	0.4131399
## cg14914422	0.25437815	0.30339207	0.3290041
## cg09473867	0.05255956	0.04322238	0.0433248
## cg25978218	0.42926544	0.17350541	0.1796035
## cg18154117	0.36635971	0.57234962	0.6567593
## cg11799650	0.29937972	0.19707495	0.1733119
##	X9376538140_R01C01	X9376538140_R04C01	
pValues_ttest_first20k			
## cg00147160	0.34446583	0.34958276	
7.371467e-05			
## cg14914422	0.23800262	0.23057053	
9.750103e-05			
## cg09473867	0.05566484	0.05299677	
1.349507e-04			
## cg25978218	0.38048423	0.37558371	
1.977524e-04			
## cg18154117	0.42184015	0.43663288	
2.490115e-04			
## cg11799650	0.32566641	0.29110664	
4.283231e-04			

```
final_wilcox_first20k <-
final_wilcox_first20k[order(final_wilcox_first20k$pValues_wilcox_first20k),]
```

head(final_wilcox_first20k) #Note that in these settings the pvalues from Mann-Whitney tests are discrete

##	X9344737127_R01C02	X9344737127_R03C02	X9344737127_R06C01
## cg00381604	0.05846309	0.09592991	0.1045477
## cg17149495	0.72517404	0.73466157	0.7742136
## cg18024856	0.12749133	0.07603291	0.1354115
## cg08939496	0.79736900	0.77656754	0.6795802
## cg05662829	0.72568997	0.71536574	0.6859303
## cg08258224	0.70397508	0.68573484	0.8200589
##	X9344737127_R02C02	X9376538140_R02C01	X9376538140_R05C01
## cg00381604	0.1000693	0.09964109	0.08688974


```
## cg17149495      0.7457946      0.69211431      0.74052050
## cg18024856      0.1640822      0.09915869      0.11421504
## cg08939496      0.7183194      0.83588818      0.82138202
## cg05662829      0.6726331      0.81876726      0.74025011
## cg08258224      0.7226379      0.36607173      0.71147781
##                X9376538140_R01C01 X9376538140_R04C01
pValues_wilcox_first20k
## cg00381604      0.3128018      0.1086204
0.02857143
## cg17149495      0.7629406      0.7606389
0.02857143
## cg18024856      0.1523937      0.1553234
0.02857143
## cg08939496      0.7367617      0.7537639
0.02857143
## cg05662829      0.6493831      0.7063788
0.02857143
## cg08258224      0.7324056      0.7703751
0.02857143
```

How many probes have a pValue<=0.05?

```
final_ttest_first20k_0.05 <-
final_ttest_first20k[final_ttest_first20k$pValues_ttest_first20k<=0.05,
]
```

```
dim(final_ttest_first20k_0.05)
```

```
## [1] 1438      9
```

```
final_wilcox_first20k_0.05 <-
final_wilcox_first20k[final_wilcox_first20k$pValues_wilcox_first20k<=0.
05,]
```

```
dim(final_wilcox_first20k_0.05)
```

```
## [1] 1181      9
```

```
intersection <-
intersect(rownames(final_ttest_first20k_0.05),rownames(final_wilcox_fir
st20k_0.05))
```

```
length(intersection)
```

```
## [1] 827
```

CHUNK 3: dmpFinder

```
?dmpFinder
```

As you can see, dmpFinder is based on limma

```
pValues_dmpFinder_first20k <-  
dmpFinder(first20k_beta_preprocessQuantile, pheno=pheno$Group, type="cate  
gorical")
```

```
length(pValues_dmpFinder_first20k)
```

```
## [1] 4
```

```
head(pValues_dmpFinder_first20k)
```

##	intercept	f	pval	qval
## cg09473867	0.04285011	149.41918	1.824650e-05	0.2382118
## cg24254842	0.15665602	113.84081	3.997762e-05	0.2609579
## cg00147160	0.41246759	94.23881	6.857942e-05	0.2722893
## cg14914422	0.31533967	87.94943	8.342705e-05	0.2722893
## cg01399319	0.88251887	76.23728	1.248398e-04	0.2880072
## cg19247124	0.13458138	73.68354	1.373597e-04	0.2880072

```
table(rownames(final_ttest_first20k)==rownames(pValues_dmpFinder_first2  
0k))
```

```
##
```

```
## FALSE TRUE  
## 19836 164
```

We will use the merge() function to merge the final_ttest_first20k with the pValues_dmpFinder_first20k dataframe

```
final_dmpFinder_first20k <-  
merge(first20k_beta_preprocessQuantile, pValues_dmpFinder_first20k, by="r  
ow.names")
```

```
head(final_dmpFinder_first20k)
```

##	Row.names	9344737127_R01C02	9344737127_R03C02	9344737127_R06C01
## 1	cg00000957	0.81640937	0.82114722	0.8155907
## 2	cg00001446	0.82185854	0.80003893	0.7962052
## 3	cg00001534	0.90610172	0.91086961	0.9029241
## 4	cg00002028	0.06068736	0.06908112	0.0640245

```
## 5 cg00002593      0.77573976      0.83001325      0.8208749
## 6 cg00002837      0.43292499      0.36323490      0.3855876
## 9344737127_R02C02 9376538140_R02C01 9376538140_R05C01
9376538140_R01C01
## 1      0.81361638      0.8119120      0.82062518
0.81896818
## 2      0.80602512      0.8489746      0.83697850
0.79149513
## 3      0.88071301      0.9082726      0.90604794
0.89457867
## 4      0.08284857      0.1015421      0.07787474
0.08182502
## 5      0.81381464      0.7930347      0.80870725
0.80847452
## 6      0.42348142      0.4284411      0.38012955
0.41017220
## 9376538140_R04C01 intercept      f      pval      qval
## 1      0.82091943 0.81752343 0.008519631 0.92946294 0.6373615
## 2      0.81782946 0.82696263 3.989291107 0.09277006 0.3672238
## 3      0.90503726 0.90782297 4.545336550 0.07699854 0.3579074
## 4      0.07764969 0.07729634 0.005218851 0.94475758 0.6408768
## 5      0.81764123 0.80187374 1.265015352 0.30367555 0.4491793
## 6      0.41277305 0.40118263 0.126789954 0.73396941 0.5852405
```

```
rownames(final_dmpFinder_first20k) <- final_dmpFinder_first20k[,1]
```

```
head(final_dmpFinder_first20k)
```

```
##      Row.names 9344737127_R01C02 9344737127_R03C02
9344737127_R06C01
## cg00000957 cg00000957      0.81640937      0.82114722
0.8155907
## cg00001446 cg00001446      0.82185854      0.80003893
0.7962052
## cg00001534 cg00001534      0.90610172      0.91086961
0.9029241
## cg00002028 cg00002028      0.06068736      0.06908112
0.0640245
## cg00002593 cg00002593      0.77573976      0.83001325
0.8208749
## cg00002837 cg00002837      0.43292499      0.36323490
0.3855876
##      9344737127_R02C02 9376538140_R02C01 9376538140_R05C01
## cg00000957      0.81361638      0.8119120      0.82062518
## cg00001446      0.80602512      0.8489746      0.83697850
## cg00001534      0.88071301      0.9082726      0.90604794
```

```
## cg00002028      0.08284857      0.1015421      0.07787474
## cg00002593      0.81381464      0.7930347      0.80870725
## cg00002837      0.42348142      0.4284411      0.38012955
##                9376538140_R01C01 9376538140_R04C01 intercept
f
## cg00000957      0.81896818      0.82091943 0.81752343
0.008519631
## cg00001446      0.79149513      0.81782946 0.82696263
3.989291107
## cg00001534      0.89457867      0.90503726 0.90782297
4.545336550
## cg00002028      0.08182502      0.07764969 0.07729634
0.005218851
## cg00002593      0.80847452      0.81764123 0.80187374
1.265015352
## cg00002837      0.41017220      0.41277305 0.40118263
0.126789954
##                pval      qval
## cg00000957 0.92946294 0.6373615
## cg00001446 0.09277006 0.3672238
## cg00001534 0.07699854 0.3579074
## cg00002028 0.94475758 0.6408768
## cg00002593 0.30367555 0.4491793
## cg00002837 0.73396941 0.5852405
```

```
final_dmpFinder_first20k <-
final_dmpFinder_first20k[order(final_dmpFinder_first20k$pval),]
```

```
head(final_dmpFinder_first20k)
```

```
##                Row.names 9344737127_R01C02 9344737127_R03C02
9344737127_R06C01
## cg09473867 cg09473867      0.04168016      0.04317311
0.05235453
## cg24254842 cg24254842      0.15872861      0.15952195
0.28559818
## cg00147160 cg00147160      0.40907489      0.40395268
0.34677567
## cg14914422 cg14914422      0.32145688      0.30750562
0.24472666
## cg01399319 cg01399319      0.88334874      0.87516261
0.85595999
## cg19247124 cg19247124      0.14159854      0.11948331
0.06122449
##                9344737127_R02C02 9376538140_R02C01 9376538140_R05C01
## cg09473867      0.05255956      0.04322238      0.0433248
```

```
## cg24254842      0.28507633      0.14961596      0.1587576
## cg00147160      0.36517947      0.42370284      0.4131399
## cg14914422      0.25437815      0.30339207      0.3290041
## cg01399319      0.86013584      0.88462509      0.8869391
## cg19247124      0.06275233      0.15164837      0.1255953
##                9376538140_R01C01 9376538140_R04C01 intercept      f
## cg09473867      0.05566484      0.05299677 0.04285011 149.41918
## cg24254842      0.26475540      0.24213309 0.15665602 113.84081
## cg00147160      0.34446583      0.34958276 0.41246759  94.23881
## cg14914422      0.23800262      0.23057053 0.31533967  87.94943
## cg01399319      0.85879701      0.85423465 0.88251887  76.23728
## cg19247124      0.05856086      0.07477696 0.13458138  73.68354
##                pval      qval
## cg09473867 1.824650e-05 0.2382118
## cg24254842 3.997762e-05 0.2609579
## cg00147160 6.857942e-05 0.2722893
## cg14914422 8.342705e-05 0.2722893
## cg01399319 1.248398e-04 0.2880072
## cg19247124 1.373597e-04 0.2880072
```

How many probes have a pValue<=0.05?

```
final_dmpFinder_first20k_0.05 <-
final_dmpFinder_first20k[final_dmpFinder_first20k$pval<=0.05,]
dim(final_dmpFinder_first20k_0.05)
```

```
## [1] 1844  13
```

Lest's visualize the intersection between the 3 approaches that we used; we will prepare a Venn Diagram using the VennDiagram function

```
#install.packages("VennDiagram")
library(VennDiagram)
```

```
## Loading required package: grid
##
```

```
## Attaching package: 'grid'
```

```
## The following object is masked from 'package:Biostrings':
##
##      pattern
```

```
## Loading required package: futile.logger
```

```
?venn.diagram
```

```
venn.diagram(list(A=rownames(final_dmpFinder_first20k_0.05),B=rownames(final_ttest_first20k_0.05),C=rownames(final_wilcox_first20k_0.05)),filename = "Venn_differentially_methylated_positions.png")
```

```
## Warning in tiff(filename = filename, height = height, width = width, units =
```

```
## units, : compression is not supported for type = "quartz"
```

```
## [1] 1
```

CHUNK 4: analysis of a large dataset

Now let's try to perform parametric and non parametric tests on a larger dataset (chr18_large_example.RData). I have already prepared an R object containing chr18 beta values for 64 subjects, 32 normal and 32 with cancer. The pheno vector is:

```
pheno_large <- factor(c(rep("normal",32),rep("cancer",32)))
```

```
pheno_large
```

```
## [1] normal normal normal normal normal normal normal normal normal normal normal
```

```
## [11] normal normal normal normal normal normal normal normal normal normal normal
```

```
## [21] normal normal normal normal normal normal normal normal normal normal normal
```

```
## [31] normal normal cancer cancer cancer cancer cancer cancer cancer cancer cancer
```

```
## [41] cancer cancer cancer cancer cancer cancer cancer cancer cancer cancer cancer
```

```
## [51] cancer cancer cancer cancer cancer cancer cancer cancer cancer cancer cancer
```

```
## [61] cancer cancer cancer cancer
```

```
## Levels: cancer normal
```

```
load("~/Dropbox/DRD_2025/5/chr18_large_example.RData")
```

```
ls()
```

```
## [1] "beta_preprocessQuantile"
```

```
"beta_preprocessQuantile_mean"
```

```
## [3] "beta_preprocessQuantile_stdev" "chr18_large_example"
```

```
## [5] "final_dmpFinder_first20k"
"final_dmpFinder_first20k_0.05"
## [7] "final_ttest_first20k"          "final_ttest_first20k_0.05"
## [9] "final_wilcox_first20k"        "final_wilcox_first20k_0.05"
## [11] "first20k_beta_preprocessQuantile"
"firstbeta_preprocessQuantile"
## [13] "intersection"                  "M_preprocessQuantile"
## [15] "M_preprocessQuantile_mean"    "M_preprocessQuantile_stdev"
## [17] "My_mannwhitney_function"      "My_ttest_function"
## [19] "pheno"                        "pheno_large"
## [21] "preprocessQuantile_results"   "pValues_dmpFinder_first20k"
## [23] "pValues_ttest"               "pValues_ttest_first20k"
## [25] "pValues_wilcox"              "pValues_wilcox_first20k"
## [27] "RGset"                       "t_test"
## [29] "wilcox"
```

```
head(chr18_large_example)
```

##	Subject_1	Subject_2	Subject_3	Subject_4
Subject_5				
## cg00002236	0.051924210	0.039619170	0.0637307800	0.048701550
	0.043449720			
## cg00005215	0.918009400	0.913728000	0.8796654000	0.904910800
	0.817326000			
## cg00008713	0.079664810	0.078984420	0.1272162000	0.077300300
	0.080132220			
## cg00009970	0.012037380	0.028638370	0.0259769700	0.027369580
	0.014679520			
## cg00012238	0.227600200	0.250835100	0.4569388000	0.292086000
	0.255555700			
## cg00014996	0.002975828	0.005993401	0.0001673069	0.004635822
	0.003584911			
##	Subject_6	Subject_7	Subject_8	Subject_9 Subject_10
Subject_11				
## cg00002236	0.03030228	0.037410770	0.034971830	0.025557060 0.03552555
	0.03621641			
## cg00005215	0.87756100	0.900757600	0.893627000	0.905713000 0.91559640
	0.93575960			
## cg00008713	0.06193672	0.065214460	0.063196320	0.052319070 0.06272060
	0.06442876			
## cg00009970	0.01828937	0.029580250	0.010760880	0.018628200 0.01705869
	0.01616332			
## cg00012238	0.28803490	0.243381600	0.265967400	0.389922300 0.26373380
	0.42958300			
## cg00014996	0.00001000	0.003213845	0.001348037	0.003494899 0.00001000
	0.01117144			

##	Subject_12	Subject_13	Subject_14	Subject_15
Subject_16				
##	cg00002236	0.0322707700	0.028632920	0.03662413 0.03488821 0.032454840
##	cg00005215	0.9422333000	0.934184200	0.90361530 0.84092860 0.900810000
##	cg00008713	0.0517122300	0.056819270	0.05754937 0.05339904 0.062249670
##	cg00009970	0.0000100000	0.009129659	0.01381849 0.02276441 0.003073726
##	cg00012238	0.2527888000	0.285876700	0.33909950 0.28959050 0.264794200
##	cg00014996	0.0002651671	0.001601406	0.00001000 0.00001000 0.000010000
##	Subject_17	Subject_18	Subject_19	Subject_20
Subject_21				
##	cg00002236	0.034979200	0.042156240	0.036207260 0.045472950 0.030814280
##	cg00005215	0.895639600	0.896318100	0.833297200 0.858381900 0.830982300
##	cg00008713	0.060076410	0.068092610	0.062181280 0.070341220 0.048015460
##	cg00009970	0.010532600	0.009991204	0.018443180 0.009958929 0.007008213
##	cg00012238	0.253625700	0.336847100	0.262692400 0.237629300 0.292530300
##	cg00014996	0.001980767	0.000010000	0.004870268 0.005224958 0.002624116
##	Subject_22	Subject_23	Subject_24	Subject_25
Subject_26				
##	cg00002236	0.041679950	0.032258160	0.045818560 0.038748400 0.03135229
##	cg00005215	0.871341800	0.891431300	0.869692300 0.863139300 0.93137040
##	cg00008713	0.061713180	0.053883100	0.066658930 0.053977770 0.05735427
##	cg00009970	0.014470510	0.015085770	0.007678816 0.015429800 0.01373107
##	cg00012238	0.398408000	0.269042600	0.285924200 0.274741200 0.25528740
##	cg00014996	0.003057802	0.003585184	0.004300429 0.003861612 0.00001000
##	Subject_27	Subject_28	Subject_29	Subject_30
Subject_31				
##	cg00002236	0.035467290	0.039747520	0.04137253 0.039519660 0.043037190


```

## cg00005215 0.813336300 0.909070400 0.91627980 0.871833800
0.879089700
## cg00008713 0.060094160 0.057323430 0.06359669 0.089519930
0.060377450
## cg00009970 0.020371340 0.013279540 0.01089359 0.013727520
0.028805400
## cg00012238 0.344056700 0.247416600 0.36280680 0.261294700
0.347804100
## cg00014996 0.002990922 0.002332495 0.01066541 0.004906817
0.007073356
##          Subject_32  Subject_33 Subject_34 Subject_35 Subject_36
Subject_37
## cg00002236 0.039326080 0.037313830 0.03991271 0.04150001 0.04405512
0.039036860
## cg00005215 0.881567200 0.898110200 0.94529980 0.89368090 0.91535390
0.916642400
## cg00008713 0.064364110 0.070062770 0.08210083 0.08146181 0.09682240
0.083534320
## cg00009970 0.013750240 0.004395185 0.01235090 0.03336070 0.01315988
0.018451110
## cg00012238 0.289041600 0.317202300 0.24862060 0.30360210 0.36201610
0.273347400
## cg00014996 0.007462094 0.003517131 0.00001000 0.01004824 0.00001000
0.005368013
##          Subject_38  Subject_39 Subject_40  Subject_41 Subject_42
## cg00002236 0.035081750 0.037269440 0.03920398 0.020849250 0.03291355
## cg00005215 0.938190300 0.871466400 0.92743480 0.919204900 0.95964120
## cg00008713 0.061542210 0.063703080 0.06126447 0.058111820 0.07647511
## cg00009970 0.020765500 0.009240599 0.02083739 0.021089590 0.02344563
## cg00012238 0.399203600 0.259957700 0.32078710 0.298697700 0.24474260
## cg00014996 0.004166369 0.012067180 0.00001000 0.005493919 0.00001000
##          Subject_43  Subject_44 Subject_45  Subject_46 Subject_47
Subject_48
## cg00002236 0.032233890 0.033343340 0.02319633 0.030031940 0.03862547
0.03757882
## cg00005215 0.875626400 0.868339000 0.89915860 0.917872200 0.92195770
0.88903900
## cg00008713 0.061745300 0.064673200 0.05927679 0.071311720 0.05319103
0.06243812
## cg00009970 0.007667073 0.016537380 0.01793953 0.018567840 0.01789618
0.01076269
## cg00012238 0.330415700 0.325757400 0.29704450 0.325560100 0.38579150
0.35349610
## cg00014996 0.000010000 0.004839076 0.00001000 0.003260774 0.00001000
0.00001000

```

##	Subject_49	Subject_50	Subject_51	Subject_52
Subject_53				
## cg00002236	0.044049100	0.042025550	0.044541880	0.04887654
0.033860990				
## cg00005215	0.912593800	0.914848400	0.936310400	0.90937550
0.896392900				
## cg00008713	0.070532150	0.071106990	0.062444550	0.06370760
0.065280940				
## cg00009970	0.014152220	0.008199104	0.012201910	0.01877461
0.010818970				
## cg00012238	0.351483900	0.321948800	0.337013400	0.34276180
0.308555000				
## cg00014996	0.002529265	0.000010000	0.004549827	0.00001000
0.006197025				
##	Subject_54	Subject_55	Subject_56	Subject_57
Subject_58				
## cg00002236	0.033130580	0.03120043	0.02856003	0.0356676900
0.048481820				
## cg00005215	0.911251100	0.89026690	0.88881330	0.9349962000
0.860565800				
## cg00008713	0.068906190	0.06311023	0.05940491	0.0617433300
0.062926880				
## cg00009970	0.015216810	0.01086817	0.02214076	0.0165400300
0.021033940				
## cg00012238	0.416942200	0.33220370	0.35992730	0.4094517000
0.271162900				
## cg00014996	0.001156038	0.00001000	0.00001000	0.0003517051
0.009857366				
##	Subject_59	Subject_60	Subject_61	Subject_62
Subject_63				
## cg00002236	0.031536300	0.049171310	0.046699040	0.044465670
0.04517000				
## cg00005215	0.867225500	0.927497800	0.921970600	0.886995500
0.89501710				
## cg00008713	0.056424860	0.057986950	0.060974510	0.072856330
0.06139426				
## cg00009970	0.014459640	0.019488790	0.014169150	0.012118910
0.01941472				
## cg00012238	0.368183800	0.316199600	0.340919100	0.270603100
0.31449200				
## cg00014996	0.003455899	0.003903026	0.003366298	0.003259631
0.00001000				
##	Subject_64			
## cg00002236	0.031395640			
## cg00005215	0.925778900			
## cg00008713	0.059795820			

```
## cg00009970 0.017467720
## cg00012238 0.296342500
## cg00014996 0.006306296
```

Build a function to perform ttest and a function to perform Mann-Whitney test

DRD_2025_Lesson_6

```
rm(list=ls())
```

```
setwd("~/Dropbox/DRD_2025/6")
```

For these last analyses, we will use the object `final_ttest_first20k`, that we previously prepared by applying the `ttest` function to the first 20000 rows of the `beta_preprocessQuantile` dataframe. If it is not present in your workspace or in Lesson 5 folder, you can download it from Lesson 6 folder:

```
load("final_ttest_first20k.RData")
```

```
str(final_ttest_first20k)
```

```
## 'data.frame':    20000 obs. of  9 variables:
##  $ X9344737127_R01C02      : num  0.4091 0.3215 0.0417 0.2026 0.6431
##  ...
##  $ X9344737127_R03C02      : num  0.404 0.3075 0.0432 0.2525 0.586 ...
##  $ X9344737127_R06C01      : num  0.3468 0.2447 0.0524 0.4507 0.3696
##  ...
##  $ X9344737127_R02C02      : num  0.3652 0.2544 0.0526 0.4293 0.3664
##  ...
##  $ X9376538140_R02C01      : num  0.4237 0.3034 0.0432 0.1735 0.5723
##  ...
##  $ X9376538140_R05C01      : num  0.4131 0.329 0.0433 0.1796 0.6568
##  ...
##  $ X9376538140_R01C01      : num  0.3445 0.238 0.0557 0.3805 0.4218
##  ...
##  $ X9376538140_R04C01      : num  0.35 0.231 0.053 0.376 0.437 ...
##  $ pValues_ttest_first20k: num  7.37e-05 9.75e-05 1.35e-04 1.98e-04
##  2.49e-04 ...
```

```
ls()
```

```
## [1] "final_ttest_first20k"
```

If the `pheno` object is not present in your workspace, you can load it again from Lesson 3 folder:

```
pheno <-
read.csv("~/Dropbox/DRD_2025/3/Input_data/SampleSheet.csv", header=T,
stringsAsFactors=T)
```

```
str(pheno)
```

```
## 'data.frame':      8 obs. of  7 variables:

## $ Sample_ID : int  21050 21060 22041 22051 21090 21100 23081 23091
## $ Sample_Well: Factor w/ 8 levels "A11","A12","E11",...: 6 1 4 7 5 2
3 8
## $ Array      : Factor w/ 8 levels "R01C01","R01C02",...: 2 5 8 4 3 7
1 6
## $ Slide      : num  9.34e+09 9.34e+09 9.34e+09 9.34e+09 9.38e+09
...
## $ Group      : Factor w/ 2 levels "A","B": 1 1 2 2 1 1 2 2
## $ Gender     : Factor w/ 2 levels "F","M": 1 1 1 1 1 1 1 2
## $ BMI        : num  21.6 20.8 26.6 26.6 17.5 ...
```

CHUNK 1: multiple test correction

We will use the function `p.adjust` to perform the multiple test correction. For the sake of simplicity, we will focus on the results of the `ttest`, but you can apply the same procedure also to the p-values resulting from the non parametric test.

```
?p.adjust
```

We will apply the Benjamini & Hochberg and the Bonferroni corrections:

```
corrected_pValues_BH <-
p.adjust(final_ttest_first20k$pValues_ttest_first20k, "BH")

corrected_pValues_Bonf <- p.adjust(final_ttest_first20k$pValues_ttest_first20k, "bonferroni")
final_ttest_first20k_corrected <- data.frame(final_ttest_first20k, corrected_pValues_BH,
corrected_pValues_Bonf)
head(final_ttest_first20k_corrected)
```

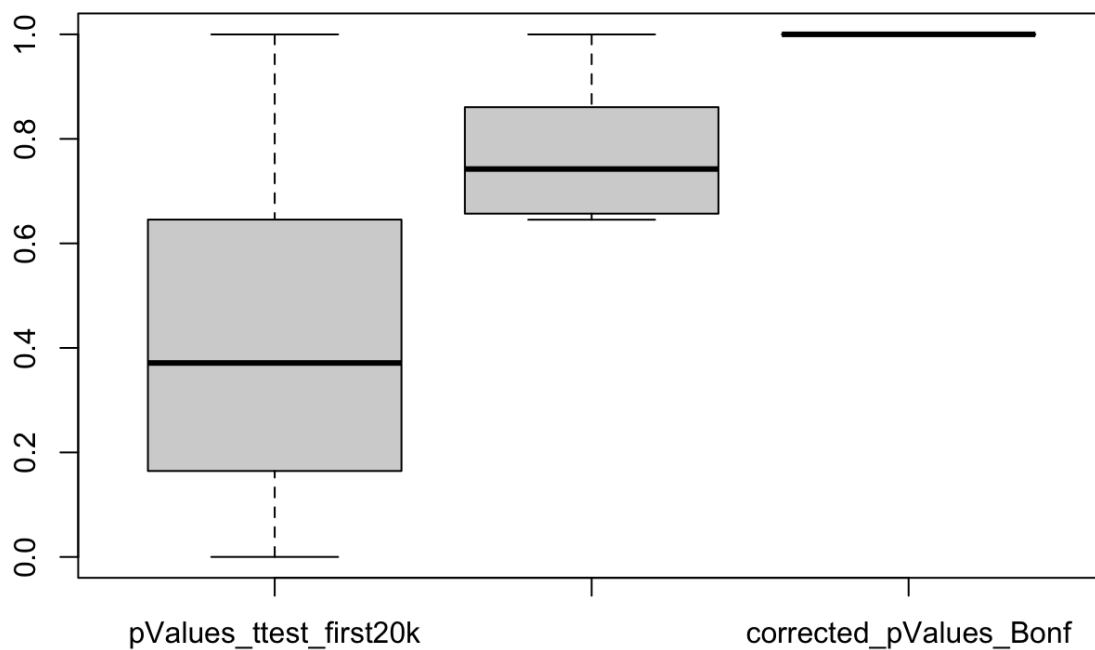
```
##           X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01

## cg00147160          0.40907489          0.40395268          0.34677567
## cg14914422          0.32145688          0.30750562          0.24472666
## cg09473867          0.04168016          0.04317311          0.05235453
## cg25978218          0.20257688          0.25247419          0.45067145
## cg18154117          0.64307543          0.58601626          0.36963643
## cg11799650          0.20758273          0.22121794          0.34532280
##           X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## cg00147160          0.36517947          0.42370284          0.4131399
## cg14914422          0.25437815          0.30339207          0.3290041
## cg09473867          0.05255956          0.04322238          0.0433248
## cg25978218          0.42926544          0.17350541          0.1796035
## cg18154117          0.36635971          0.57234962          0.6567593
```

```
## cg11799650      0.29937972      0.19707495      0.1733119
##              X9376538140_R01C01 X9376538140_R04C01
pValues_ttest_first20k
## cg00147160      0.34446583      0.34958276
7.371467e-05
## cg14914422      0.23800262      0.23057053
9.750103e-05
## cg09473867      0.05566484      0.05299677
1.349507e-04
## cg25978218      0.38048423      0.37558371
1.977524e-04
## cg18154117      0.42184015      0.43663288
2.490115e-04
## cg11799650      0.32566641      0.29110664
4.283231e-04
##              corrected_pValues_BH corrected_pValues_Bonf
## cg00147160      0.6453685      1
## cg14914422      0.6453685      1
## cg09473867      0.6453685      1
## cg25978218      0.6453685      1
## cg18154117      0.6453685      1
## cg11799650      0.6453685      1
```

We can visualize the distributions of the p-values and of the corrected p-values by boxplots:

```
boxplot(final_ttest_first20k_corrected[,9:11])
```



How many probes survive the multiple test correction?

```
dim(final_ttest_first20k_corrected[final_ttest_first20k_corrected$pValues_ttest<=0.05,])
## [1] 1438 11

dim(final_ttest_first20k_corrected[final_ttest_first20k_corrected$corrected_pValues_BH<=0.05,])
## [1] 0 11

dim(final_ttest_first20k_corrected[final_ttest_first20k_corrected$corrected_pValues_Bonf<=0.05,])
## [1] 0 11
```

CHUNK 2: Volcano plots

First of all, we have to calculate the difference between the average of group A values and the average of group B values. To this aim, we will first create two matrixes containing the beta-

values of group A and group B samples, and then we will calculate the mean within each group for each row.

```
pheno$Group
```

```
## [1] A A B B A A B B
```

```
## Levels: A B
```

```
beta_first20k <- final_ttest_first20k_corrected[,1:8]
```

```
beta_first20k_groupA <- beta_first20k[,pheno$Group=="A"]
```

```
mean_beta_first20k_groupA <- apply(beta_first20k_groupA,1,mean)
```

```
beta_first20k_groupB <- beta_first20k[,pheno$Group=="B"]
```

```
mean_beta_first20k_groupB <- apply(beta_first20k_groupB,1,mean)
```

Now we can calculate the difference between average values:

```
delta_first20k <- mean_beta_first20k_groupB-mean_beta_first20k_groupA
```

```
head(delta_first20k)
```

```
## cg00147160 cg14914422 cg09473867 cg25978218 cg18154117  
cg11799650
```

```
## -0.06096666 -0.07342018 0.01054381 0.20696121 -0.21593287  
0.11557201
```

Now we create a dataframe with two columns, one containing the delta values and the other with the -log10 of p-values

```
toVolcPlot <- data.frame(delta_first20k, -  
log10(final_ttest_first20k_corrected$pValues_ttest))
```

```
head(toVolcPlot)
```

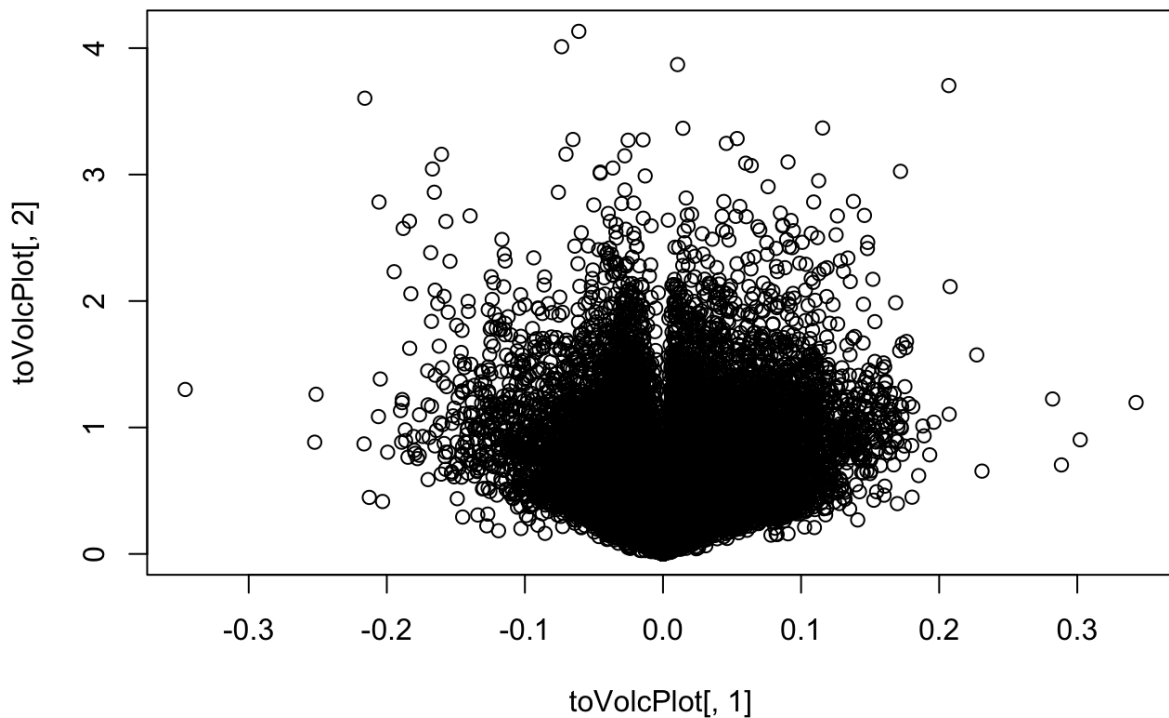
```
##          delta_first20k  
X.log10.final_ttest_first20k_corrected.pValues_ttest.
```

```
## cg00147160    -0.06096666  
4.132446  
## cg14914422    -0.07342018  
4.010991  
## cg09473867     0.01054381  
3.869825  
## cg25978218     0.20696121  
3.703878  
## cg18154117    -0.21593287  
3.603781
```



```
## cg11799650      0.11557201
3.368229
```

```
plot(toVolcPlot[,1], toVolcPlot[,2])
```



Let's improve a bit the plot: <http://www.statmethods.net/advgraphs/parameters.html> We can also add a threshold for pvalue significance (for example, 0.01) by using the abline function

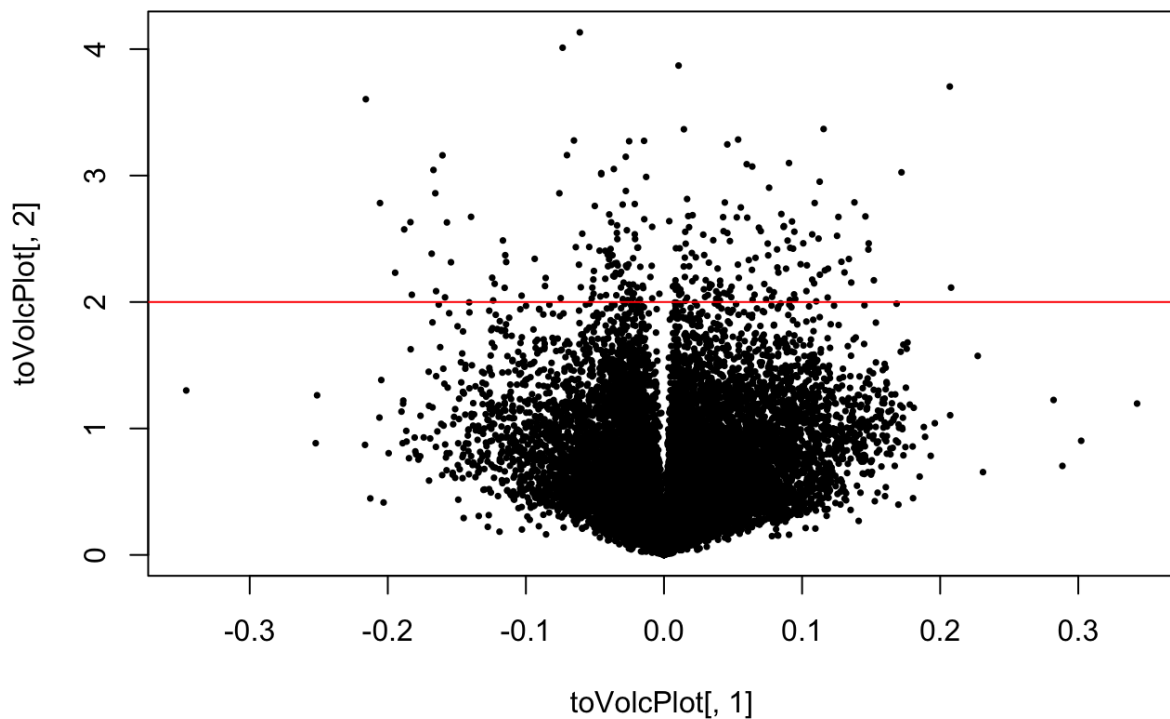
```
plot(toVolcPlot[,1], toVolcPlot[,2], pch=16, cex=0.5)
```

```
-log10(0.01)
```

```
## [1] 2
```

```
?abline
```

```
abline(h=-log10(0.01), col="red")
```



Now I want to highlight the probes (that is, the points), that have a nominal pValue<0.01 and a delta > 0.1)

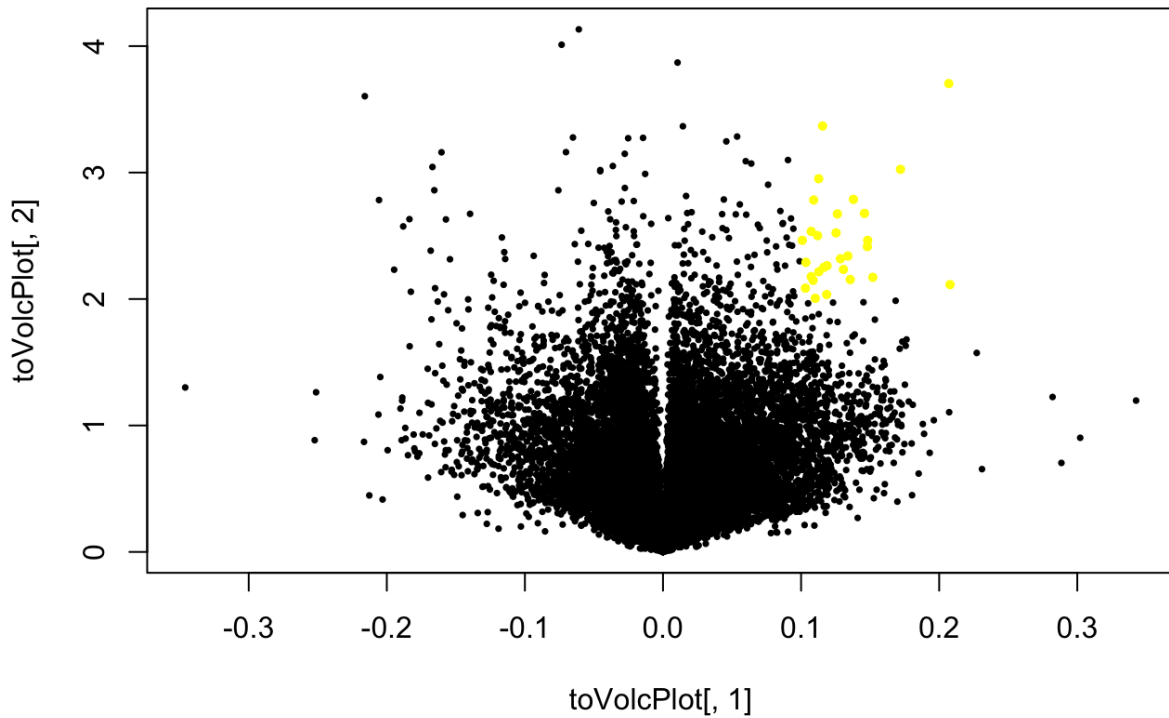
```
plot(toVolcPlot[,1], toVolcPlot[,2],pch=16,cex=0.5)
```

```
toHighlight <- toVolcPlot[toVolcPlot[,1]>0.1 & toVolcPlot[,2]>(-log10(0.01)),]
head(toHighlight)
```

```
##          delta_first20k
X.log10.final_ttest_first20k_corrected.pValues_ttest.
## cg25978218      0.2069612
3.703878
## cg11799650      0.1155720
3.368229
## cg04542977      0.1719873
3.025450
## cg24254842      0.1127347
2.951444
## cg04885881      0.1379048
2.787783
```

```
## cg02367921      0.1091145
2.783046
```

```
points(toHighlight[,1], toHighlight[,2],pch=16,cex=0.7,col="yellow")
```



If I want to highlight the points with an absolute delta > 0.01

```
plot(toVolcPlot[,1], toVolcPlot[,2],pch=16,cex=0.5)
```

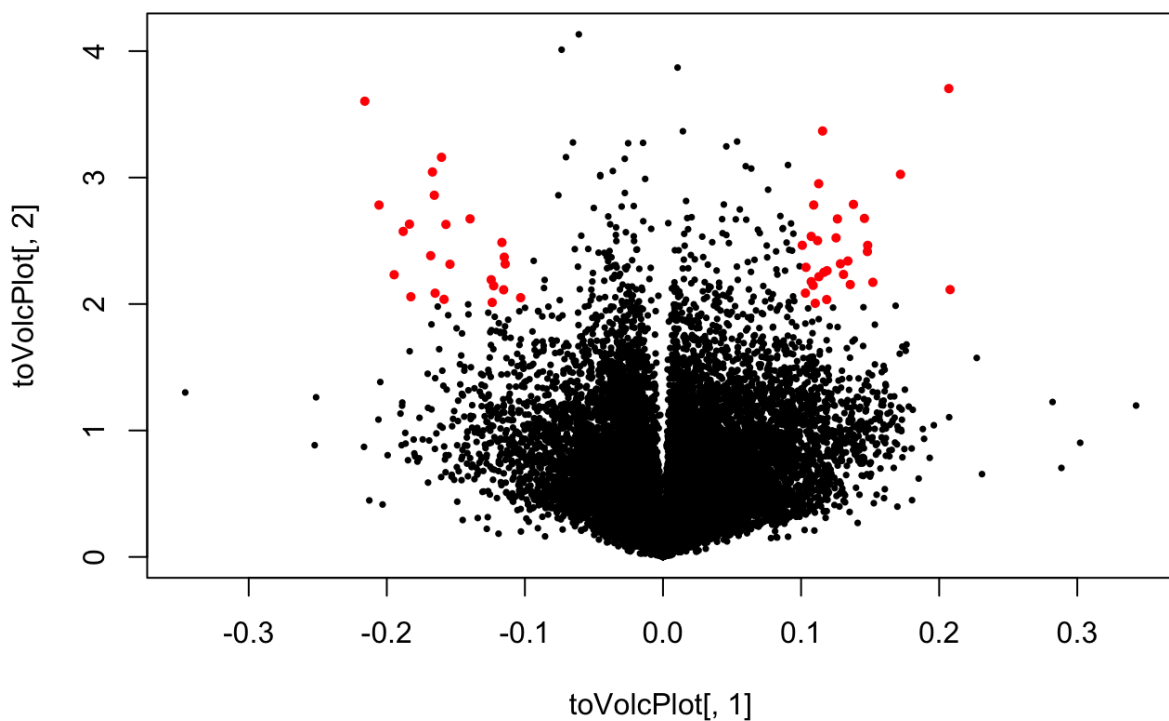
```
toHighlight <- toVolcPlot[abs(toVolcPlot[,1])>0.1 & toVolcPlot[,2]>(-log10(0.01)),]
head(toHighlight)
```

```
##          delta_first20k
X.log10.final_ttest_first20k_corrected.pValues_ttest.
```

```
## cg25978218      0.2069612
3.703878
## cg18154117     -0.2159329
3.603781
## cg11799650      0.1155720
3.368229
```

```
## cg00569500      -0.1604060
3.160202
## cg16125214      -0.1669493
3.044163
## cg04542977       0.1719873
3.025450
```

```
points(toHighlight[,1], toHighlight[,2],pch=16,cex=0.7,col="red")
```



CHUNK 3: Manhattan plots

We will use the “qqman” package, from the CRAN repository: to install it type:
`install.packages("qqman")`

```
library(qqman)

##

## For example usage please run: vignette('qqman')

##
```



```

## 1
II
## 2 ACGAAAAACAACGCACAACCTATAATAATTTTTAAAATAAATAAACCCCG
I
## 3
II
## 4
II
## 5
II
## 6 CCCAATTAACCGCAAAACTAAACAAATTATACGATCGAAAAACGTACG
I
##   Next_Base Color_Channel
## 1
## 2           A           Red
## 3
## 4
## 5
## 6           T           Red
##
Forward_Sequence
## 1
AATCCAAAGATGATGGAGGAGTGCCCGCTCATGATGTGAAGTACCTGCTCAGCTGGAAAC [CG] AATTTGA
GATAAATTCAAGGGTCTATGTGGACAAGACTGCTAGTGTCTCTCTCTGGATTG
## 2
TATCTCTGTCTGGCGAGGAGGCAACGCACAACCTGTGGTGGTTTTTGGAGTGGGTGGACCC [CG] GCCAAGA
CGGCCTGGGCTGACCAGAGACGGGAGGCAGAAAAAGTGGGCAGGTGGTTGCAG
## 3
TCAACAAATGAGAGACATTGAAGAACTAATTCACACTACTATTTGGTTACTTTATTTTCCAT [CG] AAGAAAA
CCTCTTTTTTAAAACTAACACATAAATAAAATGAACGAAGAACAACTAAACG
## 4
CTCCTGTACTTGTTTCATTAAATAATGATTCCTTGATATACCAAGTCTGGATAGCGGATT [CG] ATGGAAG
CATTTTTGTAAATATACGTTTCAGTATTTTGTGTGGAAGAACACAATCTAGCTG
## 5
AGGTGAATGAAGAGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACTCT [CG] CGCTGCC
ACAAAAAGCCAATTAACGAAGCAAACTGTTTTATCTTTTCCTCTCCCCAGAAG
## 6
CCATTGGCCCGCCCCAGTTGGCCGCAGGGACTGAGCAAGTTATGCGGTCGGAAGACGTG [CG] TTAAAGG
GCTGAAGGGGAGGGACGGAAGTACAGTCTCTGTGACAGCTCTGAGGTGGGAG
##   Genome_Build CHR   MAPINFO
SourceSeq
## 1           37   Y   8553009
AGACACTAGCAGTCTTGTCACATAGACCCTTGAATTTATCTCAAATTTCG
## 2           37   Y   9363356
CGGGGTCCACCCACTCCAAAAACCACCACAGTTGTGCGTTGCCTCCTCGC

```

3 37 Y 25314171
CGATGGAAAATAAAGTAACCAAATAGTAGTGAATTAGTTCTTCAATGTCT

4 37 Y 22741795
CGATGGAAGCATTTTTGTAAATATACGTTTCAGTATTTTGTGTGGAAGAAC

5 37 Y 21664296
AGACTAATGGGAGTGGCTTGCAAGCCAGGTACTGTGCCCTACAACCTCTCG

6 37 Y 21239348
CGCACGTCTTCCCGACCGCATAACTTGCTCAGTCCCTGCGGCCAACTGGG

Chromosome_36 Coordinate_36 Strand Probe_SNPs Probe_SNPs_10
Random_Loci

1 Y 8613009 F

NA

2 Y 9973356 R

NA

3 Y 23723559 R

NA

4 Y 21151183 F rs9341313 rs13447379

NA

5 Y 20123684 R

NA

6 Y 19698736 R

NA

Methyl27_Loci UCSC_RefGene_Name UCSC_RefGene_Accession

1 NA TTTY18 NR_001550

2 NA TSPY4;FAM197Y2 NM_001164471;NR_001553

3 NA DAZ1;DAZ4;DAZ4 NM_004081;NM_020420;NM_001005375

4 NA EIF1AY NM_004681

5 NA BCORL2 NR_002923

6 NA TTTY14 NR_001543

UCSC_RefGene_Group UCSC_CpG_Islands_Name

Relation_to_UCSC_CpG_Island Phantom

1 TSS1500

2 Body;TSS1500 chrY:9363680-9363943

N_Shore

3 Body;Body;Body

4 Body chrY:22737825-22738052

S_Shelf

5 Body chrY:21664481-21665063

N_Shore

6 TSS200 chrY:21238448-21240005

Island

DMR Enhancer HMM_Island Regulatory_Feature_Name

1 NA

2 NA Y:9973136-9976273

3 NA

4 NA

```
## 5          NA
## 6          NA Y:19697854-19699393
##  Regulatory_Feature_Group DHS
## 1          NA
## 2          NA
## 3          NA
## 4          NA
## 5          NA
## 6          NA
```

```
head(final_ttest_reduced)
```

```
##          X9344737127_R01C02 X9344737127_R03C02 X9344737127_R06C01
## cg20794042          0.07829501          0.08373973          0.1201370
## cg04340719          0.05252592          0.05310158          0.0652381
## cg12358527          0.87801191          0.87981341          0.8652472
## cg11185549          0.13628803          0.15247124          0.2752861
## cg20218135          0.26227644          0.21476548          0.8359332
## cg07324172          0.68868893          0.66228627          0.4732796
##          X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01
## cg20794042          0.11725255          0.08247280          0.07970218
## cg04340719          0.06359918          0.05204832          0.05412298
## cg12358527          0.86412648          0.87775107          0.87901299
## cg11185549          0.25399951          0.12302005          0.12177307
## cg20218135          0.84482192          0.29236695          0.23632145
## cg07324172          0.47617591          0.64406619          0.66539160
##          X9376538140_R01C01 X9376538140_R04C01 pValues_ttest
## cg20794042          0.11719196          0.12175898 5.442489e-07
## cg04340719          0.06423333          0.06396302 1.778325e-06
## cg12358527          0.86708185          0.86441144 7.820567e-06
## cg11185549          0.24949587          0.25091325 1.421267e-05
## cg20218135          0.82382573          0.84587250 1.428266e-05
## cg07324172          0.48627249          0.51804417 1.500840e-05
##          corrected_pValues_BH corrected_pValues_Bonf
## cg20794042          0.2642394          0.2642394
## cg04340719          0.4316990          0.8633979
## cg12358527          0.6568315          1.0000000
## cg11185549          0.6568315          1.0000000
## cg20218135          0.6568315          1.0000000
## cg07324172          0.6568315          1.0000000
```

We want to merge on the basis of the CpG probes, but unfortunately in the `final_ttest_corrected` object the CpG probes are stored in the rownames, not in a column. We can overcome this issue as follows:


```
final_ttest_reduced <-
data.frame(rownames(final_ttest_reduced),final_ttest_reduced)
```

```
head(final_ttest_reduced)
```

```
##                rownames.final_ttest_reduced. X9344737127_R01C02
X9344737127_R03C02

## cg20794042                cg20794042                0.07829501
0.08373973
## cg04340719                cg04340719                0.05252592
0.05310158
## cg12358527                cg12358527                0.87801191
0.87981341
## cg11185549                cg11185549                0.13628803
0.15247124
## cg20218135                cg20218135                0.26227644
0.21476548
## cg07324172                cg07324172                0.68868893
0.66228627
##                X9344737127_R06C01 X9344737127_R02C02 X9376538140_R02C01
## cg20794042                0.1201370                0.11725255                0.08247280
## cg04340719                0.0652381                0.06359918                0.05204832
## cg12358527                0.8652472                0.86412648                0.87775107
## cg11185549                0.2752861                0.25399951                0.12302005
## cg20218135                0.8359332                0.84482192                0.29236695
## cg07324172                0.4732796                0.47617591                0.64406619
##                X9376538140_R05C01 X9376538140_R01C01 X9376538140_R04C01
## cg20794042                0.07970218                0.11719196                0.12175898
## cg04340719                0.05412298                0.06423333                0.06396302
## cg12358527                0.87901299                0.86708185                0.86441144
## cg11185549                0.12177307                0.24949587                0.25091325
## cg20218135                0.23632145                0.82382573                0.84587250
## cg07324172                0.66539160                0.48627249                0.51804417
##                pValues_ttest corrected_pValues_BH corrected_pValues_Bonf
## cg20794042 5.442489e-07                0.2642394                0.2642394
## cg04340719 1.778325e-06                0.4316990                0.8633979
## cg12358527 7.820567e-06                0.6568315                1.0000000
## cg11185549 1.421267e-05                0.6568315                1.0000000
## cg20218135 1.428266e-05                0.6568315                1.0000000
## cg07324172 1.500840e-05                0.6568315                1.0000000
```

```
colnames(final_ttest_reduced)
```

```
## [1] "rownames.final_ttest_reduced." "X9344737127_R01C02"
```

```
## [3] "X9344737127_R03C02"                "X9344737127_R06C01"
```

```
## [5] "X9344737127_R02C02" "X9376538140_R02C01"
## [7] "X9376538140_R05C01" "X9376538140_R01C01"
## [9] "X9376538140_R04C01" "pValues_ttest"
## [11] "corrected_pValues_BH" "corrected_pValues_Bonf"
```

```
colnames(final_ttest_reduced)[1] <- "IlmnID"
```

```
colnames(final_ttest_reduced)
```

```
## [1] "IlmnID" "X9344737127_R01C02"
"X9344737127_R03C02"
```

```
## [4] "X9344737127_R06C01" "X9344737127_R02C02"
"X9376538140_R02C01"
## [7] "X9376538140_R05C01" "X9376538140_R01C01"
"X9376538140_R04C01"
## [10] "pValues_ttest" "corrected_pValues_BH"
"corrected_pValues_Bonf"
```

```
final_ttest_reduced_annotated <- merge(final_ttest_reduced,
Illumina450Manifest_clean,by="IlmnID")
```

```
dim(final_ttest_reduced)
```

```
## [1] 48539 12
```

```
dim(final_ttest_reduced_annotated)
```

```
## [1] 48539 44
```

```
str(final_ttest_reduced_annotated)
```

```
## 'data.frame': 48539 obs. of 44 variables:
```

```
## $ IlmnID : chr "cg00000165" "cg00000714"
"cg00001249" "cg00001593" ...
## $ X9344737127_R01C02 : num 0.269 0.174 0.559 0.791 0.844
...
## $ X9344737127_R03C02 : num 0.252 0.145 0.605 0.807 0.833
...
## $ X9344737127_R06C01 : num 0.2 0.201 0.709 0.845 0.792 ...
## $ X9344737127_R02C02 : num 0.219 0.188 0.594 0.829 0.841
...
## $ X9376538140_R02C01 : num 0.305 0.145 0.542 0.784 0.863
...
## $ X9376538140_R05C01 : num 0.266 0.128 0.492 0.783 0.845
...
## $ X9376538140_R01C01 : num 0.203 0.187 0.64 0.815 0.803
...
```

```

## $ X9376538140_R04C01      : num  0.219 0.214 0.713 0.839 0.827
...
## $ pValues_ttest            : num  0.00607 0.00689 0.0225 0.00331
0.06466 ...
## $ corrected_pValues_BH     : num  0.657 0.657 0.657 0.657 0.657
...
## $ corrected_pValues_Bonf    : num  1 1 1 1 1 1 1 1 1 1 ...
## $ Name                     : Factor w/ 485512 levels
"cg00000029","cg00000108",...: 4 12 24 34 67 68 88 89 93 94 ...
## $ AddressA_ID              : Factor w/ 485512 levels
"10600322","10600328",...: 16431 164675 30331 298508 349158 185801 39687
183067 91531 225153 ...
## $ AlleleA_ProbeSeq         : Factor w/ 485512 levels
"AAAAAAAAAAAAAAAAAACACAAAAAATCCCCACTCTAAAACCCACACA",...: 246632 396253
240100 229712 16540 307494 81897 150397 400732 226832 ...
## $ AddressB_ID              : Factor w/ 135477 levels
"", "10600313",...: 1 1 1 1 1 75826 92263 1 1 1 ...
## $ AlleleB_ProbeSeq         : Factor w/ 135477 levels
"", "AAAAAAAAAAAAAAAAAACCGCGAAAAAATCCCCGCTCTAAAACCCCGCG",...: 1 1 1 1 1
82521 24769 1 1 1 ...
## $ Infinium_Design_Type     : Factor w/ 2 levels "I","II": 2 2 2 2
2 1 1 2 2 2 ...
## $ Next_Base                : Factor w/ 4 levels "", "A", "C", "T": 1
1 1 1 1 2 2 1 1 1 ...
## $ Color_Channel            : Factor w/ 3 levels "", "Grn", "Red": 1
1 1 1 1 3 3 1 1 1 ...
## $ Forward_Sequence         : Factor w/ 485512 levels
"AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAATATGTCTGCTAAATAATCACAGAGTGAAA[CG]AAGAGG
AAGAAACTAAAGTACAGTTTGGAACATAGTACAGTGCCCCCTGGGTACTGAATCA",...: 208983
58295 482218 359611 449191 149903 67942 317126 354147 99212 ...
## $ Genome_Build             : int   37 37 37 37 37 37 37 37 37 37
...
## $ CHR                      : Factor w/ 24 levels
"1","10","11",...: 1 11 6 1 1 9 11 1 20 8 ...
## $ MAPINFO                  : int   91194674 54695678 60389786
170490434 223568700 76482353 17336525 45080600 15725862 66586745 ...
## $ SourceSeq                : Factor w/ 485343 levels
"AAAAAAAAAACCAACCCCAACAAAAACCCCTGTGCCAGAAATTGCATTCG",...: 39242 214315
8707 195071 168600 329409 364822 107635 412235 476709 ...
## $ Chromosome_36            : Factor w/ 25 levels
"1","10","11",...: 1 11 6 1 1 9 11 1 20 8 ...
## $ Coordinate_36           : Factor w/ 484738 levels
"100000057","100000691",...: 463603 367595 367839 143073 192679 418499
147676 311039 127896 387831 ...
## $ Strand                   : Factor w/ 2 levels "F","R": 2 1 1 2
1 2 2 2 2 1 ...

```

```
## $ Probe_SNPs : Factor w/ 56004 levels
"", "rs10000615", ...: 1 1 1 1 1 1 1 1 1 11925 ...
## $ Probe_SNPs_10 : Factor w/ 35790 levels
"", "rs10000804", ...: 1 1 1 1 1 1 19705 1 1 1 ...
## $ Random_Loci : logi NA NA NA NA NA NA ...
## $ Methyl27_Loci : logi NA NA NA NA NA NA ...
## $ UCSC_RefGene_Name : Factor w/ 42456 levels
"", "A1BG", "A1CF;A1CF;A1CF", ...: 1 38792 1 1 4696 10445 25904 31453 22177
7918 ...
## $ UCSC_RefGene_Accession : Factor w/ 52670 levels
"", "NM_000014", ...: 1 35023 1 1 42569 44168 35349 31035 22900 45605 ...
## $ UCSC_RefGene_Group : Factor w/ 4165 levels
"", "1stExon", "1stExon;1stExon", ...: 1 2348 1 1 746 2035 2970 2035 2 2521
...
## $ UCSC_CpG_Islands_Name : Factor w/ 27177 levels
"", "chr1:10003165-10003585", ...: 2347 13922 7236 1 1197 1 12346 1965 1
9176 ...
## $ Relation_to_UCSC_CpG_Island: Factor w/ 6 levels
"", "Island", "N_Shelf", ...: 6 6 5 1 6 1 4 3 1 6 ...
## $ Phantom : Factor w/ 11912 levels "", "high-
CpG:100009860-100010119", ...: 1 1 1 1 1 1 1 1 1 1 ...
## $ DMR : Factor w/ 4 levels
"", "CDMR", "DMR", ...: 2 1 1 1 1 1 1 2 1 ...
## $ Enhancer : logi TRUE NA NA TRUE NA NA ...
## $ HMM_Island : Factor w/ 63086 levels
"", "1:100088009-100088708", ...: 5024 1 1 1 1 27567 1 1 53857 1 ...
## $ Regulatory_Feature_Name : Factor w/ 47969 levels
"", "1:10001853-10003978", ...: 1 1 1 1 1 1 1 1 1 18615 ...
## $ Regulatory_Feature_Group : Factor w/ 9 levels
"", "Gene_Associated", ...: 1 1 1 1 1 1 1 1 1 8 ...
## $ DHS : logi NA NA NA NA NA NA ...
```

Note that the dataframe is automatically reordered on the basis of alphabetical order of the column used for merging

```
head(final_ttest_reduced_annotated)
```

```
##      IlmnID X9344737127_R01C02 X9344737127_R03C02
X9344737127_R06C01
```

```
## 1 cg00000165      0.2693781      0.2523965
0.2002441
## 2 cg00000714      0.1738167      0.1448749
0.2005353
## 3 cg00001249      0.5586365      0.6054516
0.7086758
```

## 4	cg00001593	0.7906507	0.8066051
	0.8445328		
## 5	cg00002808	0.8443084	0.8328302
	0.7924436		
## 6	cg00002809	0.7708458	0.7646070
	0.7982595		
##	X9344737127_R02C02 X9376538140_R02C01 X9376538140_R05C01 X9376538140_R01C01		
## 1	0.2190469	0.3047645	0.2656781
	0.2026381		
## 2	0.1878646	0.1448605	0.1278426
	0.1871288		
## 3	0.5942167	0.5415010	0.4919873
	0.6395910		
## 4	0.8285542	0.7835228	0.7833327
	0.8149185		
## 5	0.8412397	0.8634898	0.8452696
	0.8031069		
## 6	0.7805688	0.7594264	0.7696202
	0.7753781		
##	X9376538140_R04C01 pValues_ttest corrected_pValues_BH corrected_pValues_Bonf		
## 1	0.2194006	0.006065725	0.6568315
	1		
## 2	0.2137097	0.006890308	0.6568315
	1		
## 3	0.7134948	0.022499353	0.6568315
	1		
## 4	0.8392078	0.003306490	0.6568315
	1		
## 5	0.8268710	0.064660966	0.6568315
	1		
## 6	0.7928560	0.021671455	0.6568315
	1		
##	Name AddressA_ID		
	AlleleA_ProbeSeq		
## 1	cg00000165 12637463		
	CAAAATCTATTAATACAATAACTTTTAATAAAACAACATAAAACACACATC		
## 2	cg00000714 31749374		
	TAACCCCTAAACCTATTCTAATTTCTACTTCTTAACAACCTAACCCCTCC		
## 3	cg00001249 14610435		
	CAAAAAATAAATATCCTATTAACCAACTTATAACTACTCTCCTACTCTTC		
## 4	cg00001593 49715486		
	ATTACCTAAATACTAAATATTCACCTCAATCTAAATATACTATCTACAAC		
## 5	cg00002808 56677312		
	AAAAAATCACATTACCACATCTCCTTCTAACTACRCACCTATAAACCAAAC		

```

## 6 cg00002809      34718366
CCTAATCCTAAACATAACACCACCAAAAACCCAAAAACAAATAAACACA
##   AddressB_ID                                     AlleleB_ProbeSeq
## 1
## 2
## 3
## 4
## 5
## 6      46624378 CCTAATCCTAAACATAACGCCACCGAAAACCCAAAAACAAATAAACACG
##   Infinium_Design_Type Next_Base Color_Channel
## 1                                     II
## 2                                     II
## 3                                     II
## 4                                     II
## 5                                     II
## 6                                     I           A           Red
##
Forward_Sequence
## 1
CTAAGTGCAGTCAGGATCTGTTAGTACAGTGGCTTTTGATGGAACAGCTGAGGCACACAT [CG] CCCGTGG
CATGGACTCCGGGGCCGAACGCTCACGACCAAGACTTTTGCCCTTTTGAAATG
## 2
AGAGCGCCTTGGCAGCTGAGGCCCCGGGAGACCCGTCGTCAGGAGCTCCTGGAGAAGATTA [CG] GAGGGCC
AGGCTGCTAAGAAGCAGAACTAGAACAGGCTTCAGGGGCCAGCTCAAGCCAG
## 3
TTTGTGGGGGAAGTAGAGATGACCTTTAGCCTCTTAAATGGGCCAGCCCTGAAAGAGCTT [CG] AAGAGCA
GGAGAGTAGTTACAAGCTGGCCAACAGGACATTCACCTCTTGATGAATTCAGT
## 4
GTCAAGACTCAATTGCCTGGGTGCTGGGTATTCACTTCAGTCTGAATATGCTGTCTACAG [CG] TATGTTT
GCCAGTCTAGTCTAGTCTCCCTTAGGCTCCTCTAGCTGGCCACCAGCTTGAT
## 5
TGGTACTAGAGGCCAGCTCCGTGCCTGTGAGCAGAACAGGGGTTACTGAGAACCAAGGA [CG] CCTGGCT
TACAGTGCAGCAGCCAGAAGGAGATGTGGCAATGTGATTCTTTTGTAATCTGA
## 6
CCAGGTGCATCACCTGGTCCTGGAACATGGCGCCACCGAAGGCCCAGAAGCAGGTGAACA [CG] AAGTACA
GCTCGTACAGCTCCCTGGGGGAGTCGGGGGGCACGGTCTTCTCCGTGAGCAGG
##   Genome_Build CHR   MAPINFO
SourceSeq
## 1           37   1   91194674
AGGATCTGTTAGTACAGTGGCTTTTGATGGAACAGCTGAGGCACACATCG
## 2           37  19   54695678
CGGAGGGCCAGGCTGCTAAGAAGCAGAACTAGAACAGGCTTCAGGGGCC
## 3           37  14   60389786
AAGAGGTGAATGTCTGTGGCCAGCTTGTAATACTCTCCTGCTCTTCG
## 4           37   1  170490434
CGCTGTAGACAGCATATTCAGACTGAAGTGAATACCCAGCACCCAGGCAA

```

```

## 5          37    1 223568700
CGCCTGGCTTACAGTGCAGCCAGAAGGAGATGTGGCAATGTGATTCCT
## 6          37   17  76482353
CGTGTTCACCTGCTTCTGGGCCTTCGGTGGCGCCATGTTCCAGGACCAGG
##   Chromosome_36 Coordinate_36 Strand Probe_SNPs Probe_SNPs_10
Random_Loci
## 1          1          90967262      R
NA
## 2          19          59387490      F
NA
## 3          14          59459539      F
NA
## 4          1          168757058      R
NA
## 5          1          221635323      F
NA
## 6          17          73993948      R
NA
##   Methyl27_Loci UCSC_RefGene_Name UCSC_RefGene_Accession
UCSC_RefGene_Group
## 1          NA
## 2          NA      TSEN34;TSEN34 NM_024075;NM_001077446
Body;Body
## 3          NA
## 4          NA
## 5          NA      Clorf65;Clorf65      NM_152610;NM_152610
3'UTR;1stExon
## 6          NA          DNAH17          NM_173628
Body
##   UCSC_CpG_Islands_Name Relation_to_UCSC_CpG_Island Phantom   DMR
Enhancer
## 1   chr1:91190489-91192804          S_Shore      CDMR
TRUE
## 2   chr19:54693034-54695497          S_Shore
NA
## 3   chr14:60386271-60386750          S_Shelf
NA
## 4
TRUE
## 5   chr1:223566642-223567268          S_Shore
NA
## 6
NA
##   HMM_Island Regulatory_Feature_Name
Regulatory_Feature_Group DHS

```

```
## 1 1:90967262-90967361
NA
## 2
NA
## 3
NA
## 4
NA
## 5
NA
## 6 17:73993880-73994100
NA
```

Now we can create the input for the Manhattan plot analysis. The input object should contain 4 info: probe, chromosome, position on the chromosome and p-value. We will select these columns in the `final_ttest_corrected_annotated` object

```
input_Manhattan <-
final_ttest_reduced_annotated[colnames(final_ttest_reduced_annotated)
%in% c("IlmnID", "CHR", "MAPINFO", "pValues_ttest")]
```

```
dim(input_Manhattan)
```

```
## [1] 48539      4
```

```
head(input_Manhattan)
```

##	IlmnID	pValues_ttest	CHR	MAPINFO
## 1	cg00000165	0.006065725	1	91194674
## 2	cg00000714	0.006890308	19	54695678
## 3	cg00001249	0.022499353	14	60389786
## 4	cg00001593	0.003306490	1	170490434
## 5	cg00002808	0.064660966	1	223568700
## 6	cg00002809	0.021671455	17	76482353

```
str(input_Manhattan)
```

```
## 'data.frame': 48539 obs. of 4 variables:
```

```
## $ IlmnID      : chr  "cg00000165" "cg00000714" "cg00001249"
"cg00001593" ...
## $ pValues_ttest: num  0.00607 0.00689 0.0225 0.00331 0.06466 ...
## $ CHR         : Factor w/ 24 levels "1","10","11",...: 1 11 6 1 1 9
11 1 20 8 ...
## $ MAPINFO      : int   91194674 54695678 60389786 170490434
223568700 76482353 17336525 45080600 15725862 66586745 ...
```

```
levels(input_Manhattan$CHR)
```



```
## [1] "1" "10" "11" "12" "13" "14" "15" "16" "17" "18" "19" "2"
      "20" "21" "22"

## [16] "3" "4" "5" "6" "7" "8" "9" "X" "Y"
```

It is better to reorder the levels of the column CHR

```
order_chr <-
c("1","2","3","4","5","6","7","8","9","10","11","12","13","14","15","16",
  "17","18","19","20","21","22","X","Y")
```

```
input_Manhattan$CHR <- factor(input_Manhattan$CHR,levels=order_chr )
levels(input_Manhattan$CHR)
```

```
## [1] "1" "2" "3" "4" "5" "6" "7" "8" "9" "10" "11" "12"
      "13" "14" "15"

## [16] "16" "17" "18" "19" "20" "21" "22" "X" "Y"
```

The function that we will use is “manhattan”

```
?manhattan
```

As you see, the column “CHR” should be numeric → we will convert factors to numbers

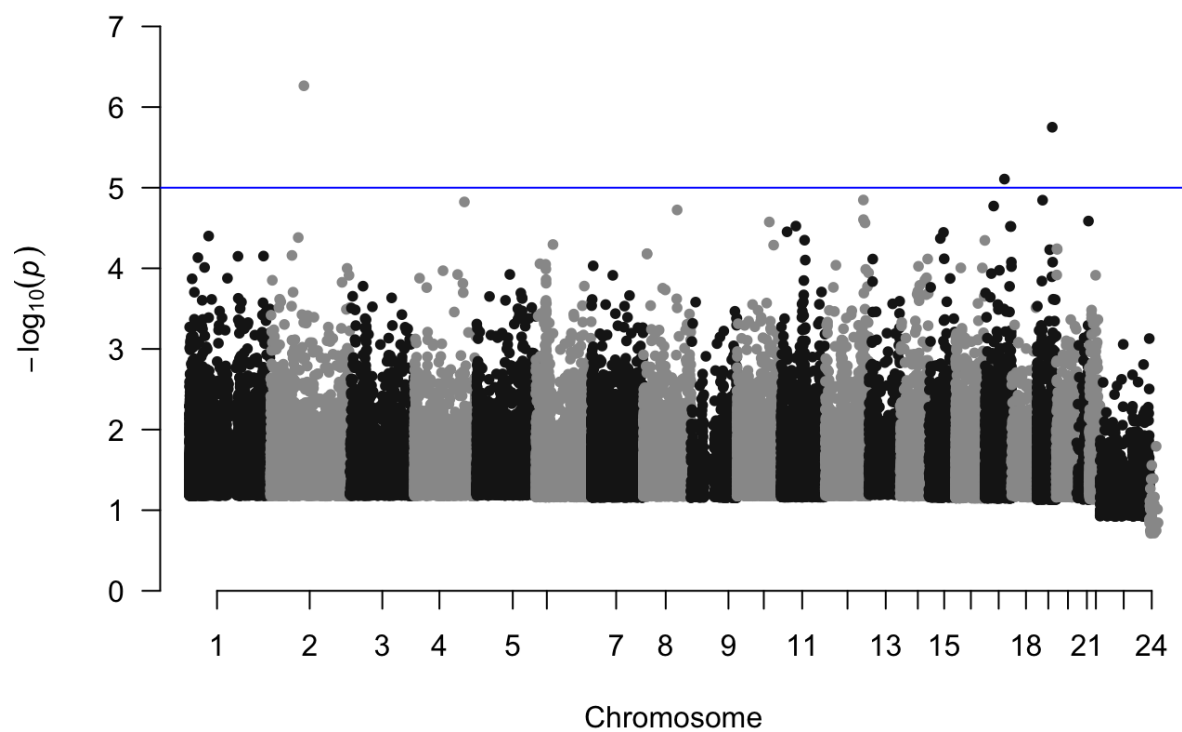
```
input_Manhattan$CHR <- as.numeric(input_Manhattan$CHR)
table(input_Manhattan$CHR)
```

```
##

##  1   2   3   4   5   6   7   8   9  10  11  12  13
14  15  16
## 4685 3481 2515 2046 2432 3661 3001 2095  986 2438 2879 2453 1228
1507 1525 2196
##  17  18  19  20  21  22  23  24
## 2787  592 2552 1037  424  855 1123  41
```

and finally we can produce our Manhattan plot

```
manhattan(input_Manhattan, snp="IlmnID",chr="CHR", bp="MAPINFO",
p="pValues_ttest" )
```

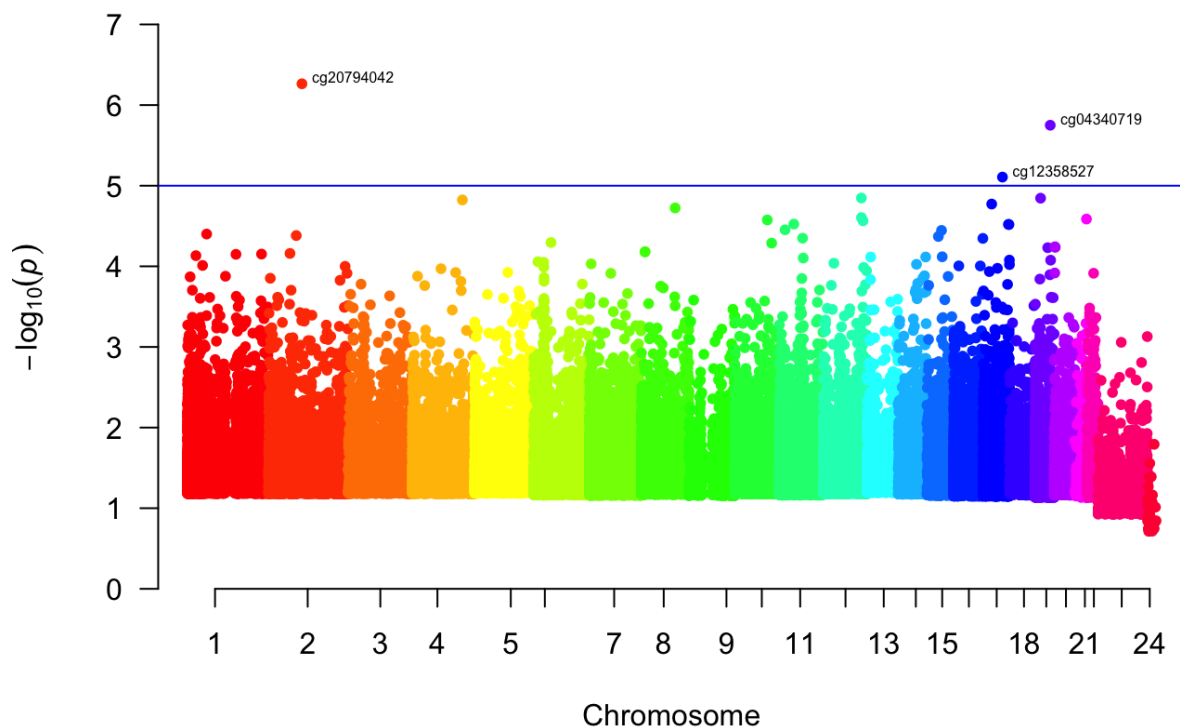


what is the blue line?

```
-log10(0.00001)
```

```
## [1] 5
```

```
manhattan(input_Manhattan, snp="IlmnID", chr="CHR", bp="MAPINFO",  
p="pValues_ttest", annotatePval = 0.00001, col=rainbow(24) )
```



CHUNK 4: heatmaps with hierarchical clustering

We need the gplots package from CRAN repository

```
# install.packages("gplots")
```

```
library(gplots)
```

```
## Warning: package 'gplots' was built under R version 4.4.1
```

```
##
```

```
## Attaching package: 'gplots'
```

```
## The following object is masked from 'package:stats':
```

```
##
```

```
##      lowess
```

We will use the function heatmap.2.

```
?heatmap.2
```

It wants as input a matrix. As an heatmap on several thousands of probes is computationally demanding, for our analysis we will use just the top 100 most significant CpG probes. We will extract only beta values for these top 100 probes and convert them in a matrix:

```
input_heatmap=as.matrix(final_ttest_first20k[1:100,1:8])
```

The default method for distance is “euclidean” (click on “dist” in distfun paragraph), while the default function for hierarchical clustering is “complete” (click on “hclust” in hclustfun paragraph)

Finally, we we create a bar of colors for Group A or Group B membership

```
pheno$Group
```

```
## [1] A A B B A A B B
```

```
## Levels: A B
```

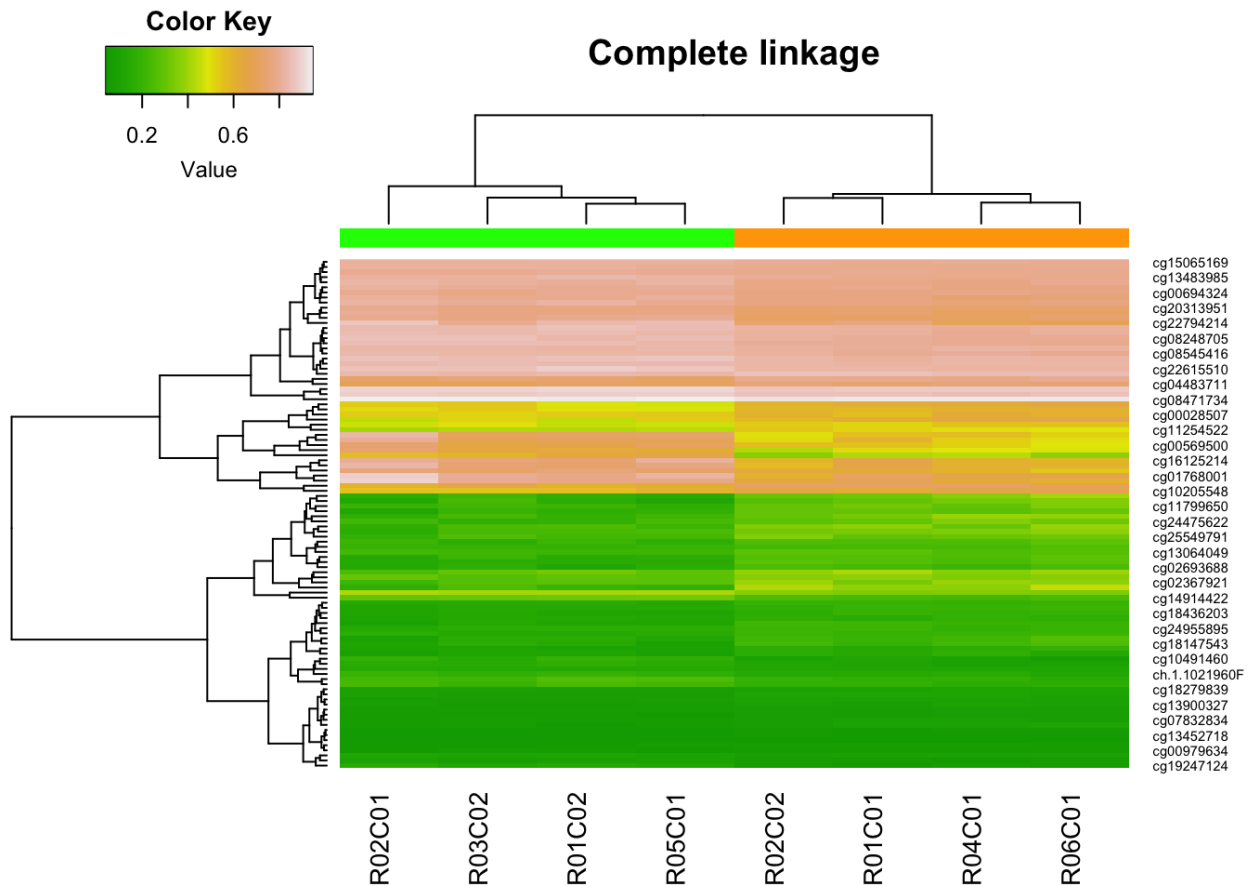
```
colorbar <-
```

```
c("green", "green", "orange", "orange", "green", "green", "orange", "orange")
```

In the following lines we will compare the results of hierchical clustering using different methods.

Complete (default options)

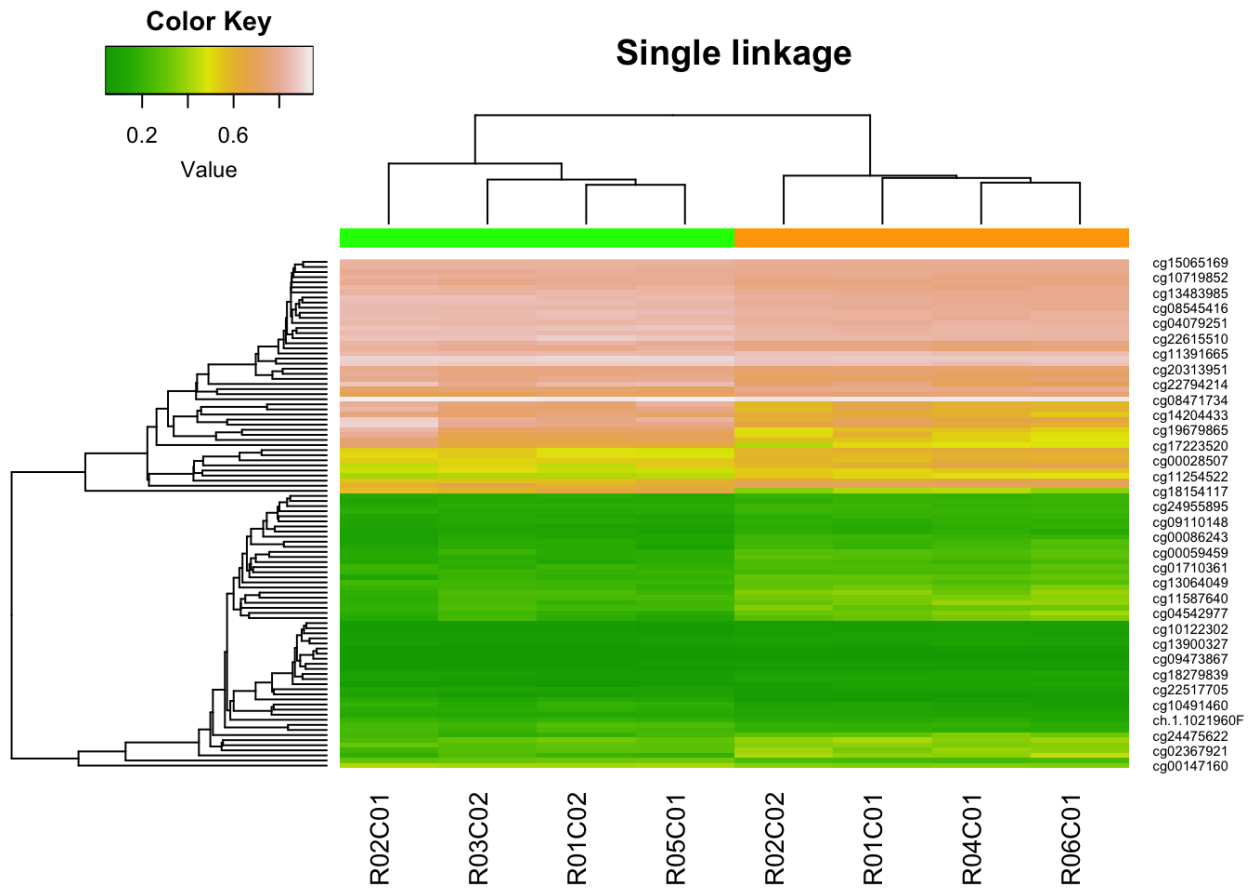
```
heatmap.2(input_heatmap,col=terrain.colors(100),Rowv=T,Colv=T,dendrogram="both",key=T,ColSideColors=colorbar,density.info="none",trace="none",scale="none",symm=F,main="Complete linkage")
```



You can see that hierarchical clustering divides well group A and group B samples; in addition, you see that some probes are hypermethylated in Group A compared to group B, others are hypomethylated.

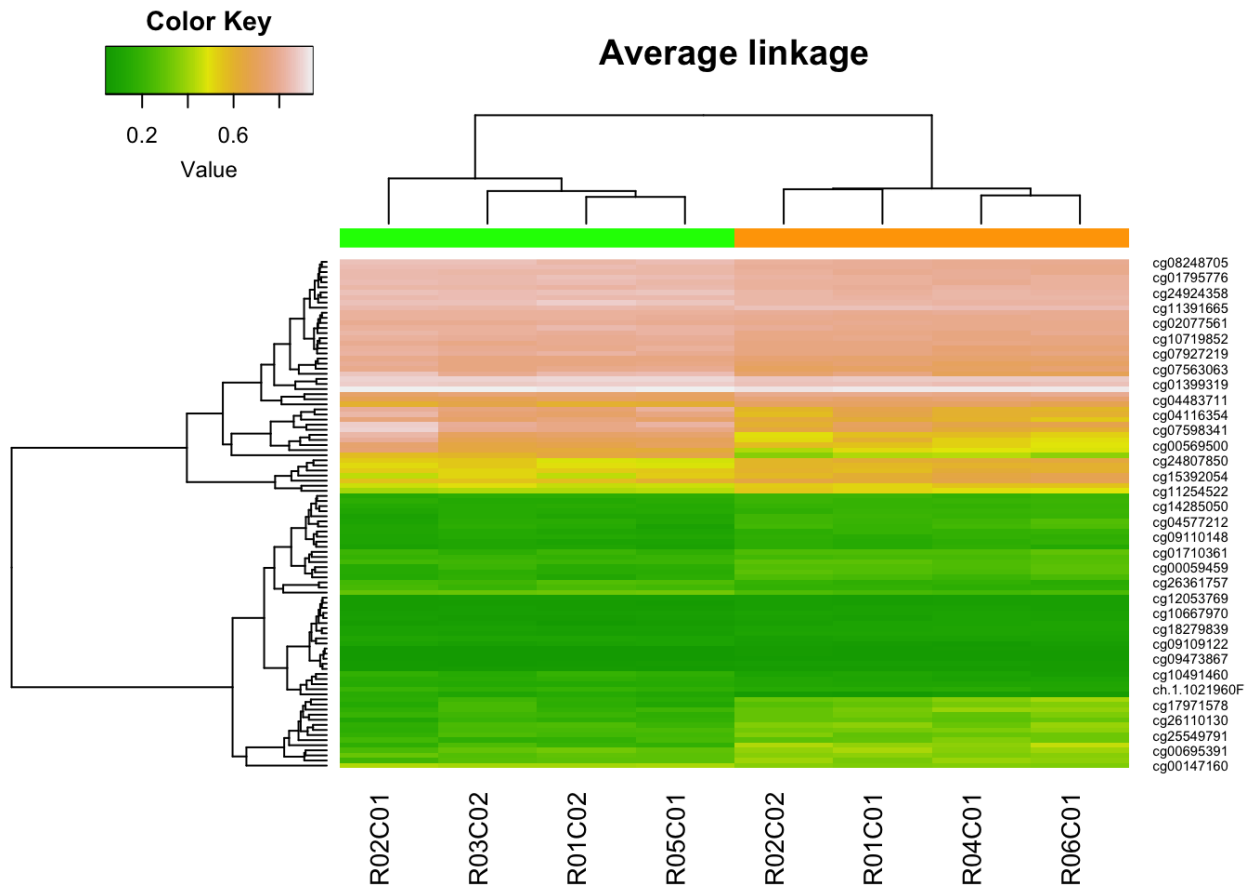
Single

```
heatmap.2(input_heatmap,col=terrain.colors(100),Rowv=T,Colv=T,hclustfun
= function(x) hclust(x,method =
'single'),dendrogram="both",key=T,ColSideColors=colorbar,density.info="
none",trace="none",scale="none",symm=F,main="Single linkage")
```



Average

```
heatmap.2(input_heatmap,col=terrain.colors(100),Rowv=T,Colv=T,hclustfun
= function(x) hclust(x,method =
'average'),dendrogram="both",key=T,ColSideColors=colorbar,density.info=
"none",trace="none",scale="none",symm=F,main="Average linkage")
```



You can set your palette of colors

```
col2=colorRampPalette(c("green", "black", "red"))(100)
```

```
heatmap.2(input_heatmap,col=col2,Rowv=T,Colv=T,dendrogram="both",key=T,ColSideColors=
colorbar,density.info="none",trace="none",scale="none",symm=F)
```

